

CAREERS360
PREPARATION Series

NEET 2026

**Answer Key with
Solutions PDF
Code-Wise Answers
for All Sets**

All Codes NEET 2026 Answer Key

<u>Code 13</u>		<u>Code 11</u>		<u>Code 12</u>		<u>Code 14</u>	
Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
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NEET 2026 Code 11 Question Paper with Answer Key & Solutions PDF

Physics

- Q. 1** The speed of light in vacuum is taken as unity. If light takes 6 min 40 s to reach the Earth from the Sun, the distance between the Sun and the Earth in new unit is :

Option 1:

$3. \times 10^8$

Option 2:

500

Option 3:

3×10^{10}

Option 4:

400

Correct Answer:

400

Solution:

Distance = speed $\times$ time

$$d = 1 \times 400 = 400 \text{ units}$$

Hence, the correct option is (4).

Q. 2 Match **List I** with **List II** :

List I		List II	
A.	Young's Modulus	I.	$\frac{\Delta d}{\Delta L} \left(\frac{L}{d} \right)$
B.	Compressibility	II.	$\frac{FL}{A(\Delta L)}$
C.	Bulk Modulus	III.	$-\frac{1}{\Delta P} \left(\frac{\Delta V}{V} \right)$
D.	Poisson's Ratio	IV.	$-P \left(\frac{V}{\Delta V} \right)$

Choose the **correct** answer from the options given below:

Option 1:

A-III, B-II, C-I, D-IV

Option 2:

A-II, B-III, C-IV, D-I

Option 3:

A-I, B-IV, C-III, D-II

Option 4:

A-IV, B-I, C-II, D-III

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

A. Young's Modulus $Y = \frac{FL}{A(\Delta L)}$

B. Compressibility $\text{Compressibility} = -\frac{1}{\Delta P} \left(\frac{\Delta V}{V} \right)$

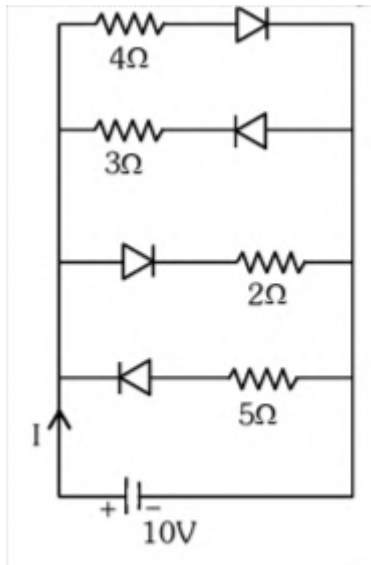
C. Bulk Modulus $K = -P \left(\frac{V}{\Delta V} \right)$

D. Poisson's Ratio

Poisson's ratio $= \frac{\Delta d}{\Delta L} \left(\frac{L}{d} \right)$

Hence, the correct option is (2).

- Q. 3** The current I in the circuit shown below is :
(All diodes are ideal and identical)



Option 1:
 $\frac{1}{3}$ A

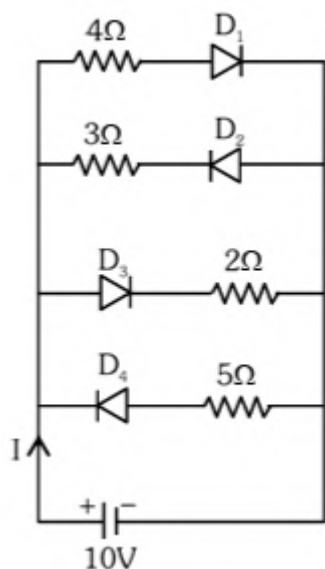
Option 2:
 $\frac{15}{2}$ A

Option 3:
 $\frac{5}{3}$ A

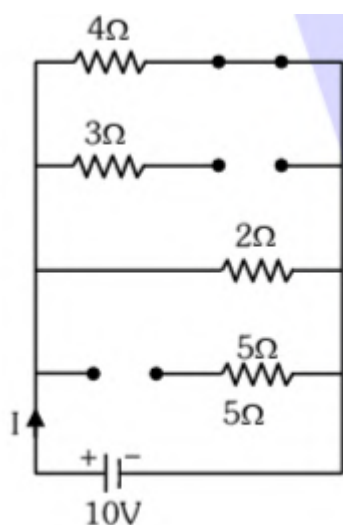
Option 4:
 $\frac{5}{9}$ A

Correct Answer:
 $\frac{15}{2}$ A

Solution:



Here D_1 and D_3 are F.B and D_2 and D_4 are R.B. then equivalent circuit



$$R_{eq} = \frac{4 \times 2}{4 + 2} = \frac{4}{3} \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{10}{4/3} = \frac{15}{2} \text{ A}$$

Hence, the correct option is (2).

- Q. 4** The angular speed of a flywheel is increased from 600 rpm to 1200 rpm in 10 s . The number of revolutions completed by the flywheel during this times is:

Option 1:

600

Option 2:

300

Option 3:

900

Option 4:

150

Correct Answer:

150

Solution:

$$\omega_1 = 600\text{rpm} = 20\pi\text{rad/sec}$$

$$\omega_f = 1200\text{rpm} = 40\pi\text{rad/sec}$$

$$t = 10\text{sec}$$

$$\theta = \left(\frac{\omega_i + \omega_f}{2} \right) t$$

$$\theta = \left(\frac{40\pi + 20\pi}{2} \right) 10$$

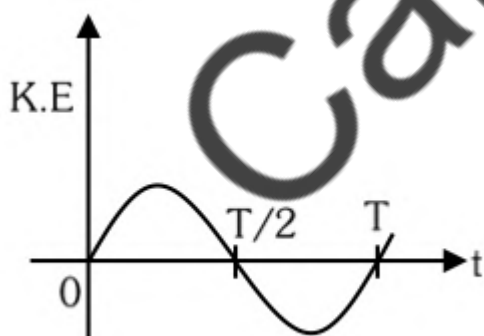
$$\theta = 300\pi$$

$$\text{number of revolution} = \frac{300\pi}{2\pi} = 150 \text{ rotation}$$

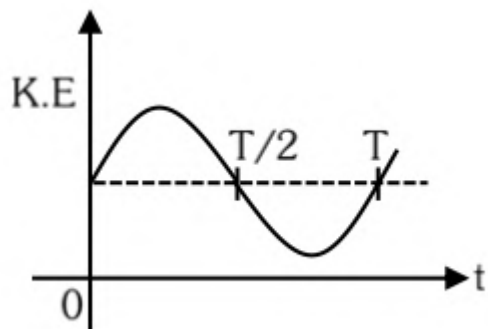
Hence, the correct option is (4).

Q. 5 For a simple pendulum, having time period T , the variation of kinetic energy (K.E.) with time (t) is represented by :

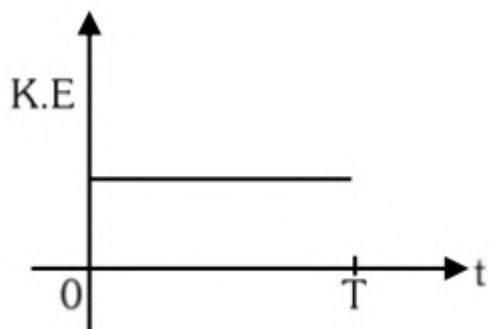
Option 1:



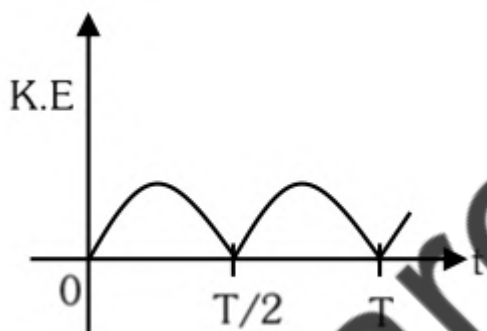
Option 2:



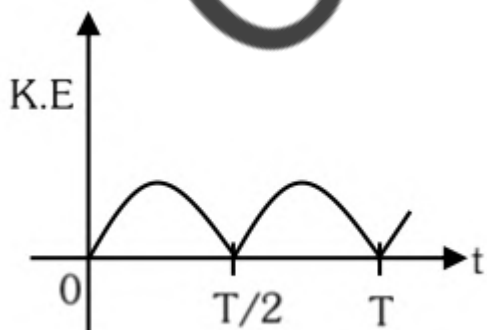
Option 3:



Option 4:



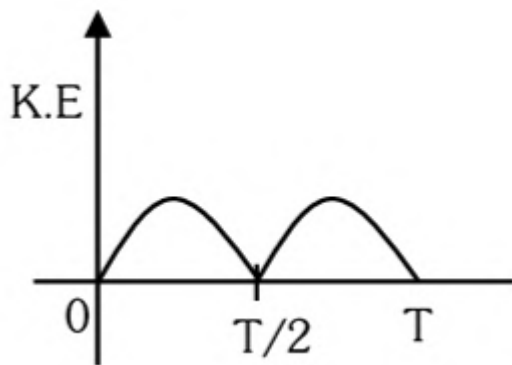
Correct Answer:



Solution:

$$K.E = \frac{1}{2} m \omega^2 A^2 \cos^2 \omega t$$

oscillation starts from extreme position



Hence, the correct answer is option (4).

Q. 6 A resistor is connected to a battery of 12 V emf and internal resistance 2Ω . If the current in the circuit is 0.6 A, the terminal voltage of the battery is :

Option 1:

12 V

Option 2:

1.2 V

Option 3:

10 V

Option 4:

10.8 V

Correct Answer:

10.8 V

Solution:

Given that:

$$\text{EMF} = 12 \text{ V}$$

$$r = 2\Omega$$

$$I = 0.6 \text{ A}$$

So terminal voltage

$$V = E - Ir$$

$$= 12 - (0.6)^2$$

$$V = 10.8 \text{ volt}$$

Hence, the correct option is (4).

- Q. 7** A flask contains argon and chlorine in the ratio of 2 : 1 by mass. The temperature of the mixture is 27°C . The ratio of root mean square speed of the molecules of the two gases $\left(\frac{v_{\text{rms}}^{\text{Ar}}}{v_{\text{rms}}^{\text{Cl}}}\right)$ is :
(Atomic mass of argon = 40.0u and molecular mass of chlorine 70.0u)

Option 1:

$$\frac{7}{4}$$

Option 2:

$$\frac{-2}{\sqrt{7}}$$

Option 3:

$$\frac{\sqrt{7}}{2}$$

Option 4:

$$\frac{7}{2}$$

Correct Answer:

$$\frac{\sqrt{7}}{2}$$

Solution:

$$\text{As } V_{\text{RMS}} = \sqrt{\frac{3RT}{M_w}}$$

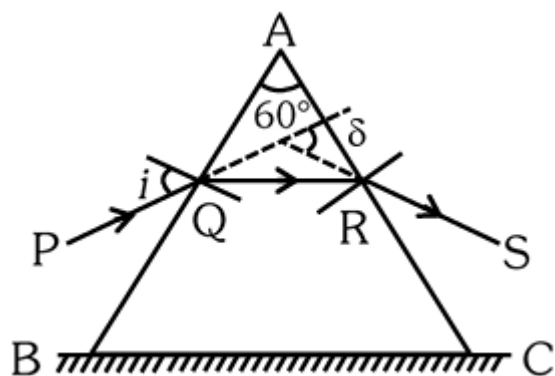
Given temperature $\rightarrow$ constant

$$\text{Then } V_{\text{RMS}} \propto \frac{1}{\sqrt{M_w}}$$

$$\Rightarrow \frac{V_{\text{RMS}}^{\text{Ar}}}{V_{\text{RMS}}^{\text{Cl}}} = \sqrt{\frac{M_w^{\text{Cl}}}{M_w^{\text{Ar}}}} = \sqrt{\frac{70}{40}} = \frac{\sqrt{7}}{2}$$

Hence, the correct option (3)

- Q. 8** A ray of monochromatic light is passing through an equilateral prism (ABC) as shown in the figure. The refracted ray (QR) is parallel to its base (BC) and the angle of incidence (i) is 50° . Then the angle of deviation (δ) is :



Option 1:
 45°

Option 2:
 55°

Option 3:
 35°

Option 4:
 40°

Correct Answer:
 40°

Solution:

For an equilateral prism, $A = 60^\circ$ since refracted ray is parallel to base so it is condition of minimum deviation.

$$r = \frac{A}{2} = \frac{60}{2} = 30^\circ$$

$$\text{and } i = e = 50^\circ$$

$$\delta = i + e - A$$

$$\begin{aligned} \text{But } \delta &= 50 + 50 - 60 \\ &= 40^\circ \end{aligned}$$

Hence, the correct option is (4).

Q. 9 Match List I with List II :

List I		List II	
A.	$E = h\nu$	I.	de Broglie wavelength
B.	Diffraction and Interference	II.	Particle nature of light
C.	$\lambda = h/p$	III.	Wave nature of light
D.	Compton effect	IV.	Energy of photon

Choose the **correct** answer from the options given below :

Option 1:

A-IV, B-III, C-II, D-I

Option 2:

A-IV, B-III, C-I, D-II

Option 3:

A-I, B-IV, C-III, D-II

Option 4:

A-IV, B-I, C-II, D-III

Correct Answer:

A-IV, B-III, C-I, D-II

Solution:

A. $E = h\nu \rightarrow$ Energy of photon $\rightarrow$ IV

B. Diffraction and Interference $\rightarrow$ Wave nature of light $\rightarrow$ III

C. $\lambda = \frac{h}{p} \rightarrow$ de Broglie wavelength $\rightarrow$ I

D. Compton effect $\rightarrow$ Particle nature of light $\rightarrow$ II

Hence, the correct option is (2).

Q. 10 In the first excited state of hydrogen atom, the energy of its electron is -3.4 eV . The radial distance of the electron from the hydrogen nucleus in this case is approximately :

(Take $1\text{eV} = 1.6 \times 10^{-19} \text{ J}$, $e = 1.6 \times 10^{-19} \text{ C}$ and $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$)

Option 1:

$2.1 \times 10^{-8} \text{ m}$

Option 2:

$$2.1 \times 10^{-10} \text{ m}$$

Option 3:

$$2.1 \times 10^{-11} \text{ m}$$

Option 4:

$$2.1 \times 10^{-9} \text{ m}$$

Correct Answer:

$$2.1 \times 10^{-10} \text{ m}$$

Solution:

$$E = \frac{kq_1q_2}{2r}$$

$$3.4 = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{2 \times r}$$

$$r = 2.1 \times 10^{-10} \text{ m}$$

Hence, the correct option is (2).

Q. 11

A box of mass 15 kg is kept on the floor of a stationary trolley. The coefficient of static friction between the box and the trolley is 0.12 . Keeping the box in stationary state over the trolley, the maximum acceleration with which the trolley can be moved horizontally in ms^{-2} is :

$$(g = 10 \text{ m/s}^2)$$

Option 1:

$$1.8$$

Option 2:

$$1.2$$

Option 3:

$$1.5$$

Option 4:

$$2.1$$

Correct Answer:

$$1.2$$

Solution:

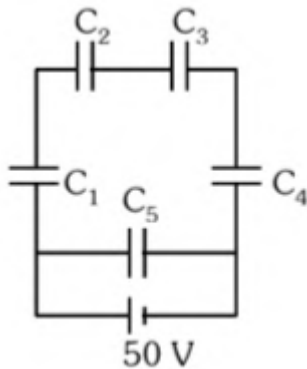
$$ma_{\max} = F_{lm}$$

$$ma_{\max} = \mu mg$$

$$a_{\max} = \mu g = 1.2 \text{ m/s}^2$$

Hence, the correct option is (2).

- Q. 12** Five capacitors of capacitances $C_1 = C_2 = C_3 = C_4 = 40\mu\text{ F}$ and $C_5 = 2 \cdot 5\mu\text{ F}$ are connected as shown, along with a battery of 50 V. The equivalent capacitance and the charges on each capacitor respectively are :



Option 1:

$4\mu\text{ F}$, $250\mu\text{C}$ on C_1 to C_4 and $125\mu\text{C}$ on C_5

Option 2:

$5\mu\text{ F}$, $250\mu\text{C}$ on all capacitors

Option 3:

$5\mu\text{ F}$, $125\mu\text{C}$ on C_1 to C_4 and $25\mu\text{C}$ on C_5

Option 4:

$5\mu\text{ F}$, $125\mu\text{C}$ on all capacitors

Correct Answer:

$5\mu\text{ F}$, $125\mu\text{C}$ on all capacitors

Solution:

Top branch: C_2 and C_3 in series

$$\frac{1}{C_{23}} = \frac{1}{10} + \frac{1}{10} \Rightarrow C_{23} = 5\mu F$$

Now C_1, C_{23}, C_4 are in series:

$$\frac{1}{C_{eq1}} = \frac{1}{10} + \frac{1}{5} + \frac{1}{10} = \frac{1+2+1}{10} = \frac{4}{10} \Rightarrow C_{eq1} = 2.5\mu F$$

Parallel with C_5

$$C_{total} = C_{eq1} + C_5 = 2.5 + 2.5 = 5\mu F$$

Total charge

$$Q = C_{total} V = 5 \times 50 = 250\mu C$$

Each branch capacitance = $2.5\mu F \rightarrow$ equal charge sharing

$$Q_{\text{each branch}} = 125\mu C$$

Hence, the correct option is (4).

Careers360

Q. 13 The amount of work done to raise a mass ' m ' from the surface of the Earth to a height equal to the radius of the Earth ' R ', will be :

Option 1:
 $mg \frac{R}{2}$

Option 2:
 mgR

Option 3:
 $mg \frac{R}{4}$

Option 4:

2 mg R

Correct Answer:

$$mg \frac{R}{2}$$

Solution:

Change in potential energy:

$$W = U_2 - U_1 = -\frac{GMm}{2R} - \left(-\frac{GMm}{R}\right)$$

$$W = \frac{GMm}{R} - \frac{GMm}{2R} = \frac{GMm}{2R}$$

$$g = \frac{GM}{R^2}$$

So,

$$W = \frac{mgR}{2}$$

Hence, the correct option is (1).

-
- Q. 14** Each side of a metallic cube of mass 5.580 kg is measured to be 9.0 cm . Keeping the significant figures in view, the density of the material of the cube can be best expressed as $X \times 10^3 \text{ kg m}^{-3}$, where the value of X is :

Option 1:

7.654

Option 2:

7 . 7

Option 3:

7.65

Option 4:

7 . 6

Correct Answer:

7 . 7

Solution:

$$\text{Density} = \frac{\text{mass}}{\text{volume}} = \frac{5.580 \text{ kg}}{(9.0 \times 10^{-2})^3}$$

$$= 7.654 \times 10^3 \text{ kg/m}^3$$

Rounding off to 2 significant digits

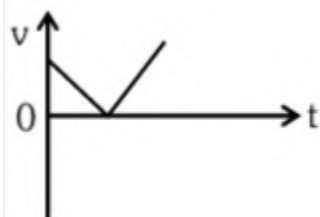
$$\text{Result} = 7.7 \times 10^3$$

$$\Rightarrow x = 7.7$$

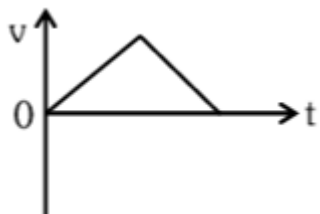
Hence, the correct option is (2).

- Q. 15** The following plots show variation of velocity (v) with time (t), of a ball thrown vertically upward, and falling back. Which of the following plots is/are correct ?

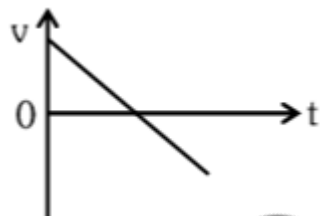
Option 1:



Option 2:



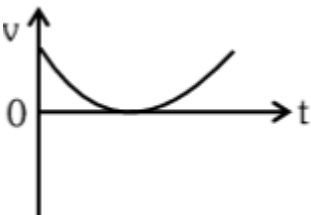
Option 3:



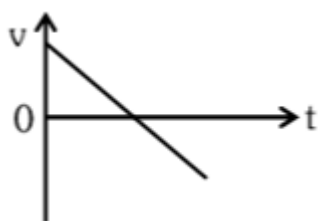
Option 4:



Option 5:



Correct Answer:



Solution:

Physics: Velocity decreases linearly to zero, then becomes negative Graph: Straight line with negative slope crossing zero

Hence, the correct option is (3).

Q. 16 The sum of kinetic energy and potential energy of a simple pendulum bob is 0.02 joule. The speed of the simple pendulum bob at equilibrium position is approximately :
(Consider mass of the bob = 20 g)

Option 1:

2.0 m/s

Option 2:

0.2 m/s

Option 3:

14 · 1 m/s

Option 4:

1.41 m/s

Correct Answer:

1.41 m/s

Solution:

$$\text{Total energy} \Rightarrow \frac{1}{2}KA^2 = 0.02 \text{ J}$$

$$\frac{1}{2} m^2 A^2 = 0.02 \text{ J}$$

$$\frac{1}{2} mV_{\max}^2 = 0.02 \text{ J}$$

$$V_{\max}^2 = \frac{0.04}{20 \times 10^{-3}} = 2$$

$$V_{\max} = 1.41 \text{ m/s}$$

Hence, the correct option is (4).

Q. 17 In Young's double slit experiment, using monochromatic light of wavelength λ , the intensity of light at a point on the screen where the path difference is λ , is K units. The intensity of light at a point where the path difference is $\frac{\lambda}{3}$ will be :

Option 1:

K

Option 2:

2K

Option 3:

$\frac{K}{2}$

Option 4:

$\frac{K}{4}$

Correct Answer:

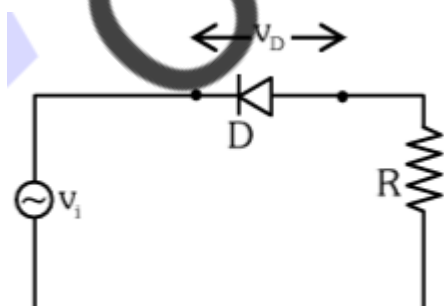
$\frac{K}{4}$

Solution:

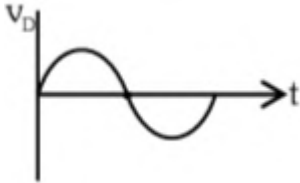
$$\begin{aligned} I_R &= K \cos^2 \frac{\Delta\phi}{2} \\ &= K \cos^2 \left(\frac{120^\circ}{2} \right) \\ &= K \cos^2 60^\circ = \frac{K}{4} \end{aligned}$$

Hence, the correct option is (4).

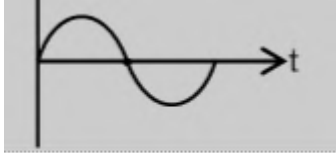
Q. 18 In the circuit shown below, the voltage appearing across the diode D will be of the form :



Option 1:

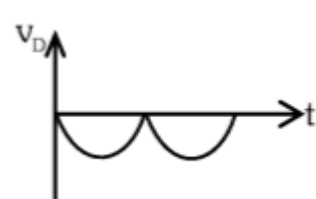


Option 2:

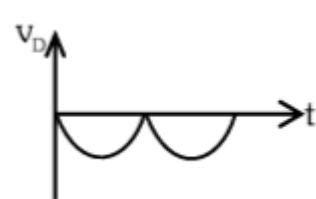


v_D

Option 3:



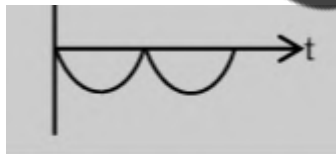
Option 4:



Correct Answer:



Solution:



v_D

Logic: In forward bias (positive half cycle), the diode conducts, and voltage across it is nearly 0 V . In reverse bias (negative half cycle), the diode does not conduct, and the entire input voltage appears across the diode.

Graph: The output V_D will show the negative half cycles.

Hence, the correct option is (4).

Q. 19 An ac circuit contains a resistance of $1\text{k}\Omega$, a capacitor of $0.1\mu\text{F}$ and an inductor of 1mH connected in series. The resonance frequency of the circuit is approximately :

Option 1:

15.9 kHz

Option 2:

20.7 kHz

Option 3:

$10 \cdot 1\text{kHz}$

Option 4:

$13 \cdot 5\text{kHz}$

Correct Answer:

15.9 kHz

Solution:

For a series LCR circuit, resonance frequency is:

$$f = \frac{1}{2\pi\sqrt{LC}}$$
$$f = \frac{1}{2\pi\sqrt{10^{-3} \times 10^{-7}}} = \frac{1}{2\pi\sqrt{10^{-10}}} = \frac{1}{2\pi \times 10^{-5}}$$
$$f \approx \frac{1}{6.28 \times 10^{-5}} \approx 1.59 \times 10^4 \text{ Hz}$$
$$f \approx 15.9\text{kHz}$$

Hence, the correct option is (1).

Q. 20 In interference and diffraction, the light energy is redistributed. If it reduces in one region, producing a dark fringe, it increases in another region, producing a bright fringe.
A. As there is no gain or loss of energy, these phenomena are consistent with the principle of conservation of energy.
B. Diffraction and interference are characteristics exhibited only by light waves.
Choose the **correct** answer from the options given below :

Option 1:

A is false, but B is true

Option 2:

A is true, but B is false

Option 3:

A is true and B is also true

Option 4:

Both A and B are false

Correct Answer:

A is true, but B is false

Solution:

Statement A: Energy is redistributed (bright & dark fringes), no net loss $\rightarrow$ consistent with conservation of energy $\rightarrow$ True

Statement B: Interference and diffraction are wave phenomena, but not only for light (also seen in sound, water waves, etc.) $\rightarrow$ False

Hence, the correct option is (2).

-
- Q. 21** For a travelling harmonic wave $y(x, t) = 2.0 \cos 2\pi(10t - 0.0080x + 0.35)$, where x and y are in cm and t in s. The phase difference between oscillatory motion of two points separated by a distance of 0.5λ is :

Option 1:

$0.08\pi\text{rad}$

Option 2:

$0.008\pi\text{rad}$

Option 3:

$0.8\pi\text{rad}$

Option 4:

$8\pi\text{rad}$

Correct Answer:

$0.8\pi\text{rad}$

Solution:

Wave Number (k) : From the equation, $k = 2\pi \times 0.0080$

Phase Difference: $\Delta\phi = k\Delta x = (2\pi \times 0.0080) \times 0.5 = 0.008\pi\text{rad}$

Hence, the correct option is (3).

-
- Q. 22** The magnitude and direction of the acceleration produced in a body of mass 5 kg when two mutually perpendicular forces 8 N and 6 N act on it, are respectively

Option 1:

2 m s^{-2} ; $\tan^{-1}(3/4)$ with 8 N force

Option 2:

2 m s^{-2} ; $\tan^{-1} 2 \text{ s}/3$) with 8 N force

Option 3:

2 m s^{-2} ; $\tan^{-1}(3/4)$ with 6 N force

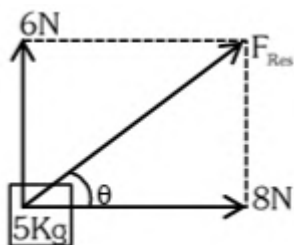
Option 4:

20 m s^{-2} ; $\tan^{-1}(4/3)$ with 8 N force

Correct Answer:

2 m s^{-2} ; $\tan^{-1}(3/4)$ with 8 N force

Solution:



$$F_{\text{Res}} = \sqrt{6^2 + 8^2}$$

$$F_{\text{Res}} = 10 \text{ N}$$

$$a = \frac{F_{\text{Res}}}{m} = \frac{10}{5} = 2 \text{ m/s}^2$$

$$\tan \theta = \frac{6}{8} = \frac{3}{4}$$

$$a = 2 \text{ m/s}^2 \tan^{-1} \left(\frac{3}{4} \right) \text{ with 8 N force}$$

Hence, the correct the option is (1).

Q. 23

Consider two uncharged capacitors of equal capacitance 200 pF . One of them is charged by a 100 V supply and disconnected. Now this capacitor is connected to the uncharged capacitor. The amount of electrostatic energy lost in the process is :

Option 1:

0.5 J

Option 2:

$1.0 \times 10^{-6} \text{ J}$

Option 3:

$$0.5 \times 10^{-6} \text{ J}$$

Option 4:

$$1.0 \text{ J}$$

Correct Answer:

$$0.5 \times 10^{-6} \text{ J}$$

Solution:

Initial energy:

$$U_i = \frac{1}{2} CV^2$$

Final voltage becomes $V/2$ (equal capacitors sharing charge)

Final energy:

$$U_f = \frac{1}{2} (2C) \left(\frac{V}{2} \right)^2 = \frac{1}{4} CV^2$$

Energy lost:

$$\Delta U = U_i - U_f = \frac{1}{2} CV^2 - \frac{1}{4} CV^2 = \frac{1}{4} CV^2$$

$$C = 200 \text{ pF} = 200 \times 10^{-12}$$

$$\begin{aligned} \Delta U &= \frac{1}{4} \times 200 \times 10^{-12} \times (100)^2 \\ &= 0.5 \times 10^{-6} \text{ J} \end{aligned}$$

Hence, the correct option is (3).

Q. 24 The power of a crane, which lifts a mass of 1000 kg to a height of 20 m in 10 s is :

$$(g = 9.8 \text{ m/s}^2)$$

Option 1:

$$19.6 \text{ kW}$$

Option 2:

$$19 \cdot 6 \text{ W}$$

Option 3:

$$39 \cdot 2 \text{ kW}$$

Option 4:

$$39 \cdot 2 \text{ W}$$

Correct Answer:
19.6 kW

Solution:

$$\begin{aligned}\text{Power of crane} &= \frac{mgh}{t} \\ &= \frac{1000 \times 9.8 \times 20}{10} \\ &= 19.6 \times 1000 \\ &= 19.6 \text{ kW}\end{aligned}$$

Hence, the correct option is (1).

Q. 25 In a vernier callipers, 20 VSD comerde 16 MSD (each division $\overline{p}$ f length 1 mm). The least count of the vernier callipers is :

Option 1:
0.2 cm

Option 2:
0.1 cm

Option 3:
0.02 cm

Option 4:
0.01 cm

Correct Answer:
0.02 cm

Solution:

$$\begin{aligned}20\text{VSD} &= 16\text{MSD} \\ \text{VSD} &= \frac{16}{20}(1 \text{ mm}) \\ &= 0.8 \text{ mm} \\ \text{LC} &= \text{MSD} - \text{VSD} \\ &= 1 - 0.8 \\ &= 0.2 \text{ mm} \\ &= 0.02 \text{ cm}\end{aligned}$$

Hence, the correct option is (3).

Q. 26 When a ruler falls vertically, 5 different person catch it with different reaction times.
($g = 9.8 \text{ m s}^{-2}$)

- A. Person A has reaction time of 0.20 s .
- B. Person B has reaction time of 0.22 s .
- C. Person C has reaction time of 0.18 s .
- D. Person D has reaction time of 0.19 s .
- E. Person E has reaction time of 0.21 s .

What is the **correct** order of the distance travelled by the ruler for each person?

Option 1:

$B > E > A > C > D$

Option 2:

$C > D > A > E > B$

Option 3:

$C > D > A > B > E$

Option 4:

$B > E > A > D > C$

Correct Answer:

$B > E > A > D > C$

Solution:

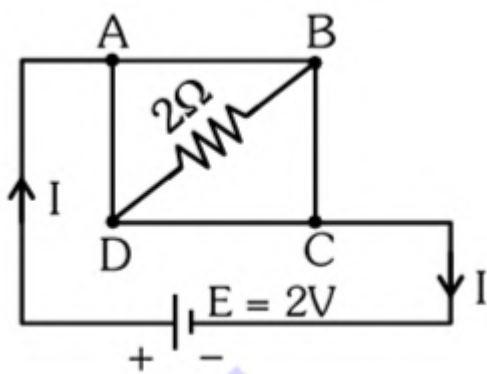
Distance fallen: $s = \frac{1}{2}gt^2$

Distance $\propto t^2$, so compare reaction times:

Largest time $\rightarrow$ largest distance:

Hence, the correct option is (4).

- Q. 27** A uniform metallic wire having resistance 4Ω is bent to form a square loop (ABCD) (see figure). A resistance of 2Ω is connected between points B and D and a battery of 2 V is connected across points A and C as shown in the figure. Now the value of current (I) is :



Option 1:

4A

Option 2:

4.5A

Option 3:

8A

Option 4:

2A

Correct Answer:

2A

Solution:

Resistance of each wire of

$$W.S.B. = \frac{4}{4} = 1\Omega$$

This is balance W.S.B.

$$\text{So, } R_{\text{Net}} = 1\Omega$$

So, current

$$I = \frac{E}{R_{\text{Net}}} = \frac{2}{1} = 2 \text{ A}$$

Hence, the correct option is (4).

- Q. 28** A room heater is rated 400 W, 220 V. If the supply voltage drops to 200 V , what will be the power consumed (approximately)?

Option 1:

400 W

Option 2:

121 W

Option 3:

331 W

Option 4:

200 W

Correct Answer:

331 W

Solution:

Given that:

Printed voltage = 220 V

Printed Power = 400 W

Supply voltage = 200 V

Resistance of heater $R_H = \frac{(220)^2}{400}$

So Actual power gained

$$\begin{aligned} P' &= \frac{V_s^2}{R_H} = \frac{(200)^2}{(220)^2} (400) \\ &= 330.57 \text{ W} \\ &\simeq 331 \text{ W} \end{aligned}$$

Hence, the correct option is (3) 331 W.

-
- Q. 29** A 100-turn closely wound circular coil of radius 5 cm has a magnetic field of $3.14 \times 10^{-3} \text{ T}$ at its centre. The current flowing through the coil, and the magnitude of the magnetic moment of this coil are, respectively :
(Take $\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$)

Option 1:

2.5 A, 20 A m²

Option 2:

2 A, 4Am²

Option 3:

2.5 A, 2 A m²

Option 4:

2 A, 10 A m²

Correct Answer:

2.5 A, 2 A m²

Solution:

$$B = \frac{\mu_0 NI}{2R} \Rightarrow I = \frac{2BR}{\mu_0 N}$$

$$\text{Current } I = \frac{2 \times 3.14 \times 10^{-3} \times 5 \times 10^{-2}}{4\pi \times 10^{-7} \times 100}$$

$$\text{Current } I = 2.5 \text{ A}$$

$$\text{Magnetic moment } M = NIA$$

$$= 100 \times 2.5 \times \pi r^2$$

$$= 100 \times 2.5 \times 3.14 \times 25 \times 10^{-4}$$

$$= 1.96$$

$$= 2 \text{ A} \times \text{m}^2$$

Hence, the correct option is (3).

Q. 30

A rectangular wire loop of sides 8 cm and 3 cm with a small cut, is moving out of a region of uniform magnetic field of magnitude 0.3 T directed normal to the plane of the loop. The emf developed across the cut, if the velocity of the loop is 2 cm s⁻¹, in a direction normal to the shorter side of the loop, will be :

Option 1:

1.8 × 10⁻⁴ volt

Option 2:

1.3 × 10⁻⁴ volt

Option 3:

1.2 × 10⁻⁴ volt

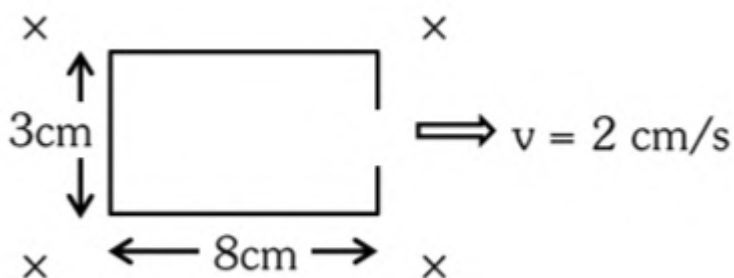
Option 4:

4.8 × 10⁻⁴ volt

Correct Answer:

1.8 × 10⁻⁴ volt

Solution:



$$e = Blv = 0.3 \times 0.03 \times 0.02 = 1.8 \times 10^{-4} \text{ V}$$

Hence, the correct option is (1).

- Q. 31** Four statements are given (A is mass number) :
- A. The volume of a nucleus is proportional to $A^{1/3}$.
 - B. The volume of a nucleus is proportional to A.
 - C. The difference in mass of an atom and its nucleus is called the mass defect.
 - D. The difference in mass of a nucleus and its constituents is called the mass defect.
- Choose the **correct** answer from the options given below :

Option 1:

A and D are true, but B and C are false

Option 2:

B and D are true, but A and C are false

Option 3:

B and C are true, but A and D are false

Option 4:

A and C are true, but B and D are false

Correct Answer:

B and D are true, but A and C are false

Solution:

Correct statements

A: Volume $\propto R^3$, and $R \propto A^{1/3} \rightarrow \text{Volume} \propto A \rightarrow \text{false}$

B: Volume $\propto A \rightarrow \text{true}$

C: Mass defect = difference between atom and nucleus $\rightarrow \text{false}$

D: Mass defect = difference between nucleus and its nucleons $\rightarrow \text{true}$

Hence, the correct option is (2).

- Q. 32** An unknown nucleus has nuclear density $2.29 \times 10^{17} \text{ kg/m}^3$ and mass of $19.926 \times 10^{-27} \text{ kg}$. Its mass number A is approximately :
(Take $R_0 = 1.2 \times 10^{-15} \text{ m}$, $4\pi = 12.56$)

Option 1:

16

Option 2:

20

Option 3:

12

Option 4:

19

Correct Answer:

12

Solution:

$$\text{Density } \rho = \frac{\text{Mass}}{\text{Volume}} = \frac{M}{\frac{4}{3}\pi R^3}$$

$$\text{Nuclear Radius: } R = R_0 A^{1/3}, \text{ so } R^3 = R_0^3 A$$

$$A = \frac{3M}{4\pi R_0^3 \rho} = \frac{3 \times 19.926 \times 10^{-27}}{12.56 \times (1.2 \times 10^{-15})^3 \times 2.29 \times 10^{17}}$$

$$A \approx \frac{59.778 \times 10^{-27}}{12.56 \times 1.728 \times 10^{-45} \times 2.29 \times 10^{17}} \approx 12$$

Hence, the correct option is (3).

- Q. 33** Savitha, a XI standard student, while conducting an experiment to determine the effective length of a simple pendulum L, notes down the data of time taken to complete 30 oscillations as 60 s and hence calculates the length of the simple pendulum as :
(Take $\pi^2 = 9.8$, and $g = 9.8 \text{ m/s}^2$)

Option 1:

0.75 m

Option 2:

1 m

Option 3:

1.5 m

Option 4:

2 m

Correct Answer:

1 m

Solution:

$$T = \frac{\text{Total Time}}{\text{Number of Oscillations}} = \frac{60 \text{ s}}{30} = 2 \text{ s}$$

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$T^2 = 4\pi^2 \left(\frac{L}{g} \right)$$

$$L = \frac{T^2 \cdot g}{4\pi^2}$$

$$L = \frac{2^2 \cdot 9.8}{4 \cdot 9.8} = \frac{4 \cdot 9.8}{4 \cdot 9.8} = 1 \text{ m}$$

Hence, the correct option is (2).

Q. 34 An electric heater supplies heat to a system at a rate of 100 W . If the system performs work at a rate of 75 J/s, then the rate at which internal energy increases will be :

Option 1:

75 W

Option 2:

25 W

Option 3:

100 W

Option 4:

125 W

Correct Answer:

25 W

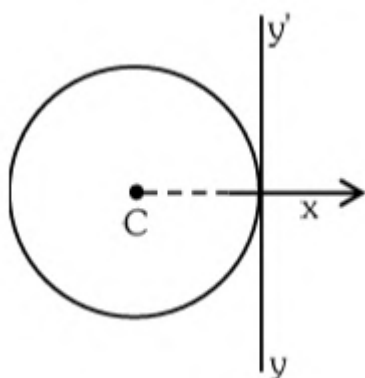
Solution:

Rate of increase of internal energy Using first law of thermodynamics:

$$\frac{dU}{dt} = \text{Heat supplied} - \text{Work done} = 100 - 75 = 25 \text{ W}$$

Hence, the correct option is (2).

- Q. 35** A thin wire of length ' L ' and linear mass density ' m ' is bent into a circular ring (in $x - y$ plane) with centre ' C ' as shown in figure. The moment of inertia of the ring about an axis yy' will be :



Option 1:

$$\frac{3 mL^3}{8\pi}$$

Option 2:

$$\frac{3 mL^2}{8\pi^2}$$

Option 3:

$$\frac{3 mL^3}{8\pi^2}$$

Option 4:

$$\frac{3 mL^2}{8}$$

Correct Answer:

$$\frac{3 mL^3}{8\pi^2}$$

Solution:

Let mass of wire is M

length = L

given m = Linear mass density

$$m = \frac{M}{L} \quad L = 2\pi R$$

$$M = mL \quad R = \frac{L}{2\pi}$$

$$I_{yy'} = \frac{MR^2}{2} + MR^2$$

$$= \frac{3}{2}MR^2 = \frac{3}{2}(mL) \left(\frac{L^2}{4\pi^2} \right)$$

$$I_{yy'} = \frac{3}{8} \frac{mL^3}{\pi^2}$$

Hence, the correct option is (3).

Q. 36 A galvanometer of resistance 100Ω gives full scale deflection for a current of 1 mA . It is converted into an ammeter of range 0 – 10 A. The shunt required is :

Option 1:
 0.001Ω

Option 2:
 0.10Ω

Option 3:
 1.0Ω

Option 4:
 0.01Ω

Correct Answer:
 0.01Ω

Solution:

Given that

$$R_g = 100\Omega$$

$$I_g = 1 \text{ mA}$$

$$I = 10 \text{ A}$$

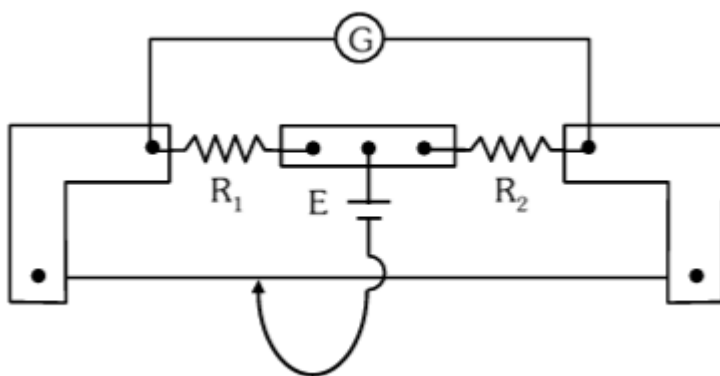
$$S = \frac{I_g R_g}{(I - I_g)} = \frac{10^{-3}(100)}{(10 - 10^{-3})}$$

Since $I_g \ll I$ so $I - I_g \simeq I$

$$S = 10^{-2} = 0.01\Omega$$

Hence, the correct option is (4).

- Q. 37** In a metre bridge experiment (see figure), the positions of the cell, E, and galvanometer, G, are interchanged. We shall observe in the galvanometer :



Option 1:

Only the right-sided deflection

Option 2:

Only the left-sided deflection

Option 3:

There will be no deflection irrespective of the position of the jockey

Option 4:

Both right-sided and left-sided deflection and at balance point, no deflection

Correct Answer:

Both right-sided and left-sided deflection and at balance point, no deflection

Solution:

In the given circuit, it is not mentioned whether the meter bridge is balanced or not. So on interchanging the positions of battery and galvanometer, deflection may or may not be zero. In case of unbalanced condition, deflection may be left or right sided in galvanometer.

Hence, the correct option is (4).

- Q. 38** The peak value of an alternating current is 5 A and frequency is 60 Hz . How long will the current, starting from zero, take to reach the peak value ?

Option 1:

$\frac{1}{120}$ s

Option 2:

$$\frac{1}{240} \text{ s}$$

Option 3:

$$\frac{1}{30} \text{ s}$$

Option 4:

$$\frac{1}{60} \text{ s}$$

Correct Answer:

$$\frac{1}{240} \text{ s}$$

Solution:

Time to reach peak current

For AC:

$$i = I_0 \sin(\omega t)$$

Peak occurs when $\omega t = \frac{\pi}{2}$

$$t = \frac{\pi/2}{\omega} = \frac{\pi/2}{2\pi f} = \frac{1}{4f}$$

$$t = \frac{1}{4 \times 60} = \frac{1}{240} \text{ s}$$

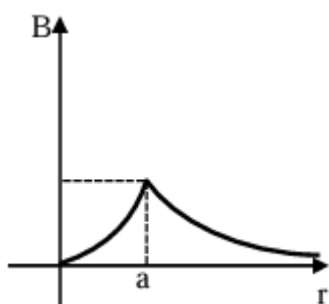
Hence, the correct option is (2).

Q. 39

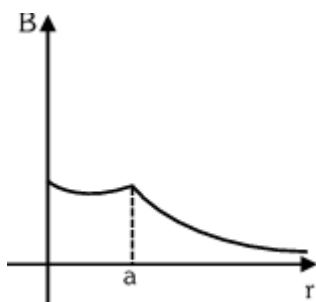
The figure given below shows a long straight solid wire of circular cross-section of radius 'a' carrying steady current I . The current I is uniformly distributed across its cross-section. The plot which correctly represents the variation of magnetic field (B) with distance (r) from the axis of the conductor in the region is :



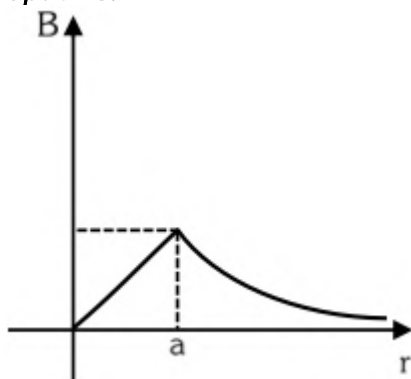
Option 1:



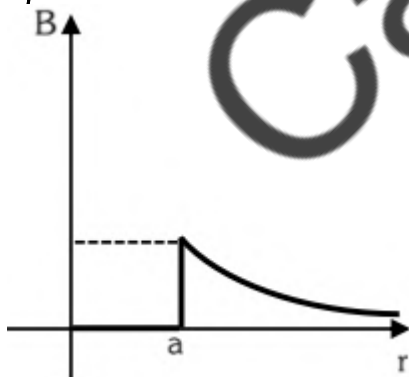
Option 2:



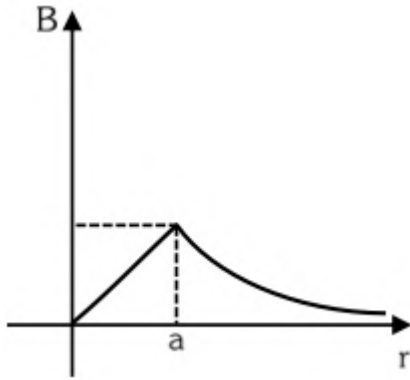
Option 3:



Option 4:



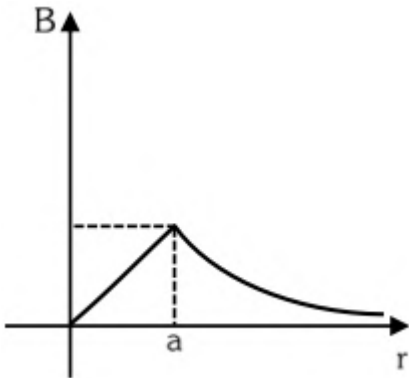
Correct Answer:



Solution:

$$B_{\text{in}} = \frac{\mu_0 I r}{2\pi R^2} \propto r$$

$$B_{\text{out}} = \frac{\mu_0 I}{2\pi r} \propto \frac{1}{r}$$



Hence, the correct option is (3).

Q. 40 Two statements are given below :

A. When the forward bias voltage across a p-n junction diode increases above a certain threshold voltage, the diode current increases significantly.

B. This current is called reverse saturation current.

Choose the **correct** answer from the options given below :

Option 1:

Both Statements A and B are true

Option 2:

Both Statements A and B are false

Option 3:

Statement *A* is true, but Statement *B* is false

Option 4:

Statement A is false, but Statement B is true

Correct Answer:

Statement *A* is true, but Statement *B* is false

Solution:

Statement A: When forward bias voltage increases above a certain threshold, the diode current increases significantly. (True)

Statement B: This current is called reverse saturation current. (False)

Analysis: Statement A is true, but Statement B is false.

Hence, the correct option is (3).

Q. 41 Which of the following statements are correct?

A. Inside a conductor, the electrostatic field is zero.

B. Electric field at the surface of a charged conductor does not depend on its surface charge density.

C. The interior of a charged conductor can have no excess charge in the static situation.

D. At the surface of a charged conductor, the electrostatic field must be normal to the surface at every point.

E. The electrostatic potential is zero everywhere inside a charged conductor.

Choose the **correct** answer from the options given below :

Option 1:

A, C and D only

Option 2:

A, C and E only

Option 3:

C, D and E only

Option 4:

A, B and D only

Correct Answer:

A, C and D only

Solution:

A. Inside a conductor, the electrostatic field is zero. (True)

B. Electric field at the surface does not depend on surface charge density. (False: $E = \sigma/\epsilon_0$)

C. The interior of a charged conductor can have no excess charge in the static situation. (True)

D. At the surface, the electrostatic field must be normal to the surface at every point. (True)

E. The electrostatic potential is zero everywhere inside a charged conductor. (False)

Analysis: Statements A, C, and D are correct.

Hence, the correct option is (1).

Q. 42 For a metal of work function 6.6 eV, which of the following wavelengths of incident radiation does not give rise to the photoelectric effect?
(Take Planck's constant as $6.6 \times 10^{-34} \text{ J s}$)

Option 1:

200 nm

Option 2:

100 nm

Option 3:

50 nm

Option 4:

150 nm

Correct Answer:

200 nm

Solution:

$$\phi = 6.6 \text{ eV}$$

$$\frac{hc}{\lambda} = 6.6 \text{ eV}$$

$$\lambda_0 = \frac{12400}{6.6}$$

$$\lambda_0 = 1878.7 \quad = 187.87 \text{ nm}$$

For photo electric effect

$$\lambda \leq \lambda_0$$

$\therefore$ The wavelength of radiation does not give rise to photoelectric effect is 200 nm .

Hence, the correct option is (1).

Q. 43 In a concave lens, a ray of light emanating from the object parallel to the principal axis of the lens, after refraction :

Option 1:

passes through the second principal focus.

Option 2:

appears to diverge from the first principal focus.

Option 3:

passes through $2F$, which is the radius of curvature of the lens.

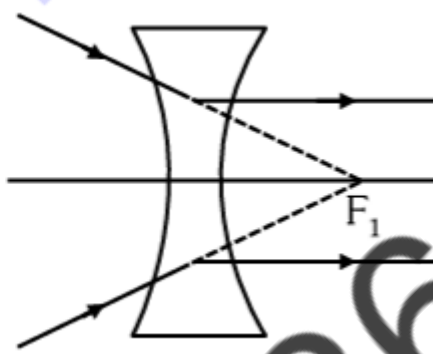
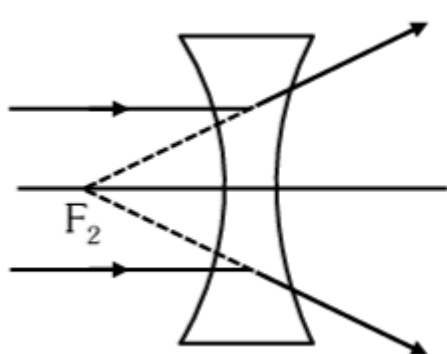
Option 4:

emerges parallel to the principal axis.

Correct Answer:

appears to diverge from the first principal focus.

Solution:



When incident light is parallel to principal axis, after refraction image will be at focus. This focus will be second principal focus (F_2).

In this question as lens is concave (diverging) after refraction light will diverge and appears to be coming from second principal focus (F_2).

Hence, the correct option is (2).

- Q. 44** A submarine is designed to withstand an absolute pressure of 100 atm . How deep can it go below the water surface?
(Consider the density of water = 1000 kg m^{-3} . $1 \text{ atm} = 1 \times 10^5 \text{ Pa}$ and gravitational acceleration $g = 10 \text{ m/s}^2$)

Option 1:

9900 m

Option 2:

990 m

Option 3:

9000 m

Option 4:

99 m

Correct Answer:

990 m

Solution:

Absolute pressure at depth h in water

$$P = P_0 + h\rho g \quad P_0 = \text{atmospheric pressure}$$

$$P - P_0 = h\rho g$$

$$h = \frac{P - P_0}{\rho g}$$
$$= \frac{(100 - 1) \times 10^5}{10^3 \times 10}$$

$$h = 990 \text{ m}$$

Hence, the correct option is (2).

Q. 45 Match List I with List II :

List I (Electromagnetic wave)		List II (Production)	
A.	Microwave	I.	Electrons in atoms emit light when they move from a higher energy level to a lower energy level
B.	Visible light	II.	Radioactive decay of nucleus
C.	Gamma rays	III.	Vibration of atoms and molecules
D.	Infra-red rays	IV.	Klystron valve or magnetron valve

Choose the **correct** answer from the options given below:

Option 1:

A-IV, B-III, C-II, D-I

Option 2:

A-III, B-IV, C-I, D-III

Option 3:

A-III, B-I, C-II, D-IV

Option 4:

A-IV, B-I, C-II, D-III

Correct Answer:

A-IV, B-I, C-II, D-III

Solution:

- A. Microwave: Produced by Klystron valve or magnetron valve
- B. Visible light: Electrons in atoms emit light moving to lower energy levels
- C. Gamma rays: Radioactive decay of nucleus
- D. Infra-red rays: Vibration of atoms and molecules

Hence, the correct option is (4).

Chemistry

Q. 1 Select the reagents that reduce nitriles to primary amines:

- A. (i) LiAlH_4 ;
(ii) H_2O
- B. $\text{Sn} + \text{HCl}$
- C. H_2/Ni
- D. $\text{Na}(\text{Hg})/\text{C}_2\text{H}_5\text{OH}$
- E. $\text{Br}_2/\text{aq. NaOH}$

Choose the correct answer from the options given below

Option 1:

A, B and C only

Option 2:

A, D and E only

Option 3:

A, C and D only

Option 4:

B, D and E only

Correct Answer:

A, C and D only

Solution:

A. $\text{LiAlH}_4/\text{H}_2\text{O}$

Strong reducing agent $\rightarrow$ converts nitrile $\rightarrow$ primary amine

B. $\text{Sn} + \text{HCl}$

Used for nitro $\rightarrow$ amine, not for nitriles $\rightarrow$ Incorrect

C. H_2/Ni

Catalytic hydrogenation $\rightarrow$ nitrile $\rightarrow$ primary amine

D. $\text{Na}(\text{Hg})/\text{C}_2\text{H}_5\text{OH}$

Mild reduction → can reduce nitriles → primary amine

E. $\text{Br}_2/\text{aq. NaOH}$

Hoffmann bromamide reaction (amide → amine with one less carbon)

Hence, the correct answer is option (3)

Q. 2 Match List - I with List - II

List - I (Transition metal /compound /complex)	List - II (Catalytic Role)
A. V_2O_5	I. Preparation of ammonia from N_2/H_2 mixture
B. Fe	II. Polymerisation of alkynes
C. PdCl_2	III. Preparation of H_2SO_4 from SO_2
D. Ni complex	IV. Oxidation of ethyne to ethanal

Choose the correct answer from the options given below:

Option 1:

A-III, B-IV, C-I, D-II

Option 2:

A-III, B-I, C-IV, D-II

Option 3:

A-II, B-I, C-IV, D-III

Option 4:

A-IV, B-I, C-III, D-II

Correct Answer:

A-III, B-I, C-IV, D-II

Solution:

A. V_2O_5 : Used in the Contact Process for the preparation of H_2SO_4 from SO_2 . (A-III)

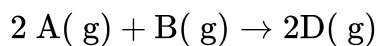
B. Fe (Fine powder): Used in the Haber's Process for the preparation of ammonia from N_2 and H_2 . (B-I)

C. PdCl_2 : Specifically used in the Wacker Process for the oxidation of ethyne (or ethene) to ethanal. (C-IV)

D. Ni complex: Specifically organometallic Nickel complexes (like Zeigler-Natta complexes) are used in the polymerisation of alkynes. (D-II)

Hence, the correct answer is option (2)

Q. 3 Consider the following reaction :



$$\Delta U^\ominus = -10 \text{ kJ mol}^{-1} \text{ and } \Delta S^\ominus = -44 \text{ J K}^{-1} \text{ at } 298 \text{ K}.$$

Identify the correct option with $\Delta G^\ominus$ for the reaction and spontaneity of the reaction at 298 K.

(Given : $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$)

Option 1:

+0.63568 kJmol⁻¹, non-spontaneous

Option 2:

-0.63568 kJ mol⁻¹, spontaneous

Option 3:

-1.635 kJ mol⁻¹ spontaneous

Option 4:

+1.635 kJ mol⁻¹, non-spontaneous

Correct Answer:

+0.63568 kJmol⁻¹, non-spontaneous

Solution:

$$\Delta S^\ominus = -44 \text{ J} = -0.044 \text{ kJ K}^{-1}$$

$$\Delta H = \Delta U + \Delta n_g RT$$

$$\Delta n_g = 2 - 3 = -1$$

$$\Delta H = -10 + (-1)(8.31 \times 298/1000)$$

$$= -10 - 2.476 \approx -12.476 \text{ kJ}$$

$$\Delta G = \Delta H - T\Delta S$$

$$= -12.476 - (298 \times -0.044)$$

$$= -12.476 + 13.112$$

$$= +0.636 \text{ kJ mol}^{-1}$$

$$\Delta G > 0 \Rightarrow \text{N-spontaneous}$$

Hence, the correct answer is option (1)

Q. 4 Match List - I with List - II

	List - I (Quantum Numbers)			List - II (Orbital)
	"n"	'l'		
A	2	1	I	3d
B	4	0	II	2p
C	5	3	III	4s
D	3	2	IV	5f

Choose the correct answer from the options given below

Option 1:

A-II, B-III, C-IV, D-I

Option 2:

A-I, B-II, C-III, D-IV

Option 3:

A-II, B-III, C-I, D-IV

Option 4:

A-IV, B-II, C-III, D-I

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

The orbital is designated as nl , where n is the principal quantum number and l corresponds to the subshell ($0 = s, 1 = p, 2 = d, 3 = f$).

A. $n = 2, l = 1 : 2p$ (A – II)

B. $n = 4, l = 0 : 4s$ (B-III)

C. $n = 5, l = 3 : 5f$ (C – IV)

D. $n = 3, l = 2 : 3d$ (D – I)

Hence, the correct answer is option (1)

- Q. 5** In a qualitative analysis, Bi^{3+} is detected by appearance of precipitate of $\text{BiO}(\text{OH})(\text{s})$. Calculate pH when the following equilibrium exists at 298 K :
- $$\text{BiO}(\text{OH})(\text{s}) \rightleftharpoons \text{BiO}^+(\text{aq}) + \text{OH}^-(\text{aq}),$$
- $$K = 4 \times 10^{-10}$$
- (Given : $\log 2 = 0.3010$)

Option 1:

4.699

Option 2:

5.286

Option 3:

8.714

Option 4:

9.301

Correct Answer:

9.301

Solution:

For the reaction: $\text{BiO}(\text{OH})(\text{s}) \rightleftharpoons \text{BiO}^+(\text{aq}) + \text{OH}^-(\text{aq})$

The equilibrium constant is $K = [\text{BiO}^+][\text{OH}^-]$.

Let $[\text{BiO}^+] = [\text{OH}^-] = s$.

$$s^2 = K = 4 \times 10^{-10}$$

$$s = [\text{OH}^-] = \sqrt{4 \times 10^{-10}} = 2 \times 10^{-5} \text{ M}$$

pOH

$$\text{pOH} = -\log[\text{OH}^-]$$

$$\text{pOH} = -\log(2 \times 10^{-5}) = 5 - \log 2$$

$$\text{pOH} = 5 - 0.3010 = 4.699$$

pH

At 298 K, $\text{pH} + \text{pOH} = 14$.

$$\text{pH} = 14 - 4.699 = \mathbf{9.301}$$

Hence, the correct answer is option (4)

Q. 6 The correct statement with regard to the secondary structure of DNA/RNA is :

Option 1:

DNA possesses a double strand helix structure and contains thymine as one of the four bases.

Option 2:

DNA possesses a single strand helix structure and contains uracil as one of the four bases.

Option 3:

RNA possesses a double strand helix structure and contains uracil as one of the four bases.

Option 4:

RNA possesses a single strand helix structure and contains thymine as one of the four bases.

Correct Answer:

DNA possesses a double strand helix structure and contains thymine as one of the four bases.

Solution:

DNA is **double-stranded (double helix)**

Bases: A, T, G, C → **thymine present**

Hence, the correct answer is option (1)

Q. 7 The pair of molecules that are metamers among the following is :

Option 1:

$\text{CH}_3\text{OCH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$

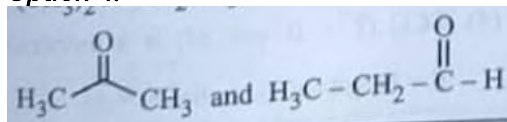
Option 2:

$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ and $\text{CH}_3 - \text{CH}(\text{OH}) - \text{CH}_3$

Option 3:

$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_3$

Option 4:



Correct Answer:

$\text{CH}_3\text{OCH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$

Solution:

Metamers are isomers that have the same molecular formula but differ in the distribution of carbon atoms (alkyl groups) on either side of a functional group (like $-\text{O}-$, $-\text{S}-$, or $-\text{NH}-$).

Option (1): $\text{CH}_3\text{OCH}_2\text{CH}_2\text{CH}_3$ (Methyl propyl ether) and $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$ (Diethyl ether). Both have the same formula ($\text{C}_4\text{H}_{10}\text{O}$) but different alkyl groups around the oxygen atom. These are metamers.

Option (2): Propan-1-ol and Propan-2-ol are position isomers.

Option (3): n-pentane and isopentane are chain isomers.

Option (4): Propanone and Propanal are functional isomers.

Hence, the correct answer is option (1)

Q. 8 Match List - I with List -II

List- I (Complex)	List- II (Type of Isomerism)
A. $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$	I. Optical
B. $[\text{Co}(\text{en})_3]^{3+}$	II. 'Solvate
C. $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$	III. Geometrical
D. $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$	IV. Linkage

Choose the correct answer from the options given below:

Option 1:

A-I, B-III, C-II, D=IV

Option 2:

A-II, B-IV, C-III, D-I

Option 3:

A-III, B-I, C-IV, D-II

Option 4:

A-III, B-I, C-II, D-IV

Correct Answer:

A-III, B-I, C-IV, D-II

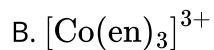
Solution:

A. $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

- Square planar complex (Pt^{2+} , d^8)

- Shows cis-trans isomerism

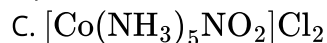
Geometrical isomerism $\rightarrow$ III



- Octahedral with bidentate ligand (en)

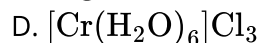
- Forms non-superimposable mirror images

Optical isomerism $\rightarrow$ I

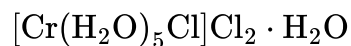


- NO_2^- can bind via N or O

Linkage isomerism $\rightarrow$ IV



- Can form hydrate/solvate isomers like:



Solvate isomerism $\rightarrow$ II

Hence, the correct answer is option (3)

Q. 9 Match List - I with List - II

List - I (order of reaction)	List - II (unit of rate constant)
A. Zero order	I. $\text{mol}^{-1} \text{L s}^{-1}$
B. First order	II. $\text{mol}^{-2} \text{L}^2 \text{s}^{-1}$
C. Second order	III. s^{-1}
D. Third order	IV. $\text{molL}^{-1} \text{s}^{-1}$

Choose the correct answer from the options given below:

Option 1:

A-IV, B-III, C- II, D- I

Option 2:

A-IV, B-III, C-I, D-II

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-IV, B-II, C-I, D-III

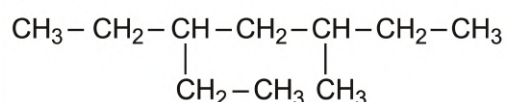
Correct Answer:
A-IV, B-III, C-I, D-II

Solution:

List -I	List - II
A. Zero order	IV. $\text{mol L}^{-1} \text{s}^{-1}$
B. First order	III. s^{-1}
C. Second order	I. $\text{mol}^{-1} \text{L s}^{-1}$
D. Third order	II. $\text{mol}^{-2} \text{L}^2 \text{s}^{-1}$

Hence, the correct answer is option 2

Q. 10 The correct IUPAC name of the following compound is :



Option 1:

3-ethyl-5-methylheptane

Option 2:

3-methyl-5-ethylheptane

Option 3:

2,4-diethylhexane

Option 4:

3,5-diethylhexane

Correct Answer:

3-ethyl-5-methylheptane

Solution:

3-ethyl-5-methylheptane

Hence, the correct answer is option (1)

Q. 11 A bulb is rated at 150 watt, converting 8% energy into light. If energy of one photon is $4.42 \times 10^{-19} \text{ J}$, how many photons are emitted by the bulb per second?

Option 1:

27.2×10^{19}

Option 2:

$$4.06 \times 10^{19}$$

Option 3:

$$1.35 \times 10^{19}$$

Option 4:

$$2.71 \times 10^{19}$$

Correct Answer:

$$2.71 \times 10^{19}$$

Solution:

The bulb is rated at 150 W, but only 8% is converted to light.

$$\text{Light Power} = 150 \times \frac{8}{100} = 12 \text{ W (or J/s)}$$

Using the formula $P = n \times E_{\text{photon}}$:

$$n = \frac{\text{Power}}{\text{Energy of one photon}}$$

$$n = \frac{12}{4.42 \times 10^{-19}} \approx 2.7149 \times 10^{19}$$

Hence, the correct answer is option (4)

Q. 12 Methane reacts with steam at 1273 K in the presence of nickel catalyst to form:

Option 1:

CO and H₂

Option 2:

CO and H₂O

Option 3:

CO₂ and H₂

Option 4:

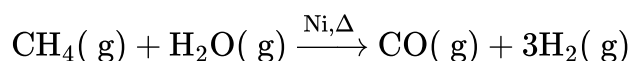
CO₂ and H₂O

Correct Answer:

CO and H₂

Solution:

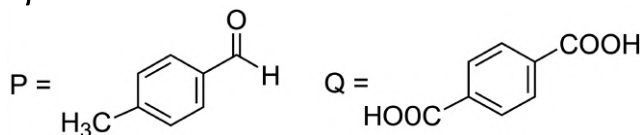
When methane reacts with steam at high temperatures (1273 K) in the presence of a nickel catalyst, it undergoes a process called steam reforming to produce Syngas (Synthesis Gas).



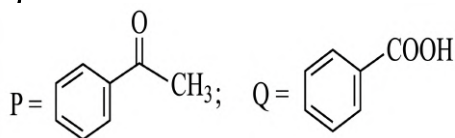
Hence, the correct answer is option (1)

Q. 13 Compound P (C_8H_8O) gives a red orange precipitate with 2,4-DNP reagent and it does not reduce Fehling's reagent. On drastic oxidation with chromic acid, P gives an aromatic product Q that produces effervescence on treating with aq. $NaHCO_3$. Compounds P and Q, respectively, are:

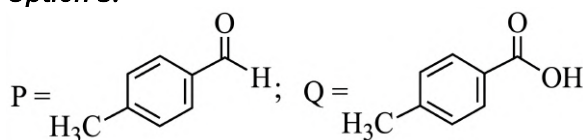
Option 1:



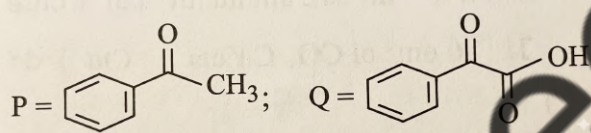
Option 2:



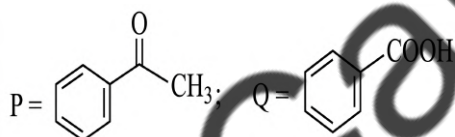
Option 3:



Option 4:



Correct Answer:



Solution:

1. Identify the functional group of P (2,4-DNP Test)

Observation: P gives a red-orange precipitate with 2,4-DNP reagent.

Inference: P contains a carbonyl group ($>C=O$), meaning it is either an aldehyde or a ketone.

2. Narrow down the carbonyl type (Fehling's Test)

Observation: P does not reduce Fehling's reagent.

Inference: P is a ketone (specifically an aromatic ketone here) or an aromatic aldehyde (which also don't reduce Fehling's), but the formula C_8H_8O and the next step point toward a ketone.

3. Analyze Product Q ($NaHCO_3$ Test)

Observation: Drastic oxidation of P with chromic acid yields Q, which produces effervescence with aq. $NaHCO_3$.

Inference: Q is a carboxylic acid (effervescence is due to CO_2 gas release).

4. Final Structure Identification

For P (C_8H_8O) : Acetophenone ($C_6H_5COCH_3$) fits the formula. It is a ketone, so it gives a positive 2,4-DNP test but a negative Fehling's test.

Drastic Oxidation: When alkyl benzenes or aromatic ketones like acetophenone are treated with strong oxidising agents (like chromic acid or alkaline $KMnO_4$), the entire side chain is oxidised down to a carboxyl

group ($-COOH$) attached directly to the ring.

Product Q : Benzoic acid (C_6H_5COOH).

Hence, the correct answer is option (2)

Q. 14 Match List- I with List - II:

List-I	List -II
A. C_2H_4	I. 3σ bonds, 2π bonds
B. C_2H_2	II. 3σ bonds, one lone pair
C. CH_4	III. 4σ bonds
D. NH_3	IV. 5σ bonds, 1π bond

Choose the correct answer from the options given below :

Option 1:

A-IV, B-I, C-III, D-II

Option 2:

A-III, B-IV, C-II, D-I

Option 3:

A-II, B-III, C-I, D-IV

Option 4:

A-I, B-II, C-IV, D-III

Correct Answer:

A-IV, B-I, C-III, D-II

Solution:

A. C_2H_4 (Ethene): Contains a $C = C$ double bond and four $C - H$ single bonds. Total: 5σ bonds, 1π bond. (A-IV)

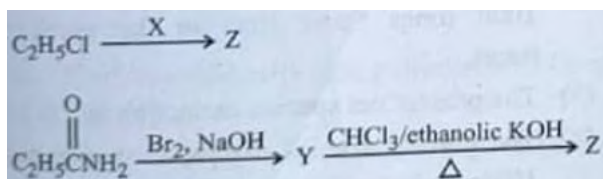
B. C_2H_2 (Ethyne): Contains a $C \equiv C$ triple bond and two $C - H$ single bonds. Total: 3σ bonds, 2π bonds. (B-I)

C. CH_4 (Methane): Contains four $C - H$ single bonds. Total: 4σ bonds. (C-III)

D. NH_3 (Ammonia): Contains three $N - H$ single bonds and one lone pair on Nitrogen. Total: 3 σ bonds, one lone pair. (D-II)

Hence, the correct answer is option (1).

Q. 15 The following two reactions give e^x the same foul smelling product Z .



X and Z respectively, are:

Option 1:

X = AgCN; Z = $\text{C}_2\text{H}_5\text{CN}$

Option 2:

X = AgCN; Z = $\text{C}_2\text{H}_5\text{NC}$

Option 3:

X = KCN; Z = $\text{C}_2\text{H}_5\text{CN}$

Option 4:

X = KCN; Z = $\text{C}_2\text{H}_5\text{NC}$

Correct Answer:

X = AgCN; Z = $\text{C}_2\text{H}_5\text{NC}$

Solution:

$\text{C}_2\text{H}_5\text{Cl} + \text{AgCN} \rightarrow \text{C}_2\text{H}_5\text{NC}$ (Ethyl isocyanide).

AgCN is covalent, so the nitrogen atom acts as the nucleophile, forming an isocyanide.

$\text{C}_2\text{H}_5\text{CONH}_2 \xrightarrow{\text{Br}_2, \text{NaOH}} \text{C}_2\text{H}_5\text{NH}_2$ (Hofmann Bromamide Degradation).

Hence, the correct answer is option (2).

Q. 16 The number of hydrogen atoms present in 5.4 g of urea is :

(Given : Molar mass of urea : 60 g mol^{-1} ,

$N_A : 6.022 \times 10^{23} \text{ particles mol}^{-1}$)

Option 1:

2.168×10^{22}

Option 2:

2.168×10^{23}

Option 3:

$$1.084 \times 10^{22}$$

Option 4:

$$1.084 \times 10^{23}$$

Correct Answer:

$$2.168 \times 10^{23}$$

Solution:

$$\text{Moles} = \frac{\text{Given Mass}}{\text{Molar Mass}} = \frac{5.4 \text{ g}}{60 \text{ g/mol}} = 0.09 \text{ moles}$$

The formula for urea is NH_2CONH_2 .

Each molecule of urea contains 4 hydrogen atoms.

$$\text{Total H atoms} = \text{Moles} \times \text{H atoms per molecule} \times N_A$$

$$\text{Total H atoms} = 0.09 \times 4 \times 6.022 \times 10^{23}$$

$$\text{Total H atoms} = 0.36 \times 6.022 \times 10^{23} \approx \mathbf{2.168 \times 10^{23}}$$

Hence, the correct answer is option (2)

Q. 17 Identify the incorrect statement from the following:

Option 1:

Phosphorus, arsenic and antimony show catenation property.

Option 2:

$\text{P}(\text{C}_2\text{H}_5)_3$ and $\text{As}(\text{C}_6\text{H}_5)_3$ form $d\pi - d\pi$ bond with transition metals.

Option 3:

Nitrogen can form $d\pi - p\pi$ bond with oxygen.

Option 4:

Nitrogen can form $\pi - p\pi$ multiple bonds with itself.

Correct Answer:

Nitrogen can form $d\pi - p\pi$ bond with oxygen.

Solution:

(1) Correct: Catenation (the linking of atoms of the same element into chains) is a property shown by Phosphorus, Arsenic, and Antimony. Phosphorus shows it most significantly (as seen in P_4 or red phosphorus chains). While Nitrogen's catenation is limited due to high inter-electronic repulsion between lone pairs, the heavier elements do exhibit it.

(2) INCORRECT: Organophosphorus and organoarsenic compounds (like phosphines and arsines) act as ligands with transition metals. They form bonds through π -backbonding, but the specific interaction involves the donation of electrons from the filled d-orbitals of the transition metal into the vacant d-orbitals of Phosphorus or Arsenic. Therefore, the bond is a $d\pi(\text{metal}) - d\pi(\text{ligand})$ interaction. However, the statement is often considered incorrect in standard textbooks because the p -block elements use their vacant d-orbitals to accept electrons, but the primary sigma bond is formed by the lone pair. More importantly, in many competitive contexts, the focus is on the fact that Nitrogen cannot do this because it lacks d-orbitals.

(3) Correct: Nitrogen forms $p\pi - p\pi$ multiple bonds with Oxygen (as seen in oxides like NO_2 or NO_3^-). Since both N and O are in the second period, their p -orbitals are small enough to overlap effectively sideways.

(4) Correct: Nitrogen has a very small atomic size and high electronegativity, allowing it to form very strong $p\pi - p\pi$ multiple bonds with itself, resulting in the stable $\text{N} \equiv \text{N}$ triple bond. Heavier elements like Phosphorus cannot do this effectively because their p -orbitals are too large and diffuse to have effective sideways overlap.

Hence, the correct answer is option (3)

Q. 18 Which one of the following is an ambidentate ligand?

Option 1:

Oxalate

Option 2:

Ethane-1,2-diamine

Option 3:

Thiocyanate

Option 4:

Ethylenediaminetetraacetate ion

Correct Answer:

Thiocyanate

Solution:

An ambidentate ligand is a ligand that can coordinate to a central metal atom through two different atoms but only one at a time.

Thiocyanate (SCN^-) can bind through the Sulfur (S) or the Nitrogen (N) atom.

Hence, the correct answer is option (3)

Q. 19 The correct order of increasing metallic character of Na, Be, P, Mg and Si is :

Option 1:

$\text{Be} < \text{Si} < \text{P} < \text{Mg} < \text{Na}$

Option 2:

$\text{P} < \text{Si} < \text{Na} < \text{Mg} < \text{Be}$

Option 3:

$\text{P} < \text{Si} < \text{Be} < \text{Mg} < \text{Na}$

Option 4:

$\text{P} < \text{Mg} < \text{Be} < \text{Si} < \text{Na}$

Correct Answer:

$\text{P} < \text{Si} < \text{Be} < \text{Mg} < \text{Na}$

Solution:

Left to Right metallic character decreases

Top to Bottom metallic character increases

Na, Mg, Si, P: These are all in Period

$\text{P} < \text{Si} < \text{Mg} < \text{Na}$

- Be is in Group 2, located directly above Magnesium. Since metallic character increases downward, Be is less metallic than Mg.
- Beryllium is a metal, while Silicon is a metalloid. Therefore, Si is less metallic than Be.

$\text{P} < \text{Si} < \text{Be} < \text{Mg} < \text{Na}$

Hence, correct answer is option (3)

Q. 20 Match the LIST - I with LIST - II

List - I	List - II
A. $\text{H}_3\text{C}-\text{CH}(\text{C}_6\text{H}_5)-\text{CH}_3 \rightarrow \text{C}_6\text{H}_5\text{OH}$	I. (i) oleum; (ii) NaOH, Δ; (iii) H^+
B. $\text{CH}_3\text{COOH} \rightarrow \text{CH}_3\text{CH}_2\text{OH}$	II. (i) O_2; (ii) $\text{H}_2\text{O}/\text{H}^+$
C. $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \rightarrow \text{CH}_3-\text{CH}(\text{OH})-\text{CH}_3$	III. (i) CH_3OH, H^+; (ii) H_2, catalyst
D. $\text{C}_6\text{H}_6 \rightarrow \text{C}_6\text{H}_5\text{OH}$	IV. (i) conc. H_2SO_4, Δ; (ii) $\text{H}^+/\text{H}_2\text{O}$

Choose the correct answer from the options given below :

Option 1:

A-I, B-III, C-IV, D-II

Option 2:

A-II, B-IV, C-III, D-I

Option 3:

A-II, B-III, C-IV, D-I

Option 4:

A-II, B-III, C-I, D-IV

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

A. Cumene to Phenol: This is the industrial preparation of phenol via cumene hydroperoxide. It requires oxidation with air (O_2) followed by acid hydrolysis.

A $\rightarrow$ II

B. Ethanoic Acid to Ethanol: This is a reduction of a carboxylic acid to a primary alcohol. While LiAlH_4 is common, another route involves esterification with methanol (CH_3OH , H^+) followed by catalytic hydrogenation.

B → III.

C. Propan-1-ol to Propan-2-ol: This requires dehydration to form propene (using conc. H_2SO_4 , Δ) followed by hydration (acid-catalyzed addition of water) to form the more stable secondary alcohol.

C → IV.

D. Benzene to Phenol: This follows the classical sulfonating route: sulfonation with oleum to form benzene sulfonic acid, fusion with NaOH, and final acidification.

D → I

Hence, the correct answer is option (3)

Q. 21 Although +3 oxidation state is most common in lanthanoids, cerium still shows +4 oxidation state because :

Option 1:

After losing one, more electron, it acquires $4f^{14}$ electronic configuration.

Option 2:

Its nearest inert gas is Radon.

Option 3:

After losing one more electron, it acquires $4f^0$ electronic configuration.

Option 4:

Its atomic number is 61 .

Correct Answer:

After losing one more electron, it acquires $4f^0$ electronic configuration.

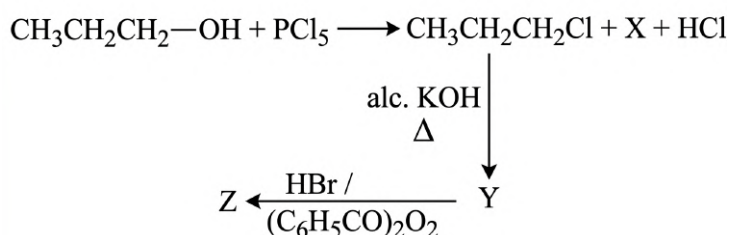
Solution:

Cerium (Ce) has the atomic number 58 . Its ground-state electronic configuration is $[Xe]4f^15d^16s^2$. By losing 4 electrons (to reach the +4 state), it achieves the configuration of the noble gas Xenon ($[Xe]4f^0$).

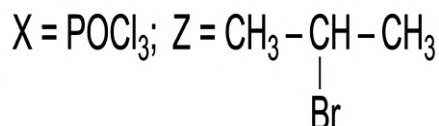
This empty $4f^0$ orbital provides extra stability to the +4 state.

Hence, the correct answer is option (3)

Q. 22 In the following reaction sequence, X and Z , respectively are :



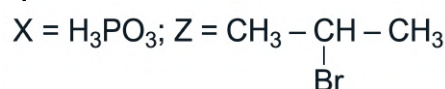
Option 1:



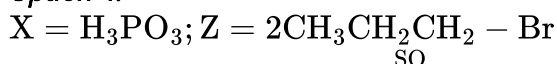
Option 2:



Option 3:



Option 4:



Correct Answer:



Solution:

1. Formation of **X** (Reaction with PCl_5)

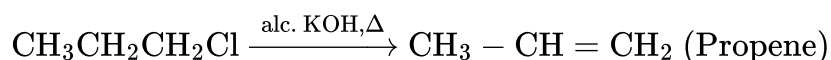
The reaction of propan-1-ol with phosphorus pentachloride (PCl_5) is as follows:



- The byproduct **X** is phosphorus oxychloride (POCl_3).

2. Formation of **Y** (Dehydrohalogenation)

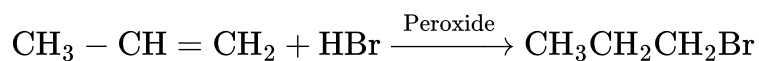
Treating the 1-chloropropane with alcoholic KOH and heat (Δ) causes an **E2** elimination reaction:



- Product **Y** is Propene.

3. Formation of **Z** (Peroxide Effect)

The final step is the addition of HBr in the presence of benzoyl peroxide $(\text{C}_6\text{H}_5\text{CO})_2\text{O}_2$. This is the AntiMarkovnikov addition (also known as the Kharasch effect or Peroxide effect).



The bromine atom attaches to the terminal carbon (the one with more hydrogens).

Product **Z** is 1-bromopropane.

Hence, the correct answer is option (2)

Q. 23 Match List I with List II

List - I (Complex/ion)	List -II (Shape/geometry)
A. $[\text{Pt}(\text{Cl}_2)(\text{NH}_3)_2]$	I. Octahedral
B. $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$	II. Trigonal bipyramidal
C. $[\text{NiCl}_4]^{2-}$	III. Square planar
D. $[\text{Fe}(\text{CO})_5]$	IV. Tetrahedral

Choose the **correct** answer from the options given below:

Option 1:

A-I, B-III, C-IV, D-II

Option 2:

A-III, B-IV, C-I, D-IX

Option 3:

A-III, B-I, C-IV, D-IV-

Option 4:

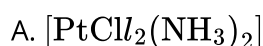
A-IV, B-I, C-III, D-IX

Correct Answer:

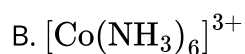
A-III, B-I, C-IV, D-IV-

Solution:

(3) A-III, B-I, C-IV, D-II



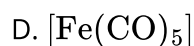
Pt^{2+} is a d^8 system, which typically forms square planar complexes $\rightarrow$ III



Co^{3+} with 6 ligands gives coordination number 6, so octahedral $\rightarrow$ I



Ni^{2+} with weak field ligand Cl^- forms a tetrahedral complex $\rightarrow$ IV



This is a classic trigonal bipyramidal structure $\rightarrow$ II

Hence, the correct answer is option (3)

Q. 24 The functional group that can be identified through phthalein dye test is :

Option 1:

Carboxylic acid

Option 2:

Alcohol

Option 3:

Aldehyde

Option 4:

Phenolic

Correct Answer:

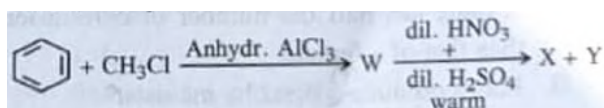
Phenolic

Solution:

The phthalein dye test is used to identify the phenolic group. When a phenol is heated with phthalic anhydride in the presence of concentrated H_2SO_4 , it forms a phthalein dye (like phenolphthalein) which turns pink/red in alkaline solution.

Hence, the correct answer is option (4)

Q. 25 Two products X and Y are formed in the following reaction sequence.



The suitable method that can be used for the separation of products X and Y is :

Option 1:

Continuous extraction

Option 2:

Differential extraction

Option 3:
Sublimation

Option 4:
Fractional distillation

Correct Answer:
Fractional distillation

Solution:

Benzene + CH_3Cl with $\text{AlCl}_3 \rightarrow$ Toluene (W).

Toluene + nitrating mixture ($\text{HNO}_3 + \text{H}_2\text{SO}_4$) $\rightarrow$ o-nitrotoluene (X) and p-nitrotoluene (Y).

Separation Method:

X and Y are structural isomers with different boiling points. While steam distillation is commonly used for similar separations (like o- and p-nitrophenols), fractional distillation is the most suitable general industrial method for separating these two components based on their boiling point difference.

Hence, the correct answer is option (4)

Q. 26 Identify the correct statements :

A. The molality of 2.5 g of ethanoic acid (Molar mass : 60 g mol^{-1}) in 75 g of benzene solution is 0.556.

B. The molarity of a solution containing 5 g of NaOH (molar mass : 40 g mol^{-1}) in 450 mL of solution is 0.278 M at 298 K .

C. Aquatic species are more comfortable in cold water.

D. The solubility of gas increases with decrease in pressure.

E. For a binary mixture of A and B , the number of moles of A and B are n_A and n_B respectively. The mole fraction of B will be $x_B = \frac{n_B}{n_A + n_B} =$

Choose the correct answer from the options given below:

Option 1:
A, B and C only

Option 2:
A, D and E only

Option 3:
A and B only

Option 4:
A and C only

Correct Answer:
A, B and C only

Solution:

Statement A (True): Molality (m) = $\frac{\text{moles of solute}}{\text{mass of solvent in kg}}$.

$$\text{Moles of ethanoic acid} = \frac{2.5}{60} = 0.0417.$$

$$m = \frac{0.0417}{0.075 \text{ kg}} = 0.556 \text{ m}$$

Statement B (True): Molarity (M) = $\frac{\text{moles of solute}}{\text{Volume of solution in L}}$.

$$\text{Moles of NaOH} = \frac{5}{40} = 0.125.$$

$$M = \frac{0.125}{0.450 \text{ L}} = 0.278 \text{ M}$$

Statement C (True): The solubility of gases (like O_2) increases as temperature decreases. Therefore, aquatic species are more comfortable in cold water due to higher dissolved oxygen levels.

Statement D (False): According to Henry's Law, the solubility of a gas increases with an increase in pressure.

Statement E (False): The mole fraction of B is $x_B = \frac{n_B}{n_A + n_B}$. The provided formula is for x_A .

Hence, the correct answer is option (1)

Q. 27 During Lassaigne's test, the elements present in an organic compound are converted from :

Option 1:

ionic form to ionic form

Option 2:

covalent form to ionic form

Option 3:

ionic form to covalent form

Option 4:

covalent form to covalent form

Correct Answer:

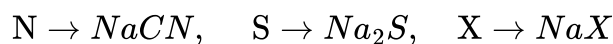
covalent form to ionic form

Solution:

In organic compounds, elements like nitrogen, sulphur and halogens are present in covalent form, so they are not easily detectable by ordinary chemical tests.

During Lassaigne's test, the organic compound is fused with sodium metal. Sodium is highly reactive and converts these covalently bonded elements into ionic (water-soluble) compounds, which can be easily detected.

Examples:



Thus, in Lassaigne's test, elements are converted from:

Covalent form $\rightarrow$ Ionic form

Hence, the correct answer is option (2)

- Q. 28** A solution of copper sulphate is electrolysed for 10 minutes with a current of 1.5 amperes. The mass of copper deposited at cathode is :
(Given : Molar mass of Cu = 63 g mol⁻¹; 1 F = 96487 C mol⁻¹)

Option 1:

0.2938 g

Option 2:

0.5876 g

Option 3:

2.4036 g

Option 4:

1.7018 g

Correct Answer:

0.2938 g

Solution:

79: Total charge (Q) :

$$Q = I \times t = 1.5 \text{ A} \times (10 \times 60) \text{ s} = 900 \text{ C}$$

The reaction at the cathode is $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}(s)$.

$$\text{Mass deposited } (m) = \frac{\text{Molar mass} \times Q}{n \times F}$$

$$m = \frac{63 \times 900}{2 \times 96487} \approx \mathbf{0.2938 \text{ g}}$$

Hence, the correct answer is option (1)

- Q. 29** At a certain temperature, T(K), during a process, 500 J is absorbed by the system and work of 200 J is done by the system. Then change in internal energy of the system is :

Option 1:

700 J

Option 2:

400 J

Option 3:

300 J

Option 4:

500 J

Correct Answer:

300 J

Solution:

$$\Delta U = q - w$$

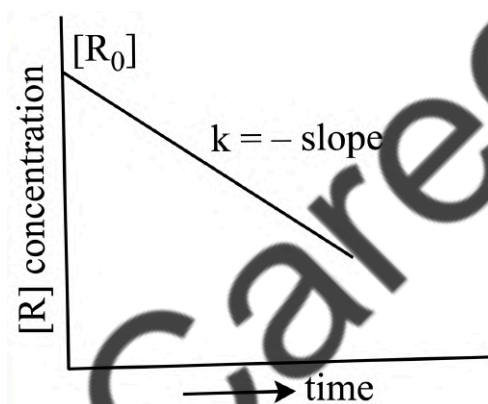
$q = +500J$ (heat absorbed $\rightarrow$ positive)

$w = +200J$ (work done by system)

$$\Delta U = 500 - 200 = 300 \text{ J}$$

Hence, the correct answer is option (3)

Q. 30 For a certain reaction $R \rightarrow \text{Product}$, the plot of concentration $[R]$ vs time has a negative slope as shown. The order of reaction is :



Option 1:

1

Option 2:

2.5

Option 3:

2

Option 4:

0

Correct Answer:

0

Solution:

Zero order: $[R] = [R_0] - kt \rightarrow$ straight line (slope = $-k$)

Hence, the correct answer is option (4)

Q. 31 Identify the correct statement about ClF_3 from the following options :

Option 1:

It has T-shaped geometry with three lone pairs on Cl atom.

Option 2:

It has T-shaped geometry with two lone pairs on Cl atom.

Option 3:

It has a trigonal pyramidal geometry with two lone pairs on Cl atom.

Option 4:

It has a planar trigonal geometry with two lone pairs on Cl atom.

Correct Answer:

It has T-shaped geometry with two lone pairs on Cl atom.

Solution:

- Central atom: Cl (7 valence electrons)
- Bonded atoms: 3 F atoms
- Total electron pairs = 5
- Electron geometry $\rightarrow$ Trigonal bipyramidal
- Two lone pairs occupy equatorial positions (to minimise repulsion)
- T-shaped geometry
- Lone pairs = 2
- Bond pairs = 3

It has T-shaped geometry with two lone pairs on the Cl atom

Hence, the correct answer is option (2)

Q. 32 In a test tube containing a salt, a few drops of dilute H_2SO_4 was added, which gave colourless vapours having the smell of vinegar. The vapours turned the blue litmus paper red.

Identify the correct anion from the following :

Option 1:

Carbonate, CO_3^{2-}

Option 2:

Sulphide, S^{2-}

Option 3:

Acetate, CH_3COO^-

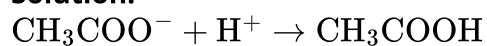
Option 4:

Sulphate, SO_4^{2-}

Correct Answer:

Acetate, CH_3COO^-

Solution:



Acetic acid vapours:

- Are **colourless**
- Have **vinegar smell**
- Turn **blue litmus red (acidic nature)**

Hence, the correct answer is option (3)

Q. 33 At 298 K, a certain buffer solution contains equal concentrations of X^- and HX , K_b for X^- is 10^{-10} . What is the pH of this buffer solution?

Option 1:

10

Option 2:

4

Option 3:

2

Option 4:

6

Correct Answer:

4

Solution:

For a basic buffer, we use:

$$pOH = pK_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

Since concentrations are equal:

$$\log \frac{[\text{salt}]}{[\text{base}]} = \log 1 = 0$$

So,

$$\text{pOH} = \text{p}K_b$$

Given:

$$K_b = 10^{-10} \Rightarrow \text{p}K_b = 10$$

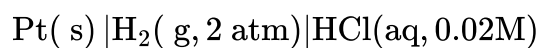
$$\text{pOH} = 10$$

Now:

$$\text{pH} = 14 - \text{pOH} = 14 - 10 = 4$$

Hence, the correct answer is option (2)

Q. 34 Calculate emf of the half cell given below :



$$E_{\text{H}_2/\text{H}^+}^\circ = 0 \text{ V}$$

$$\left(\text{Given : } \frac{2.303RT}{F} 0.059, \right.$$

$$\left. \log 2 = 0.3010 \right)$$

Option 1:

$$-0.109 \text{ V}$$

Option 2:

$$0.109 \text{ V}$$

Option 3:

$$0.035 \text{ V}$$

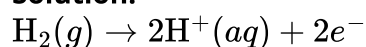
Option 4:

$$-0.035 \text{ V}$$

Correct Answer:

$$0.109 \text{ V}$$

Solution:



$$E_{\text{ox}} = E_{\text{ox}}^{\circ} - \frac{0.059}{n} \log Q$$

$$E_{\text{ox}} = 0 - \frac{0.059}{2} \log \left(\frac{(0.02)^2}{2} \right)$$

$$E_{\text{ox}} = -0.0295 \log \left(\frac{0.0004}{2} \right)$$

$$E_{\text{ox}} = -0.0295 \log(0.0002)$$

$$\log(2 \times 10^{-4}) = \log 2 + \log 10^{-4}$$

$$\log(2 \times 10^{-4}) = 0.3010 - 4 = -3.699$$

$$E_{\text{ox}} = -0.0295 \times (-3.699)$$

$$E_{\text{ox}} \approx +0.109 \text{ V}$$

Hence, the correct answer is option (2)

Q. 35 The calculated 'spin-only' magnetic moment of Ti^{2+} ($3d^2$) is :

Option 1:

3.87 BM

Option 2:

4.90 BM

Option 3:

2.84 BM

Option 4:

5.92 BM

Correct Answer:

2.84 BM

Solution:

The formula for 'spin-only' magnetic moment (μ) is:

$$\mu = \sqrt{n(n+2)} \text{ BM}$$

where n is the number of unpaired electrons.

For Ti^{2+} (configuration $3d^2$), there are 2 unpaired electrons ($n = 2$).

$$\mu = \sqrt{2(2 + 2)} = \sqrt{2 \times 4} = \sqrt{8}$$

$$\mu \approx 2.83 \text{ BM}$$

Hence, the correct answer is option (3)

Q. 36 Identify the incorrect statement from the following :

Option 1:

Oxygen exhibits only -2 oxidation state.

Option 2:

The order of catenation property of Group 14 elements is $\text{C} \gg \text{Si} > \text{Ge} \approx \text{Sn}$.

Option 3:

Carbon has the ability to form $p\pi - p\pi$ multiple bond with itself.

Option 4:

ECl_3 ($\text{E} = \text{B}$ and Al) is a monomer when $\text{E} = \text{B}$ and a dimer when $\text{E} = \text{Al}$.

Correct Answer:

Oxygen exhibits only -2 oxidation state.

Solution:

Statement (1) is Incorrect: Oxygen does not exhibit only the -2 oxidation state. It shows +2 in OF_2 , +1 in O_2F_2 , -1 in peroxides (H_2O_2), and $-1/2$ in superoxides (KO_2).

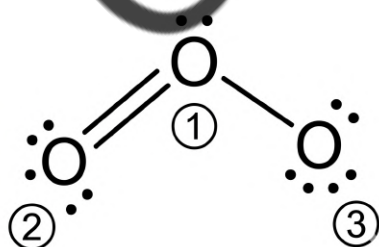
Statement (2) is Correct: Catenation ability decreases down the group as bond enthalpy decreases.

Statement (3) is Correct: Carbon frequently forms $p\pi - p\pi$ multiple bonds ($\text{C} = \text{C}$, $\text{C} \equiv \text{C}$).

Statement (4) is Correct: BCl_3 is a monomer (trigonal planar), while AlCl_3 achieves stability by forming a dimer (Al_2Cl_6).

Hence, the correct answer is option (1)

Q. 37



The correct formal charges on oxygen atoms numbered 2, 1 and 3 respectively are :

Option 1:

0, 0, 0

Option 2:

-1, 0, +1

Option 3:

+1, 0, -1

Option 4:

0, +1, -1

Correct Answer:

0, +1, -1

Solution:

O(2) (double bonded terminal O):

- Lone pairs = $4e^-$

- Bonding = $4e^-$

$$FC = 6 - 4 - 2 = 0$$

O(1) (central O):

- Lone pairs = $2e^-$

- Bonding = $6e^-$

$$FC = 6 - 2 - 3 = +1$$

O(3) (single bonded terminal O):

Lone pairs = $6e^-$

Bonding = $2e^-$

$$FC = 6 - 6 - 1 = -1$$

Hence, the correct answer is option (4)

Q. 38 Phenolphthalein is used as an indicator for the titration of sodium hydroxide solution against a standard solution of oxalic acid. The colour change that is observed at an alkaline pH close to the equivalence point during this titration is :

Option 1:

pink to colourless

Option 2:

pinkish red to yellow

Option 3:

colourless to pink

Option 4:

yellow to pinkish red

Correct Answer:
colourless to pink

Solution:

In a titration of Oxalic acid (standard solution in the flask) against Sodium hydroxide (base in the burette). The solution starts as acidic (colorless). As the equivalence point is reached and the solution becomes slightly alkaline, the phenolphthalein changes color. The color change observed is colourless to pink.

Hence, the correct answer is option (3)

Q. 39 When 1dm^3 of CO_2 gas is passed over hot coke, the volume of gaseous mixture after complete reaction at STP becomes 1.4dm^3 . The composition of the gaseous mixture at STP is :

Option 1:

0.6dm^3 of CO, 0.8dm^3 of CO_2

Option 2:

0.8dm^3 of CO, 0.8dm^3 of CO_2

Option 3:

0.6dm^3 of CO, 0.4dm^3 of CO_2

Option 4:

0.8dm^3 of CO, 0.6dm^3 of CO_2

Correct Answer:

0.8dm^3 of CO, 0.6dm^3 of CO_2

Solution:

Reaction of CO_2 with hot coke:



Let:

$x\text{dm}^3$ of CO_2 reacts

Initial $\text{CO}_2 = 1\text{dm}^3$

After reaction:

Remaining $\text{CO}_2 = 1 - x$

CO formed $= 2x$

Total final volume:

$$(1 - x) + 2x = 1 + x$$

Given:

$$1 + x = 1.4 \Rightarrow x = 0.4$$

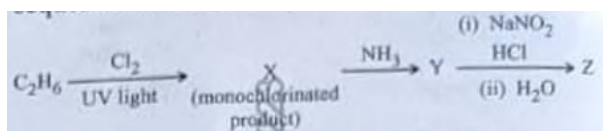
Now calculate composition:

$$\text{CO}_2 = 1 - 0.4 = 0.6\text{dm}^3$$

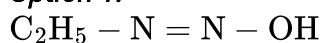
$$\text{CO} = 2 \times 0.4 = 0.8\text{dm}^3$$

Hence, the correct answer is option (4)

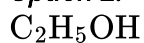
Q. 40 The major product Z formed in the following sequence of reactions is :



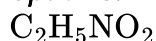
Option 1:



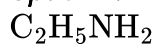
Option 2:



Option 3:



Option 4:



Correct Answer:



Solution:

Step 1: Ethane (C_2H_6) undergoes free radical chlorination with $\text{Cl}_2/\text{UV light}$ to form Chloroethane (X): $\text{C}_2\text{H}_5\text{Cl}$.

Step 2: Chloroethane reacts with Ammonia (NH_3) via nucleophilic substitution ($\text{S}_\text{N}2$) to form Ethylamine (Y) : $\text{C}_2\text{H}_5\text{NH}_2$.

Step 3: Ethylamine reacts with nitrous acid ($\text{NaNO}_2 + \text{HCl}$) to form an unstable diazonium salt, which immediately reacts with water (H_2O) to form Ethanol (Z): $\text{C}_2\text{H}_5\text{OH}$.

Hence, the correct answer is option (2)

Q. 41 Given below is an expression for the rate constant of a first order reaction occurring at a certain temperature, T(K).

$$\ln k = 14.34 - \frac{1.25 \times 10^4}{T}$$

The energy of activation in kcal mol^{-1} for the reaction is :
(Given : k in s^{-1} , $R = 1.987 \text{ cal mol}^{-1} \text{ K}^{-1}$)

Option 1:

12.42

Option 2:

18.63

Option 3:

14.34

Option 4:

24.84

Correct Answer:

24.84

Solution:

$$52. \ln k = \ln A - \frac{E_a}{RT}$$

$$\frac{E_a}{R} = 1.25 \times 10^4$$

$$E_a = (1.25 \times 10^4) \times R$$

$$R = 1.987 \text{ cal mol}^{-1} \text{ K}^{-1}$$

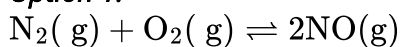
$$E_a = 12500 \times 1.987 = 24837.5 \text{ cal mol}^{-1}$$

$$E_a = \frac{24837.5}{1000} \approx 24.84 \text{ kcal mol}^{-1}$$

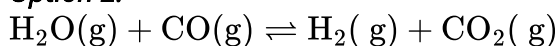
Hence, the correct answer is option (4)

Q. 42 Given below are certain reactions. Identify the reaction for which $K_p \neq K_c$.

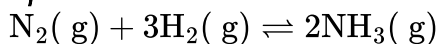
Option 1:



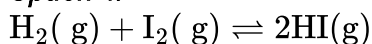
Option 2:



Option 3:



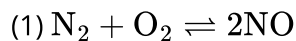
Option 4:



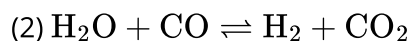
Correct Answer:



Solution:



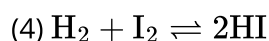
$$\Delta n = 2 - (1 + 1) = 0 \rightarrow K_p = K_c$$



$$\Delta n = 2 - 2 = 0 \rightarrow K_p = K_c$$



$$\Delta n = 2 - 4 = -2 \rightarrow K_p \neq K_c$$



$$\Delta n = 2 - 2 = 0 \rightarrow K_p = K_c$$

Hence, the correct answer is option (3)

Q. 43 Identify the incorrect statement from the following:

Option 1:

The largest and the smallest species among Mg , Mg^{2+} , Al and Al^{3+} are Al and Mg^{2+} , respectively.

Option 2:

The IUPAC name of the element with atomic number 107 is Unnilseptium.

Option 3:

The similarity in behaviour of Li with Mg is referred to as 'diagonal relationship'.

Option 4:

The oxidation state and covalency of Al in $[\text{AlCl}(\text{H}_2\text{O})_5]^{2+}$ are 3 and 6 respectively.

Correct Answer:

The largest and the smallest species among Mg , Mg^{2+} , Al and Al^{3+} are Al and Mg^{2+} , respectively.

Solution:

Statement (1) is Incorrect: Between Mg and Al , Mg is larger because atomic size decreases across a period. Between Mg^{2+} and Al^{3+} , Al^{3+} is the smallest because it has more protons pulling the same number of electrons. Therefore, the largest is Mg and the smallest is Al^{3+} .

Statement (2) is Correct: Atomic number 107 (1 = Un, 0=nil, 7=sept) is Unnilseptium.
Statement (3) is Correct: Li and Mg show diagonal relationship due to similar ionic sizes and electronegativities.

Statement (4) is Correct: In $[\text{AlCl}(\text{H}_2\text{O})_5]^{2+}$, let x be the oxidation state of Al :
 $x + (-1) + 5(0) = +2 \Rightarrow x = +3$. The coordination number (covalency) is 6 (1Cl + 5H₂O ligands).

Hence, the correct answer is option (1)

Q. 44 Mixture of chloroform and acetone forms a solution with negative deviation from Raoult's law due to :

Option 1:

stronger intermolecular forces between chloroform molecules than those between chloroform and acetone molecules.

Option 2:

formation of hydrogen bonding between acetone and chloroform.

Option 3:

repulsive forces.

Option 4:

increase in escaping tendency of molecules of each component.

Correct Answer:

formation of hydrogen bonding between acetone and chloroform.

Solution:

- Chloroform (CHCl₃) has a slightly acidic hydrogen
- Acetone (CH₃COCH₃) has a lone pair on oxygen

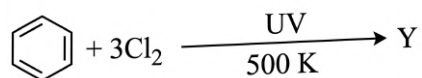
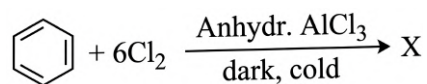
This leads to hydrogen bonding between chloroform and acetone molecules which results:

- Strong A-B interactions → molecules are held more tightly
- Vapour pressure decreases → negative deviation

Hydrogen bonding between chloroform and acetone

Hence, the correct answer is option (2)

Q. 45 The number of chlorine atoms present in the organic products X and Y of the following reactions, respectively, are :



Option 1:

3 and 3

Option 2:

6 and 6

Option 3:

6 and 3

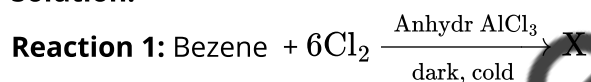
Option 4:

3 and 6

Correct Answer:

6 and 6

Solution:



Conditions: Lewis acid (AlCl_3) + dark $\rightarrow$ Electrophilic substitution

Benzene undergoes complete chlorination $\rightarrow$ Hexachlorobenzene (C_6Cl_6)

Number of Cl atoms in **X** = **6**



High temperature ($\sim 500 \text{ K}$) $\rightarrow$ Addition reaction

Forms Benzene hexachloride (BHC/ $\text{C}_6\text{H}_6\text{Cl}_6$)

Number of Cl atoms in **Y** = **6**

Hence, the correct answer is option (2)

Biology

Q. 1 In angiosperms, root hairs arise from which one of the following regions of the root?

Option 1:

The region of elongation

Option 2:

The region of meristematic activity

Option 3:

The region of maturation

Option 4:

The root cap zone

Correct Answer:

The region of maturation

Solution:

Root hairs develop from epidermal cells in the zone of maturation (differentiation).

This greatly increases the absorptive surface area of roots, helping in the efficient uptake of water and minerals. They are absent in the meristematic and elongation zones, which are primarily for cell division and growth.

Hence, the correct answer is option 3) The region of maturation.

Q. 2

In which one of the following is the ovules are not enclosed by an ovary wall and remains exposed?

Option 1:

Pinus

Option 2:

Wolffia

Option 3:

Funaria

Option 4:

Selaginella

Correct Answer:

Pinus

Solution:

In gymnosperms like Pinus, ovules are naked (not enclosed in an ovary) because they are borne directly on megasporophylls without being enclosed by an ovary wall.

This is a key distinction between gymnosperms and angiosperms, where ovules are protected inside the ovary. The exposed ovules of Pinus later develop into seeds after fertilisation.

Hence, the correct answer is option 1) Pinus.

Q. 3 In the lac operon, the *z* gene codes for :

Option 1:

the repressor of lac operon

Option 2:

transacetylase

Option 3:

permease

Option 4:

beta-galactosidase

Correct Answer:

beta-galactosidase

Solution:

The lacZ gene encodes β -galactosidase, which hydrolyses lactose into glucose and galactose.

This enzymatic activity enables bacteria to utilise lactose as a carbon source when glucose is scarce. Together with lacY (permease) and lacA (transacetylase), lacZ forms the structural gene set of the lac operon.

Hence, the correct answer is option 4) beta-galactosidase.

Q. 4 Exploring molecular, genetic and species-level diversity for products of economic importance is called :

Option 1:

Bioprospecting

Option 2:

Biofortification

Option 3:

Biomagnification

Option 4:

Bioremediation

Correct Answer:
Bioprospecting

Solution:

Bioprospecting involves searching biological diversity at molecular, genetic, and species levels to identify useful products of economic importance.

Term	Definition	Example Use
Bioprospecting	Exploring biodiversity for economic products	Drugs from medicinal plants
Biomagnification	Toxin concentration increases in the food chain	Mercury in fish
Bioremediation	Microbes clean pollutants	Oil spill cleanup
Biofortification	Nutrient-rich crop breeding	Iron-fortified rice

Hence, the correct answer is option 1) Bioprospecting.

Q. 5 Match List I with List II :

	List I		List II
A.	Genetically modified organism	I.	Agrobacterium tumefaciens
B.	Thermostable DNA polymerase	II.	Bt cotton
C.	Ti plasmid	III.	Thermus aquaticus
D.	pBR322	IV.	Escherichia coli

Choose the correct answer from the options given below:

Option 1:

A-II, B-I, C-I , D-III

Option 2:

A-II, B-III, G-I, D-IV

Option 3:

A-I, B-IV, C-III, D-II

Option 4:

A-I, B-II, C-IV, D-III

Correct Answer:

A-II, B-III, G-I, D-IV

Solution:

- Bt cotton is a genetically modified organism (A → II)
- Thermostable DNA polymerase comes from Thermus aquaticus (B → III)
- Ti plasmid is from Agrobacterium tumefaciens (C → I)

- pBR322 is a cloning vector (D → IV)

Hence, the correct answer is option 2) A-II, B-III, G-I, D-IV.

Q. 6 Match List I with List II :

	List I		List II
A.	Productivity	I.	Gross primary productivity minus respiration losses
B.	Net primary productivity	II.	Rate of formation of new organic matter by consumers
C.	Gross primary productivity	III.	Rate of biomass production
D.	Secondary productivity	IV.	The rate of production of organic matter during photosynthesis

Choose the correct answer from the options given below :

Option 1:

A-III, B-I, C-II, D-IV

Option 2:

A-I, B-II, C-III, D-IV

Option 3:

A-III, B-I, C-IV, D-II

Option 4:

A-I, B-III, C-IV, D-II

Correct Answer:

A-III, B-I, C-IV, D-II

Solution:

- Productivity = biomass production rate (III)
- $NPP = GPP - R$ (I)
- GPP is the total photosynthetic production (IV)
- Secondary productivity is production by consumers (II).

Hence, the correct answer is option 3) A-III, B-I, C-IV, D-II.

Q. 7

Since the origin and diversification of life on Earth, there have been five episodes of mass extinction of species. How is the sixth extinction, which is in progress, different from the previous episodes?

Option 1:

The current species extinction rates are far lower than those in previous episodes.

Option 2:

The present species extinction rates are 100 to 1000 times faster than in pre-human times.

Option 3:

The present net species extinction rate is zero.

Option 4:

The current species extinction rate is nearly 10 times faster than that in previous episodes.

Correct Answer:

The present species extinction rates are 100 to 1000 times faster than in pre-human times.

Solution:

Earth has witnessed five major mass extinctions in the past, caused by natural events (volcanic eruptions, asteroid impacts, climate shifts).

The ongoing (sixth) extinction is largely human-driven, and the extinction rate is estimated to be 100–1000× higher than the natural background rate. It is human-driven due to deforestation, habitat loss, pollution, overexploitation, and climate change.

Hence, the correct answer is option 2) The present species extinction rates are 100 to 1000 times faster than in pre-human times.

Q. 8 The alpha-helix is found in which level of protein structure?

Option 1:

Secondary structure

Option 2:

Primary structure

Option 3:

Tertiary structure

Option 4:

Quaternary structure

Correct Answer:

Secondary structure

Solution:

The alpha helix is a regular folding pattern stabilised by hydrogen bonds between the backbone atoms of the polypeptide chain. This makes it a hallmark of the secondary structure of proteins.

- Primary - Amino acid sequence

- Secondary - alpha-helix & beta-sheet (local folding)
- Tertiary - Overall 3D folding of one polypeptide
- Quaternary - Association of multiple polypeptides

Hence, the correct answer is option 1) Secondary structure.

Q. 9 The main function of bulliform cells in grasses is:

Option 1:

to perform photosynthesis.

Option 2:

to minimise water loss during water stress.

Option 3:

to make the leaf impermeable to fungal spores.

Option 4:

to transport water.

Correct Answer:

to minimise water loss during water stress.

Solution:

Bulliform cells help in leaf rolling during water stress, reducing transpiration and conserving water. During water stress, they lose turgidity, causing the leaf to roll inward and reduce the exposed surface area. This rolling mechanism helps minimise transpiration and conserve water.

They do not perform photosynthesis or transport water, but act as a protective adaptation against drought.

Hence, the correct answer is option 2) to minimise water loss during water stress.

Q. 10 Identify the correct sequence of steps in each cycle of Polymerase Chain Reaction :

Option 1:

Denaturation → Extension → Annealing

Option 2:

Denaturation → Annealing → Extension

Option 3:

Annealing → Denaturation → Extension

Option 4:

Extension → Annealing → Denaturation

Correct Answer:

Denaturation → Annealing → Extension

Solution:

PCR follows: denaturation (strand separation) → annealing (primer binding) → extension (DNA synthesis).

These together amplify DNA exponentially. This cycle is repeated multiple times (typically 25–35 cycles) to generate millions of copies of the target DNA sequence.

Hence, the correct answer is option 2) Denaturation → Annealing → Extension.

Q. 11 Match List I with List II :

	List I		List II
	(Phase of cell cycle)		(Activity)
A.	G_1 phase	I.	Actual cell division occurs
B.	S phase	II.	The cell is metabolically active and continuously grows, but does not replicate its DNA
C.	G_2 phase	III.	Synthesis of DNA occurs, and the amount of DNA per cell doubles
D.	M phase	IV.	Proteins are synthesised while cell growth continues

Choose the correct answer from the options given below:

Option 1:

A-III, B-IV, C-I, D-II

Option 2:

A-IV, B-I, C-II, D-III

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-II, B-III, C-IV, D-1

Correct Answer:

A-II, B-III, C-IV, D-1

Solution:

- G_1 is growth without DNA replication (*II*)
- S phase involves DNA synthesis (*III*)
- G_2 prepares for division with protein synthesis (*IV*)
- M phase is the actual cell division (*I*).

Hence, the correct answer is option 4) A-II, B-III, C-IV, D-1.

Q. 12 Which of the following statements are correct?

- A. The Amazon rainforest being cut and cleared for the cultivation of soybeans is an example of habitat loss.
- B. Steller's sea cow and passenger pigeon became extinct due to over-exploitation by humans.
- C. The Nile perch introduced into Lake Victoria in East Africa helped in the population growth of cichlid fish in the lake.
- D. Water hyacinth is an invasive species.
- E. When a species becomes extinct, the plant and animal species associated with it are not affected.

Choose the correct answer from the options given below:

Option 1:

A, B and D only

Option 2:

B, C and D only

Option 3:

A, B and E only

Option 4:

C, D and E only

Correct Answer:

A, B and D only

Solution:

- The Amazon rainforest example highlights habitat loss and fragmentation, the most severe driver of biodiversity decline.
- Species like Steller's sea cow and passenger pigeon vanished due to human over-exploitation.
- The introduction of alien species, such as water hyacinth, disrupts native ecosystems and reduces biodiversity.

- Nile perch actually caused the decline of cichlid fish, and co-extinctions occur when dependent species are lost together.

A, B, and D are correct; Nile perch reduced cichlid populations (C false), and extinction affects associated species (E false).

Hence, the correct answer is option 1) A, B and D only.

- Q. 13** Which of the following statements are correct with reference to a transcription unit?
- A. A transcription unit in DNA is defined primarily by three regions: promoter, structural gene and terminator
 - B. The promoter is said to be located towards the 5'-end of the structural gene
 - C. The promoter is a DNA sequence that provides binding site for RNA polymerase
 - D. The promoter defines the template and coding strands
 - E. The terminator is located towards the 3'-end of the coding strand and it defines the end of the process of transcription
- Choose the correct answer from the options given below:

Option 1:

B, C, D and E only

Option 2:

A, B, C, D and E

Option 3:

A, B, C and D only

Option 4:

A, C, D and E only

Correct Answer:

A, B, C, D and E

Solution:

A transcription unit has a promoter, a structural gene, and a terminator.

The promoter lies upstream at the 5' end and binds RNA polymerase. It determines template/coding strands, and the terminator marks the transcription end. The promoter region ensures correct initiation of transcription by RNA polymerase, while the terminator signals its end. Together with the structural gene, these elements define the functional transcription unit in both prokaryotes and eukaryotes.

Hence, the correct answer is option 2) A, B, C, D and E.

- Q. 14** Which one of the following statements is not true about the universal rules of binomial nomenclature?

Option 1:

The first word in the biological name represents the specific epithet, while the second component denotes the genus.

Option 2:

The specific epithet in the biological name starts with a small letter.

Option 3:

Both the words in a biological name, when handwritten, are separately underlined or printed in italics.

Option 4:

Biological names are generally in Latin.

Correct Answer:

The first word in the biological name represents the specific epithet, while the second component denotes the genus.

Solution:

According to the universal rules, the first word denotes the genus and always begins with a capital letter, while the second word is the specific epithet written in small letters. Both words are either italicised in print or underlined separately when handwritten to maintain uniformity. Biological names are generally derived from Latin or are Latinized, ensuring global recognition.

The first word represents the genus and the second is the specific epithet, so statement (1) is incorrect.

Hence, the correct answer is option 1) The first word in the biological name represents the specific epithet, while the second component denotes the genus.

Q. 15 Match List I with List II :

	List I		List II
A.	Decomposition	I.	Accumulation of dark Coloured amorphous colloidal substance
B.	Detritus	II.	Release of inorganic nutrients by the activity of microbes in soil
C.	Mineralisation	III.	Breaking down complex organic matter into inorganic substances
D.	Humification	IV.	Dead remains of plants and animals, including faecal matter

Choose the correct answer from the options given below:

Option 1:

A-III, B-II, C-I, D-IV

Option 2:

A-IV, B-III, C-I, D-II

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-III, B-IV, C-II, D-I

Correct Answer:

A-III, B-IV, C-II, D-I

Solution:

- Decomposition = breakdown (III)
- Detritus = dead remains (IV)
- Mineralisation = release of inorganic nutrients (II)
- Humification = formation of humus (I).

This matching shows the sequential processes of decomposition in soil ecology. Dead organic matter (detritus) is broken down, nutrients are mineralised, and humus is formed, enriching soil fertility.

Hence, the correct answer is option 4) A-III, B-IV, C-II, D-I.

Q. 16 Which one of the following is the site for active ribosomal RNA synthesis?

Option 1:

Nucleolus

Option 2:

Kinetochore

Option 3:

Centrosome

Option 4:

Chromatin

Correct Answer:

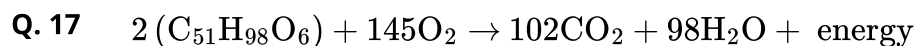
Nucleolus

Solution:

rRNA is synthesised in the nucleolus, where ribosome assembly begins. It also serves as the site for assembly of ribosomal subunits, which later move to the cytoplasm for protein synthesis.

Other nuclear structures, like the kinetochore, centrosome, and chromatin, have different roles and are not involved in rRNA synthesis.

Hence, the correct answer is option 1) Nucleolus.



The Respiratory Quotient (RQ) of a biomolecule used for respiration, as per the above equation, would be :

Option 1:

Less than 0.5

Option 2:

Between 1.25 and 2

Option 3:

1.0

Option 4:

Between 0.5 and 0.95

Correct Answer:

Between 0.5 and 0.95

Solution:

$$RQ = \frac{CO_2 \text{ produced}}{O_2 \text{ consumed}} = \frac{102}{145} \approx 0.7, \text{ thus lying between 0.5 and 0.95.}$$

Since the RQ value is less than 1, it clearly indicates the substrate is a fat rather than a carbohydrate. Fats require more oxygen for complete oxidation, hence their RQ typically lies between 0.7 and 0.8, matching the calculated value here.

Hence, the correct answer is option 4) Between 0.5 and 0.95.

Q. 18 Match List I with List II :

List - I	List - II
A. Incomplete dominance	I. Human skin colour
B. Co-dominance	II. Inheritance of flower colour in Antirrhinum sp.
C. Pleiotropy	III. Phenylketonuria disease in humans
D. Polygenic inheritance	IV. ABO blood groups

Choose the correct answer from the options given below :

Option 1:

A-I, B-IV, C-III, D-II

Option 2:

A-I, B-III, C-II, D-IV

Option 3:

A-II, B-IV, C-III, D-I

Option 4:

A-II, B-I, C-III, D-IV

Correct Answer:

A-II, B-IV, C-III, D-I

Solution:

Incomplete dominance → II; Co-dominance → IV; Pleiotropy → III; Polygenic inheritance → I.

- In incomplete dominance (Antirrhinum), the heterozygote shows an intermediate phenotype, e.g., pink flowers from red × white.
- Co-dominance (ABO blood groups) allows both alleles to be expressed equally, such as I^A and I^B producing AB blood type.
- Pleiotropy (Phenylketonuria) occurs when a single gene affects multiple traits, leading to diverse symptoms.
- Polygenic inheritance (human skin colour) involves multiple genes contributing additively to a continuous variation.

Hence, the correct answer is option 3) A-II, B-IV, C-III, D-I.

Q. 19 Arrange the following steps of DNA fingerprinting in a correct sequence.

- A. Isolation of DNA and its digestion by restriction endonucleases
- B. Hybridisation using a labelled VNTR probe
- C. Transferring of separated DNA fragments to synthetic membranes
- D. Detection of hybridised DNA fragments by autoradiography
- E. Separation of DNA fragments by electrophoresis.

Choose the correct answer from the options given below :

Option 1:

A, E, B, C, D

Option 2:

A,D,B,E,C

Option 3:

A, B, D, C, E

Option 4:

A, E, C, B, D

Correct Answer:

A, E, C, B, D

Solution:

In DNA fingerprinting, DNA is isolated and cut, then fragments are separated by electrophoresis, transferred to a membrane, hybridised with a probe, and finally detected by autoradiography.

This technique relies on Variable Number Tandem Repeats (VNTRs), which are unique to individuals, making DNA fingerprinting highly specific. The autoradiography step finally reveals the distinct banding pattern, serving as a genetic identity marker.

Hence, the correct answer is option 4) A, E, C, B, D.

Q. 20 Which of the following statements are correct with reference to the packaging of the DNA helix?

- A. Histones are organised to form a unit of eight molecules, called a histone octamer
- B. Histones are negatively charged basic proteins
- C. Histones are rich in the basic amino acid residues - lysine and arginine.
- D. The positively charged DNA is wrapped around the histone octamer to form a nucleosome.
- E. The packaging of chromatin at higher levels requires an additional set of proteins called non-histone chromosomal proteins.

Choose the correct answer from the options given below :

Option 1:

B, D and E only

Option 2:

A, B and D only

Option 3:

C, D and E only

Option 4:

A, C and E only

Correct Answer:

A, C and E only

Solution:

Histones are positively charged proteins due to lysine and arginine residues, which help them bind tightly to the negatively charged DNA. They form an octamer core around which DNA wraps, creating nucleosomes - the fundamental units of chromatin.

Higher-order chromatin packaging requires non-histone chromosomal proteins for further compaction. Statements B and D are incorrect because histones are not negatively charged, and DNA is negatively charged, not positive.

Hence, the correct answer is option 4) A, C and E only.

Q. 21 Find the incorrect statement(s) about photosynthesis from the following:

- A. The water splitting complex is associated with PS I.
- B. C_4 plants use the C_3 pathway of CO_2 fixation as the main biosynthetic pathway.
- C. In C_4 plants, photorespiration does not occur.
- D. C_3 plants exhibit 'Kranz' anatomy.
- E. ATP synthesis in chloroplasts occurs through chemiosmosis.

Choose the answer from the options given below :

Option 1:

B and C only

Option 2:

B and E only

Option 3:

B only

Option 4:

A and D only

Correct Answer:

A and D only

Solution:

Water splitting is linked to PS II (A false) and Kranz anatomy is seen in C_4 plants, not C_3 (D false).

- The water splitting complex is linked to PS II, not PS I, and it provides electrons by splitting water into oxygen and protons.
- Kranz anatomy is a special leaf structure found only in C_4 plants, where bundle sheath cells surround vascular bundles.
- C_4 plants still use the C_3 cycle as the main biosynthetic pathway, but photorespiration is minimised due to spatial separation of steps.
- ATP synthesis in chloroplasts indeed occurs via chemiosmosis, similar to mitochondria.

Hence, the correct answer is option 4) A and D only.

Q. 22 Arrange the following steps of somatic hybridisation in a correct sequence.

- A. Digestion of cell walls.
- B. Isolation of naked protoplasts.
- C. Fusion of protoplasts to get a hybrid protoplast.
- D. Isolation of single cells from two different varieties of plants.
- E. Growing of hybrid protoplast to form a new plant.

Choose the correct answer from the options given below :

Option 1:

D, B, A, E, C

Option 2:

E, A, B, C, D

Option 3:

E, B, A, D, C

Option 4:

D, A, B, C, E

Correct Answer:

D, A, B, C, E

Solution:

First cells are isolated → cell walls digested → protoplasts obtained → fusion → regeneration into a plant.

This process allows the fusion of protoplasts from different plant varieties, bypassing sexual reproduction barriers.

The regenerated hybrid protoplast develops into a new plant with combined genetic traits of both parents. Thus, somatic hybridisation is a powerful tool in plant biotechnology and crop improvement.

Hence, the correct answer is option 4) D, A, B, C, E

Q. 23 Match List I with List II :

	List I		List II
A.	Conjunctive tissue	I.	Specialised cells in the vicinity of guard cells
B.	Casparian strips	II.	Endodermal cells are rich in starch
C.	Subsidiary cells	III.	Tissue between xylem and phloem
D.	Starch sheath	IV.	Endodermal cells with suberin deposition

Choose the correct answer from the options given below :

Option 1:

A-IV, B-III, C-II, D-I

Option 2:

A-III, B-IV, C-I, D-II

Option 3:

A-IV, B-III, C-I, D-II

Option 4:

A-III, B-IV, G-II, D-I

Correct Answer:

A-III, B-IV, C-I, D-II

Solution:

- Conjunctive tissue lies between xylem & phloem (III)
- Casparian strips have suberin (IV)
- Subsidiary cells surround guard cells (I)
- The starch sheath has starch-rich endodermis (II).

Hence, the correct answer is option 2) A-III, B-IV, C-I, D-II

Q. 24 Which one of the following is not a characteristic of plant cells in the phase of elongation?

Option 1:

New cell wall deposition

Option 2:

Cell enlargement

Option 3:

Large conspicuous nuclei

Option 4:

Increased vacuolation

Correct Answer:

Large conspicuous nuclei

Solution:

During the elongation phase, cells primarily focus on increasing size through vacuolation and deposition of new cell wall material.

The nuclei become less conspicuous as the cytoplasm stretches, unlike meristematic cells, where nuclei are large and prominent.

Thus, elongation is marked by growth in volume and structure, not active nuclear division.

Hence, the correct answer is option 3) Large conspicuous nuclei.

Q. 25 Match List I with List II :

	List I		List II
	(Growth Regulator)		(Function/Effect)
A.	2,4-D	I.	Brewing industry
B.	GA ₃	II.	Stimulation of stomatal closure
C.	Kinetin	III.	Herbicide
D.	ABA	IV.	Nutrient mobilisation

Choose the correct answer from the options given below:

Option 1:

A-I, B-II, C-IV, D-III

Option 2:

A-I, B-IV, C-III, D-II

Option 3:

A-IV, B-III, C-II, D-I

Option 4:

A-III, B-I, C-IV, D-II

Correct Answer:

A-III, B-I, C-IV, D-II

Solution:

- 2, 4-*D* acts as a herbicide ($A \rightarrow III$)
- GA_3 is used in the brewing industry ($B \rightarrow I$)
- Kinetin promotes nutrient mobilisation ($C \rightarrow IV$)
- ABA induces stomatal closure ($D \rightarrow II$)

Hence, the correct answer is option 4) A-III, B-I, C-IV, D-II.

Q. 26 The enzyme required for carboxylation in the Calvin cycle is :

Option 1:

PEP carboxylase

Option 2:

RuBP carboxylase - oxygenase

Option 3:

Carboxypeptidase

Option 4:

Hexokinase

Correct Answer:

RuBP carboxylase - oxygenase

Solution:

Carboxylation in the Calvin cycle is catalysed by RuBisCO, which fixes CO_2 to RuBP, forming 3-PGA.

It fixes CO_2 to RuBP, producing 3-phosphoglycerate (3-PGA) as the first stable compound. This step is the key entry point of inorganic carbon into the biosphere, driving photosynthetic productivity.

Hence, the correct answer is option 2) RuBP carboxylase - oxygenase.

Q. 27 How many ATP and NADPH molecules are required to make one molecule of glucose through the Calvin pathway?

Option 1:

12 ATP and 18 NADPH

Option 2:

18 ATP and 12 NADPH

Option 3:

6 ATP and 12 NADPH

Option 4:

24 ATP and 18 NADPH

Correct Answer:

18 ATP and 12 NADPH

Solution:

For the synthesis of one glucose, the Calvin cycle consumes 18 *ATP* and 12 *NADPH*.

To fix one molecule of CO_2 , the Calvin cycle requires 3 ATP and 2 NADPH. Since glucose formation needs 6 CO_2 molecules, the total requirement becomes 18 ATP and 12 NADPH.

ATP provides the energy, while NADPH supplies the reducing power for carbon fixation. This highlights the high energy demand of photosynthesis, linking the light reactions to the dark reactions.

Hence, the correct answer is option 2) 18 ATP and 12 NADPH.

Q. 28 Which of the following floral formula is the correct floral formula of Solanaceae family ?

Option 1:

$\oplus K_{(5)} C_{(5)} A_5 \underline{G}_{(2)}$

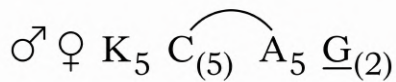
Option 2:

$\oplus K_5 C_5 A_5 \underline{G}_{(2)}$

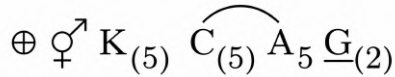
Option 3:

$\oplus \overset{\nearrow}{\underset{\nwarrow}{\text{♀}}} K_{(5)} \overbrace{C_{(5)}} A_5 \underline{G}_{(2)}$

Option 4:



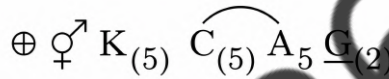
Correct Answer:



Solution:

Flowers of the Solanaceae family are actinomorphic and bisexual. Their calyx consists of five fused sepals, and the corolla has five fused petals. There are five stamens with a bicarpellary gynoecium and a superior ovary.

Hence, the correct answer is option 3)



Q. 29 Which of the following is an in situ conservation method?

Option 1:

Seed Banks

Option 2:

Sacred Groves

Option 3:

Botanical Gardens

Option 4:

Wildlife Safari Parks

Correct Answer:

Sacred Groves

Solution:

In situ conservation means protecting organisms in their natural habitat. Sacred groves are protected forest patches where biodiversity is preserved due to cultural and religious beliefs.

Unlike seed banks or botanical gardens (ex situ), this method allows species to evolve naturally in their environment. It is considered the most effective way to conserve endangered plants and animals.

Hence, the correct answer is option 2) Sacred Groves.

Q. 30 Which of the following statements are not true regarding restriction endonucleases?

- A. They are called molecular scissors.
 - B. These are the enzymes responsible for restricting the growth of bacteriophages in *E. coli*.
 - C. They cut the DNA only at the centre of the palindromic sites.
 - D. They remove nucleotides only from the ends of DNA fragments.
 - E. They recognise specific palindromic base-pair sequences.
- Choose the answer from the options given below :

Option 1:

C and D only

Option 2:

A and E only

Option 3:

D and E only

Option 4:

A and B only

Correct Answer:

C and D only

Solution:

Restriction enzymes cut at specific sites but not necessarily at the centre (*C false*), and they do not remove nucleotides from ends (*D false*).

Restriction endonucleases cut DNA at specific recognition sites, usually palindromic sequences, making them vital tools in genetic engineering. Unlike exonucleases, they do not remove nucleotides from the ends, but act within the DNA strand at defined positions.

Hence, the correct answer is option 1) C and D only.

Q. 31 In racemose inflorescence, ____ .

Option 1:

The growth is limited

Option 2:

Flowers are solitary

Option 3:

Flowers are borne in an acropetal succession

Option 4:

The main axis terminates in a flower

Correct Answer:

Flowers are borne in an acropetal succession

Solution:

In a racemose inflorescence, the main axis shows unlimited growth and flowers develop in acropetal succession (older at base, younger towards apex).

Option 1: The growth is limited - False, because in a racemose inflorescence, the main axis shows indefinite growth.

Option 2: Flowers are solitary - False, because racemose inflorescence has multiple flowers arranged along the axis, not solitary.

Option 3: Flowers are borne in an acropetal succession - True, because in a racemose inflorescence, younger flowers are at the apex, older at the base.

Option 4: The main axis terminates in a flower - False, because that is a feature of cymose inflorescence, not racemose.

Hence, the correct answer is option 3) Flowers are borne in an acropetal succession.

Q. 32 Arrange the following in the correct developmental sequence related to microsporogenesis :

- A. Microspore tetrads
- B. Sporogenous tissue
- C. Pollen grains
- D. Pollen mother cells

Choose the correct answer from the options given below:

Option 1:

A, D, C, B

Option 2:

D, A, C, B

Option 3:

B, D, C, A

Option 4:

B, D, A, C

Correct Answer:

B, D, A, C

Solution:

Sporogenous tissue pollen mother cells → microspore tetrads → pollen grains.

This sequence highlights the progression from diploid sporogenous tissue to haploid pollen grains through meiosis.

The pollen mother cells act as the key transitional stage, giving rise to microspore tetrads. Finally, each microspore matures into a functional pollen grain

Hence, the correct answer is option 4) B, D, A, C.

Q. 33 Identify the correct statements about biomolecules.

- A. Lipids are generally water-soluble.
- B. Proteins are polypeptides
- C. Polysaccharides are long chains of sugars.
- D. Adenine and guanine are substituted pyrimidines.
- E. Almost all enzymes are proteins.

Choose the correct answer from the options given below:

Option 1:

A, B and C only

Option 2:

B, D and E only

Option 3:

B, C and E only

Option 4:

C, D and E only

Correct Answer:

B, C and E only

Solution:

Lipids are water-insoluble (A false), and adenine and guanine are purines (D false). Statements B, C and E are correct.

- Proteins are long chains of amino acids linked by peptide bonds, hence called polypeptides.
- Polysaccharides such as starch, glycogen, and cellulose are complex carbohydrates made of repeated sugar units.
- Almost all enzymes are proteins, functioning as biological catalysts that speed up reactions.
- Lipids, however, are water-insoluble molecules, and adenine and guanine are purines, not pyrimidines.

Hence, the correct answer is option 3) B, C and E only.

- Q. 34** Which of the following statements are true with reference to sex determination in honeybees?
- A. An offspring formed from the union of a sperm and an egg develops as a female (queen or worker).
 - B. An unfertilized egg develops as a male by parthenogenesis.
 - C. A male has half the number of chromosomes as a female.
 - D. Males produce sperm by meiosis
 - E. Honeybees have a haplodiploid sex-determination system.
- Choose the correct answer from the options given below:

Option 1:

A, B, C and E only

Option 2:

A, B, C and D only

Option 3:

B, C, D and E only

Option 4:

A, B, D and E only

Correct Answer:

A, B, C and E only

Solution:

In honeybees, fertilised eggs ($2n$) form females, unfertilised eggs (n) form males. Males are haploid and produce sperm by mitosis, not meiosis, under haplodiploidy.

- A. True - Fertilised eggs (sperm + egg) develop into females (queen or worker).
- B. True - Unfertilized eggs develop into males (drones) by parthenogenesis.
- C. True - Males (haploid) have half the number of chromosomes compared to females (diploid).
- D. False - Males are haploid, so they cannot undergo meiosis to produce gametes. They produce sperm by mitosis.
- E. True - Honeybees indeed have a haplodiploid sex-determination system.

Hence, the correct answer is option 1) A, B, C and E only.

- Q. 35** Heterophyllous development in response to the environment is an example of which of the following phenomena?

Option 1:

Redifferentiation

Option 2:

Dedifferentiation

Option 3:

Elasticity

Option 4:

Plasticity

Correct Answer:

Plasticity

Solution:

Heterophylly shows that the same genotype can produce different leaf forms under different environmental conditions. This is called phenotypic plasticity. In heterophylly, plants like Cotton, Coriander, and Buttercup show varied leaf shapes in response to light, water, or temperature.

This adaptability helps plants optimise photosynthesis and survival under changing environments. It is not due to genetic change but due to flexible expression of the same genetic makeup. Thus, heterophylly is a classic example of plasticity in plants.

Hence, the correct answer is option 4) Plasticity.

Q. 36 Which of the following statements are correct regarding amino acids?

- A. They are substituted methanes.
- B. Serine is an aromatic amino acid.
- C. Valine is a neutral amino acid.
- D. Lysine is an acidic amino acid.

Choose the correct answer from the options given below:

Option 1:

A and B only

Option 2:

C and D only

Option 3:

B and C only

Option 4:

A and C only

Correct Answer:
A and C only

Solution:

Amino acids are considered substituted methane derivatives because the central carbon is bonded to an amino, carboxyl, hydrogen, and variable side chains.

Amino acids are substituted methane derivatives (A correct); valine is neutral (C correct). Serine is not aromatic, and lysine is basic, not acidic.

Hence, the correct answer is option 4) A and C only.

Q. 37 "The Evil Quartet" of biodiversity loss includes which of the following ?

Option 1:

Over-exploitation; Alien species invasions; Soil pollution; Co-extinctions

Option 2:

Habitat loss and fragmentation; Air pollution; Water pollution; Co-extinctions

Option 3:

Habitat loss and fragmentation; over-exploitation; Alien species invasions; Co-extinctions

Option 4:

Over-exploitation; Alien species invasions; Air pollution; Co-extinctions

Correct Answer:

Habitat loss and fragmentation; over-exploitation; Alien species invasions; Co-extinctions

Solution:

The "evil quartet" consists of habitat loss and fragmentation, over-exploitation, alien species invasions, and co-extinctions. These are the major causes of biodiversity decline.

- Habitat loss and fragmentation reduce living space and isolate populations.
- Over-exploitation depletes species faster than they can recover.
- Alien species invasions disrupt native ecosystems and outcompete local species.
- Co-extinctions occur when the extinction of one species leads to the loss of another dependent species.

Hence, the correct answer is option 3) Habitat loss and fragmentation; over-exploitation; Alien species invasions; Co-extinctions.

Q. 38 Match List I with List II :

List - I (Process)	List -II (Location)
A. Glycolysis	I. Inner mitochondrial membrane
B. ETS	II. Mitochondrial matrix
C. Accumulation of protons	III. Cytoplasm
D. Krebs' cycle	IV. Intermembrane space

Choose the correct answer from the options given below:

Option 1:

A-IV, B-II, C-I, D-III

Option 2:

A-I, B-IV, C-III, D-II

Option 3:

A-II, B-III, C-IV, D-I

Option 4:

A-III, B-I, C-IV, D-II

Correct Answer:

A-III, B-I, C-IV, D-II

Solution:

- Glycolysis occurs in the cytoplasm (III)
- ETS on inner membrane (I)
- Protons accumulate in the intermembrane space (IV)
- Krebs cycle in matrix (II).

Hence, the correct answer is option 4) A-III, B-I, C-IV, D-II

Q. 39 Which one of the following is a triploid cell?

Option 1:

Synergid

Option 2:

Primary endosperm cell

Option 3:

Central cell

Option 4:

Zygote

Correct Answer:

Primary endosperm cell

Solution:

The primary endosperm cell is triploid ($3n$), formed by the fusion of one male gamete with two polar nuclei ($n + n + n$).

This triploid cell develops into the endosperm, which nourishes the developing embryo in angiosperms. Thus, it is the only triploid structure among the given options.

Hence, the correct answer is option 2) Primary endosperm cell.

Q. 40 Which one of the following types of pollination brings genetically different types of pollen grains to the stigma?

Option 1:

Geitonogamy

Option 2:

Xenogamy

Option 3:

Cleistogamy

Option 4:

Autogamy

Correct Answer:

Xenogamy

Solution:

In xenogamy, pollen is transferred between flowers of different plants of the same species. Xenogamy ensures maximum genetic variation because pollen comes from a different plant of the same species. This type of pollination enhances the adaptability and evolutionary fitness of the population.

Unlike autogamy or geitonogamy, it prevents inbreeding and maintains healthy gene pools. It is often facilitated by agents like insects, wind, or water, depending on the plant species. Thus, xenogamy is the only pollination type that guarantees genetically different pollen grains on the stigma.

Hence, the correct answer is option 2) Xenogamy.

Q. 41 Match List I with List II :

	List I (Placentation)		List II (Example)
A.	Marginal	I.	Mustard
B.	Axile	II.	Pea
C.	Parietal	III.	Marigold
D.	Basal	IV.	Lemon

Choose the correct answer from the options given below :

Option 1:

A-II, B-IV, C-I, D-III

Option 2:

A-IV, B-II, C-I, D-III

Option 3:

A-III, B-I, C-IV, D-II

Option 4:

A-I, B-III, C-II, D-IV

Correct Answer:

A-II, B-IV, C-I, D-III

Solution:

Placentation refers to the arrangement of ovules within the ovary, and different plants exhibit distinct types.

- In pea, ovules are borne along the margin (marginal placentation).
- In a lemon, ovules are attached to the central axis (axile placentation).
- In mustard, ovules develop on the inner wall (parietal placentation).
- In marigolds, ovules are attached at the base of the ovary (basal placentation).

Marginal → Pea (II); Axile → Lemon (IV); Parietal → Mustard (I); Basal → Marigold (III).

Hence, the correct answer is option 1) A-II, B-IV, C-I, D-III.

- Q. 42** The main criteria used for Five Kingdom Classification proposed by R.H. Whittaker (1969) included :
- A. Cell structure
 - B. Body organisation
 - C. Presence of flagellum
 - D. Reproductions
 - E. Phylogenetic relationships

Choose the correct answer from the options given below :

Option 1:

A, B and E only

Option 2:

A, B, C, D and E

Option 3:

B, C and D only

Option 4:

A, B, D and E only

Correct Answer:

A, B, D and E only

Solution:

R. H. Whittaker used cell structure, thallus/body organisation, mode of reproduction, and phylogenetic relationships. The presence of a flagellum was not one of the main criteria. This system was revolutionary as it combined morphological, reproductive, and evolutionary criteria for classification. It resolved the limitations of earlier two-kingdom and three-kingdom systems by recognising fungi separately and including Monera and Protista.

Hence, the correct answer is option 4) A, B, D and E only.

Q. 43 Match the List - I with List - II

List - I	List - II
A. Trypsin	I. Intercellular ground substance
B. Morphine	II. Lectin
C. Concanavalin	III. Enzyme
D. Collagen	IV. Alkaloid

Choose the correct answer from the options given below:

Option 1:

A-III, B-IV, C-II, D-I

Option 2:

A-IV, B-III, C-II, D-I

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-III, B-II, C-IV, D-I

Correct Answer:

A-III, B-IV, C-II, D-I

Solution:

Trypsin → enzyme (III); Morphine → alkaloid (IV); Concanavalin A → lectin (II); Collagen → intercellular ground substance (I).

- Trypsin is a digestive enzyme that hydrolyses proteins into peptides.
- Morphine is a plant-derived alkaloid used medicinally as an analgesic.
- Concanavalin A is a lectin that binds specifically to certain sugars, often used in biochemical studies.
- Collagen is the main structural protein of the intercellular ground substance, providing strength and elasticity to connective tissues.
- **Hence, the correct answer is option 1) A-III, B-IV, C-II, D-I.**

- Q. 44** Which of the following statements are correct with respect to DNA separation, isolation and visualisation?
- A. The cutting of DNA is done by molecular scissors.
 - B. The DNA fragments separate according to their size in an agarose gel upon electrophoresis
 - C. The separated DNA fragments can be seen without staining when exposed to UV light
 - D. The separated PNA fragments, when stained with ethidium bromide, can be seen in visible light.

Choose the correct answer from the options given below:

Option 1:

A and B only

Option 2:

B and D only

Option 3:

A and D only

Option 4:

B and C only

Correct Answer:

A and B only

Solution:

Restriction enzymes act as molecular scissors (A true), and electrophoresis separates DNA by size (B true). DNA needs staining and is visualised under UV, not visible light (C, D false).

- Restriction endonucleases act as molecular scissors, cutting DNA at specific recognition sites.
- During agarose gel electrophoresis, DNA fragments migrate according to their size, with smaller fragments moving faster.
- DNA cannot be seen directly; it must be stained (commonly with ethidium bromide) and visualised under UV light.

Hence, the correct answer is option 1) A and B only.

-
- Q. 45** Which one of the following disorders is caused by the substitution of Glutamic acid (Glu) by Valine (Val) at the sixth position of the beta globin chain of the haemoglobin molecule?

Option 1:

Phenylketomuria

Option 2:

Haemophilif

Option 3:
Sickle-cell anaemia

Option 4:
Thalassemia

Correct Answer:
Sickle-cell anaemia

Solution:

Sickle-cell anaemia is caused by a point mutation where *Glu* → *Val* at position 6 of β -globin.

This point mutation changes the hydrophilic glutamic acid to hydrophobic valine, altering the haemoglobin structure. As a result, red blood cells become sickle-shaped under low oxygen conditions, leading to blockages in blood vessels.

The disorder causes anaemia, pain crises, and organ damage due to impaired oxygen transport. It is a classic example of a genetic disease caused by a substitution mutation.

Hence, the correct answer is option 3) Sickle-cell anaemia.

Q. 46 Match **List I** with **List II** :

List I		List II	
A	Cortisol	I	Stimulates the formation of alveoli in mammary glands
B	Aldosterone	II	Produces anti-inflammatory reactions
C	Cholecystokinin	III	Stimulates reabsorption of Na^+ and water from renal tubule
D	Progesterone	IV	Stimulates secretion of pancreatic enzymes and bile juice

Choose the **correct** answer from the options given below:

Option 1:
A-II, B-III, C-IV, D-I

Option 2:
A-II, B-III, C-I, D-IV

Option 3:

A-IV, B-II, C-I, D-III

Option 4:

A-III, B-II, C-IV, D-I

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

Cortisol shows anti-inflammatory action (II), aldosterone increases Na^+ reabsorption (III), cholecystokinin stimulates pancreatic enzyme and bile secretion (IV), and progesterone helps in mammary gland development (I).

Hence, the correct answer is option 1) A-II, B-III, C-IV, D-I

Q. 47 Arrange the following events occurring in Renin-Angiotensin mechanism in the correct order :

- A. Increase in blood pressure and Glomerular filtration rate.
- B. Reabsorption of Na^+ and water from distal parts of tubule due to Aldosterone.
- C. Fall in Glomerular filtration rate.
- D. Vasoconstriction by Angiotensin II and release of Aldosterone.
- E. Renin converts Angiotensinogen into Angiotensin I, followed by Angiotensin II.

Choose the **correct** answer from the options given below:

Option 1:

A, C, E, B, D

Option 2:

C, A, B, D, E

Option 3:

A, D, B, E, C

Option 4:

C, E, D, B, A

Correct Answer:

C, E, D, B, A

Solution:

The glomerular filtration rate usually refers to the amount of filtrate formed by both kidneys per minute.

Fall in GFR → renin release → Angiotensin II formation → vasoconstriction & aldosterone → Na⁺/water reabsorption → increased BP & GFR.

Hence, the correct answer is option 4) C, E, D, B, A

Q. 48 In humans, respiration occurs in the following steps. Arrange these steps in the correct order.

- A. Diffusion of O₂ and CO₂ between blood and tissues
- B. Diffusion of O₂ and CO₂ across alveolar membrane
- C. Pulmonary ventilation by which atmospheric air is drawn in and CO₂ rich alveolar air is released out
- D. Cellular respiration
- E. Transport of gases by the blood

Choose the **correct** answer from the options given below:

Option 1:

A, B, C, D, E

Option 2:

C, A, B, E, D

Option 3:

C, B, E, A, D

Option 4:

E, A, C, D, B

Correct Answer:

C, B, E, A, D

Solution:

Respiration is the process of obtaining energy from the breakdown of food molecules. It usually starts with pulmonary ventilation (C), followed by diffusion across alveoli (B), transport via blood (E), diffusion into tissues (A), and finally cellular respiration (D).

Hence, the correct answer is option 3) C, B, E, A, D

Q. 49 The following are the stages of life cycle of Plasmodium. Arrange the stages in the proper order.

- A. The parasites reproduce asexually in RBCs, bursting the cells.
- B. The parasites reproduce asexually in liver cells, bursting the cells and releasing into blood.
- C. Gametocytes develop in RBCs.
- D. Sporozoites reach the liver through the blood
- E. Female mosquito injects sporozoites into humans during bite (1)

Choose the **correct** answer from the options given below :

Option 1:

A, B, C, D, E

Option 2:

E, C, D, B, A

Option 3:

E, D, B, A, C

Option 4:

C, A, B, D, E

Correct Answer:

E, D, B, A, C

Solution:

Sequence: mosquito injects sporozoites → reach liver → multiply in liver → infect RBCs → gametocytes form. The Plasmodium sporozoites multiply in the liver cells and then infect RBCs. They reproduce further there and develop into gametocytes.

Hence, the correct answer is option 3) E, D, B, A, C

Q. 50 Insertion of a foreign DNA at BamHI site in an E. coli cloning vector pBR322 results in the loss of antibiotic resistance towards :

Option 1:

Ampicillin and tetracycline

Option 2:
Ampicillin

Option 3:
Tetracycline

Option 4:
Gentamycin

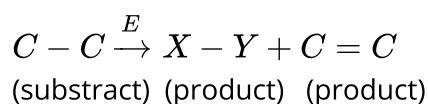
Correct Answer:
Tetracycline

Solution:

The BamHI site lies within the tetracycline resistance gene; insertion disrupts it, causing loss of tetracycline resistance. The insertion normally disrupts it because it interrupts the normal coding sequence of the tetR gene. This turns it non-functional and causes the transformed bacteria to lose tetracycline resistance.

Hence, the correct answer is option 3) Tetracycline

Q. 51 The following reaction depicts the activity of a particular class of enzymes:



Identify the enzyme class 'E' from the following options:

Option 1:
Isomerases

Option 2:
Ligases

Option 3:
Transferases

Option 4:
Lyases

Correct Answer:
Lyases

Solution:

Lyases catalyze the breaking of bonds without hydrolysis or ATP, often forming double bonds (like $C = C$).

Hence, the correct answer is option 4) Lyases

Q. 52 The specific receptors for neurotransmitters synapse are present on?

Option 1:

Pre-synaptic membrane

Option 2:

Post-synaptic membrane

Option 3:

Myelin sheath

Option 4:

Schwann cell

Correct Answer:

Post-synaptic membrane

Solution:

The membranes of post-synaptic neurons contain receptor proteins which can bind neurotransmitters. Therefore, the Neurotransmitters bind to receptors located on the post-synaptic membrane to transmit nerve impulses after diffusing through the synaptic cleft. This binding is what initiates new electrical impulses in the next neuron.

Hence, the correct answer is option 2) Post-synaptic membrane

Q. 53 What is the probability of having children with 'O' blood group, where both mother and father are heterozygous for 'A' and 'B' blood group, respectively?

Option 1:

0%

Option 2:

50%

Option 3:
25%

Option 4:
75%

Correct Answer:
25%

Solution:

Cross: $I^A i \times I^B i \Rightarrow$ offspring genotypes: $I^A I^B, I^A i, I^B i, ii$.

Only ii gives blood group O, so the probability = $\frac{1}{4} = 25$.

Hence, the correct answer is option 3) 25%

Q. 54 Match List I with List II :

List I (Respiratory Volume)		List II (Capacity)	
A	ERV (Expiratory Reserve Volume)	I	2500 - 3000 mL
B	RV (Residual Volume)	II	500 mL
C	IRV (Inspiratory Reserve Volume)	III	1000-1100mL
D	TV (Tidal Volume)	IV	1100-1200mL

Choose the **correct** answer from the options below:

Option 1:
A-III, B-IV, C-I, D-II

Option 2:
A-III, B-I, C-IV, D-II

Option 3:
A-I, B-II, C-III, D-IV

Option 4:
A-I, B-III, C-II, D-IV

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:

ERV $\approx$ 1000–1100, RV $\approx$ 1100–1200, IRV $\approx$ 2500–3000, TV $\approx$ 500 mL.

TV = Tidal Volume is the amount of air inhaled or exhaled during normal breathing.

IRV, = IRV on the other hand, is the extra amount of air a person can inhale forcibly after a normal inspiration.

ERV = It is the extra amount of air that a person can exhale forcibly after a normal expiration.

Hence, the correct answer is option 1. A-III, B-IV, C-I, D-II

Q. 55 Which of the following is not an example of convergent evolution?

Option 1:

Fore limbs of whales and bats .

Option 2:

Flippers of penguins and dolphins

Option 3:

Eyes of octopuses and mammals

Option 4:

Wings of butterflies and birds

Correct Answer:

Fore limbs of whales and bats .

Solution:

Forelimbs of whales and bats are homologous (divergent evolution), not convergent.

Hence, the correct answer is option 1) Fore limbs of whales and bats.

Q. 56 Male frogs can be distinguished from female frogs due to the presence of :

- A. Bulging eyes
- B. Vocal sacs
- C. Webbed digits in feet
- D. Copulatory pad on first digit of fore limbs
- E. Olive green-coloured skin with dark irregular spots

Choose the **correct** answer from the options given below:

Option 1:

A and B only

Option 2:

C and E only

Option 3:

B and D only

Option 4:

B and C only

Correct Answer:

B and D only

Solution:

Male frogs possess vocal sacs for croaking and copulatory (nuptial) pads on forelimbs. Other features that are common in both male and female frogs are bulging eyes, webbed feet, and skin color. Therefore, we can not use these features to distinguish the sexes.

Hence, the correct answer is option 3) B and D only

Q. 57 A group of researchers procured some fish-like animals and upon investigation the following characters were observed :

- A. Endoskeleton was made of cartilage.
- B. Ectoparasitic; as they were found attached on fish skin with their circular sucking mouth.
- C. Paired fins and scales were absent, but 7 pairs of gill slits were present.

Which of the following species of animals did they consider to fit best with these characters?

Option 1:
Scoliodon sp.

Option 2:
Exocoetus sp

Option 3:
Petromyzon sp

Option 4:
Branchiostoma sp

Correct Answer:
Petromyzon sp

Solution:

Petromyzon (lamprey) shows cartilaginous skeleton, ectoparasitic habit, circular mouth, no paired fins, and multiple gill slits.

Hence, the correct answer is option 3) Petromyzon sp

Q. 58 Match **List I** with **List II** with respect to chronology of evolution of life forms :

List I		List II	
A	About 65 mya	I	Jawless fish probably evolved
B	About 500 mya	II	The dinosaurs suddenly disappeared from the earth
C	About 350 mya	III	Seaweeds and few plants probably existed
D	About 320 mya	IV	Invertebrates were formed and became active

Choose the **correct** answer from the options given below :

Option 1:

A-III, B-IV, C-I, D-II

Option 2:

A-II, B-IV, C-III, D-I

Option 3:

A-II, B-IV, C-I, D-III

Option 4:

A-I, B-II, C-III, D-IV

Correct Answer:

A-II, B-IV, C-I, D-III

Solution:

About 65 million years ago, the dinosaurs disappeared from the Earth. Around 500 million years ago, the invertebrates were active in the oceans. Jawless fishes probably evolved about 350 million years ago, while seaweeds and a few plants existed somewhere around 320 million years ago.

Hence, the correct answer is option 3) A-II, B-IV, C-I, D-III

Q. 59 Match List I with List II :

List I		List II	
A	Progestasert	I	Barrier made of rubber used by females
B	Multiload 375	II	Oral contraceptive
C	Diaphragm	III	Hormone releasing IUD
D	Saheli	IV	Copper releasing IUD

Choose the **correct** answer from the options given below :

Option 1:

A-IV, B-II, C-I, D-III

Option 2:

A-IV, B-III, C-I, D-II

Option 3:

A-III, B-IV, C-II, D-I

Option 4:

A-III, B-IV, C-I, D-II

Correct Answer:

A-III, B-IV, C-I, D-II

Solution:

Progestasert → hormone IUD (III), Multiload 375 → copper IUD (IV), Diaphragm → barrier (I), Saheli → oral pill (II).

Hence, the correct answer is option 4) A-III, B-IV, C-I, D-II

Q. 60 The WBC count of a person's blood sample 8000/cu. mm. How many eosinophils and lymphocytes would be in the same blood sample approximately?

Option 1:

300 – 500/cu. mm and 500 – 700/cu. mm, respectively

Option 2:

300 – 500/cu. mm and 1200 – 1500/cu. mm, respectively

Option 3:

100 – 120/cu. mm and 160 – 200/cu. mm, respectively

Option 4:

160 – 240/cu. mm and 1600 – 2000/cu. mm, respectively

Correct Answer:

160 – 240/cu. mm and 1600 – 2000/cu. mm, respectively

Solution:

Eosinophils $\approx 2-3$ of 8000 $\Rightarrow 160-240$.

Lymphocytes $\approx 20-25$ of 8000 $\Rightarrow 1600-2000$.

Eosinophils constitute about 2-3% and lymphocytes about 20-25% of total WBCs.

Hence, the correct answer is option 4) 160 – 240/cu. mm and 1600 – 2000/cu. mm, respectively

Q. 61 Match **List I** and **List II**

List I (Drug)		List II (Effect)	
A	Nicotine	I	Causes sense of euphoria and increased energy
B	Morphine	II	Stimulates adrenal gland to release catecholamines into blood circulation
C	Herois	III	Effective sedative and painkiller
D	Cocaine	IV	A depressant; slows down body functio

Choose the **correct** answer from the options given below :

Option 1:

A-II, B-III, C-I, D-IV

Option 2:

A-III, B-II, C-IV, D-I

Option 3:

A-III, B-II, C-I, D-IV

Option 4:

A-II, B-III, C-IV, D-I

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

Nicotine → II (stimulates catecholamine release), Morphine → III (sedative), Heroin → IV (depressant), Cocaine → I (euphoria).

Hence, the correct answer is option 4) A-II, B-III, C-IV, D-I

Q. 62 The human protein named α -1-antitrypsin, obtained from transgenic animals, is used for the treatment of ____

Option 1:

Alzheimer's disease

Option 2:

Rheumatoid arthritis

Option 3:

Emphysema

Option 4:

Cystic fibrosis

Correct Answer:

Emphysema

Solution:

α -1-antitrypsin deficiency leads to lung damage due to unchecked protease activity, causing emphysema. Transgenic animals produce this protein for therapy. Alpha antitrypsin is a human protein that is produced by using transgenic animals. It is used to treat emphysema. Emphysema is a lung disease caused by a deficiency of this protein. Alpha antitrypsin protects lung tissues from damage by certain enzymes.

Hence, the correct answer is option 3).Emphysema

Q. 63 Select the set of fishes which belong to class Osteichthyes?

Option 1:

Saw fish, Fighting fish and Dog fish

Option 2:

Devil fish, Cuttlefish and Hagfish

Option 3:

Starfish, Hagfish and Cuttlefish

Option 4:

Flying fish, Angel fish and Fighting fish

Correct Answer:

Flying fish, Angel fish and Fighting fish

Solution:

Osteichthyes are bony fishes; Flying fish, Angel fish, and fighting fish belong to this class, while others include cartilaginous or non-fish organisms. This is because the Flying fish, angel fish, and fighting fish are true bony fish. Osteichthyes are bony fishes with a skeleton that is made mainly of bone. Sharks and rays, on the other hand, belong to Class Chondrichthyes.

Hence, the correct answer is option 4) Flying fish, Angel fish, and Fighting fish

Q. 64 Select the **incorrect** statements from the following :

- A. Digestive system in Platyhelminthes is incomplete.
- B. Bilateral symmetry is a characteristic feature of adult Echinoderms?
- C. Pseudocoelom is possessed by Aschelminthes.
- D. Notochord is persistent throughout life in the class Chondrichthyes.
- E. Members of class Reptilia maintain a constant body temperature.

Choose the answer from the options given below :

Option 1:

A and C only

Option 2:

B and E only

Option 3:

B and D only

Option 4:

C and D only

Correct Answer:

B and E only

Solution:

Echinoderms are radially symmetrical (not bilateral), and reptiles are ectothermic (not constant temperature). Others are correct. The adult echinoderms have radial symmetry; during their larval stage, they possess bilateral symmetry. Reptiles, being cold-blooded animals, can not maintain a constant body temperature.

Hence, the correct answer is option 2) B and E only

Q. 65 Non-membrane bound cell organelles found in both prokaryotic and eukaryotic cells are _____ .

Option 1:
Lysosomes

Option 2:
Centrosomes

Option 3:
Mitochondria x

Option 4:
Ribosomes

Correct Answer:
Ribosomes

Solution:

Ribosomes are cellular organelles that synthesize proteins. These are non-membrane-bound and present in both prokaryotic and eukaryotic cells.

Hence, the correct answer is option 4) Ribosomes

Q. 66 Which of the following equations depicts Verhulst–Pearl logistic population growth?

Option 1:
$$\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$$

Option 2:
$$\frac{dN}{dt} = rN \left(\frac{K}{K-N} \right)$$

Option 3:
$$\frac{dN}{dt} = rN \left(\frac{K-N}{N} \right)$$

Option 4:
$$\frac{dN}{dt} = rN \left(\frac{K+N}{K} \right)$$

Correct Answer:

$$\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$$

Solution:

The logistic growth equation explains that the population grows rapidly at first when resources are provided in abundance.

The logistic growth equation is $\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$, where K is the carrying capacity and growth slows as the population approaches K .

Hence, the correct answer is option 1) $\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$

Q. 67 Select the **incorrect** statements with reference to Rh grouping.

- A. Erythroblastosis foetalis is a condition observed having foetus with Rh^{-ve} blood and mother with Rh^{+ve} blood.
- B. Rh antigen is observed on RBCs in the majority of human beings.
- C. Before blood transfusion, Rh group should also be matched.
- D. Rh incompatibility is observed when a pregnant mother is Rh^{-ve} and the foetus is Rh^{+ve} .
- E. Erythroblastosis foetalis can be avoided by administering anti-Rh antibodies to the mother immediately after the delivery of the second child.

Choose the answer from the options given below :

Option 1:

A and B only

Option 2:

C and D only

Option 3:

A and E only

Option 4:

B and C only

Correct Answer:

A and E only

Solution:

Erythroblastosis occurs when mother is Rh⁻ and foetus Rh⁺, not vice versa. Anti-Rh antibodies are given after the first delivery, not the second. These Anti-RH antibodies destroy any RH-positive foetal cells in the blood of the mother before her immune system can react and form permanent antibodies.

Hence, the correct answer is option 3) A and E only

Q. 68 Match **List I** with **List II**:

List I Bioactive molecules)		List II (Importance)	
A	Streptokinase	I	Immunosuppressive agent
B	Statins	II	Removal of clots from the blood vessels
C	Lipases	III	Blood cholesterol-lowering agent
D	Cyclosporin A	IV	Detergent formulations

Choose the **correct** answer from the options given below:

Option 1:

A-II, B-III, C-I, D-IV

Option 2:

A-III, B-II, C-IV, D-I

Option 3:

A-II, B-III, C-IV, D-I

Option 4:

A-IV, B-III, C-II, D-I

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

Streptokinase dissolves clots and is used during heart attacks (II), statins lower cholesterol (III), lipases are used in detergents (IV), and cyclosporin A is an immunosuppressant that is used during organ transplantation (I).

Hence, the correct answer is option 3) A-II, B-III, C-IV, D-I

Q. 69 Match **List I** with **List II** :

List I		List II	
A	Molluscs	I	Pulmonary respiration only
B	Reptiles	II	Branchial respiration
C	Adult amphibians	III	Cellular respiration
D	Amoeba	IV	Pulmonary and Cutaneous respiration amphibians

Choose the **correct** answer from the options given below :

Option 1:

A-III, B-II, C-I, D-IV

Option 2:

A-II, B-I, C-III, D-IV

Option 3:

A-I, B-II, C-IV, D-III

Option 4:

A-II, B-I, C-IV, D-III

Correct Answer:

A-II, B-I, C-IV, D-III

Solution:

Molluscs → gills (II); Reptiles → lungs only (I); Adult amphibians → lungs + skin (IV); Amoeba → cellular respiration (III).

Hence, the correct answer is option (4).

Q. 70 The sixth mutant codon of beta globin gene causing polymerization of Haemoglobin and change in RBC shape is ____

Option 1:
CAG

Option 2:
AUG

Option 3:
GUG

Option 4:
GAG

Correct Answer:
GUG

Solution:

In Sickle cell anemia disease, the 6th codon of the beta globin gene mutates from GAG to GUG. This substitution results in the production of Hemoglobin S. This Hemoglobin S polymerizes at low oxygen levels and is responsible for causing sickle-shaped RBCs.

Hence, the correct answer is option 3) GUG

Q. 71 Choose the correct statements regarding muscle contraction.

- A. A motor neuron carries a signal sent by the Central Nervous System (CNS) to the sarcolemma of the muscle fibre.
- B. The neural signal generates an action potential which cause the release of Ca^{++} into sarcoplasm.
- C. Increase in Ca^{++} inactivates the actin for breaking cross bridges.
- D. Actin binds to the myosin head to form a cross bridge.
- E. Shortening of sarcomere takes place, by pulling actin filaments towards the centre of 'A' band.

Choose the **correct** answer from the options given below :

Option 1:

A and B only

Option 2:

C and E only

Option 3:

C and D only

Option 4:

A, B, D and E only

Correct Answer:

A, B, D and E only

Solution:

Ca^{2+} activates actin by removing troponin inhibition, not inactivates it. Cross-bridge formation and sliding of actin toward A-band center cause contraction.

Hence, the correct answer is option 4).A, B, D and E only

Q. 72 Which of the following statements are correct with reference to human endoskeleton?

- A. Human skull is monocondylic.
- B. The joint between any two adjoining vertebrae is a cartilaginous joint.
- C. In human beings, the number of cervical vertebrae is seven
- D. All ribs except the last 2 pairs are bicephalic.
- E. The occipital bone of skull is articulated with atlas vertebra.

Choose the **correct** answer from the options given below :

Option 1:

C, D and E only

Option 2:

B, C and E only

Option 3:

A, B and D only

Option 4:

B and E only

Correct Answer:

B, C and E only

Solution:

Humans are dicondylic (not monocondylic). Vertebrae joints are cartilaginous, cervical vertebrae = 7, and the occipital bone articulates with the atlas. The joints between the adjacent vertebrae are cartilaginous because of the presence of the intervertebral disc. Humans have seven cervical vertebrae in the neck region.

Hence, the correct answer is option 2) B, C, and E only

Q. 73 Spermatogonia undergo a series of cell divisions to produce sperms. Select the correct statements from the following:

- A. Spermatogonia always undergo meiotic cell division.
- B. Primary spermatocytes divide mitotically to produce secondary spermatocytes
- C. Secondary spermatocytes, through their second meiotic division, produce haploid spermatids.
- D. Spermatids produce spermatozoa through mitosis.
- E. Spermatids transform into spermatozoa by spermiogenesis

Choose the **correct** answer from the options given below :

Option 1:

A, C and E only

Option 2:

C and E only

Option 3:

A and E only

Option 4:

B, C and D only

Correct Answer:

C and E only

Solution:

Spermatogonia divide mitotically, not always meiotically. Meiosis II forms haploid spermatids, which transform into spermatozoa by spermiogenesis. Spermatids are immature sperm cells, while the spermatozoa are the mature and functional sperm.

Hence, the correct answer is option 2) C and E only

Q. 74 The JGA (Juxta Glomerular Apparatus) is a special sensitive region formed by cellular modifications in _____ related to the same nephron.

Option 1:

Proximal convoluted tubule and efferent renal arteriole

Option 2:

Distal convoluted tubule and efferent renal arteriole

Option 3:

Distal convoluted tubule and afferent renal arteriole

Option 4:

Proximal convoluted tubule and afferent renal arteriole

Correct Answer:

Distal convoluted tubule and afferent renal arteriole

Solution:

JGA is formed by modified cells of the distal convoluted tubule (macula densa) and the afferent arteriole, regulating blood pressure and filtration. Together, both the distal convoluted tubule and afferent arteriole regulate blood pressure and glomerular filtration rate.

Hence, the correct answer is option 3) Distal convoluted tubule and afferent renal arteriole

Q. 75 Which one of the following is an appropriate example of 'sexual deceit'?

Option 1:

Sea anemone and clown fish

Option 2:

Ophrys and bumblebee

Option 3:

Female wasp and fig

Option 4:

Cuckoo and crow

Correct Answer:

Ophrys and bumblebee

Solution:

In sexual deceit, the orchid Ophrys mimics female insects to attract male pollinators, leading to pollination without reward. It is because the male insects try to mate with the flower, during which the pollen transfer and pollination occur without getting any reward.

Hence, the correct answer is option 2) Ophrys and bumblebee

Q. 76 Choose the correct statements regarding frogs anatomy :

- A. Hepatic portal system is the special venous connection between liver and intestine. .
- B. There are twelve pairs of cranial nerves arising from the brain.
- C. The ureters and oviducts open separately into the cloaca in female frogs ×
- D. Hind-brain consists of cerebellum, medulla oblongata and optic lobes.
- E. Sinus venosus joins the right atrium of heart.

Choose the **correct** answer from the options given below:

Option 1:

B and D only

Option 2:

A, B and C only

Option 3:

B and C only

Option 4:

A, C and E only

Correct Answer:

A, C and E only

Solution:

Hepatic portal system connects intestine to liver (A true), ureters and oviducts open separately into cloaca (C true), sinus venosus joins right atrium (E true). Frogs have 10 pairs of cranial nerves (B false), and optic lobes belong to the midbrain (D false).

Hence, the correct answer is option 4) A, C, and E only

Q. 77 Match List I and List II related to embryonic development at various months of pregnancy :

List I		List II	
A	The foetus movement starts and hair appears on the head	I	24 weeks of pregnancy
B	The foetus develops limbs and digits	II	20 weeks of pregnancy
C	The foetus develops external genital organs	III	8 weeks of pregnancy
D	The foetus body is covered with fine hair; eyelids N separate and eyelashes are formed	IV	12 weeks of pregnancy

Choose the **correct** answer from the options given below:

Option 1:

A-II, B-IV, C-III, D-I

Option 2:

A-II, B-III, C-IV, D-I

Option 3:

A-IV, B-II, C-III, D-I

Option 4:

A-III, B-II, C-IV, D-I

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

20 weeks → movement & hair, 8 weeks → limbs, 12 weeks → genital organs, 24 weeks → eyelashes & fine hair.

During the development of a foetus, the limbs and digits appear by 8 weeks, external genital organs appear by 12 weeks of pregnancy. In the same way, foetal movement starts around 20 weeks.

Hence, the correct answer is option 2) A-II, B-III, C-IV, D-I

Q. 78 In a population of a grasshopper species, the chromosome number of some members is 23 and some other members possess 24 chromosomes. The 23 and 24 chromosome-bearing members in this species are ____.

Option 1:

females and males, respectively

Option 2:

males and females, respectively

Option 3:

all males

Option 4:

all females

Correct Answer:

males and females, respectively

Solution:

Grasshoppers show XO type sex determination: males = 23 (XO), females = 24 (XX).

The grasshopper shows XO-type sex determination. In the XO type, females have two X chromosomes, and males have only one X chromosome. Both males and females have the same number of autosomes. Therefore, grasshoppers with 24 chromosomes are females, and those with 23 chromosomes are males.

Hence, the correct answer is option 2) males and females, respectively

Q. 79 In which animal do haploid cells mitotically to produce gametes ?

Option 1:

Male honeybees

Option 2:

Male grasshoppers

Option 3:

Male earthworms

Option 4:

Male frogs

Correct Answer:

Male honeybees

Solution:

Male honeybees are haploid organisms formed from unfertilized eggs. Since they are already haploid, gamete formation occurs by mitosis instead of meiosis. Meiosis reduces chromosome number by half, which is why it becomes an unnecessary process in drones.

Hence, the correct answer is option 1) Male honeybees

Q. 80 Arrange the following cell layers/structures around the female gamete, from outer to inner side:

- A. Zona pellucida
- B. Perivitelline space
- C. Corona radiata
- D. Plasma membrane of ovum

Choose the **correct** answer from the options given below:

Option 1:

D, B, A, C

Option 2:

A, C, B, D

Option 3:

C, A, D, B

Option 4:

C, A, B, D

Correct Answer:

C, A, B, D

Solution:

From outer to inner: corona radiata → zona pellucida → perivitelline space → plasma membrane of ovum.

- Corona radiata - the outermost layer of the ovum
- Zona pellucida - protective glycoprotein layer.
- Perivitelline space - is the plasma membrane of the ovum

Hence, the correct answer is option 4) C, A, B, D

Q. 81 What is the reason behind production of large holes in 'Swiss Cheese'?

Option 1:

The production of large amount of CO_2 and H_2 by Trichoderma polysporum

Option 2:

The production of large amount of CO_2 and H_2 by lactiges acid bacteria called Lactobacillus

Option 3:

The production' of large amount of CO_2 by Propionibacterium sharmanii

Option 4:

The production of large amount of CO_2 by Clostridium butylicum

Correct Answer:

The production' of large amount of CO_2 by Propionibacterium sharmanii

Solution:

Large holes are due to CO₂ production by *Propionibacterium sharmanii* during fermentation. This normally happens because the gas can not escape easily from the semi-solid cheese. And hence, creates large holes in the cheese.

Hence, the correct answer is option 3) The production' of large amount of CO₂ by *Propionibacterium sharmanii*

Q. 82 The toxin proteins isolated from *Bacillus huringiensis*, coded by which of the following genes would control cotton bollworms and corn borerer, respectively ?

Option 1:

cryIAc and *cryIIB*

Option 2:

cryIAc and *cryIIIAb*

Option 3:

cryIAc and *cryIAB*

Option 4:

cryIIAb and *cryIAc*

Correct Answer:

cryIAc and *cryIAB*

Solution:

The *cryIAc* gene produces a toxin against the cotton bollworms, whereas on the other hand, the *cryIAb* gene produces a toxin that controls corn borers.

Hence, the correct answer is option 3) *cryIAc* and *cryIAB*

Q. 83 Ecological pyramids represent the relationship between the organisms at different trophic levels and they are generally inverted for :

Option 1:

Pyramid of number in grassland

Option 2:

Pyramid of energy in pond ecosystem

Option 3:

Pyramid of biomass in grassland

Option 4:

Pyramid of biomass in sea

Correct Answer:

Pyramid of biomass in sea

Solution:

In aquatic ecosystems, the biomass pyramid is inverted as phytoplankton biomass is lower than that of consumers.

Hence, the correct answer is option 4) Pyramid of biomass in sea

Q. 84 Choose the correct statement regarding GIFT to overcome infertility.

Option 1:

Ova collected from a female donor are transferred to the uterus of an infertile female.

Option 2:

Early embryos with up to 8 blastomeres are transferred into the fallopian tube of an infertile female

Option 3:

Early embryos with up to 8 blastomeres are transferred to the uterus of an infertile female.

Option 4:

It is the transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce ovum but can provide suitable environment for fertilization and development.

Correct Answer:

It is the transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce ovum but can provide suitable environment for fertilization and development.

Solution:

GIFT is Gamete Intra-Fallopian Transfer. GIFT involves transfer of an ovum into the fallopian tube for in vivo fertilization, not embryo transfer (which is IVF-ET). Embryo transfer is a technique in which the fertilised egg or embryo in the early stage is transferred into the female reproductive tract.

Hence, the correct answer is option 4). It is the transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce ovum but can provide suitable environment for fertilization and development.

Q. 85 Choose the correct statements regarding cell organelles and their inclusions.

- A. The endomembrane system includes Golgi complex, endoplasmic reticulum and mitochondria.
- B. Rough endoplasmic reticulum bears ribosomes on its surface.
- C. Both mitochondria and plastids have circular DNA.
- D. A network of microtubules, microfilaments and intermediate filaments present in the cytoplasm is called cytoskeleton.
- E. Mitochondrion is a single membrane-bound structure.

Choose the **correct** answer from the options given below :

Option 1:

A, B and C only

Option 2:

C, D and E only

Option 3:

A and B only

Option 4:

B, C and D only

Correct Answer:

B, C and D only

Solution:

Mitochondria are not part of the endomembrane system (A false). RER has ribosomes (B true), mitochondria & plastids have circular DNA (C true), cytoskeleton definition is correct (D true), mitochondria are double membrane-bound (E false).

Hence, the correct answer is option 4) B, C, and D only

Q. 86 Select the correct statements regarding cell membrane in eukaryotic cell.

- A. Membrane of human , RBCs has approximately 52% protein.
- B. Major phospholipids are arranged in a bilayer.
- C. Extensions of the plasma membrane into the cell form mesosomes.
- D. Tails towards the inner part of lipids are hydrophobic and - thus protected from aqueous medium
- E. Glycocalyx is present on the outer surface of the plasma membrane.

Choose the **correct** answer from the options given below :

Option 1:

C, D and E only

Option 2:

B, C and E only

Option 3:

A, B and D only

Option 4:

A, C and E only

Correct Answer:

A, B and D only

Solution:

RBC membrane has 52% protein (A true), phospholipids form a bilayer (B true), hydrophobic(water fearing) tails face inward (D true). Mesosomes are found in prokaryotes (C false), glycocalyx(a carbohydrate-rich covering) is present, but statement combos make (3) correct.

Hence, the correct answer is option 3) A, B, and D only

Q. 87 Match List I with List II related to muscular/skeletal system :

List I		List II	
A	Tetany	I	Inflammation of joints
B	Arthritis	II	Autoimmune disorder affecting neuromuscular Junction
C	Myasthenia gravis	III	Wild contraction in muscle due to low Ca^{++} in body fluid
D	Muscular dystrophy	IV	Progressive degeneration of skeletal muscle

Choose the **correct** answer from the options given below:

Option 1:

A-IV, B-III, C-II, D-I

Option 2:

A-III, B-I, C-II, D-IV

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-III, B-II, C-I, D-IV

Correct Answer:

A-III, B-I, C-II, D-IV

Solution:

Tetany is due to low Ca^{2+} causing spasms (III), arthritis causes inflammation of joints (I), myasthenia gravis is an autoimmune disease and affects the muscles (II), and muscular dystrophy shows degeneration (IV).

Hence, the correct answer is option 2) A-III, B-I, C-II, D-IV

Q. 88

Evolution of human appears parallel to the progressive development of brain and language skills. As such, the evolution of individual species in the order of sequence of their appearance is :

Option 1:

Ramapithecus→Homo habilis →Homo erectus→ Neanderthal→Homo sapiens

Option 2:

Neanderthal → Ramapithecus →

Homo habilis Homo erectus →

Homo sapiens

Option 3:

Homo habilis → Homo erectus →

Ramapithecus → Neanderthal →

Homo sapiens

Option 4:

Homo sapiens → Ramapithecus →

Homo habilis → Neanderthal

Homo erectus

Correct Answer:

Ramapithecus→Homo habilis →Homo erectus→ Neanderthal→Homo sapiens

Solution:

Correct evolutionary sequence: Ramapithecus → Homo habilis → Homo erectus → Neanderthal → Homo sapiens.

- Ramapithecus was the earliest ape-man ancestor.
- Homo habilis - first tool maker
- Homo erectus - upright man
- Neanderthal man - these were more advanced
- Homo sapiens - modern humans

Hence, the correct answer is option 1) Ramapithecus→Homo habilis →Homo erectus→ Neanderthal→Homo sapiens

Q. 89 The flightless bird with forelimbs modified as paddle-like structures suited for swimming is known as :

Option 1:

Psittacula

Option 2:

Aptenodytes

Option 3:
Neophron

Option 4:
Struthio

Correct Answer:
Aptenodytes

Solution:

Aptenodytes (penguins) are flightless birds with flipper-like wings adapted for swimming.

Hence, the correct answer is option 2) Aptenodytes

Q. 90 Choose the correct statements regarding population interactions between two species.

- A. In both parasitism and commensalism, only one species benefits and the other species is harmed.
- B. Both species benefit in mutualism
- C. Both species benefit in commensalism.
- D. In parasitism, only one species benefits and the other species is harmed.
- E. In amensalism, one species is harmed and the other is unaffected.

Choose the **correct** answer from the options given below:

Option 1:
A and B only

Option 2:
B and E only

Option 3:
B, D and E only

Option 4:
A and D only X

Correct Answer:
B, D and E only

Solution:

In mutualism, both species benefit (B true). In parasitism, one benefits and the other is harmed (D true). In amensalism, one species is harmed while the other remains unaffected (E true). Statement A is incorrect because in commensalism, the other species is unaffected, not harmed. Statement C is incorrect as only one species benefits in commensalism.

Hence, the correct answer is option 3) B, D, and E only

Careers360

Code 12 NEET 2026 Question Paper with Answer Key & Solutions

- Q. 1** A 100-turn closely wound circular coil of radius 5 cm has a magnetic field of $3.14 \times 10^{-3} \text{ T}$ at its centre. The current flowing through the coil, and the magnitude of the magnetic moment of this coil are, respectively :
(Take $\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$)

Option 1:

2.5 A, 20 A m²

Option 2:

2 A, 4 A m²

Option 3:

2.5 A, 2 A m²

Option 4:

2 A, 10 A m²

Correct Answer:

2.5 A, 2 A m²

Solution:

$$B = \frac{\mu_0 NI}{2R} \Rightarrow I = \frac{2BR}{\mu_0 N}$$

$$\text{Current } I = \frac{2 \times 3.14 \times 10^{-3} \times 5 \times 10^{-2}}{4\pi \times 10^{-7} \times 100}$$

$$\text{Current } I = 2.5 \text{ A}$$

$$\text{Magnetic moment } M = NIA$$

$$= 100 \times 2.5 \times \pi r^2$$

$$= 100 \times 2.5 \times 3.14 \times 25 \times 10^{-4}$$

$$= 1.96$$

$$= 2 \text{ A} \times \text{m}^2$$

Hence, the correct option is (3).

LIST I		LIST II	
A.	$E = h\nu$	I.	de Broglie wavelength
B.	Diffraction and Interference	II.	Particle nature of light
C.	$\lambda = h/p$	III.	Wave nature of light
D.	Compton effect	IV.	Energy of photon

Choose the **correct** answer from the options given below :

Option 1:

A-IV, B-III, C-II, D-I

Option 2:

A-IV, B-III, C-I, D-II

Option 3:

A-I, B-IV, C-III, D-II

Option 4:

A-IV, B-I, C-II, D-III

Correct Answer:

A-IV, B-III, C-I, D-II

Solution:

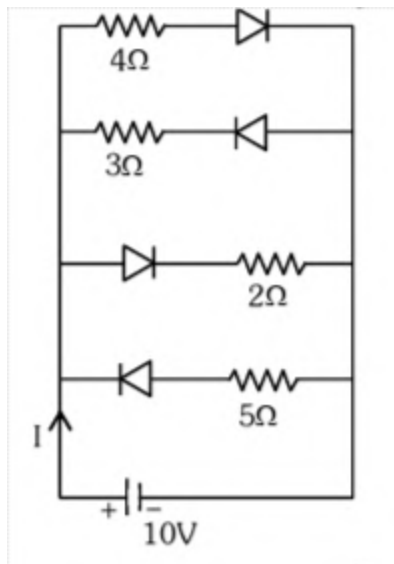
A. $E = h\nu \rightarrow$ Energy of photon $\rightarrow$ IV

B. Diffraction and Interference $\rightarrow$ Wave nature of light $\rightarrow$ III

C. $\lambda = \frac{h}{p} \rightarrow$ de Broglie wavelength $\rightarrow$ I

D. Compton effect $\rightarrow$ Particle nature of light $\rightarrow$ II

Hence, the correct option is (2).



Option 1:
 $\frac{1}{3}$ A

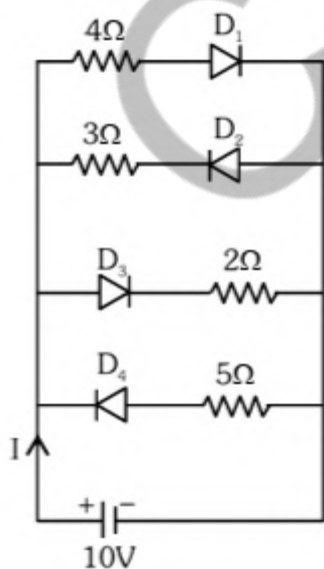
Option 2:
 $\frac{15}{2}$ A

Option 3:
 $\frac{5}{3}$ A

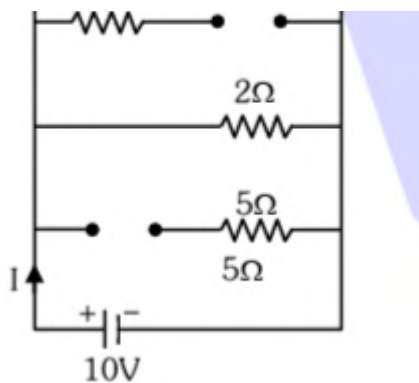
Option 4:
 $\frac{5}{9}$ A

Correct Answer:
 $\frac{15}{2}$ A

Solution:



Here D_1 and D_3 are F.B and D_2 and D_4 are R.B. then equivalent circuit



$$R_{eq} = \frac{4 \times 2}{4 + 2} = \frac{4}{3} \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{10}{4/3} = \frac{15}{2} \text{ A}$$

Hence, the correct option is (2).

Q. 4 The speed of light in vacuum is taken as unity. If light takes 6 min 40 s to reach the Earth from the Sun, the distance between the Sun and the Earth in new unit is :

Option 1:

$$3. \times 10^8$$

Option 2:

$$500$$

Option 3:

$$3 \times 10^{10}$$

Option 4:

$$400$$

Correct Answer:

$$400$$

Solution:

Distance = speed $\times$ time

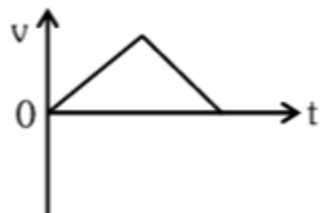
$$d = 1 \times 400 = 400 \text{ units}$$

Hence, the correct option is (4).

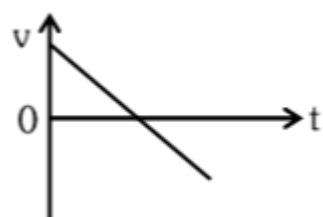
Q. 5 The following plots show variation of velocity (v) with time (t), of a ball thrown vertically upward, and falling back. Which of the following plots is/are correct ?



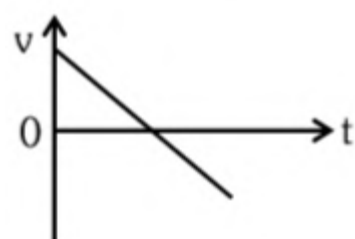
Option 2:



Option 3:



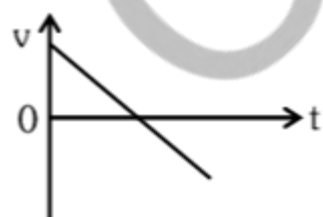
Option 4:



Option 5:



Correct Answer:



Solution:

Physics: Velocity decreases linearly to zero, then becomes negative Graph: Straight line with negative slope crossing zero

Hence, the correct option is (3).

Option 1:

0.2 cm

Option 2:

0.1 cm

Option 3:

0.02 cm

Option 4:

0.01 cm

Correct Answer:

0.02 cm

Solution:

$$20\text{VSD} = 16\text{MSD}$$

$$\text{VSD} = \frac{16}{20} (1 \text{ mm})$$

$$= 0.8 \text{ mm}$$

$$\text{LC} = \text{MSD} - \text{VSD}$$

$$= 1 - 0.8$$

$$= 0.2 \text{ mm}$$

$$= 0.02 \text{ cm}$$

Hence, the correct option is (3).

Q. 7

An ac circuit contains a resistance of $1\text{k}\Omega$, a capacitor of $0.1\mu\text{F}$ and an inductor of 1 mH connected in series. The resonance frequency of the circuit is approximately :

Option 1:

15.9 kHz

Option 2:

20.7 kHz

Option 3:

$10 \cdot 1\text{kHz}$

Option 4:

$13 \cdot 5\text{kHz}$

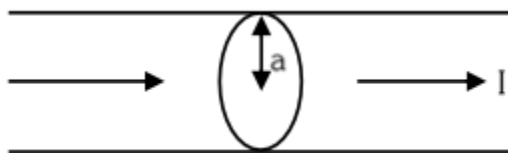
Solution:

For a series LCR circuit, resonance frequency is:

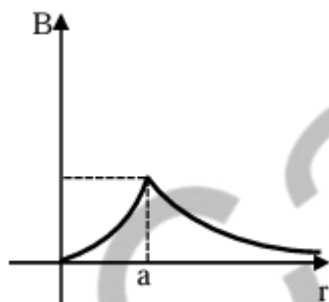
$$f = \frac{1}{2\pi\sqrt{LC}}$$
$$f = \frac{1}{2\pi\sqrt{10^{-3} \times 10^{-7}}} = \frac{1}{2\pi\sqrt{10^{-10}}} = \frac{1}{2\pi \times 10^{-5}}$$
$$f \approx \frac{1}{6.28 \times 10^{-5}} \approx 1.59 \times 10^4 \text{ Hz}$$
$$f \approx 15.9 \text{ kHz}$$

Hence, the correct option is (1).

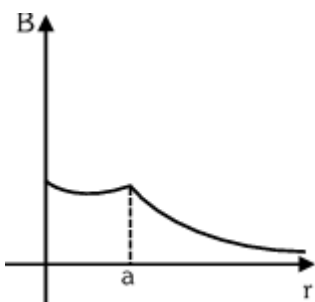
- Q. 8** The figure given below shows a long straight solid wire of circular cross-section of radius 'a' carrying steady current I . The current I is uniformly distributed across its cross-section. The plot which correctly represents the variation of magnetic field (B) with distance (r) from the axis of the conductor in the region is :

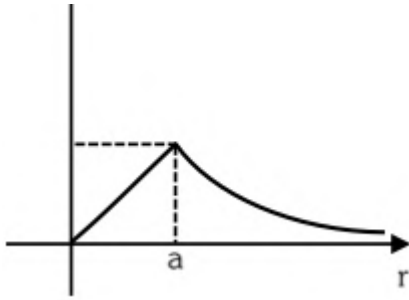


Option 1:

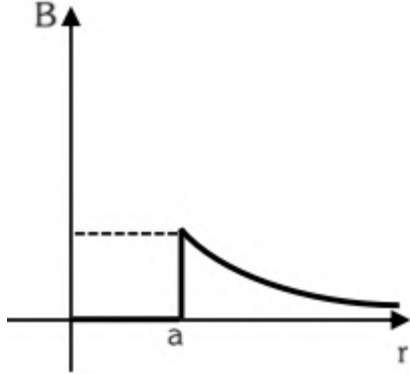


Option 2:

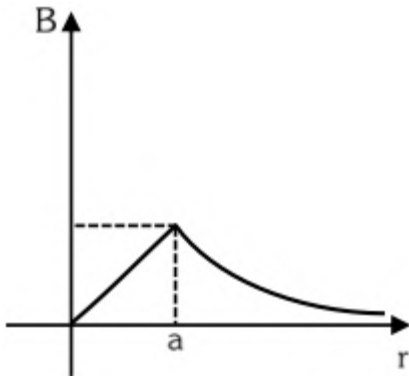




Option 4:



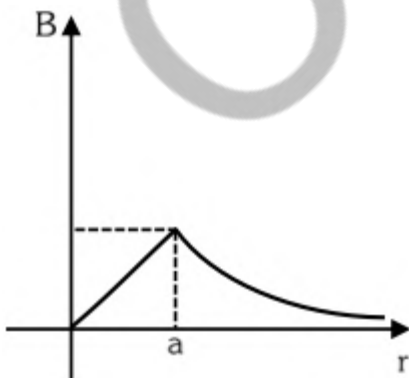
Correct Answer:



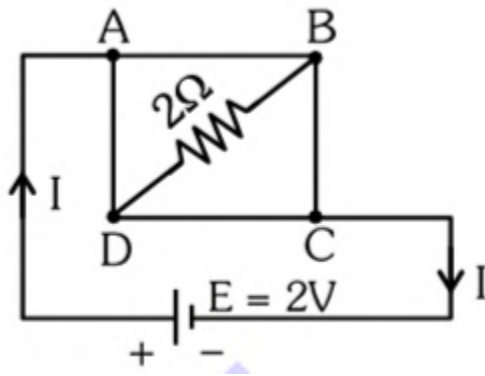
Solution:

$$B_{\text{in}} = \frac{\mu_0 I r}{2\pi R^2} \propto r$$

$$B_{\text{out}} = \frac{\mu_0 I}{2\pi r} \propto \frac{1}{r}$$



Hence, the correct option is (3).



Option 1:

4A

Option 2:

4.5A

Option 3:

8A

Option 4:

2A

Correct Answer:

2A

Solution:

Resistance of each wire of

$$W.S.B. = \frac{4}{4} = 1\Omega$$

This is balance W.S.B.

$$\text{So, } R_{\text{Net}} = 1\Omega$$

So, current

$$I = \frac{E}{R_{\text{Net}}} = \frac{2}{1} = 2 \text{ A}$$

Hence, the correct option is (4).

Q. 10 An unknown nucleus has nuclear density $2.29 \times 10^{17} \text{ kg/m}^3$ and mass of $19.926 \times 10^{-27} \text{ kg}$. Its mass number A is approximately :
(Take $R_0 = 1.2 \times 10^{-15} \text{ m}$, $4\pi = 12.56$)

Option 1:

16

Option 3:

12

Option 4:

19

Correct Answer:

12

Solution:

$$\text{Density } \rho = \frac{\text{Mass}}{\text{Volume}} = \frac{M}{\frac{4}{3}\pi R^3}$$

$$\text{Nuclear Radius: } R = R_0 A^{1/3}, \text{ so } R^3 = R_0^3 A$$

$$A = \frac{3M}{4\pi R_0^3 \rho} = \frac{3 \times 19.926 \times 10^{-27}}{12.56 \times (1.2 \times 10^{-15})^3 \times 2.29 \times 10^{17}}$$

$$A \approx \frac{59.778 \times 10^{-27}}{12.56 \times 1.728 \times 10^{-45} \times 2.29 \times 10^{17}} \approx 12$$

Hence, the correct option is (3).

Q. 11 A rectangular wire loop of sides 8 cm and 3 cm with a small cut, is moving out of a region of uniform magnetic field of magnitude 0.3 T directed normal to the plane of the loop. The emf developed across the cut, if the velocity of the loop is 2 cm s^{-1} , in a direction normal to the shorter side of the loop, will be :

Option 1:

$$1.8 \times 10^{-4} \text{ volt}$$

Option 2:

$$1.3 \times 10^{-4} \text{ volt}$$

Option 3:

$$1.2 \times 10^{-4} \text{ volt}$$

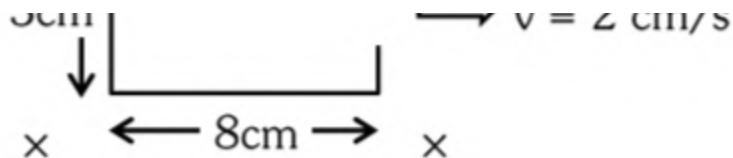
Option 4:

$$4.8 \times 10^{-4} \text{ volt}$$

Correct Answer:

$$1.8 \times 10^{-4} \text{ volt}$$

Solution:



$$e = Blv = 0.3 \times 0.08 \times 0.02 = 1.8 \times 10^{-4} \text{ V}$$

Hence, the correct option is (1).

Q. 12 A galvanometer of resistance 100Ω gives full scale deflection for a current of 1 mA . It is converted into an ammeter of range $0 - 10 \text{ A}$. The shunt required is :

Option 1:
 0.001Ω

Option 2:
 0.10Ω

Option 3:
 1.0Ω

Option 4:
 0.01Ω

Correct Answer:
 0.01Ω

Solution:

Given that

$$R_g = 100\Omega$$

$$I_g = 1 \text{ mA}$$

$$I = 10 \text{ A}$$

$$S = \frac{I_g R_g}{(I - I_g)} = \frac{10^{-3}(100)}{(10 - 10^{-3})}$$

Since $I_g \ll I$ so $I - I_g \simeq I$

$$S = 10^{-2} = 0.01\Omega$$

Hence, the correct option is (4).

Option 1:

K

Option 2:

2K

Option 3:

$\frac{K}{2}$

Option 4:

$\frac{K}{4}$

Correct Answer:

$\frac{K}{4}$

Solution:

$$\begin{aligned} I_R &= K \cos^2 \frac{\Delta\phi}{2} \\ &= K \cos^2 \left(\frac{120^\circ}{2} \right) \\ &= K \cos^2 60^\circ = \frac{K}{4} \end{aligned}$$

Hence, the correct option is (4).

Q. 14 The magnitude and direction of the acceleration produced in a body of mass 5 kg when two mutually perpendicular forces 8 N and 6 N act on it, are respectively

Option 1:

2 m s^{-2} ; $\tan^{-1}(3/4)$ with 8 N force

Option 2:

2 ms^{-2} ; $\tan^{-1} 2$ s/3) with 8 N force

Option 3:

2 m s^{-2} ; $\tan^{-1}(3/4)$ with 6 N force

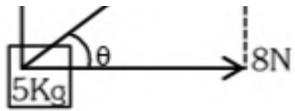
Option 4:

20 m s^{-2} ; $\tan^{-1}(4/3)$ with 8 N force

Correct Answer:

2 m s^{-2} ; $\tan^{-1}(3/4)$ with 8 N force

Solution:



$$F_{\text{Res}} = \sqrt{6^2 + 8^2}$$

$$F_{\text{Res}} = 10 \text{ N}$$

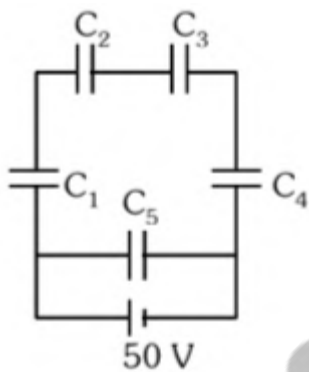
$$a = \frac{F_{\text{Res}}}{m} = \frac{10}{5} = 2 \text{ m/s}^2$$

$$\tan \theta = \frac{6}{8} = \frac{3}{4}$$

$$a = 2 \text{ m/s}^2 \tan^{-1} \left(\frac{3}{4} \right) \text{ with } 8 \text{ N force}$$

Hence, the correct the option is (1).

- Q. 15** Five capacitors of capacitances $C_1 = C_2 = C_3 = C_4 = 40 \mu\text{F}$ and $C_5 = 2 \cdot 5 \mu\text{F}$ are connected as shown, along with a battery of 50 V .The equivalent capacitance and the charges on each capacitor respectively are :



Option 1:

$4 \mu\text{F}$, $250 \mu\text{C}$ on C_1 to C_4 and $125 \mu\text{C}$ on C_5

Option 2:

$5 \mu\text{F}$, $250 \mu\text{C}$ on all capacitors

Option 4:

$5\mu\text{ F}$, $125\mu\text{C}$ on all capacitors

Correct Answer:

$5\mu\text{ F}$, $125\mu\text{C}$ on all capacitors

Solution:

Top branch: C_2 and C_3 in series

$$\frac{1}{C_{23}} = \frac{1}{10} + \frac{1}{10} \Rightarrow C_{23} = 5\mu\text{F}$$

Now C_1 , C_{23} , C_4 are in series:

$$\frac{1}{C_{eq1}} = \frac{1}{10} + \frac{1}{5} + \frac{1}{10} = \frac{1+2+1}{10} = \frac{4}{10} \Rightarrow C_{eq1} = 2.5\mu\text{F}$$

Parallel with C_5

$$C_{total} = C_{eq1} + C_5 = 2.5 + 2.5 = 5\mu\text{F}$$

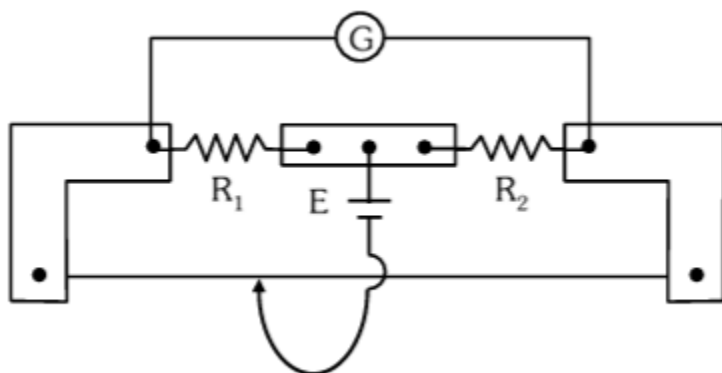
Total charge

$$Q = C_{total} V = 5 \times 50 = 250\mu\text{C}$$

Each branch capacitance = $2.5\mu\text{ F}$ $\rightarrow$ equal charge sharing

$$Q_{\text{each branch}} = 125\mu\text{C}$$

Hence, the correct option is (4).



Option 1:

Only the right-sided deflection

Option 2:

Only the left-sided deflection

Option 3:

There will be no deflection irrespective of the position of the jockey

Option 4:

Both right-sided and left-sided deflection and at balance point, no deflection

Correct Answer:

Both right-sided and left-sided deflection and at balance point, no deflection

Solution:

In the given circuit, it is not mentioned whether the meter bridge is balanced or not. So on interchanging the positions of battery and galvanometer, deflection may or may not be zero. In case of unbalanced condition, deflection may be left or right sided in galvanometer.

Hence, the correct option is (4).

Q. 17 The power of a crane, which lifts a mass of 1000 kg to a height of 20 m in 10 s is :

$$(g = 9.8 \text{ m/s}^2)$$

Option 1:

19.6 kW

Option 2:

19 · 6 W

Option 4:

$$39 \cdot 2 \text{ W}$$

Correct Answer:

$$19.6 \text{ kW}$$

Solution:

$$\begin{aligned}\text{Power of crane} &= \frac{mgh}{t} \\ &= \frac{1000 \times 9.8 \times 20}{10} \\ &= 19.6 \times 1000 \\ &= 19.6 \text{ kW}\end{aligned}$$

Hence, the correct option is (1).

Q. 18 Match **List I** with **List II** :

List I		List II	
A.	Young's Modulus	I.	$\frac{\Delta d}{\Delta L} \left(\frac{L}{d} \right)$
B.	Compressibility	II.	$\frac{FL}{A(\Delta L)}$
C.	Bulk Modulus	III.	$-\frac{1}{\Delta P} \left(\frac{\Delta V}{V} \right)$
D.	Poisson's Ratio	IV.	$-P \left(\frac{V}{\Delta V} \right)$

Choose the **correct** answer from the options given below:

Option 1:

A-III, B-II, C-I, D-IV

Option 2:

A-II, B-III, C-IV, D-I

Option 3:

A-I, B-IV, C-III, D-II

Option 4:

A-IV, B-I, C-II, D-III

Correct Answer:

A-II, B-III, C-IV, D-I

C. Bulk Modulus $K = -P \left(\frac{V}{\Delta V} \right)$

D. Poisson's Ratio

$$\text{Poisson's ratio} = \frac{\Delta d}{\Delta L} \left(\frac{L}{d} \right)$$

Hence, the correct option is (2).

Q. 19 In a concave lens, a ray of light emanating from the object parallel to the principal axis of the lens, after refraction :

Option 1:

passes through the second principal focus.

Option 2:

appears to diverge from the first principal focus.

Option 3:

passes through $2F$, which is the radius of curvature of the lens.

Option 4:

emerges parallel to the principal axis.

Correct Answer:

appears to diverge from the first principal focus.

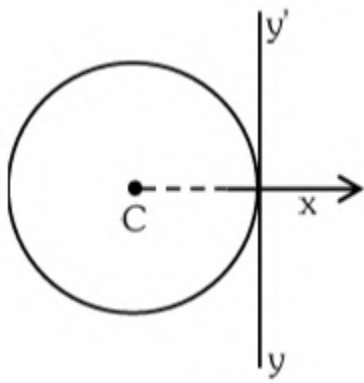
Solution:



When incident light is parallel to principal axis, after refraction image will be at focus. This focus will be second principal focus (F_2).

In this question as lens is concave (diverging) after refraction light will diverge and appears to be coming from second principal focus (F_2).

Hence, the correct option is (2).



Option 1:

$$\frac{3 mL^3}{8\pi}$$

Option 2:

$$\frac{3 mL^2}{8\pi^2}$$

Option 3:

$$\frac{3 mL^3}{8\pi^2}$$

Option 4:

$$\frac{3 mL^2}{8}$$

Correct Answer:

$$\frac{3 mL^3}{8\pi^2}$$

Solution:

Let mass of wire is M

length = L

given m = Linear mass density

$$m = \frac{M}{L} \quad L = 2\pi R$$

$$M = mL \quad R = \frac{L}{2\pi}$$

$$I_{yy'} = \frac{MR^2}{2} + MR^2$$

$$= \frac{3}{2} MR^2 = \frac{3}{2} (mL) \left(\frac{L^2}{4\pi^2} \right)$$

$$I_{yy'} = \frac{3}{8} \frac{mL^3}{\pi^2}$$

Hence, the correct option is (3).

- Q. 21** Each side of a metallic cube of mass 5.580 kg is measured to be 9.0 cm . Keeping the significant figures in view, the density of the material of the cube can be best expressed as $X \times 10^3 \text{ kg m}^{-3}$, where the value of X is :

Option 2:

7 · 7

Option 3:

7.65

Option 4:

7 · 6

Correct Answer:

7 · 7

Solution:

$$\text{Density} = \frac{\text{mass}}{\text{volume}} = \frac{5.580 \text{ kg}}{(9.0 \times 10^{-2})^3}$$

$$= 7.654 \times 10^3 \text{ kg/m}^3$$

Rounding off to 2 significant digits

$$\text{Result} = 7.7 \times 10^3$$

$$\Rightarrow x = 7.7$$

Hence, the correct option is (2).

Q. 22 For a travelling harmonic wave $y(x, t) = 2.0 \cos 2\pi(10t - 0.0080x + 0.35)$, where x and y are in cm and t in s . The phase difference between oscillatory motion of two points separated by a distance of $0.5\lambda/B$:

Option 1:

$0.08\pi\text{rad}$

Option 2:

$0.008\pi\text{rad}$

Option 3:

$0.8\pi\text{rad}$

Option 4:

$8\pi\text{rad}$

Correct Answer:

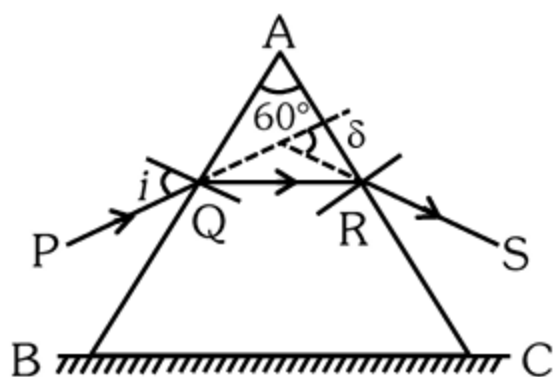
$0.8\pi\text{rad}$

Solution:

Wave Number (k) : From the equation, $k = 2\pi \times 0.0080$

Phase Difference: $\Delta\phi = k\Delta x = (2\pi \times 0.0080) \times 0.5 = 0.008\pi\text{rad}$

Hence, the correct option is (3).



Option 1:

45°

Option 2:

55°

Option 3:

35°

Option 4:

40°

Correct Answer:

40°

Solution:

For an equilateral prism, $A = 60^\circ$ since refracted ray is parallel to base so it is condition of minimum deviation.

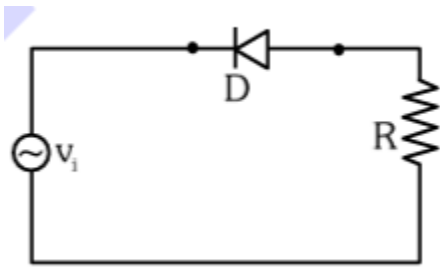
$$r = \frac{A}{2} = \frac{60}{2} = 30^\circ$$

$$\text{and } i = e = 50^\circ$$

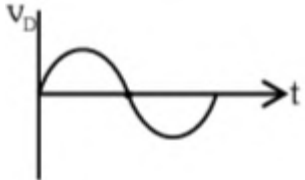
$$\delta = i + e - A$$

$$\begin{aligned} \text{But } \delta &= 50 + 50 - 60 \\ &= 40^\circ \end{aligned}$$

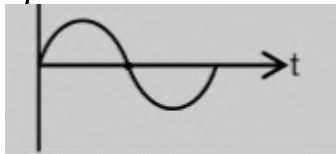
Hence, the correct option is (4).



Option 1:



Option 2:

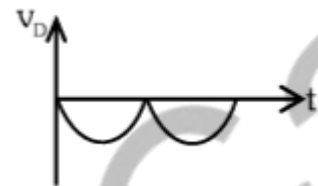


v_D

Option 3:



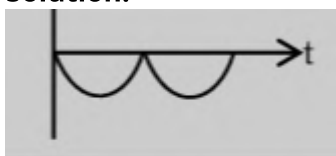
Option 4:



Correct Answer:



Solution:

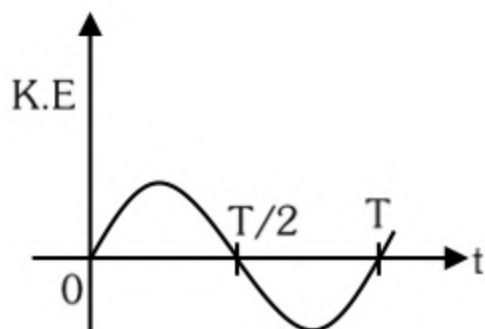


v_D

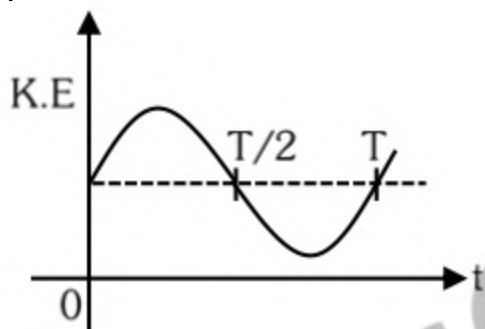
Hence, the correct option is (4).

- Q. 25** For a simple pendulum, having time period T , the variation of kinetic energy (K.E.) with time (t) is represented by :

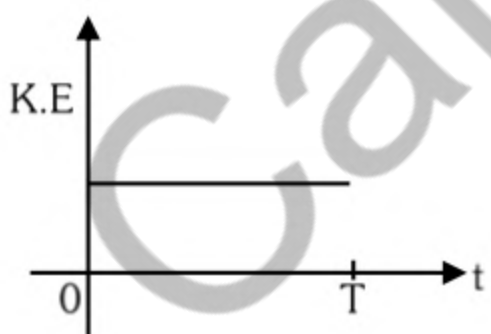
Option 1:

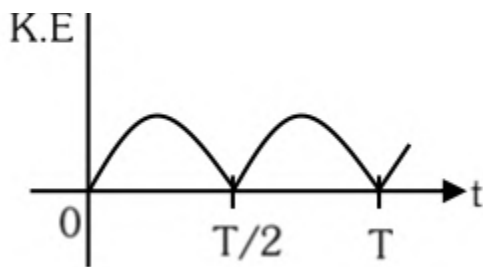


Option 2:

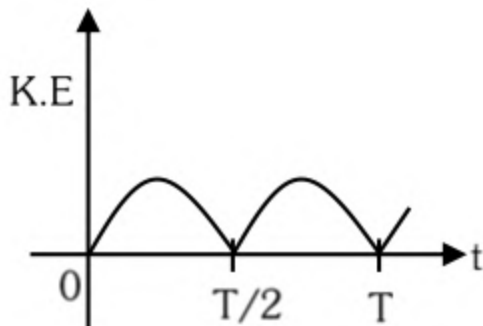


Option 3:





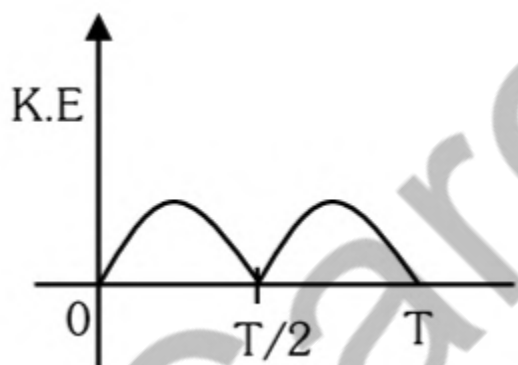
Correct Answer:



Solution:

$$K.E = \frac{1}{2} m \omega^2 A^2 \cos^2 \omega t$$

oscillation starts from extreme position



Hence, the correct answer is option (4).

Q. 26 A resistor is connected to a battery of 12 V emf and internal resistance 2Ω . If the current in the circuit is 0.6 A, the terminal voltage of the battery is :

Option 1:

12 V

Option 2:

1.2 V

Option 4:

10.8 V

Correct Answer:

10.8 V

Solution:

Given that:

$$\text{EMF} = 12 \text{ V}$$

$$r = 2\Omega$$

$$I = 0.6 \text{ A}$$

So terminal voltage

$$V = E - Ir$$

$$= 12 - (.6)^2$$

$$V = 10.8 \text{ volt}$$

Hence, the correct option is (4).

Q. 27 The amount of work done to raise a mass 'm' from the surface of the Earth to a height equal to the radius of the Earth 'R', will be :

Option 1:

$$mg \frac{R}{2}$$

Option 2:

$$mgR$$

Option 3:

$$mg \frac{R}{4}$$

Option 4:

$$2 mg R$$

Correct Answer:

$$mg \frac{R}{2}$$

Solution:

Change in potential energy:

$$W = U_2 - U_1 = -\frac{GMm}{2R} - \left(-\frac{GMm}{R}\right)$$

$$W = \frac{GMm}{R} - \frac{GMm}{2R} = \frac{GMm}{2R}$$

$$g = \frac{GM}{R^2}$$

So,

Q. 28 An electric heater supplies heat to a system at a rate of 100 W . If the system performs work at a rate of 75 J/s, then the rate at which internal energy increases will be :

Option 1:

75 W

Option 2:

25 W

Option 3:

100 W

Option 4:

125 W

Correct Answer:

25 W

Solution:

Rate of increase of internal energy Using first law of thermodynamics:

$$\frac{dU}{dt} = \text{Heat supplied} - \text{Work done} = 100 - 75 = 25 \text{ W}$$

Hence, the correct option is (2).

Q. 29 A room heater is rated 400 W, 220 V. If the supply voltage drops to 200 V , what will be the power consumed (approximately)?

Option 1:

400 W

Option 2:

121 W

Option 3:

331 W

Option 4:

200 W

Correct Answer:

331 W

Solution:

Supply voltage = 200 V

$$\text{Resistance of heater } R_H = \frac{(220)^2}{400}$$

So Actual power gained

$$\begin{aligned} P' &= \frac{V_s^2}{R_H} = \frac{(200)^2}{(220)^2} (400) \\ &= 330.57 \text{ W} \\ &\simeq 331 \text{ W} \end{aligned}$$

Hence, the correct option is (3) 331 W.

Q. 30 When a ruler falls vertically, 5 different person catch it with different reaction times. ($g = 9.8 \text{ m s}^{-2}$)

- A. Person A has reaction time of 0.20 s .
- B. Person B has reaction time of 0.22 s .
- C. Person C has reaction time of 0.18 s .
- D. Person D has reaction time of 0.19 s .
- E. Person E has reaction time of 0.21 s .

What is the **correct** order of the distance travelled by the ruler for each person?

Option 1:

$$B > E > A > C > D$$

Option 2:

$$C > D > A > E > B$$

Option 3:

$$C > D > A > B > E$$

Option 4:

$$B > E > A > D > C$$

Correct Answer:

$$B > E > A > D > C$$

Solution:

$$\text{Distance fallen: } s = \frac{1}{2}gt^2$$

Distance $\propto t^2$, so compare reaction times:

Largest time $\rightarrow$ largest distance:

Hence, the correct option is (4).

Option 1:

0.5 J

Option 2:

1.0×10^{-6} J

Option 3:

0.5×10^{-6} J

Option 4:

1.0 J

Correct Answer:

0.5×10^{-6} J

Solution:

Initial energy:

$$U_i = \frac{1}{2}CV^2$$

Final voltage becomes $V/2$ (equal capacitors sharing charge)

Final energy:

$$U_f = \frac{1}{2}(2C)\left(\frac{V}{2}\right)^2 = \frac{1}{4}CV^2$$

Energy lost:

$$\Delta U = U_i - U_f = \frac{1}{2}CV^2 - \frac{1}{4}CV^2 = \frac{1}{4}CV^2$$

$$C = 200\text{pF} = 200 \times 10^{-12}$$

$$\begin{aligned}\Delta U &= \frac{1}{4} \times 200 \times 10^{-12} \times (100)^2 \\ &= 0.5 \times 10^{-6} \text{ J}\end{aligned}$$

Hence, the correct option is (3).

- Q. 32** Savitha, a XI standard student, while conducting an experiment to determine the effective length of a simple pendulum L, notes down the data of time taken to complete 30 oscillations as 60 s and hence calculates the length of the simple pendulum as :
(Take $\pi^2 = 9.8$, and $g = 9.8 \text{ m/s}^2$)

Option 1:

0.75 m

Option 2:

1 m

Option 4:

2 m

Correct Answer:

1 m

Solution:

$$T = \frac{\text{Total Time}}{\text{Number of Oscillations}} = \frac{60 \text{ s}}{30} = 2 \text{ s}$$

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$T^2 = 4\pi^2 \left(\frac{L}{g} \right)$$

$$L = \frac{T^2 \cdot g}{4\pi^2}$$

$$L = \frac{2^2 \cdot 9.8}{4 \cdot 9.8} = \frac{4 \cdot 9.8}{4 \cdot 9.8} = 1 \text{ m}$$

Hence, the correct option is (2).

Q. 33 The peak value of an alternating current is 5 A and frequency is 60 Hz . How long will the current, starting from zero, take to reach the peak value ?

Option 1:

$$\frac{1}{120} \text{ s}$$

Option 2:

$$\frac{1}{240} \text{ s}$$

Option 3:

$$\frac{1}{30} \text{ s}$$

Option 4:

$$\frac{1}{60} \text{ s}$$

Correct Answer:

$$\frac{1}{240} \text{ s}$$

Solution:

Time to reach peak current

For AC:

$$i = I_0 \sin(\omega t)$$

Peak occurs when $\omega t = \frac{\pi}{2}$

Hence, the correct option is (2).

- Q. 34** In interference and diffraction, the light energy is redistributed. If it reduces in one region, producing a dark fringe, it increases in another region, producing a bright fringe.
 A. As there is no gain or loss of energy, these phenomena are consistent with the principle of conservation of energy.
 B. Diffraction and interference are characteristics exhibited only by light waves.
 Choose the **correct** answer from the options given below :

Option 1:

A is false, but B is true

Option 2:

A is true, but B is false

Option 3:

A is true and B is also true

Option 4:

Both A and B are false

Correct Answer:

A is true, but B is false

Solution:

Statement A-Energy is redistributed (bright & dark fringes), no net loss → consistent with conservation of energy → True

Statement B:Interference and diffraction are wave phenomena, but not only for light (also seen in sound, water waves, etc.) → False

Hence, the correct option is (2).

- Q. 35** A box of mass 15 kg is kept on the floor of a stationary trolley. The coefficient of static friction between the box and the trolley is 0.12 . Keeping the box in stationary state over the trolley, the maximum acceleration with which the trolley can be moved horizontally in ms^{-2} is :

$$(g = 10 \text{ m/s}^2)$$

Option 1:

1.8

Option 2:

1.2

Option 4:

2.1

Correct Answer:

1.2

Solution:

$$ma_{\max} = F_{lm}$$

$$ma_{\max} = \mu mg$$

$$a_{\max} = \mu g = 1.2 \text{ m/s}^2$$

Hence, the correct option is (2).

Q. 36 The sum of kinetic energy and potential energy of a simple pendulum bob is 0.02 joule. The speed of the simple pendulum bob at equilibrium position is approximately :
(Consider mass of the bob = 20 g)

Option 1:

2.0 m/s

Option 2:

0.2 m/s

Option 3:

14 · 1 m/s

Option 4:

1.41 m/s

Correct Answer:

1.41 m/s

Solution:

$$\text{Total energy} \Rightarrow \frac{1}{2} KA^2 = 0.02 \text{ J}$$

$$\frac{1}{2} m^2 A^2 = 0.02 \text{ J}$$

$$\frac{1}{2} mV_{\max}^2 = 0.02 \text{ J}$$

$$V_{\max}^2 = \frac{0.04}{20 \times 10^{-3}} = 2$$

$$V_{\max} = 1.41 \text{ m/s}$$

Hence, the correct option is (4).

∴ the volume of a nucleus is proportional to A .

C. The difference in mass of an atom and its nucleus is called the mass defect.

D. The difference in mass of a nucleus and its constituents is called the mass defect.

Choose the **correct** answer from the options given below :

Option 1:

A and D are true, but B and C are false

Option 2:

B and D are true, but A and C are false

Option 3:

B and C are true, but A and D are false

Option 4:

A and C are true, but B and D are false

Correct Answer:

B and D are true, but A and C are false

Solution:

Correct statements

A: Volume $\propto R^3$, and $R \propto A^{1/3} \rightarrow \text{Volume} \propto A \rightarrow \text{false}$

B: Volume $\propto A \rightarrow \text{true}$

C: Mass defect = difference between atom and nucleus $\rightarrow \text{false}$

D: Mass defect = difference between nucleus and its nucleons $\rightarrow \text{true}$

Hence, the correct option is (2).

Q. 38 The angular speed of a flywheel is increased from 600 rpm to 1200 rpm in 10 s . The number of revolutions completed by the flywheel during this times is:

Option 1:

600

Option 2:

300

Option 3:

900

Option 4:

150

Correct Answer:

150

$$\omega = 10\pi \text{ rad/s}$$

$$\theta = \left(\frac{\omega_i + \omega_f}{2} \right) t$$

$$\theta = \left(\frac{40\pi + 20\pi}{2} \right) 10$$

$$\theta = 300\pi$$

$$\text{number of revolution} = \frac{300\pi}{2\pi} = 150 \text{ rotation}$$

Hence, the correct option is (4).

Q. 39 A submarine is designed to withstand an absolute pressure of 100 atm . How deep can it go below the water surface?

(Consider the density of water = 1000 kg m^{-3} . $1 \text{ atm} = 1 \times 10^5 \text{ Pa}$ and gravitational acceleration $g = 10 \text{ m/s}^2$)

Option 1:

9900 m

Option 2:

990 m

Option 3:

9000 m

Option 4:

99 m

Correct Answer:

990 m

Solution:

Absolute pressure at depth h in water

$$P = P_0 + h\rho g \quad P_0 = \text{atmospheric pressure}$$

$$P - P_0 = h\rho g$$

$$h = \frac{P - P_0}{\rho g}$$

$$= \frac{(100 - 1) \times 10^5}{10^3 \times 10}$$

$$h = 990 \text{ m}$$

Hence, the correct option is (2).

**LIST I
(Electromagnetic
wave)**

LIST II (Production)

A.	Microwave	I.	Electrons in atoms emit light when they move from a higher energy level to a lower energy level
B.	Visible light	II.	Radioactive decay of nucleus
C.	Gamma rays	III.	Vibration of atoms and molecules
D.	Infra-red rays	IV.	Klystron valve or magnetron valve

Choose the **correct** answer from the options given below:

Option 1:

A-IV, B-III, C-II, D-I

Option 2:

A-III, B-IV, C-I, D-IIX

Option 3:

A-III, B-I, C-II, D-IV

Option 4:

A-IV, B-I, C-II, D-III

Correct Answer:

A-IV, B-I, C-II, D-III

Solution:

- A. Microwave: Produced by Klystron valve or magnetron valve
- B. Visible light: Electrons in atoms emit light moving to lower energy levels
- C. Gamma rays: Radioactive decay of nucleus
- D. Infra-red rays: Vibration of atoms and molecules

Hence, the correct option is (4).

Q. 41 Which of the following statements are correct?

- A. Inside a conductor, the electrostatic field is zero.
- B. Electric field at the surface of a charged conductor does not depend on its surface charge density.
- C. The interior of a charged conductor can have no excess charge in the static situation.
- D. At the surface of a charged conductor, the electrostatic field must be normal to the surface at every point
- E. The electrostatic potential is zero everywhere inside a charged conductor.

Choose the **correct** answer from the options given below :

Option 2:

A, C and E only

Option 3:

C, D and E only

Option 4:

A, B and D only

Correct Answer:

A, C and D only

Solution:

A. Inside a conductor, the electrostatic field is zero. (True)

B. Electric field at the surface does not depend on surface charge density. (False: $E = \sigma/\epsilon_0$)

C. The interior of a charged conductor can have no excess charge in the static situation. (True)

D. At the surface, the electrostatic field must be normal to the surface at every point. (True)

E. The electrostatic potential is zero everywhere inside a charged conductor. (False)

Analysis: Statements A, C, and D are correct.

Hence, the correct option is (1).

Q. 42 For a metal of work function 6.6 eV, which of the following wavelengths of incident radiation does not give rise to the photoelectric effect?
(Take Planck's constant as 6.6×10^{-34} J s)

Option 1:

200 nm

Option 2:

100 nm

Option 3:

50 nm

Option 4:

150 nm

Correct Answer:

200 nm

Solution:

$$\phi = 6.6\text{eV}$$

$$\frac{hc}{\lambda} = 6.6\text{eV}$$

$$\lambda_0 = \frac{12400}{6.6}$$

$$\lambda_0 = 1878.7 \quad = 187.87 \text{ nm}$$

$\lambda \geq \lambda_0$

$\therefore$ The wavelength of radiation does not give rise to photoelectric effect is 200 nm .

Hence, the correct option is (1).

Q. 43 In the first excited state of hydrogen atom, the energy of its electron is -3.4 eV . The radial distance of the electron from the hydrogen nucleus in this case is approximately :

(Take $1\text{eV} = 1.6 \times 10^{-19} \text{ J}$, $e = 1.6 \times 10^{-19} \text{ C}$ and $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$)

Option 1:

$$2.1 \times 10^{-8} \text{ m}$$

Option 2:

$$2.1 \times 10^{-10} \text{ m}$$

Option 3:

$$2.1 \times 10^{-11} \text{ m}$$

Option 4:

$$2.1 \times 10^{-9} \text{ m}$$

Correct Answer:

$$2.1 \times 10^{-10} \text{ m}$$

Solution:

$$E = \frac{kq_1q_2}{2r}$$

$$3.4 = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{2 \times r}$$

$$r = 2.1 \times 10^{-10} \text{ m}$$

Hence, the correct option is (2).

Q. 44 Two statements are given below :

A. When the forward bias voltage across a p-n junction diode increases above a certain threshold voltage, the diode current increases significantly.

B. This current is called reverse saturation current.

Choose the **correct** answer from the options given below :

Option 1:

Both Statements A and B are true

Option 2:

Both Statements A and B are false

Option 4:

Statement A is false, but Statement B is true

Correct Answer:

Statement *A* is true, but Statement *B* is false

Solution:

Statement A: When forward bias voltage increases above a certain threshold, the diode current increases significantly. (True)

Statement B: This current is called reverse saturation current. (False)

Analysis: Statement A is true, but Statement B is false.

Hence, the correct option is (3).

- Q. 45** A flask contains argon and chlorine in the ratio of 2 : 1 by mass. The temperature of the mixture is 27°C . The ratio of root mean square speed of the molecules of the two gases $\left(\frac{v_{\text{rms}}^{\text{Ar}}}{v_{\text{rms}}^{\text{Cl}}}\right)$ is :
(Atomic mass of argon = 40.0u and molecular mass of chlorine 70.0u)

Option 1:

$$\frac{7}{4}$$

Option 2:

$$\frac{-2}{\sqrt{7}}$$

Option 3:

$$\frac{\sqrt{7}}{2}$$

Option 4:

$$\frac{7}{2}$$

Correct Answer:

$$\frac{\sqrt{7}}{2}$$

Solution:

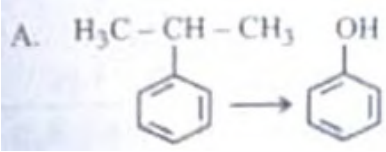
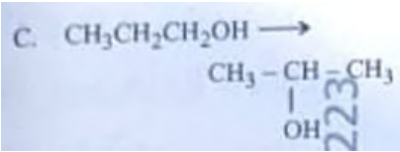
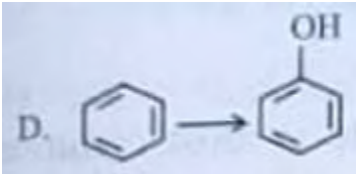
$$\text{As } V_{\text{RMS}} = \sqrt{\frac{3RT}{M_w}}$$

Given temperature $\rightarrow$ constant

$$\text{Then } V_{\text{RMS}} \propto \frac{1}{\sqrt{M_w}}$$

$$\Rightarrow \frac{V_{\text{RMS}}^{\text{Ar}}}{V_{\text{RMS}}^{\text{Cl}}} = \sqrt{\frac{M_w^{\text{Cl}}}{M_w^{\text{Ar}}}} = \sqrt{\frac{70}{40}} = \frac{\sqrt{7}}{2}$$

Hence, the correct option (3)

List - I	List - II
A. 	I. (i) oleum; (ii) NaOH , Δ ; (iii) H^+
B. $\text{CH}_3\text{COOH} \longrightarrow \text{CH}_3\text{CH}_2\text{OH}$	II. (i) O_2 ; (ii) $\text{H}_2\text{O}/\text{H}^+$
C. 	III. (i) CH_3OH , H^+ ; (ii) H_2 , catalyst
D. 	IV. (i) conc. H_2SO_4 , Δ ; (ii) $\text{H}^+/\text{H}_2\text{O}$

Choose the correct answer from the options given below :

Option 1:

A-I, B-III, C-IV, D-II

Option 2:

A-II, B-IV, C-III, D-I

Option 3:

A-II, B-III, C-IV, D-I

Option 4:

A-II, B-III, C-I, D-IV

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

A. Cumene to Phenol: This is the industrial preparation of phenol via cumene hydroperoxide. It requires oxidation with air (O_2) followed by acid hydrolysis.

A $\rightarrow$ II

B. Ethanoic Acid to Ethanol: This is a reduction of a carboxylic acid to a primary alcohol. While LiAlH_4 is common, another route involves esterification with methanol (CH_3OH , H^+) followed by catalytic hydrogenation.

B $\rightarrow$ III.

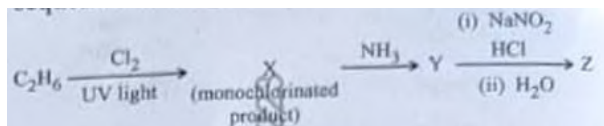
C -> IV.

D. Benzene to Phenol: This follows the classical sulfonating route: sulfonation with oleum to form benzene sulfonic acid, fusion with NaOH , and final acidification.

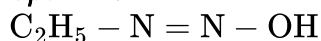
D -> I

Hence, the correct answer is option (3)

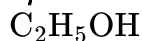
Q. 2 The major product Z formed in the following sequence of reactions is :



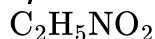
Option 1:



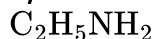
Option 2:



Option 3:



Option 4:



Correct Answer:



Solution:

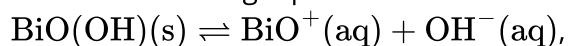
Step 1: Ethane (C_2H_6) undergoes free radical chlorination with Cl_2/UV light to form Chloroethane (X): $\text{C}_2\text{H}_5\text{Cl}$.

Step 2: Chloroethane reacts with Ammonia (NH_3) via nucleophilic substitution ($\text{S}_\text{N}2$) to form Ethylamine (Y): $\text{C}_2\text{H}_5\text{NH}_2$.

Step 3: Ethylamine reacts with nitrous acid ($\text{NaNO}_2 + \text{HCl}$) to form an unstable diazonium salt, which immediately reacts with water (H_2O) to form Ethanol (Z): $\text{C}_2\text{H}_5\text{OH}$.

Hence, the correct answer is option (2)

Q. 3 In a qualitative analysis, Bi^{3+} is detected by appearance of precipitate of $\text{BiO}(\text{OH})(\text{s})$. Calculate pH when the following equilibrium exists at 298 K :



$$K = 4 \times 10^{-10}$$

(Given : $\log 2 = 0.3010$)

Option 2:

5.286

Option 3:

8.714

Option 4:

9.301

Correct Answer:

9.301

Solution:

For the reaction: $\text{BiO}(\text{OH})(s) \rightleftharpoons \text{BiO}^+(aq) + \text{OH}^-(aq)$

The equilibrium constant is $K = [\text{BiO}^+][\text{OH}^-]$.

Let $[\text{BiO}^+] = [\text{OH}^-] = s$.

$$s^2 = K = 4 \times 10^{-10}$$

$$s = [\text{OH}^-] = \sqrt{4 \times 10^{-10}} = 2 \times 10^{-5} \text{ M}$$

pOH

$$\text{pOH} = -\log[\text{OH}^-]$$

$$\text{pOH} = -\log(2 \times 10^{-5}) = 5 - \log 2$$

$$\text{pOH} = 5 - 0.3010 = 4.699$$

pH

At 298 K, $\text{pH} + \text{pOH} = 14$.

$$\text{pH} = 14 - 4.699 = \mathbf{9.301}$$

Hence, the correct answer is option (4)

Q. 4 When 1dm^3 of CO_2 gas is passed over hot coke, the volume of gaseous mixture after complete reaction at STP becomes 1.4dm^3 . The composition of the gaseous mixture at STP is :

Option 1:

0.6dm^3 of CO, 0.8dm^3 of CO_2

Option 2:

0.8dm^3 of CO, 0.8dm^3 of CO_2

Option 3:

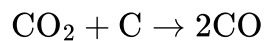
0.6dm^3 of CO, 0.4dm^3 of CO_2

Correct Answer:

0.8dm³ of CO 0.6dm³ of CO₂

Solution:

Reaction of CO₂ with hot coke:



Let:

dm³ of CO₂ reacts

Initial CO₂ = 1dm³

After reaction:

Remaining CO₂ = 1 - x

CO formed = 2x

Total final volume:

$$(1 - x) + 2x = 1 + x$$

Given:

$$1 + x = 1.4 \Rightarrow x = 0.4$$

Now calculate composition:

$$\text{CO}_2 = 1 - 0.4 = 0.6\text{dm}^3$$

$$\text{CO} = 2 \times 0.4 = 0.8\text{dm}^3$$

Hence, the correct answer is option (4)

Q. 5 Match List - I with List - II

	List - I (Quantum Numbers)			List - II (Orbital)
	"n"	"l"		
A	2	1	I	3d
B	4	0	II	2p
C	5	3	III	4s
D	3	2	IV	5f

Choose the correct answer from the options given below

Option 2:

A-I, B-II, C-III, D-IV

Option 3:

A-II, B-III, C-I, D-IV

Option 4:

A-IV, B-II, C-III, D-I

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

The orbital is designated as nl , where n is the principal quantum number and l corresponds to the subshell ($0 = s, 1 = p, 2 = d, 3 = f$).

A. $n = 2, l = 1 : 2p$ (A – II)

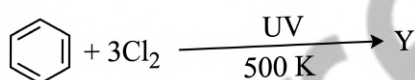
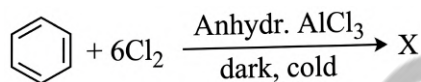
B. $n = 4, l = 0 : 4s$ (B-III)

C. $n = 5, l = 3 : 5f$ (C – IV)

D. $n = 3, l = 2 : 3d$ (D – I)

Hence, the correct answer is option (1)

Q. 6 The number of chlorine atoms present in the organic products X and Y of the following reactions, respectively, are :



Option 1:

3 and 3

Option 2:

6 and 6

Option 3:

6 and 3

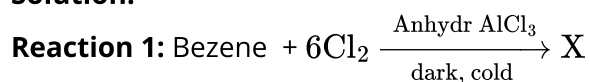
Option 4:

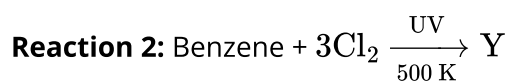
3 and 6

Correct Answer:

6 and 6

Solution:





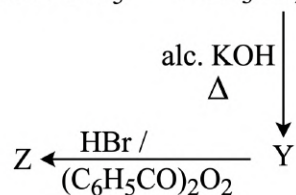
High temperature ($\sim 500\text{ K}$) $\rightarrow$ Addition reaction

Forms Benzene hexachloride (BHC / $\text{C}_6\text{H}_6\text{Cl}_6$)

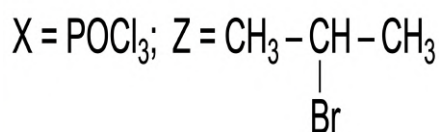
Number of Cl atoms in Y = 6

Hence, the correct answer is option (2)

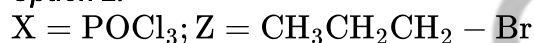
Q. 7 In the following reaction sequence, X and Z , respectively are :



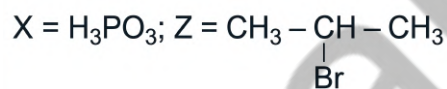
Option 1:



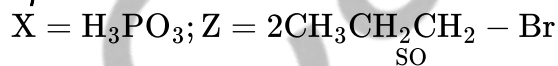
Option 2:



Option 3:



Option 4:



Correct Answer:



Solution:

1. Formation of **X** (Reaction with PCl_5)

The reaction of propan-1-ol with phosphorus pentachloride (PCl_5) is as follows:



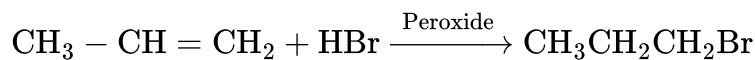
- The byproduct **X** is phosphorus oxychloride (POCl_3).

2. Formation of Y (Dehydrohalogenation)

- Product Y is Propene.

3. Formation of **Z** (Peroxide Effect)

The final step is the addition of HBr in the presence of benzoyl peroxide $(C_6H_5CO)_2O_2$. This is the AntiMarkovnikov addition (also known as the Kharasch effect or Peroxide effect).



The bromine atom attaches to the terminal carbon (the one with more hydrogens).

Product **Z** is 1-bromopropane.

Hence, the correct answer is option (2)

Q. 8 Match List - I with List - II

List - I (Transition metal /compound /complex)	List - II (Catalytic Role)
A. V_2O_5	I. Preparation of ammonia from N_2/H_2 mixture
B. Fe	II. Polymerisation of alkynes
C. $PdCl_2$	III. Preparation of H_2SO_4 from SO_2
D. Ni complex	IV. Oxidation of ethyne to ethanal

Choose the correct answer from the options given below:

Option 1:

A-III, B-IV, C-I, D-II

Option 2:

A-III, B-I, C-IV, D-II

Option 3:

A-II, B-I, C-IV, D-III

Option 4:

A-IV, B-I, C-III, D-II

Correct Answer:

A-III, B-I, C-IV, D-II

Solution:

A. V_2O_5 : Used in the Contact Process for the preparation of H_2SO_4 from SO_2 . (A-III)

B. Fe (Fine powder): Used in the Haber's Process for the preparation of ammonia from N_2 and H_2 . (B-I)

C. $PdCl_2$: Specifically used in the Wacker Process for the oxidation of ethyne (or ethene) to ethanal. (C-IV)

D. Ni complex: Specifically organometallic Nickel complexes (like Zeigler-Natta relatives) are used in the polymerisation of alkynes. (D-II)

Q. 9 Identify the correct statement about ClF_3 from the following options :

Option 1:

It has T-shaped geometry with three lone pairs on Cl atom.

Option 2:

It has T-shaped geometry with two lone pairs on Cl atom.

Option 3:

It has a trigonal pyramidal geometry with two lone pairs on Cl atom.

Option 4:

It has a planar trigonal geometry with two lone pairs on Cl atom.

Correct Answer:

It has T-shaped geometry with two lone pairs on Cl atom.

Solution:

- Central atom: Cl (7 valence electrons)
- Bonded atoms: 3 F atoms
- Total electron pairs = 5
- Electron geometry $\rightarrow$ Trigonal bipyramidal
- Two lone pairs occupy equatorial positions (to minimise repulsion)
- T-shaped geometry
- Lone pairs = 2
- Bond pairs = 3

It has T-shaped geometry with two lone pairs on the Cl atom

Hence, the correct answer is option (2)

Q. 10 Calculate emf of the half cell given below :



$$E^\circ_{\text{H}_2/\text{H}^+} = 0 \text{ V}$$

$$\left(\text{Given : } \frac{2.303RT}{F} = 0.059, \right. \\ \left. \log 2 = 0.3010 \right)$$

Option 1:

$$-0.109 \text{ V}$$

Option 2:

$$0.109 \text{ V}$$

Option 3:

$$0.035 \text{ V}$$

Correct Answer:

0.109 V

Solution:



$$E_{\text{ox}} = E_{\text{ox}}^{\circ} - \frac{0.059}{n} \log Q$$

$$E_{\text{ox}} = 0 - \frac{0.059}{2} \log \left(\frac{(0.02)^2}{2} \right)$$

$$E_{\text{ox}} = -0.0295 \log \left(\frac{0.0004}{2} \right)$$

$$E_{\text{ox}} = -0.0295 \log(0.0002)$$

$$\log(2 \times 10^{-4}) = \log 2 + \log 10^{-4}$$

$$\log(2 \times 10^{-4}) = 0.3010 - 4 = -3.699$$

$$E_{\text{ox}} = -0.0295 \times (-3.699)$$

$$E_{\text{ox}} \approx +0.109 \text{ V}$$

Hence, the correct answer is option (2)

Q. 11 Match List - I with List - II

List - I (order of reaction)	List - II (unit of rate constant)
A. Zero order	I. $\text{mol}^{-1} \text{L s}^{-1}$
B. First order	II. $\text{mol}^{-2} \text{L}^2 \text{s}^{-1}$
C. Second orde	III. s^{-1}
D. Third order	IV. $\text{molL}^{-1} \text{s}^{-1}$

Choose the correct answer from the options given below:

Option 1:

A-IV, B-III, C- II,D- I

Option 2:

A-IV, B-III, C-I, D-II

Option 4:

A-IV, B-II, C-I, D-III

Correct Answer:

A-IV, B-III, C-I, D-II

Solution:

List - I	List - II
A. Zero order	IV. $\text{molL}^{-1} \text{s}^{-1}$
B. First order	III. s^{-1}
C. Second order	I. $\text{mol}^{-1} \text{L s}^{-1}$
D. Third order	II. $\text{mol}^{-2} \text{L}^2 \text{s}^{-1}$

Hence, the correct answer is option 2

Q. 12 The calculated 'spin-only' magnetic moment of Ti^{2+} ($3d^2$) is :

Option 1:

3.87 BM

Option 2:

4.90 BM

Option 3:

2.84 BM

Option 4:

5.92 BM

Correct Answer:

2.84 BM

Solution:

The formula for 'spin-only' magnetic moment (μ) is:

$$\mu = \sqrt{n(n+2)}\text{BM}$$

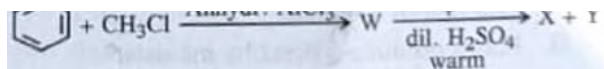
where n is the number of unpaired electrons.

For Ti^{2+} (configuration $3d^2$), there are 2 unpaired electrons ($n = 2$).

$$\mu = \sqrt{2(2+2)} = \sqrt{2 \times 4} = \sqrt{8}$$

$$\mu \approx 2.83\text{BM}$$

Hence, the correct answer is option (3)



The suitable method that can be used for the separation of products X and Y is :

Option 1:

Continuous extraction

Option 2:

Differential extraction

Option 3:

Sublimation

Option 4:

Fractional distillation

Correct Answer:

Fractional distillation

Solution:

Benzene + CH3Cl with AlCl3 → Toluene (W).

Toluene + nitrating mixture (HNO3 + H2SO4) → o-nitrotoluene (X) and p-nitrotoluene (Y).

Separation Method:

X and Y are structural isomers with different boiling points. While steam distillation is commonly used for similar separations (like o- and p-nitrophenols), fractional distillation is the most suitable general industrial method for separating these two components based on their boiling point difference.

Hence, the correct answer is option (4)

Q. 14 A bulb is rated at 150 watt, converting 8% energy into light. If eneergy of one photon is 4.42×10^{-19} J, how many photons are emitted by the bulb per second?

Option 1:

27.2×10^{19}

Option 2:

4.06×10^{19}

Option 3:

1.35×10^{19}

Option 4:

2.71×10^{19}

Correct Answer:

2.71×10^{19}

Using the formula $E = n \times E_{\text{photon}}$.

$$n = \frac{\text{Power}}{\text{Energy of one photon}}$$

$$n = \frac{12}{4.42 \times 10^{-19}} \approx 2.7149 \times 10^{19}$$

Hence, the correct answer is option (4)

- Q. 15** In a test tube containing a salt, a few drops of dilute H_2SO_4 was added, which gave colourless vapours having the smell of vinegar. The vapours turned the blue litmus paper red. Identify the correct anion from the following :

Option 1:

Carbonate, CO_3^{2-}

Option 2:

Sulphide, S^{2-}

Option 3:

Acetate, CH_3COO^-

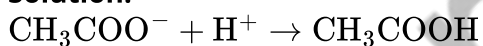
Option 4:

Sulphate, SO_4^{2-}

Correct Answer:

Acetate, CH_3COO^-

Solution:



Acetic acid vapours:

- Are **colourless**
- Have **vinegar smell**
- Turn **blue litmus red (acidic nature)**

Hence, the correct answer is option (3)

- Q. 16** Select the reagents that reduce nitriles to primary amines:

- A. (i) LiAlH_4 ;
(ii) H_2O
- B. $\text{Sn} + \text{HCl}$
- C. H_2/Ni
- D. $\text{Na(Hg)}/\text{C}_2\text{H}_5\text{OH}$
- E. $\text{Br}_2/\text{aq. NaOH}$

Choose the correct answer from the options given below

Option 2:

A, D and E only

Option 3:

A, C and D only

Option 4:

B, D and E only

Correct Answer:

A, C and D only

Solution:

A. $\text{LiAlH}_4/\text{H}_2\text{O}$

Strong reducing agent $\rightarrow$ converts nitrile $\rightarrow$ primary amine

B. $\text{Sn} + \text{HCl}$

Used for nitro $\rightarrow$ amine, not for nitriles $\rightarrow$ Incorrect

C. H_2/Ni

Catalytic hydrogenation $\rightarrow$ nitrile $\rightarrow$ primary amine

D. $\text{Na(Hg)}/\text{C}_2\text{H}_5\text{OH}$

Mild reduction $\rightarrow$ can reduce nitriles $\rightarrow$ primary amine

E. $\text{Br}_2/\text{aq. NaOH}$

Hoffmann bromamide reaction (amide $\rightarrow$ amine with one less carbon)

Hence, the correct answer is option (3)

Q. 17 Identify the incorrect statement from the following :

Option 1:

Oxygen exhibits only -2 oxidation state.

Option 2:

The order of catenation property of Group 14 elements is $\text{C} \gg \text{Si} > \text{Ge} \approx \text{Sn}$.

Option 3:

Carbon has the ability to form $p\pi - p\pi$ multiple bond with itself.

Option 4:

ECl_3 ($\text{E} = \text{B}$ and Al) is a monomer when $\text{E} = \text{B}$ and a dimer when $\text{E} = \text{Al}$.

Correct Answer:

Oxygen exhibits only -2 oxidation state.

Solution:

Statement (1) is Incorrect: Oxygen does not exhibit only the -2 oxidation state. It shows +2 in OF_2 , +1 in O_2F_2 , -1 in peroxides (H_2O_2), and $-1/2$ in superoxides (KO_2).

statement (4) is correct. BO_3 is a monomer (trigonal planar), while AlO_3 achieves stability by forming a dimer (Al_2O_6).

Hence, the correct answer is option (1)

Q. 18 Although +3 oxidation state is most common in lanthanoids, cerium still shows +4 oxidation state because :

Option 1:

After losing one, more electron, it acquires $4f^{14}$ electronic configuration.

Option 2:

Its nearest inert gas is Radon.

Option 3:

After losing one more electron, it acquires $4f^0$ electronic configuration.

Option 4:

Its atomic number is 61 .

Correct Answer:

After losing one more electron, it acquires $4f^0$ electronic configuration.

Solution:

Cerium (Ce) has the atomic number 58 . Its ground-state electronic configuration is $[\text{Xe}]4f^15d^16s^2$.

By losing 4 electrons (to reach the +4 state), it achieves the configuration of the noble gas Xenon ($[\text{Xe}]4f^0$).

This empty $4f^0$ orbital provides extra stability to the +4 state.

Hence, the correct answer is option (3)

Q. 19 During Lassaigne's test, they elements present in an organic compound are converted from :

Option 1:

ionic form to ionic form

Option 2:

covalent form to ionio form

Option 3:

ionic form to covalent form

Option 4:

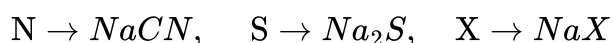
covalent form to covalent form

Correct Answer:

covalent form to ionio form

During Lassaigne's test, the organic compound is fused with sodium metal. Sodium is highly reactive and converts these covalently bonded elements into ionic (water-soluble) compounds, which can be easily detected.

Examples:



Thus, in Lassaigne's test, elements are converted from:

Covalent form $\rightarrow$ Ionic form

Hence, the correct answer is option (2)

- Q. 20** The number of hydrogen atoms present in 5.4 g of urea is :
(Given : Molar mass of urea : 60 g mol^{-1} ,
 $N_A : 6.022 \times 10^{23} \text{ particles mol}^{-1}$)

Option 1:

$$2.168 \times 10^{22}$$

Option 2:

$$2.168 \times 10^{23}$$

Option 3:

$$1.084 \times 10^{22}$$

Option 4:

$$1.084 \times 10^{23}$$

Correct Answer:

$$2.168 \times 10^{23}$$

Solution:

$$\text{Moles} = \frac{\text{Given Mass}}{\text{Molar Mass}} = \frac{5.4 \text{ g}}{60 \text{ g/mol}} = 0.09 \text{ moles}$$

The formula for urea is NH_2CONH_2 .

Each molecule of urea contains 4 hydrogen atoms.

$$\text{Total H atoms} = \text{Moles} \times \text{H atoms per molecule} \times N_A$$

$$\text{Total H atoms} = 0.09 \times 4 \times 6.022 \times 10^{23}$$

$$\text{Total H atoms} = 0.36 \times 6.022 \times 10^{23} \approx 2.168 \times 10^{23}$$

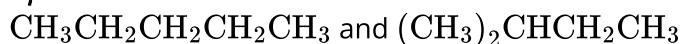
Hence, the correct answer is option (2)

- Q. 21** The pair of molecules that are metamers among the following is :

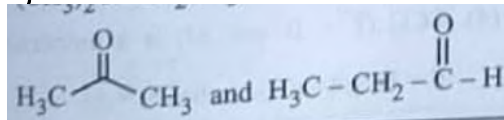
Option 2:



Option 3:



Option 4:



Correct Answer:



Solution:

Metamers are isomers that have the same molecular formula but differ in the distribution of carbon atoms (alkyl groups) on either side of a functional group (like $-\text{O}-$, $-\text{S}-$, or $-\text{NH}-$).

Option (1): $\text{CH}_3\text{OCH}_2\text{CH}_2\text{CH}_3$ (Methyl propyl ether) and $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$ (Diethyl ether). Both have the same formula ($\text{C}_4\text{H}_{10}\text{O}$) but different alkyl groups around the oxygen atom. These are metamers.

Option (2): Propan-1-ol and Propan-2-ol are position isomers.

Option (3): n-pentane and isopentane are chain isomers.

Option (4): Propanone and Propanal are functional isomers.

Hence, the correct answer is option (1)

Q. 22 Identify the incorrect statement from the following:

Option 1:

Phosphorus, arsenic and antimony show catenation property.

Option 2:

$\text{P}(\text{C}_2\text{H}_5)_3$ and $\text{As}(\text{C}_6\text{H}_5)_3$ form $d\pi - d\pi$ bond with transition metals.

Option 3:

Nitrogen can form $d\pi - p\pi$ bond with oxygen.

Option 4:

Nitrogen can form $\pi - p\pi$ multiple bonds with itself.

Correct Answer:

Nitrogen can form $d\pi - p\pi$ bond with oxygen.

Solution:

(1) Correct: Catenation (the linking of atoms of the same element into chains) is a property shown by Phosphorus, Arsenic, and Antimony. Phosphorus shows it most significantly (as seen in P_4 or red phosphorus chains). While Nitrogen's catenation is limited due to high inter-electronic repulsion between lone pairs, the heavier elements do

(2) INCORRECT: Organophosphorus and organoarsenic compounds (like phosphines and arsines) act as ligands with transition metals. They form bonds through π -backbonding, but the specific interaction involves the donation of electrons from the filled d-orbitals of the transition metal into the vacant d-orbitals of Phosphorus or Arsenic. Therefore, the bond is a $d\pi(\text{metal}) - d\pi(\text{ligand})$ interaction. However, the statement is often considered incorrect in standard textbooks because the p -block elements use their vacant d-orbitals to accept electrons, but the primary sigma bond is formed by the lone pair. More importantly, in many competitive contexts, the focus is on the fact that Nitrogen cannot do this because it lacks d-orbitals.

(3) Correct: Nitrogen forms $p\pi - p\pi$ multiple bonds with Oxygen (as seen in oxides like NO_2 or NO_3^-). Since both N and O are in the second period, their p-orbitals are small enough to overlap effectively sideways.

(4) Correct: Nitrogen has a very small atomic size and high electronegativity, allowing it to form very strong $p\pi - p\pi$ multiple bonds with itself, resulting in the stable $\text{N} \equiv \text{N}$ triple bond. Heavier elements like Phosphorus cannot do this effectively because their p-orbitals are too large and diffuse to have effective sideways overlap.

Hence, the correct answer is option (3)

Q. 23 Phenolphthalein is used as an indicator for the titration of sodium hydroxide solution against a standard solution of oxalic acid. The colour change that is observed at an alkaline pH close to the equivalence point during this titration is :

Option 1:

pink to colourless

Option 2:

pinkish red to yellow

Option 3:

colourless to pink

Option 4:

yellow to pinkish red

Correct Answer:

colourless to pink

Solution:

In a titration of Oxalic acid (standard solution in the flask) against Sodium hydroxide (base in the burette). The solution starts as acidic (colorless). As the equivalence point is reached and the solution becomes slightly alkaline, the phenolphthalein changes color. The color change observed is colourless to pink.

Hence, the correct answer is option (3)

LIST-I	LIST-II
A. C_2H_4	I. 3σ bonds, 2π bonds
B. C_2H_2	II. 3σ bonds, one lone pair
C. CH_4	III. 4σ bonds
D. NH_3	IV. 5σ bonds, 1π bond

Choose the correct answer from the options given below :

Option 1:

A-IV, B-I, C-III, D-II

Option 2:

A-III, B-IV, C-II, D-I

Option 3:

A-II, B-III, C-I, D-IV

Option 4:

A-I, B-II, C-IV, D-III

Correct Answer:

A-IV, B-I, C-III, D-II

Solution:

A. C_2H_4 (Ethene): Contains a $C = C$ double bond and four $C - H$ single bonds. Total: 5 σ bonds, 1 π bond. (A-IV)

B. C_2H_2 (Ethyne): Contains a $C \equiv C$ triple bond and two $C - H$ single bonds. Total: 3 σ bonds, 2 π bonds. (B-I)

C. CH_4 (Methane): Contains four $C - H$ single bonds. Total: 4 σ bonds. (C-III)

D. NH_3 (Ammonia): Contains three $N - H$ single bonds and one lone pair on Nitrogen. Total: 3 σ bonds, one lone pair. (D-II)

Hence, the correct answer is option (1).

Q. 25 At a certain temperature, $T(K)$, during a process, 500 J is absorbed by the system and work of 200 J is done by the system. Then change in internal energy of the system is :

Option 1:

700 J

Option 2:

400 J

Option 4:

500 J

Correct Answer:

300 J

Solution:

$$\Delta U = q - w$$

$q = +500J$ (heat absorbed $\rightarrow$ positive)

$w = +200J$ (work done by system)

$$\Delta U = 500 - 200 = 300 \text{ J}$$

Hence, the correct answer is option (3)

Q. 26 Methane reacts with steam at 1273 K in the presence of nickel catalyst to form:

Option 1:

CO and H_2

Option 2:

CO and H_2O

Option 3:

CO_2 and H_2

Option 4:

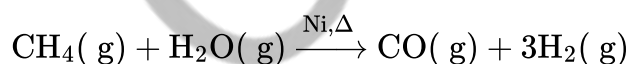
CO_2 and H_2O

Correct Answer:

CO and H_2

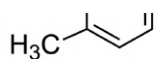
Solution:

When methane reacts with steam at high temperatures (1273 K) in the presence of a nickel catalyst, it undergoes a process called steam reforming to produce Syngas (Synthesis Gas).

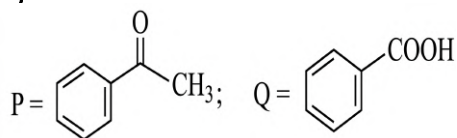


Hence, the correct answer is option (1)

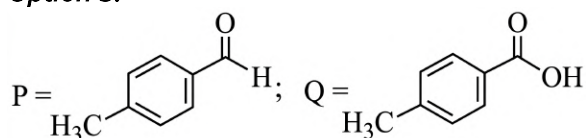
Q. 27 Compound P (C_8H_8O) gives a red orange precipitate with 2,4-DNP reagent and it does not reduce Fehling's reagent. On drastic oxidation with chromic acid, P gives an aromatic product Q that produces effervescence on treating with aq. $NaHCO_3$. Compounds P and Q, respectively, are:



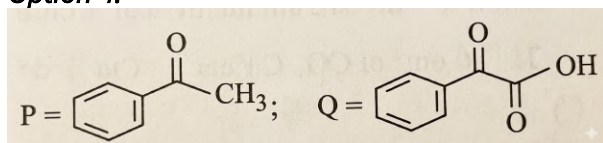
Option 2:



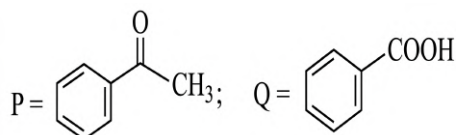
Option 3:



Option 4:



Correct Answer:



Solution:

1. Identify the functional group of *P* (2,4-DNP Test)

Observation: *P* gives a red-orange precipitate with 2,4-DNP reagent.

Inference: *P* contains a carbonyl group ($>C=O$), meaning it is either an aldehyde or a ketone.

2. Narrow down the carbonyl type (Fehling's Test)

Observation: *P* does not reduce Fehling's reagent.

Inference: *P* is a ketone (specifically an aromatic ketone here) or an aromatic aldehyde (which also don't reduce Fehling's), but the formula $\text{C}_8\text{H}_8\text{O}$ and the next step point toward a ketone.

3. Analyze Product *Q* (NaHCO_3 Test)

Observation: Drastic oxidation of *P* with chromic acid yields *Q*, which produces effervescence with aq. NaHCO_3 .

Inference: *Q* is a carboxylic acid (effervescence is due to CO_2 gas release).

4. Final Structure Identification

For *P* ($\text{C}_8\text{H}_8\text{O}$): Acetophenone ($\text{C}_6\text{H}_5\text{COCH}_3$) fits the formula. It is a ketone, so it gives a positive 2,4-DNP test but a negative Fehling's test.

Drastic Oxidation: When alkyl benzenes or aromatic ketones like acetophenone are treated with strong oxidising agents (like chromic acid or alkaline KMnO_4), the entire side chain is oxidised down to a carboxyl group ($-\text{COOH}$) attached directly to the ring.

Product *Q*: Benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$).

Hence, the correct answer is option (2)

Option 1:

0.2938 g

Option 2:

0.5876 g

Option 3:

$2 \cdot 4036$ g

Option 4:

1.7018 g

Correct Answer:

0.2938 g

Solution:

79: Total charge (Q) :

$$Q = I \times t = 1.5 \text{ A} \times (10 \times 60)\text{s} = 900\text{C}$$

The reaction at the cathode is $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}(s)$.

$$\text{Mass deposited } (m) = \frac{\text{Molar mass} \times Q}{n \times F}$$

$$m = \frac{63 \times 900}{2 \times 96487} \approx \mathbf{0.2938 \text{ g}}$$

Hence, the correct answer is option (1)

Q. 29 The functional group that can be identified through phthalein dye test is :

Option 1:

Carboxylic acid

Option 2:

Alcohol

Option 3:

Aldehyde

Option 4:

Phenolic

Correct Answer:

Phenolic

Solution:

Hence, the correct answer is option (4)

Q. 30 The correct statement with regard to the secondary structure of DNA/RNA is :

Option 1:

DNA possesses a double strand helix structure and contains thymine as one of the four bases.

Option 2:

DNA possesses a single strand helix structure and contains uracil as one of the four bases.

Option 3:

RNA possesses a double strand helix structure and contains uracil as one of the four bases.

Option 4:

RNA possesses a single strand helix structure and contains thymine as one of the four bases.

Correct Answer:

DNA possesses a double strand helix structure and contains thymine as one of the four bases.

Solution:

DNA is **double-stranded (double helix)**

Bases: A, T, G, C → **thymine present**

Hence, the correct answer is option (1)

Q. 31 Identify the correct statements :

- A. The molality of 2.5 g of ethanoic acid (Molar mass : 60 g mol^{-1}) in 75 g of benzene solution is 0.556.
- B. The molarity of a solution containing 5 g of NaOH (molar mass : 40 g mol^{-1}) in 450 mL of solution is 0.278 M at 298 K .
- C. Aquatic species are more comfortable in cold water.
- D. The solubility of gas increases with decrease in pressure.
- E. For a binary mixture of A and B , the number of moles of A and B are n_A and n_B respectively. The mole fraction of B will be $x_B = \frac{n_A}{n_A + n_B} =$

Choose the correct answer from the options given below:

Option 1:

A, B and C only

Option 2:

A, D and E only

Option 3:

A and B only

Correct Answer:

A, B and C only

Solution:

Statement A (True): Molality (m) = $\frac{\text{moles of solute}}{\text{mass of solvent in kg}}$.

Moles of ethanoic acid = $\frac{2.5}{60} = 0.0417$.

$$m = \frac{0.0417}{0.075 \text{ kg}} = 0.556 \text{ m}$$

Statement B (True): Molarity (M) = $\frac{\text{moles of solute}}{\text{Volume of solution in L}}$.

Moles of NaOH = $\frac{5}{40} = 0.125$.

$$M = \frac{0.125}{0.450 \text{ L}} = 0.278 \text{ M}$$

Statement C (True): The solubility of gases (like O_2) increases as temperature decreases. Therefore, aquatic species are more comfortable in cold water due to higher dissolved oxygen levels.

Statement D (False): According to Henry's Law, the solubility of a gas increases with an increase in pressure.

Statement E (False): The mole fraction of B is $x_B = \frac{n_B}{n_A + n_B}$. The provided formula is for x_A .

Hence, the correct answer is option (1)

Q. 32 Mixture of chloroform and acetone forms a solution with negative deviation from Raoult's law due to :

Option 1:

stronger intermolecular forces between chloroform molecules than those between chloroform and acetone molecules.

Option 2:

formation of hydrogen bonding between acetone and chloroform.

Option 3:

repulsive forces.

Option 4:

increase in escaping tendency of molecules of each component.

Correct Answer:

formation of hydrogen bonding between acetone and chloroform.

Solution:

- Chloroform (CHCl_3) has a slightly acidic hydrogen
- Acetone (CH_3COCH_3) has a lone pair on oxygen

This leads to hydrogen bonding between chloroform and acetone molecules which results:

- Strong A-B interactions $\rightarrow$ molecules are held more tightly
- Vapour pressure decreases $\rightarrow$ negative deviation

Hydrogen bonding between chloroform and acetone

Q. 33 At 298 K, a certain buffer solution contains equal concentrations of X^- and HX , K_b for X^- is 10^{-10} . What is the pH of this buffer solution?

Option 1:
10

Option 2:
4

Option 3:
2

Option 4:
6

Correct Answer:
4

Solution:

For a basic buffer, we use:

$$pOH = pK_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

Since concentrations are equal:

$$\log \frac{[\text{salt}]}{[\text{base}]} = \log 1 = 0$$

So,

$$pOH = pK_b$$

Given:

$$K_b = 10^{-10} \Rightarrow pK_b = 10$$

$$pOH = 10$$

Now:

$$pH = 14 - pOH = 14 - 10 = 4$$

Hence, the correct answer is option (2)

Q. 34 Identify the incorrect statement from the following:

Option 1:

The largest and the smallest species among Mg , Mg^{2+} , Al and Al^{3+} are Al and Mg^{2+} , respectively.

Option 3:

The similarity in behaviour of Li with Mg is referred to as 'diagonal relationship'.

Option 4:

The oxidation state and covalency of Al in $[\text{AlCl}(\text{H}_2\text{O})_5]^{2+}$ are 3 and 6 respectively.

Correct Answer:

The largest and the smallest species among Mg, Mg^{2+} , Al and Al^{3+} are Al and Mg^{2+} , respectively.

Solution:

Statement (1) is Incorrect: Between Mg and Al, Mg is larger because atomic size decreases across a period. Between Mg^{2+} and Al^{3+} , Al^{3+} is the smallest because it has more protons pulling the same number of electrons. Therefore, the largest is Mg and the smallest is Al^{3+} .

Statement (2) is Correct: Atomic number 107 (1 = Un, 0 = nil, 7 = sept) is Unnilseptium.

Statement (3) is Correct: Li and Mg show diagonal relationship due to similar ionic sizes and electronegativities.

Statement (4) is Correct: In $[\text{AlCl}(\text{H}_2\text{O})_5]^{2+}$, let x be the oxidation state of Al : $x + (-1) + 5(0) = +2 \Rightarrow x = +3$. The coordination number (covalency) is 6 (1Cl + 5 H_2O ligands).

Hence, the correct answer is option (1)

Q. 35 The correct order of increasing metallic character of Na, Be, P, Mg and Si is :

Option 1:

$\text{Be} < \text{Si} < \text{P} < \text{Mg} < \text{Na}$

Option 2:

$\text{P} < \text{Si} < \text{Na} < \text{Mg} < \text{Be}$

Option 3:

$\text{P} < \text{Si} < \text{Be} < \text{Mg} < \text{Na}$

Option 4:

$\text{P} < \text{Mg} < \text{Be} < \text{Si} < \text{Na}$

Correct Answer:

$\text{P} < \text{Si} < \text{Be} < \text{Mg} < \text{Na}$

Solution:

Left to Right metallic character decreases

Top to Bottom metallic character increases

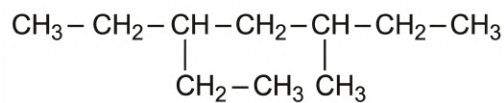
Na, Mg, Si, P: These are all in Period

$\text{P} < \text{Si} < \text{Mg} < \text{Na}$

Be is in Group 2, located directly above Magnesium. Since metallic character increases downward, Be is less metallic than Mg.

Hence, correct answer is option (3)

Q. 36 The correct IUPAC name of the following compound is :



Option 1:

3-ethyl-5-methylheptane

Option 2:

3-methyl-5-ethylheptane

Option 3:

2,4-diethylhexane

Option 4:

3,5-diethylhexane

Correct Answer:

3-ethyl-5-methylheptane

Solution:

3-ethyl-5-methylheptane

Hence, the correct answer is option (1)

Q. 37 Match List I with List II

List - I (Complex/ion)	List -II (Shape/geometry)
A. $[\text{Pt}(\text{Cl}_2)(\text{NH}_3)_2]$	I. Octahedral
B. $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$	II. Trigonal bipyramidal
C. $[\text{NiCl}_4]^{2-}$	III. Square planar
D. $[\text{Fe}(\text{CO})_5]$	IV. Tetrahedral

Choose the **correct** answer from the options given below:

Option 2:

A-III, B-IV, C-I, D-IIX

Option 3:

A-III, B-I, C-IV, D-IV-

Option 4:

A-IV, B-I, C-III, D-IIX

Correct Answer:

A-III, B-I, C-IV, D-IV-

Solution:

(3) A-III, B-I, C-IV, D-II

A. $[\text{PtCl}_2(\text{NH}_3)_2]$

Pt^{2+} is a d^8 system, which typically forms square planar complexes $\rightarrow$ III

B. $[\text{Co}(\text{NH}_3)_6]^{3+}$

Co^{3+} with 6 ligands gives coordination number 6, so octahedral $\rightarrow$ I

C. $[\text{NiCl}_4]^{2-}$

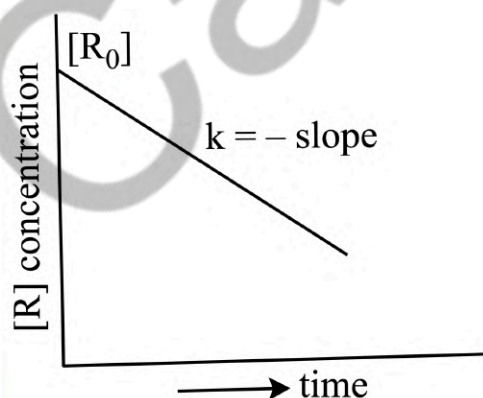
Ni^{2+} with weak field ligand Cl^- forms a tetrahedral complex $\rightarrow$ IV

D. $[\text{Fe}(\text{CO})_5]$

This is a classic trigonal bipyramidal structure $\rightarrow$ II

Hence, the correct answer is option (3)

Q. 38 For a certain reaction $\text{R} \rightarrow \text{Product}$, the plot of concentration $[\text{R}]$ vs time has a negative slope as shown. The order of reaction is :



Option 1:

Option 3:

2

Option 4:

0

Correct Answer:

0

Solution:

Zero order: $[R] = [R_0] - kt \rightarrow$ straight line (slope = $-k$)

Hence, the correct answer is option (4)

Q. 39 Which one of the following is an ambidentate ligand?

Option 1:

Oxalate

Option 2:

Ethane-1,2-diamine

Option 3:

Thiocyanate

Option 4:

Ethylenediaminetetraacetate ion

Correct Answer:

Thiocyanate

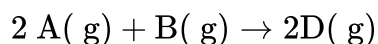
Solution:

An ambidentate ligand is a ligand that can coordinate to a central metal atom through two different atoms but only one at a time.

Thiocyanate (SCN^-) can bind through the Sulfur (S) or the Nitrogen (N) atom.

Hence, the correct answer is option (3)

Q. 40 Consider the following reaction :



$$\Delta U^\ominus = -10 \text{ kJ mol}^{-1} \text{ and } \Delta S^\ominus = -44 \text{ J K}^{-1} \text{ at } 298 \text{ K}.$$

Identify the correct option with $\Delta G^\ominus$ for the reaction and spontaneity of the reaction at 298 K .

(Given : $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$)

Option 2:

$-0.63568 \text{ kJ mol}^{-1}$, spontaneous

Option 3:

$-1.635 \text{ kJ mol}^{-1}$ spontaneous

Option 4:

$+1.635 \text{ kJ mol}^{-1}$, non-spontaneous

Correct Answer:

$+0.63568 \text{ kJ mol}^{-1}$, non-spontaneous

Solution:

$$\Delta S^\circ = -44 \text{ J} = -0.044 \text{ kJ K}^{-1}$$

$$\Delta H = \Delta U + \Delta n_g RT$$

$$\Delta n_g = 2 - 3 = -1$$

$$\Delta H = -10 + (-1)(8.31 \times 298/1000)$$

$$= -10 - 2.476 \approx -12.476 \text{ kJ}$$

$$\Delta G = \Delta H - T\Delta S$$

$$= -12.476 - (298 \times -0.044)$$

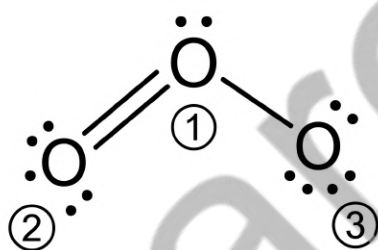
$$= -12.476 + 13.112$$

$$= +0.636 \text{ kJ mol}^{-1}$$

$$\Delta G > 0 \Rightarrow \text{N-spontaneous}$$

Hence, the correct answer is option (1)

Q. 41



The correct formal charges on oxygen atoms numbered 2, 1 and 3 respectively are :

Option 1:

0, 0, 0

Option 2:

-1, 0, +1

Option 3:

+1, 0, -1

Option 4:

0, +1, -1

Solution:

O(2) (double bonded terminal O):

- Lone pairs = $4e^-$ - Bonding = $4e^-$

$$FC = 6 - 4 - 2 = 0$$

O(1) (central O):

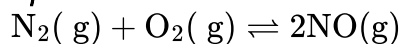
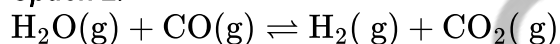
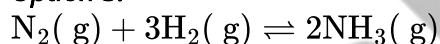
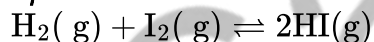
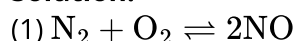
- Lone pairs = $2e^-$ - Bonding = $6e^-$

$$FC = 6 - 2 - 3 = +1$$

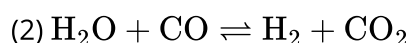
O(3) (single bonded terminal O):

Lone pairs = $6e^-$ Bonding = $2e^-$

$$FC = 6 - 6 - 1 = -1$$

Hence, the correct answer is option (4)**Q. 42** Given below are certain reactions. Identify the reaction for which $K_p \neq K_c$.**Option 1:****Option 2:****Option 3:****Option 4:****Correct Answer:****Solution:**

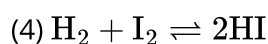
$$\Delta n = 2 - (1 + 1) = 0 \rightarrow K_p = K_c$$



$$\Delta n = 2 - 2 = 0 \rightarrow K_p = K_c$$



$$\Delta n = 2 - 4 = -2 \rightarrow K_p \neq K_c$$



$$\Delta n = 2 - 2 = 0 \rightarrow K_p = K_c$$

Given below is an expression for the rate constant of a first order reaction occurring at a certain temperature, T(K).

$$\ln k = 14.34 - \frac{1.25 \times 10^4}{T}$$

The energy of activation in kcal mol^{-1} for the reaction is :
(Given : k in s^{-1} , $R = 1.987 \text{ cal mol}^{-1} \text{ K}^{-1}$)

Option 1:

12.42

Option 2:

18.63

Option 3:

14.34

Option 4:

24.84

Correct Answer:

24.84

Solution:

$$52. \ln k = \ln A - \frac{E_a}{RT}$$

$$\frac{E_a}{R} = 1.25 \times 10^4$$

$$E_a = (1.25 \times 10^4) \times R$$

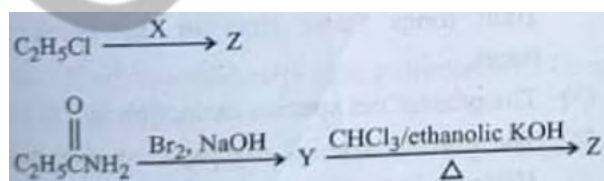
$$R = 1.987 \text{ cal mol}^{-1} \text{ K}^{-1}$$

$$E_a = 12500 \times 1.987 = 24837.5 \text{ cal mol}^{-1}$$

$$E_a = \frac{24837.5}{1000} \approx 24.84 \text{ kcal mol}^{-1}$$

Hence, the correct answer is option (4)

Q. 44 The following two reactions give e^x the same foul smelling product Z .

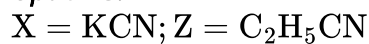


X and Z respectively, are:

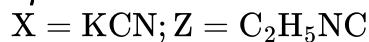
Option 1:

$\text{X} = \text{AgCN}$; $\text{Z} = \text{C}_2\text{H}_5\text{CN}$

Option 3:



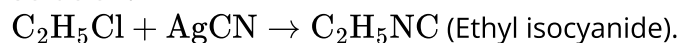
Option 4:



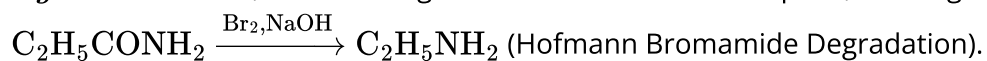
Correct Answer:



Solution:



AgCN is covalent, so the nitrogen atom acts as the nucleophile, forming an isocyanide.



Hence, the correct answer is option (2).

Q. 45 Match List - I with List -II

List- I (Complex)	List- II (Type of isomerism)
A. $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$	I. Optical
B. $[\text{Co}(\text{en})_3]^{3+}$	II. 'Solvate
C. $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$	III. Geometrical
D. $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$	IV. Linkage

Choose the correct answer from the options given below:

Option 1:

A-I, B-III, C-II, D=IV

Option 2:

A-II, B-IV, C-III, D-I

Option 3:

A-III, B-I, C-IV, D-II

Option 4:

A-III, B-I, C-II, D-IV

Correct Answer:

A-III, B-I, C-IV, D-II

Solution:

Geometrical isomerism → III

B. $[\text{Co}(\text{en})_3]^{3+}$

- Octahedral with bidentate ligand (en)
- Forms non-superimposable mirror images

Optical isomerism → I

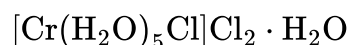
C. $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$

- NO_2^- can bind via N or O

Linkage isomerism → IV

D. $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$

- Can form hydrate/solvate isomers like:



Solvate isomerism → II

Hence, the correct answer is option (3)

Biology

Q. 1 "The Evil Quartet" of biodiversity loss includes which of the following ?

Option 1:

Over-exploitation; Alien species invasions; Soil pollution; Co-extinctions

Option 2:

Habitat loss and fragmentation; Air pollution; Water pollution; Co-extinctions

Option 3:

Habitat loss and fragmentation; over-exploitation; Alien species invasions; Co-extinctions

Option 4:

Over-exploitation; Alien species invasions; Air pollution; Co-extinctions

Correct Answer:

Habitat loss and fragmentation; over-exploitation; Alien species invasions; Co-extinctions

Solution:

The "evil quartet" consists of habitat loss and fragmentation, over-exploitation, alien species invasions, and co-extinctions. These are the major causes of biodiversity decline.

- Habitat loss and fragmentation reduce living space and isolate populations.
- Over-exploitation depletes species faster than they can recover.
- Alien species invasions disrupt native ecosystems and outcompete local species.
- Co-extinctions occur when the extinction of one species leads to the loss of another dependent species.

Hence, the correct answer is option 3) Habitat loss and fragmentation; over-exploitation; Alien species invasions; Co-extinctions.

Option 1:

Nucleolus

Option 2:

Kinetochores

Option 3:

Centrosome

Option 4:

Chromatin

Correct Answer:

Nucleolus

Solution:

rRNA is synthesised in the nucleolus, where ribosome assembly begins. It also serves as the site for assembly of ribosomal subunits, which later move to the cytoplasm for protein synthesis.

Other nuclear structures, like the kinetochores, centrosome, and chromatin, have different roles and are not involved in rRNA synthesis.

Hence, the correct answer is option 1) Nucleolus.

Q. 3 Match List I with List II :

	List I		List II
	(Phase of cell cycle)		(Activity)
A.	G_1 phase	I.	Actual cell division occurs
B.	S phase	II.	The cell is metabolically active and continuously grows, but does not replicate its DNA
C.	G_2 phase	III.	Synthesis of DNA occurs, and the amount of DNA per cell doubles
D.	M phase	IV.	Proteins are synthesised while cell growth continues

Choose the correct answer from the options given below:

Option 1:

A-III, B-IV, C-I, D-II

Option 2:

A-IV, B-I, C-II, D-III

Option 4:

A-II, B-III, C-IV, D-1

Correct Answer:

A-II, B-III, C-IV, D-1

Solution:

- G_1 is growth without DNA replication (*II*)
- S phase involves DNA synthesis (*III*)
- G_2 prepares for division with protein synthesis (*IV*)
- M phase is the actual cell division (*I*).

Hence, the correct answer is option 4) A-II, B-III, C-IV, D-1.

Q. 4 Match List I with List II :

	List I		List II
A.	Productivity	I.	Gross primary productivity minus respiration losses
B.	Net primary productivity	II.	Rate of formation of new organic matter by consumers
C.	Gross primary productivity	III.	Rate of biomass production
D.	Secondary productivity	IV.	The rate of production of organic matter during photosynthesis

Choose the correct answer from the options given below :

Option 1:

A-III, B-I, C-II, D-IV

Option 2:

A-I, B-II, C-III, D-IV

Option 3:

A-III, B-I, C-IV, D-II

Option 4:

A-I, B-III, C-IV, D-II

Correct Answer:

A-III, B-I, C-IV, D-II

Solution:

- Productivity = biomass production rate (III)
- $NPP = GPP - R$ (I)
- GPP is the total photosynthetic production (IV)

Q. 5 Which of the following statements are correct?

- A. The Amazon rainforest being cut and cleared for the cultivation of soybeans is an example of habitat loss.
- B. Steller's sea cow and passenger pigeon became extinct due to over-exploitation by humans.
- C. The Nile perch introduced into Lake Victoria in East Africa helped in the population growth of cichlid fish in the lake.
- D. Water hyacinth is an invasive species.
- E. When a species becomes extinct, the plant and animal species associated with it are not affected.

Choose the correct answer from the options given below:

Option 1:

A, B and D only

Option 2:

B, C and D only

Option 3:

A, B and E only

Option 4:

C, D and E only

Correct Answer:

A, B and D only

Solution:

- The Amazon rainforest example highlights habitat loss and fragmentation, the most severe driver of biodiversity decline.
- Species like Steller's sea cow and passenger pigeon vanished due to human over-exploitation.
- The introduction of alien species, such as water hyacinth, disrupts native ecosystems and reduces biodiversity.
- Nile perch actually caused the decline of cichlid fish, and co-extinctions occur when dependent species are lost together.

A, B, and D are correct; Nile perch reduced cichlid populations (C false), and extinction affects associated species (E false).

Hence, the correct answer is option 1) A, B and D only.

-
- C. Polysaccharides are long chains of sugars.
 - D. Adenine and guanine are substituted pyrimidines.
 - E. Almost all enzymes are proteins.

Choose the correct answer from the options given below:

Option 1:

A, B and C only

Option 2:

B, D and E only

Option 3:

B, C and E only

Option 4:

C, D and E only

Correct Answer:

B, C and E only

Solution:

Lipids are water-insoluble (A false), and adenine and guanine are purines (D false). Statements B, C and E are correct.

- Proteins are long chains of amino acids linked by peptide bonds, hence called polypeptides.
- Polysaccharides such as starch, glycogen, and cellulose are complex carbohydrates made of repeated sugar units.
- Almost all enzymes are proteins, functioning as biological catalysts that speed up reactions.
- Lipids, however, are water-insoluble molecules, and adenine and guanine are purines, not pyrimidines.

Hence, the correct answer is option 3) B, C and E only.

Q. 7 How many ATP and NADPH molecules are required to make one molecule of glucose through the Calvin pathway?

Option 1:

12 ATP and 18 NADPH

Option 2:

18 ATP and 12 NADPH

Option 3:

6 ATP and 12 NADPH

Option 4:

24 ATP and 18 NADPH

Solution:

For the synthesis of one glucose, the Calvin cycle consumes 18 *ATP* and 12 *NADPH*.

To fix one molecule of CO_2 , the Calvin cycle requires 3 *ATP* and 2 *NADPH*. Since glucose formation needs 6 CO_2 molecules, the total requirement becomes 18 *ATP* and 12 *NADPH*.

ATP provides the energy, while *NADPH* supplies the reducing power for carbon fixation. This highlights the high energy demand of photosynthesis, linking the light reactions to the dark reactions.

Hence, the correct answer is option 2) 18 *ATP* and 12 *NADPH*.

Q. 8 Which of the following statements are not true regarding restriction endonucleases?

- A. They are called molecular scissors.
- B. These are the enzymes responsible for restricting the growth of bacteriophages in *E. coli*.
- C. They cut the DNA only at the centre of the palindromic sites.
- D. They remove nucleotides only from the ends of DNA fragments.
- E. They recognise specific palindromic base-pair sequences.

Choose the answer from the options given below :

Option 1:

C and D only

Option 2:

A and E only

Option 3:

D and E only

Option 4:

A and B only

Correct Answer:

C and D only

Solution:

Restriction enzymes cut at specific sites but not necessarily at the centre (*C false*), and they do not remove nucleotides from ends (*D false*).

Restriction endonucleases cut DNA at specific recognition sites, usually palindromic sequences, making them vital tools in genetic engineering. Unlike exonucleases, they do not remove nucleotides from the ends, but act within the DNA strand at defined positions.

Hence, the correct answer is option 1) C and D only.

	LIST I		LIST II
A.	Decomposition	I.	Accumulation of dark Coloured amorphous colloidal substance
B.	Detritus	II.	Release of inorganic nutrients by the activity of microbes in soil
C.	Mineralisation	III.	Breaking down complex organic matter into inorganic substances
D.	Humification	IV.	Dead remains of plants and animals, including faecal matter

Choose the correct answer from the options given below:

Option 1:

A-III, B-II, C-I, D-IV

Option 2:

A-IV, B-III, C-I, D-II

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-III, B-IV, C-II, D-I

Correct Answer:

A-III, B-IV, C-II, D-I

Solution:

- Decomposition = breakdown (III)
- Detritus = dead remains (IV)
- Mineralisation = release of inorganic nutrients (II)
- Humification = formation of humus (I).

This matching shows the sequential processes of decomposition in soil ecology. Dead organic matter (detritus) is broken down, nutrients are mineralised, and humus is formed, enriching soil fertility.

Hence, the correct answer is option 4) A-III, B-IV, C-II, D-I.

Q. 10 In which one of the following is the ovules are not enclosed by an ovary wall and remains exposed?

Option 1:

Pinus

Option 2:

Wolffia

Option 4:
Selaginella

Correct Answer:
Pinus

Solution:

In gymnosperms like Pinus, ovules are naked (not enclosed in an ovary) because they are borne directly on megasporophylls without being enclosed by an ovary wall.

This is a key distinction between gymnosperms and angiosperms, where ovules are protected inside the ovary. The exposed ovules of Pinus later develop into seeds after fertilisation.

Hence, the correct answer is option 1) Pinus.

Q. 11 Match List I with List II :

	List I (Placentation)		List II (Example)
A.	Marginal	I.	Mustard
B.	Axile	II.	Pea
C.	Parietal	III.	Marigold
D.	Basal	IV.	Lemon

Choose the correct answer from the options given below :

Option 1:
A-II, B-IV, C-I, D-III

Option 2:
A-IV, B-II, C-I, D-III

Option 3:
A-III, B-I, C-IV, D-II

Option 4:
A-I, B-III, C-II, D-IV

Correct Answer:
A-II, B-IV, C-I, D-III

Solution:

Placentation refers to the arrangement of ovules within the ovary, and different plants exhibit distinct types.

Marginal → Pea (II); Axile → Lemon (IV); Parietal → Mustard (I); Basal → Marigold (III).

Hence, the correct answer is option 1) A-II, B-IV, C-I, D-III.

Q. 12 In angiosperms, root hairs arise from which one of the following regions of the root?

Option 1:

The region of elongation

Option 2:

The region of meristematic activity

Option 3:

The region of maturation

Option 4:

The root cap zone

Correct Answer:

The region of maturation

Solution:

Root hairs develop from epidermal cells in the zone of maturation (differentiation).

This greatly increases the absorptive surface area of roots, helping in the efficient uptake of water and minerals. They are absent in the meristematic and elongation zones, which are primarily for cell division and growth.

Hence, the correct answer is option 3) The region of maturation.

Q. 13 Which one of the following is not a characteristic of plant cells in the phase of elongation?

Option 1:

New cell wall deposition

Option 2:

Cell enlargement

Option 3:

Large conspicuous nuclei

Option 4:

Increased vacuolation

Correct Answer:

Large conspicuous nuclei

The nuclei become less conspicuous as the cytoplasm stretches, unlike meristematic cells, where nuclei are large and prominent.

Thus, elongation is marked by growth in volume and structure, not active nuclear division.

Hence, the correct answer is option 3) Large conspicuous nuclei.

- Q. 14** Which of the following statements are correct with reference to a transcription unit?
- A. A transcription unit in DNA is defined primarily by three regions: promoter, structural gene and terminator
 - B. The promoter is said to be located towards the 5'-end of the structural gene
 - C. The promoter is a DNA sequence that provides binding site for RNA polymerase
 - D. The promoter defines the template and coding strands
 - E. The terminator is located towards the 3'-end of the coding strand and it defines the end of the process of transcription
- Choose the correct answer from the options given below:

Option 1:

B, C, D and E only

Option 2:

A, B, C, D and E

Option 3:

A, B, C and D only

Option 4:

A, C, D and E only

Correct Answer:

A, B, C, D and E

Solution:

A transcription unit has a promoter, a structural gene, and a terminator.

The promoter lies upstream at the 5' end and binds RNA polymerase. It determines template/coding strands, and the terminator marks the transcription end. The promoter region ensures correct initiation of transcription by RNA polymerase, while the terminator signals its end. Together with the structural gene, these elements define the functional transcription unit in both prokaryotes and eukaryotes.

Hence, the correct answer is option 2) A, B, C, D and E.

- Q. 15** The alpha-helix is found in which level of protein structure?

Option 1:

Secondary structure

Option 3:

Tertiary structure

Option 4:

Quaternary structure

Correct Answer:

Secondary structure

Solution:

The alpha helix is a regular folding pattern stabilised by hydrogen bonds between the backbone atoms of the polypeptide chain. This makes it a hallmark of the secondary structure of proteins.

- Primary - Amino acid sequence
- Secondary - alpha-helix & beta-sheet (local folding)
- Tertiary - Overall 3D folding of one polypeptide
- Quaternary - Association of multiple polypeptides

Hence, the correct answer is option 1) Secondary structure.

Q. 16 Which of the following statements are correct regarding amino acids?

- A. They are substituted methanes.
- B. Serine is an aromatic amino acid.
- C. Valine is a neutral amino acid.
- D. Lysine is an acidic amino acid.

Choose the correct answer from the options given below:

Option 1:

A and B only

Option 2:

C and D only

Option 3:

B and C only

Option 4:

A and C only

Correct Answer:

A and C only

Solution:

Amino acids are considered substituted methane derivatives because the central carbon is bonded to an amino, carboxyl, hydrogen, and variable side chains.

Amino acids are substituted methane derivatives (A correct); valine is neutral (C correct). Serine is not aromatic, and lysine is basic, not acidic.

Q. 17 The main function of bulliform cells in grasses is:

Option 1:
to perform photosynthesis.

Option 2:
to minimise water loss during water stress.

Option 3:
to make the leaf impermeable to fungal spores.

Option 4:
to transport water.

Correct Answer:
to minimise water loss during water stress.

Solution:

Bulliform cells help in leaf rolling during water stress, reducing transpiration and conserving water. During water stress, they lose turgidity, causing the leaf to roll inward and reduce the exposed surface area. This rolling mechanism helps minimise transpiration and conserve water.

They do not perform photosynthesis or transport water, but act as a protective adaptation against drought.

Hence, the correct answer is option 2) to minimise water loss during water stress.

Q. 18 Find the incorrect statement(s) about photosynthesis from the following:

- A. The water splitting complex is associated with PS I.
- B. C_4 plants use the C_3 pathway of CO_2 fixation as the main biosynthetic pathway.
- C. In C_4 plants, photorespiration does not occur.
- D. C_3 plants exhibit 'Kranz' anatomy.
- E. ATP synthesis in chloroplasts occurs through chemiosmosis.

Choose the answer from the options given below :

Option 1:
B and C only

Option 2:
B and E only

Option 3:
B only

Correct Answer:

A and D only

Solution:

Water splitting is linked to PS II (A false) and Kranz anatomy is seen in C_4 plants, not C_3 (D false).

- The water splitting complex is linked to PS II, not PS I, and it provides electrons by splitting water into oxygen and protons.
- Kranz anatomy is a special leaf structure found only in C_4 plants, where bundle sheath cells surround vascular bundles.
- C_4 plants still use the C_3 cycle as the main biosynthetic pathway, but photorespiration is minimised due to spatial separation of steps.
- ATP synthesis in chloroplasts indeed occurs via chemiosmosis, similar to mitochondria.

Hence, the correct answer is option 4) A and D only.

Q. 19 Match List I with List II :

	List I		List II
A.	Conjunctive tissue	I.	Specialised cells in the vicinity of guard cells
B.	Casparian strips	II.	Endodermal cells are rich in starch
C.	Subsidiary cells	III.	Tissue between xylem and phloem
D.	Starch sheath	IV.	Endodermal cells with suberin deposition

Choose the correct answer from the options given below :

Option 1:

A-IV, B-III, C-II, D-I

Option 2:

A-III, B-IV, C-I, D-II

Option 3:

A-IV, B-III, C-I, D-II

Option 4:

A-III, B-IV, G-II, D-I

Correct Answer:

A-III, B-IV, C-I, D-II

Solution:

- Conjunctive tissue lies between xylem & phloem (III)

Hence, the correct answer is option 2) A-III, B-IV, C-I, D-II

Q. 20 Match List I with List II :

	List I		List II
A.	Genetically modified organism	I.	Agrobacterium tumefaciens
B.	Thermostable DNA polymerase	II.	Bt cotton
C.	Ti plasmid	III.	Thermus aquaticus
D.	pBR322	IV.	Escherichia coli

Choose the correct answer from the options given below:

Option 1:

A-II, B-I, C-I , D-III

Option 2:

A-II, B-III, G-I, D-IV

Option 3:

A-I, B-IV, C-III, D-II

Option 4:

A-I, B-II, C-IV, D-III

Correct Answer:

A-II, B-III, G-I, D-IV

Solution:

- Bt cotton is a genetically modified organism (A → II)
- Thermostable DNA polymerase comes from Thermus aquaticus (B → III)
- Ti plasmid is from Agrobacterium tumefaciens (C → I)
- pBR322 is a cloning vector (D → IV)

Hence, the correct answer is option 2) A-II, B-III, G-I, D-IV.

Q. 21 Heterophyllous development in response to the environment is an example of which of the following phenomena?

Option 1:

Redifferentiation

Option 3:

Elasticity

Option 4:

Plasticity

Correct Answer:

Plasticity

Solution:

Heterophylly shows that the same genotype can produce different leaf forms under different environmental conditions. This is called phenotypic plasticity. In heterophylly, plants like Cotton, Coriander, and Buttercup show varied leaf shapes in response to light, water, or temperature.

This adaptability helps plants optimise photosynthesis and survival under changing environments. It is not due to genetic change but due to flexible expression of the same genetic makeup. Thus, heterophylly is a classic example of plasticity in plants.

Hence, the correct answer is option 4) Plasticity.

Q. 22 In racemose inflorescence, ____ .

Option 1:

The growth is limited

Option 2:

Flowers are solitary

Option 3:

Flowers are borne in an acropetal succession

Option 4:

The main axis terminates in a flower

Correct Answer:

Flowers are borne in an acropetal succession

Solution:

In a racemose inflorescence, the main axis shows unlimited growth and flowers develop in acropetal succession (older at base, younger towards apex).

Option 1: The growth is limited - False, because in a racemose inflorescence, the main axis shows indefinite growth.

Option 2: Flowers are solitary - False, because racemose inflorescence has multiple flowers arranged along the axis, not solitary.

Option 3: Flowers are borne in an acropetal succession -True, because in a racemose inflorescence, younger flowers are at the apex, older at the base.

Hence, the correct answer is option 3) flowers are borne in an acropetal succession.

Q. 23 Which one of the following disorders is caused by the substitution of Glutamic acid (Glu) by Valine (Val) at the sixth position of the beta globin chain of the haemoglobin molecule?

Option 1:
Phenylketomuria

Option 2:
Haemophilif

Option 3:
Sickle-cell anaemia

Option 4:
Thalassemia

Correct Answer:
Sickle-cell anaemia

Solution:

Sickle-cell anaemia is caused by a point mutation where *Glu* → *Val* at position 6 of β-globin.

This point mutation changes the hydrophilic glutamic acid to hydrophobic valine, altering the haemoglobin structure. As a result, red blood cells become sickle-shaped under low oxygen conditions, leading to blockages in blood vessels.

The disorder causes anaemia, pain crises, and organ damage due to impaired oxygen transport. It is a classic example of a genetic disease caused by a substitution mutation.

Hence, the correct answer is option 3) Sickle-cell anaemia.

Q. 24 Match List I with List II :

List - I	List - II
A. Incomplete dominance	I. Human skin colour
B. Co-dominance	II. Inheritance of flower colour in Antirrhinum sp.
C. Pleiotropy	III. Phenylketonuria disease in humans
D. Polygenic inheritance	IV. ABO blood groups

Choose the correct answer from the options given below :

Option 1:
A-I, B-IV, C-III, D-II

Option 3:

A-II, B-IV, C-III, D-I

Option 4:

A-II, B-I, C-III, D-IV

Correct Answer:

A-II, B-IV, C-III, D-I

Solution:

Incomplete dominance → II; Co-dominance → IV; Pleiotropy → III; Polygenic inheritance → I.

- In incomplete dominance (Antirrhinum), the heterozygote shows an intermediate phenotype, e.g., pink flowers from red × white.
- Co-dominance (ABO blood groups) allows both alleles to be expressed equally, such as I^A and I^B producing AB blood type.
- Pleiotropy (Phenylketonuria) occurs when a single gene affects multiple traits, leading to diverse symptoms.
- Polygenic inheritance (human skin colour) involves multiple genes contributing additively to a continuous variation.

Hence, the correct answer is option 3) A-II, B-IV, C-III, D-I.

Q. 25 Arrange the following in the correct developmental sequence related to microsporogenesis :

- A. Microspore tetrads
- B. Sporogenous tissue
- C. Pollen grains
- D. Pollen mother cells

Choose the correct answer from the options given below:

Option 1:

A, D, C, B

Option 2:

D, A, C, B

Option 3:

B, D, C, A

Option 4:

B, D, A, C

Correct Answer:

B, D, A, C

Solution:

Sporogenous tissue pollen mother cells → microspore tetrads → pollen grains.

This sequence highlights the progression from diploid sporogenous tissue to haploid pollen grains through meiosis.

Hence, the correct answer is option 4) A, E, C, B, D.

Q. 26 Arrange the following steps of DNA fingerprinting in a correct sequence.

- A. Isolation of DNA and its digestion by restriction endonucleases
- B. Hybridisation using a labelled VNTR probe
- C. Transferring of separated DNA fragments to synthetic membranes
- D. Detection of hybridised DNA fragments by autoradiography
- E. Separation of DNA fragments by electrophoresis.

Choose the correct answer from the options given below :

Option 1:

A, E, B, C, D

Option 2:

A,D,B,E,C

Option 3:

A, B, D, C, E

Option 4:

A, E, C, B, D

Correct Answer:

A, E, C, B, D

Solution:

In DNA fingerprinting, DNA is isolated and cut, then fragments are separated by electrophoresis, transferred to a membrane, hybridised with a probe, and finally detected by autoradiography.

This technique relies on Variable Number Tandem Repeats (VNTRs), which are unique to individuals, making DNA fingerprinting highly specific. The autoradiography step finally reveals the distinct banding pattern, serving as a genetic identity marker.

Hence, the correct answer is option 4) A, E, C, B, D.

Q. 27 Exploring molecular, genetic and species-level diversity for products of economic importance is called :

Option 1:

Bioprospecting

Option 2:

Biofortification

Option 3:

Biomagnification

Correct Answer:

Bioprospecting

Solution:

Bioprospecting involves searching biological diversity at molecular, genetic, and species levels to identify useful products of economic importance.

Term	Definition	Example Use
Bioprospecting	Exploring biodiversity for economic products	Drugs from medicinal plants
Biomagnification	Toxin concentration increases in the food chain	Mercury in fish
Bioremediation	Microbes clean pollutants	Oil spill cleanup
Biofortification	Nutrient-rich crop breeding	Iron-fortified rice

Hence, the correct answer is option 1) Bioprospecting.

- Q. 28** Which of the following statements are true with reference to sex determination in honeybees?
- A. An offspring formed from the union of a sperm and an egg develops as a female (queen or worker).
 - B. An unfertilized egg develops as a male by parthenogenesis.
 - C. A male has half the number of chromosomes as a female.
 - D. Males produce sperm by meiosis
 - E. Honeybees have a haplodiploid sex-determination system.
- Choose the correct answer from the options given below:

Option 1:

A, B, C and E only

Option 2:

A, B, C and D only

Option 3:

B, C, D and E only

Option 4:

A, B, D and E only

Correct Answer:

A, B, C and E only

Solution:

In honeybees, fertilised eggs (2n) form females, unfertilised eggs (n) form males. Males are haploid and produce sperm by mitosis, not meiosis, under haplodiploidy.

- A. True - Fertilised eggs (sperm + egg) develop into females (queen or worker).
- B. True - Unfertilized eggs develop into males (drones) by parthenogenesis.
- C. True - Males (haploid) have half the number of chromosomes compared to females (diploid).
- D. False - Males are haploid, so they cannot undergo meiosis to produce gametes. They produce sperm by mitosis.
- E. True - Honeybees indeed have a haplodiploid sex-determination system.

Option 1:

Denaturation → Extension → Annealing

Option 2:

Denaturation → Annealing → Extension

Option 3:

Annealing → Denaturation → Extension

Option 4:

Extension → Annealing → Denaturation

Correct Answer:

Denaturation → Annealing → Extension

Solution:

PCR follows: denaturation (strand separation) → annealing (primer binding) → extension (DNA synthesis).

These together amplify DNA exponentially. This cycle is repeated multiple times (typically 25–35 cycles) to generate millions of copies of the target DNA sequence.

Hence, the correct answer is option 2) Denaturation → Annealing → Extension.

Q. 30 Which of the following statements are correct with respect to DNA separation, isolation and visualisation?

- A. The cutting of DNA is done by molecular scissors.
- B. The DNA fragments separate according to their size in an agarose gel upon electrophoresis
- C. The separated DNA fragments can be seen without staining when exposed to UV light
- D. The separated PNA fragments, when stained with ethidium bromide, can be seen in visible light.

Choose the correct answer from the options given below:

Option 1:

A and B only

Option 2:

B and D only

Option 3:

A and D only

Option 4:

B and C only

Solution:

Restriction enzymes act as molecular scissors (A true), and electrophoresis separates DNA by size (B true). DNA needs staining and is visualised under UV, not visible light (C, D false).

- Restriction endonucleases act as molecular scissors, cutting DNA at specific recognition sites.
- During agarose gel electrophoresis, DNA fragments migrate according to their size, with smaller fragments moving faster.
- DNA cannot be seen directly; it must be stained (commonly with ethidium bromide) and visualised under UV light.

Hence, the correct answer is option 1) A and B only.

- Q. 31** The main criteria used for Five Kingdom Classification proposed by R.H. Whittaker (1969) included :
- A. Cell structure
 - B. Body organisation
 - C. Presence of flagellum
 - D. Reproductions
 - E. Phylogenetic relationships

Choose the correct answer from the options given below :

Option 1:

A, B and E only

Option 2:

A, B, C, D and E

Option 3:

B, C and D only

Option 4:

A, B, D and E only

Correct Answer:

A, B, D and E only

Solution:

R. H. Whittaker used cell structure, thallus/body organisation, mode of reproduction, and phylogenetic relationships. The presence of a flagellum was not one of the main criteria. This system was revolutionary as it combined morphological, reproductive, and evolutionary criteria for classification. It resolved the limitations of earlier two-kingdom and three-kingdom systems by recognising fungi separately and including Monera and Protista.

Hence, the correct answer is option 4) A, B, D and E only.

- Q. 32** Which one of the following is a triploid cell?

Option 1:

Synergid

Option 3:
Central cell

Option 4:
Zygote

Correct Answer:
Primary endosperm cell

Solution:
The primary endosperm cell is triploid ($3n$), formed by the fusion of one male gamete with two polar nuclei ($n + n + n$).
This triploid cell develops into the endosperm, which nourishes the developing embryo in angiosperms. Thus, it is the only triploid structure among the given options.

Hence, the correct answer is option 2) Primary endosperm cell.

- Q. 33** Which of the following statements are correct with reference to the packaging of the DNA helix?
- A. Histones are organised to form a unit of eight molecules, called a histone octamer
 - B. Histones are negatively charged basic proteins
 - C. Histones are rich in the basic amino acid residues - lysine and arginine.
 - D. The positively charged DNA is wrapped around the histone octamer to form a nucleosome.
 - E. The packaging of chromatin at higher levels requires an additional set of proteins called non-histone chromosomal proteins.
- Choose the correct answer from the options given below :

Option 1:
B, D and E only

Option 2:
A, B and D only

Option 3:
C, D and E only

Option 4:
A, C and E only

Correct Answer:
A, C and E only

Solution:
Histones are positively charged proteins due to lysine and arginine residues, which help them bind tightly to the negatively charged DNA. They form an octamer core around which DNA wraps, creating nucleosomes - the fundamental units of chromatin.

Higher-order chromatin packaging requires non-histone chromosomal proteins for further compaction. Statements B and D are incorrect because histones are not negatively charged, and DNA is negatively charged, not positive.

Q. 34 Which of the following is an in situ conservation method?

Option 1:
Seed Banks

Option 2:
Sacred Groves

Option 3:
Botanical Gardens

Option 4:
Wildlife Safari Parks

Correct Answer:
Sacred Groves

Solution:

In situ conservation means protecting organisms in their natural habitat. Sacred groves are protected forest patches where biodiversity is preserved due to cultural and religious beliefs.

Unlike seed banks or botanical gardens (ex situ), this method allows species to evolve naturally in their environment. It is considered the most effective way to conserve endangered plants and animals.

Hence, the correct answer is option 2) Sacred Groves.

Q. 35 In the lac operon, the z gene codes for :

Option 1:
the repressor of lac operon

Option 2:
transacetylase

Option 3:
permease

Option 4:
beta-galactosidase

Correct Answer:
beta-galactosidase

lacY (permease) and lacA (transacetylase), lacZ forms the structural gene set of the lac operon.

Hence, the correct answer is option 4) beta-galactosidase.

Q. 36 Match List I with List II :

	List I		List II
	(Growth Regulator)		(Function/Effect)
A.	2,4-D	I.	Brewing industry
B.	GA ₃	II.	Stimulation of stomatal closure
C.	Kinetin	III.	Herbicide
D.	ABA	IV.	Nutrient mobilisation

Choose the correct answer from the options given below:

Option 1:

A-I, B-II, C-IV, D-III

Option 2:

A-I, B-IV, C-III, D-II

Option 3:

A-IV, B-III, C-II, D-I

Option 4:

A-III, B-I, C-IV, D-II

Correct Answer:

A-III, B-I, C-IV, D-II

Solution:

- 2, 4-D acts as a herbicide ($A \rightarrow III$)
- GA₃ is used in the brewing industry ($B \rightarrow I$)
- Kinetin promotes nutrient mobilisation ($C \rightarrow IV$)
- ABA induces stomatal closure ($D \rightarrow II$)

Hence, the correct answer is option 4) A-III, B-I, C-IV, D-II.

-
- C. Fusion of protoplasts to get a hybrid protoplast.
 - D. Isolation of single cells from two different varieties of plants.
 - E. Growing of hybrid protoplast to form a new plant.
- Choose the correct answer from the options given below :

Option 1:

D, B, A, E, C

Option 2:

E, A, B, C, D

Option 3:

E, B, A, D, C

Option 4:

D, A, B, C, E

Correct Answer:

D, A, B, C, E

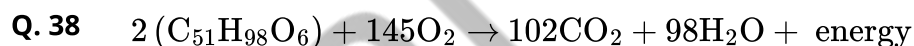
Solution:

First cells are isolated → cell walls digested → protoplasts obtained → fusion → regeneration into a plant.

This process allows the fusion of protoplasts from different plant varieties, bypassing sexual reproduction barriers.

The regenerated hybrid protoplast develops into a new plant with combined genetic traits of both parents. Thus, somatic hybridisation is a powerful tool in plant biotechnology and crop improvement.

Hence, the correct answer is option 4) D, A, B, C, E



The Respiratory Quotient (RQ) of a biomolecule used for respiration, as per the above equation, would be :

Option 1:

Less than 0.5

Option 2:

Between 1.25 and 2

Option 3:

1 · 0

Option 4:

Between 0.5 and 0.95

Solution:

$$RQ = \frac{CO_2 \text{ produced}}{O_2 \text{ consumed}} = \frac{102}{145} \approx 0.7, \text{ thus lying between } 0.5 \text{ and } 0.95.$$

Since the RQ value is less than 1, it clearly indicates the substrate is a fat rather than a carbohydrate. Fats require more oxygen for complete oxidation, hence their RQ typically lies between 0.7 and 0.8, matching the calculated value here.

Hence, the correct answer is option 4) Between 0.5 and 0.95.

Q. 39 Since the origin and diversification of life on Earth, there have been five episodes of mass extinction of species. How is the sixth extinction, which is in progress, different from the previous episodes?

Option 1:

The current species extinction rates are far lower than those in previous episodes.

Option 2:

The present species extinction rates are 100 to 1000 times faster than in pre-human times.

Option 3:

The present net species extinction rate is zero.

Option 4:

The current species extinction rate is nearly 10 times faster than that in previous episodes.

Correct Answer:

The present species extinction rates are 100 to 1000 times faster than in pre-human times.

Solution:

Earth has witnessed five major mass extinctions in the past, caused by natural events (volcanic eruptions, asteroid impacts, climate shifts).

The ongoing (sixth) extinction is largely human-driven, and the extinction rate is estimated to be $100\text{--}1000\times$ higher than the natural background rate. It is human-driven due to deforestation, habitat loss, pollution, overexploitation, and climate change.

Hence, the correct answer is option 2) The present species extinction rates are 100 to 1000 times faster than in pre-human times.

LIST - I	LIST - II
A. Trypsin	I. Interacellular ground substance
B. Morphine	II. Lectin
C. Concanavalin	III. Enzyme
D. Collagen	IV. Alkaloid

Choose the correct answer from the options given below:

Option 1:

A-III, B-IV, C-II, D-I

Option 2:

A-IV, B-III, C-II, D-I

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-III, B-II, C-IV, D-I

Correct Answer:

A-III, B-IV, C-II, D-I

Solution:

Trypsin → enzyme (III); Morphine → alkaloid (IV); Concanavalin A → lectin (II); Collagen → intercellular ground substance (I).

- Trypsin is a digestive enzyme that hydrolyses proteins into peptides.
- Morphine is a plant-derived alkaloid used medicinally as an analgesic.
- Concanavalin A is a lectin that binds specifically to certain sugars, often used in biochemical studies.
- Collagen is the main structural protein of the intercellular ground substance, providing strength and elasticity to connective tissues.
- **Hence, the correct answer is option 1) A-III, B-IV, C-II, D-I.**

Q. 41 Which one of the following statements is not true about the universal rules of binomial nomenclature?

Option 1:

The first word in the biological name represents the specific epithet, while the second component denotes the genus.

Option 2:

The specific epithet in the biological name starts with a small letter.

Option 4:

Biological names are generally in Latin.

Correct Answer:

The first word in the biological name represents the specific epithet, while the second component denotes the genus.

Solution:

According to the universal rules, the first word denotes the genus and always begins with a capital letter, while the second word is the specific epithet written in small letters. Both words are either italicised in print or underlined separately when handwritten to maintain uniformity. Biological names are generally derived from Latin or are Latinized, ensuring global recognition.

The first word represents the genus and the second is the specific epithet, so statement (1) is incorrect.

Hence, the correct answer is option 1) The first word in the biological name represents the specific epithet, while the second component denotes the genus.

Q. 42 The enzyme required for carboxylation in the Calvin cycle is :

Option 1:

PEP carboxylase

Option 2:

RuBP carboxylase - oxygenase

Option 3:

Carboxypeptidase

Option 4:

Hexokinase

Correct Answer:

RuBP carboxylase - oxygenase

Solution:

Carboxylation in the Calvin cycle is catalysed by RuBisCO, which fixes CO_2 to RuBP, forming 3-PGA.

It fixes CO_2 to RuBP, producing 3-phosphoglycerate (3-PGA) as the first stable compound.

This step is the key entry point of inorganic carbon into the biosphere, driving photosynthetic productivity.

Hence, the correct answer is option 2) RuBP carboxylase - oxygenase.

Q. 43 Which of the following floral formula is the correct floral formula of Solanaceae family ?

Option 2:

$$\oplus K_5 C_5 A_5 \underline{G}_{(2)}$$

Option 3:

$$\oplus \overset{\nearrow}{\underset{\searrow}{\text{♀}}} K_{(5)} \overset{\frown}{C_{(5)}} A_5 \underline{G}_{(2)}$$

Option 4:

$$\overset{\nearrow}{\underset{\searrow}{\text{♂}}} \underset{\searrow}{\text{♀}} K_5 \overset{\frown}{C_{(5)}} A_5 \underline{G}_{(2)}$$

Correct Answer:

$$\oplus \overset{\nearrow}{\underset{\searrow}{\text{♀}}} K_{(5)} \overset{\frown}{C_{(5)}} A_5 \underline{G}_{(2)}$$

Solution:

Flowers of the Solanaceae family are actinomorphic and bisexual. Their calyx consists of five fused sepals, and the corolla has five fused petals. There are five stamens with a bicarpellary gynoecium and a superior ovary.

Hence, the correct answer is option 3)

$$\oplus \overset{\nearrow}{\underset{\searrow}{\text{♀}}} K_{(5)} \overset{\frown}{C_{(5)}} A_5 \underline{G}_{(2)}$$

Q. 44 Which one of the following types of pollination brings genetically different types of pollen grains to the stigma?

Option 1:

Geitonogamy

Option 2:

Xenogamy

Option 3:

Cleistogamy

Option 4:

Autogamy

Correct Answer:

Xenogamy

Solution:

Unlike autogamy or geitonogamy, it prevents inbreeding and maintains healthy gene pools. It is often facilitated by agents like insects, wind, or water, depending on the plant species. Thus, xenogamy is the only pollination type that guarantees genetically different pollen grains on the stigma.

Hence, the correct answer is option 2) Xenogamy.

Q. 45 Match List I with List II :

List - I (Process)	List -II (Location)
A. Glycolysis	I. Inner mitochondrial membrane
B. ETS	II. Mitochondrial matrix
C. Accumulation of protons	III. Cytoplasm
D. Krebs' cycle	IV. Intermembrane space

Choose the correct answer from the options given below:

Option 1:

A-IV, B-II, C-I, D-III

Option 2:

A-I, B-IV, C-III, D-II

Option 3:

A-II, B-III, C-IV, D-I

Option 4:

A-III, B-I, C-IV, D-II

Correct Answer:

A-III, B-I, C-IV, D-II

Solution:

- Glycolysis occurs in the cytoplasm (III)
- ETS on inner membrane (I)
- Protons accumulate in the intermembrane space (IV)
- Krebs cycle in matrix (II).

Hence, the correct answer is option 4) A-III, B-I, C-IV, D-II

Option 1:

Ampicillin and tetracycline

Option 2:

Ampicillin

Option 3:

Tetracycline

Option 4:

Gentamycin

Correct Answer:

Tetracycline

Solution:

The BamHI site lies within the tetracycline resistance gene; insertion disrupts it, causing loss of tetracycline resistance. The insertion normally disrupts it because it interrupts the normal coding sequence of the tetR gene. This turns it non-functional and causes the transformed bacteria to lose tetracycline resistance.

Hence, the correct answer is option 3) Tetracycline

Q. 47 The sixth mutant codon of beta globin gene causing polymerization of Haemoglobin and change in RBC shape is ____

Option 1:

CAG

Option 2:

AUG

Option 3:

GUG

Option 4:

GAG

Correct Answer:

GUG

Solution:

Hence, the correct answer is option 3) GUG

Q. 48 Choose the correct statement regarding GIFT to overcome infertility.

Option 1:

Ova collected from a female donor are transferred to the uterus of an infertile female.

Option 2:

Early embryos with up to 8 blastomeres are transferred into the fallopian tube of an infertile female

Option 3:

Early embryos with up to 8 blastomeres are transferred to the uterus of an infertile female.

Option 4:

It is the transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce ovum but can provide suitable environment for fertilization and development.

Correct Answer:

It is the transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce ovum but can provide suitable environment for fertilization and development.

Solution:

GIFT is Gamete Intra-Fallopian Transfer. GIFT involves transfer of an ovum into the fallopian tube for in vivo fertilization, not embryo transfer (which is IVF-ET). Embryo transfer is a technique in which the fertilised egg or embryo in the early stage is transferred into the female reproductive tract.

Hence, the correct answer is option 4). It is the transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce ovum but can provide suitable environment for fertilization and development.

Q. 49 Which one of the following is an appropriate example of 'sexual deceit'?

Option 1:

Sea anemone and clown fish

Option 2:

Ophrys and bumblebee

Option 4:

Cuckoo and crow

Correct Answer:

Ophrys and bumblebee

Solution:

In sexual deceit, the orchid Ophrys mimics female insects to attract male pollinators, leading to pollination without reward. It is because the male insects try to mate with the flower, during which the pollen transfer and pollination occur without getting any reward.

Hence, the correct answer is option 2) Ophrys and bumblebee

Q. 50 Evolution of human appears parallel to the progressive development of brain and language skills. As such, the evolution of individual species in the sequence of their appearance is :

Option 1:

Ramapithecus → Homo habilis → Homo erectus → Neanderthal → Homo sapiens

Option 2:

Neanderthal → Ramapithecus →
Homo habilis → Homo erectus →
Homo sapiens

Option 3:

Homo habilis → Homo erectus →
Ramapithecus → Neanderthal →
Homo sapiens

Option 4:

Homo sapiens → Ramapithecus →
Homo habilis → Neanderthal
Homo erectus

Correct Answer:

Ramapithecus → Homo habilis → Homo erectus → Neanderthal → Homo sapiens

Solution:

Correct evolutionary sequence: Ramapithecus → Homo habilis → Homo erectus → Neanderthal → Homo sapiens.

- Ramapithecus was the earliest ape-man ancestor.
- Homo habilis - first tool maker
- Homo erectus - upright man
- Neanderthal man - these were more advanced
- Homo sapiens - modern humans

Q. 51 Match List I and List II related to embryonic development at various months of pregnancy :

List I		List II	
A	The foetus movement starts and hair appears on the head	I	24 weeks of pregnancy
B	The foetus develops limbs and digits	II	20 weeks of pregnancy
C	The foetus develops external genital organs	III	8 weeks of pregnancy
D	The foetus body is covered with fine hair; eyelids N separate and eyelashes are formed	IV	12 weeks of pregnancy

Choose the **correct** answer from the options given below:

Option 1:

A-II, B-IV, C-III, D-I

Option 2:

A-II, B-III, C-IV, D-I

Option 3:

A-IV, B-II, C-III, D-I

Option 4:

A-III, B-II, C-IV, D-I

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

20 weeks → movement & hair, 8 weeks → limbs, 12 weeks → genital organs, 24 weeks → eyelashes & fine hair.

During the development of a foetus, the limbs and digits appear by 8 weeks, external genital organs appear by 12 weeks of pregnancy. In the same way, foetal movement starts around 20 weeks.

Hence, the correct answer is option 2) A-II, B-III, C-IV, D-I

- A. Endoskeleton was made of cartilage.
- B. Ectoparasitic; as they were found attached on fish skin with their circular sucking mouth.
- C. Paired fins and scales were absent, but 7 pairs of gill slits were present.

Which of the following species of animals did they consider to fit best with these characters?

Option 1:

Scoliodon sp.

Option 2:

Exocoetus sp

Option 3:

Petromyzon sp

Option 4:

Branchiostoma sp

Correct Answer:

Petromyzon sp

Solution:

Petromyzon (lamprey) shows cartilaginous skeleton, ectoparasitic habit, circular mouth, no paired fins, and multiple gill slits.

Hence, the correct answer is option 3) Petromyzon sp

Q. 53 In which animal do haploid cells mitotically to produce gametes ?

Option 1:

Male honeybees

Option 2:

Male grasshoppers

Option 3:

Male earthworms

Correct Answer:
Male honeybees

Solution:

Male honeybees are haploid organisms formed from unfertilized eggs. Since they are already haploid, gamete formation occurs by mitosis instead of meiosis. Meiosis reduces chromosome number by half, which is why it becomes an unnecessary process in drones.

Hence, the correct answer is option 1) Male honeybees

- Q. 54** Spermatogonia undergo a series of cell divisions to produce sperms. Select the correct statements from the following:
- A. Spermatogonia always undergo meiotic cell division.
 - B. Primary spermatocytes divide mitotically to produce secondary spermatocytes
 - C. Secondary spermatocytes, through their second meiotic division, produce haploid spermatids.
 - D. Spermatids produce spermatozoa through mitosis.
 - E. Spermatids transform into spermatozoa by spermiogenesis

Choose the **correct** answer from the options given below :

Option 1:

A, C and E only

Option 2:

C and E only

Option 3:

A and E only

Option 4:

B, C and D only

Correct Answer:
C and E only

Solution:

Spermatogonia divide mitotically, not always meiotically. Meiosis II forms haploid spermatids, which transform into spermatozoa by spermiogenesis. Spermatids are immature sperm cells, while the spermatozoa are the mature and functional sperm.

Hence, the correct answer is option 2) C and E only

Option 1:

0%

Option 2:

50%

Option 3:

25%

Option 4:

75%

Correct Answer:

25%

Solution:

Cross: $I^A i \times I^B i \Rightarrow$ offspring genotypes: $I^A I^B, I^A i, I^B i, ii$.

Only ii gives blood group O, so the probability = $\frac{1}{4} = 25$.

Hence, the correct answer is option 3) 25%

Q. 56 Arrange the following events occurring in Renin-Angiotensin mechanism in the correct order :

- A. Increase in blood pressure and Glomerular filtration rate.
- B. Reabsorption of Na^+ and water from distal parts of tubule due to Aldosterone.
- C. Fall in Glomerular filtration rate.
- D. Vasoconstriction by Angiotensin II and release of Aldosterone.
- E. Renin converts Angiotensinogen into Angiotensin I, followed by Angiotensin II.

Choose the **correct** answer from the options given below:

Option 1:

A, C, E, B, D

Option 2:

C, A, B, D, E

Option 3:

A, D, B, E, C

Option 4:

C, E, D, B, A

Correct Answer:

C, E, D, B, A

Solution:

The glomerular filtration rate usually refers to the amount of filtrate formed by both kidneys per minute.

Q. 57 Match List I with List II :

List I (Respiratory Volume)		List II (Capacity)	
A	ERV (Expiratory Reserve Volume)	I	2500 - 3000 mL
B	RV (Residual Volume)	II	500 mL
C	IRV (Inspiratory Reserve Volume)	III	1000-1100mL
D	TV (Tidal Volume)	IV	1100-1200mL

Choose the **correct** answer from the options below:

Option 1:

A-III, B-IV, C-I, D-II

Option 2:

A-III, B-I, C-IV, D-II

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-I, B-III, C-II, D-IV

Correct Answer:

A-III, B-IV, C-I, D-II

Solution:

ERV $\approx$ 1000-1100, RV $\approx$ 1100-1200, IRV $\approx$ 2500-3000, TV $\approx$ 500 mL.

TV = Tidal Volume is the amount of air inhaled or exhaled during normal breathing.

IRV, = IRV on the other hand, is the extra amount of air a person can inhale forcibly after a normal inspiration.

ERV = It is the extra amount of air that a person can exhale forcibly after a normal expiration.

Hence, the correct answer is option 1. A-III, B-IV, C-I, D-II

LIST I		LIST II	
A	Progestasert	I	Barrier made of rubber used by females
B	Multiload 375	II	Oral contraceptive
C	Diaphragm	III	Hormone releasing IUD
D	Saheli	IV	Copper releasing IUD

Choose the **correct** answer from the options given below :

Option 1:

A-IV, B-II, C-I, D-III

Option 2:

A-IV, B-III, C-I, D-II

Option 3:

A-III, B-IV, C-II, D-I

Option 4:

A-III, B-IV, C-I, D-II

Correct Answer:

A-III, B-IV, C-I, D-II

Solution:

Progestasert → hormone IUD (III), Multiload 375 → copper IUD (IV), Diaphragm → barrier (I), Saheli → oral pill (II).

Hence, the correct answer is option 4) A-III, B-IV, C-I, D-II

Q. 59 Non-membrane bound cell organelles found in both prokaryotic and eukaryotic cells are _____.

Option 1:

Lysosomes

Option 2:

Centrosomes

Option 4:
Ribosomes

Correct Answer:
Ribosomes

Solution:

Ribosomes are cellular organelles that synthesize proteins. These are non-membrane-bound and present in both prokaryotic and eukaryotic cells.

Hence, the correct answer is option 4) Ribosomes

Q. 60 Ecological pyramids represent the relationship between the organisms at different trophic levels and they are generally inverted for :

Option 1:
Pyramid of number in grassland

Option 2:
Pyramid of energy in pond ecosystem

Option 3:
Pyramid of biomass in grassland

Option 4:
Pyramid of biomass in sea

Correct Answer:
Pyramid of biomass in sea

Solution:

In aquatic ecosystems, the biomass pyramid is inverted as phytoplankton biomass is lower than that of consumers.

Hence, the correct answer is option 4) Pyramid of biomass in sea

Q. 61 The flightless bird with forelimbs modified as paddle-like structures suited for swimming is known as :

Option 1:
Psittacula

Option 3:
Neophron

Option 4:
Struthio

Correct Answer:
Aptenodytes

Solution:
Aptenodytes (penguins) are flightless birds with flipper-like wings adapted for swimming.

Hence, the correct answer is option 2) Aptenodytes

Q. 62 Match **List I** with **List II**:

List I Bioactive molecules)		List II (Importance)	
A	Streptokinase	I	Immunosuppressive agent
B	Statins	II	Removal of clots from the blood vessels
C	Lipases	III	Blood cholesterol-lowering agent
D	Cyclosporin A	IV	Detergent formulations

Choose the **correct** answer from the options given below:

Option 1:
A-II, B-III, C-I, D-IV

Option 2:
A-III, B-II, C-IV, D-I

Option 4:

A-IV, B-III, C-II, D-I

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

Streptokinase dissolves clots and is used during heart attacks (II), statins lower cholesterol (III), lipases are used in detergents (IV), and cyclosporin A is an immunosuppressant that is used during organ transplantation (I).

Hence, the correct answer is option 3) A-II, B-III, C-IV, D-I

Q. 63 Choose the correct statements regarding cell organelles and their inclusions.

- A. The endomembrane system includes Golgi complex, endoplasmic reticulum and mitochondria.
- B. Rough endoplasmic reticulum bears ribosomes on its surface.
- C. Both mitochondria and plastids have circular DNA.
- D. A network of microtubules, microfilaments and intermediate filaments present in the cytoplasm is called cytoskeleton.
- E. Mitochondrion is a single membrane-bound structure.

Choose the **correct** answer from the options given below :

Option 1:

A, B and C only

Option 2:

C, D and E only

Option 3:

A and B only

Option 4:

B, C and D only

Correct Answer:

B, C and D only

Solution:

Mitochondria are not part of the endomembrane system (A false). RER has ribosomes (B true), mitochondria & plastids have circular DNA (C true), cytoskeleton definition is correct (D true), mitochondria are double membrane-bound (E false).

Option 1:

Saw fish, Fighting fish and Dog fish

Option 2:

Devil fish, Cuttlefish and Hagfish

Option 3:

Starfish, Hagfish and Cuttlefish

Option 4:

Flying fish, Angel fish and Fighting fish

Correct Answer:

Flying fish, Angel fish and Fighting fish

Solution:

Osteichthyes are bony fishes; Flying fish, Angel fish, and fighting fish belong to this class, while others include cartilaginous or non-fish organisms. This is because the Flying fish, angel fish, and fighting fish are true bony fish. Osteichthyes are bony fishes with a skeleton that is made mainly of bone. Sharks and rays, on the other hand, belong to Class Chondrichthyes.

Hence, the correct answer is option 4) Flying fish, Angel fish, and Fighting fish

Q. 65 In a population of a grasshopper species, the chromosome number of some members is 23 and some other members possess 24 chromosomes. The 23 and 24 chromosome-bearing members in this species are ____.

Option 1:

females and males, respectively

Option 2:

males and females, respectively

Option 3:

all males

Option 4:

all females

Solution:

Grasshoppers show XO type sex determination: males = 23 (XO), females = 24 (XX).

The grasshopper shows XO-type sex determination. In the XO type, females have two X chromosomes, and males have only one X chromosome. Both males and females have the same number of autosomes. Therefore, grasshoppers with 24 chromosomes are females, and those with 23 chromosomes are males.

Hence, the correct answer is option 2) males and females, respectively

Q. 66 The WBC count of a person's blood sample 8000/cu. mm. How many eosinophils and lymphocytes would be in the same blood sample approximately ?

Option 1:

300 – 500/cu. mm and 500 – 700/cu. mm, respectively

Option 2:

300 – 500/cu. mm and 1200 – 1500/cu. mm, respectively

Option 3:

100 – 120/cu. mm and 160 – 200/cu. mm, respectively

Option 4:

160 – 240/cu. mm and 1600 – 2000/cu. mm, respectively

Correct Answer:

160 – 240/cu. mm and 1600 – 2000/cu. mm, respectively

Solution:

Eosinophils $\approx$ 2–3% of 8000 $\Rightarrow$ 160–240.

Lymphocytes $\approx$ 20–25% of 8000 $\Rightarrow$ 1600–2000.

Eosinophils constitute about 2-3% and lymphocytes about 20-25% of total WBCs.

Hence, the correct answer is option 4) 160 – 240/cu. mm and 1600 – 2000/cu. mm, respectively

Q. 67 The toxin proteins isolated from *Bacillus thuringiensis*, coded by which of the following genes would control cotton bollworms and corn borer, respectively ?

Option 1:

cryIAc and *cryIIb*

Option 3:
cryIAc and *cryIAB*

Option 4:
cryIIAb and *cryIAc*

Correct Answer:
cryIAc and *cryIAB*

Solution:
The *cryIAc* gene produces a toxin against the cotton bollworms, whereas on the other hand, the *cryIAb* gene produces a toxin that controls corn borers.

Hence, the correct answer is option 3) *cryIAc* and *cryIAB*

Q. 68 Match **List I** and **List II**

List I (Drug)		List II (Effect)	
A	Nicotine	I	Causes sense of euphoria and increased energy
B	Morphine	II	Stimulates adrenal gland to release catecholamines into blood circulation
C	Herois	III	Effective sedative and painkiller
D	Cocaine	IV	A depressant; slows down body functio

Choose the **correct** answer from the options given below :

Option 1:
A-II, B-III, C-I, D-IV

Option 2:
A-III, B-II, C-IV, D-I

Option 3:
A-III, B-II, C-I, D-IV

Option 4:
A-II, B-III, C-IV, D-I

Solution:

Nicotine → II (stimulates catecholamine release), Morphine → III (sedative), Heroin → IV (depressant), Cocaine → I (euphoria).

Hence, the correct answer is option 4) A-II, B-III, C-IV, D-I

Q. 69 Match List I with List II related to muscular/skeletal system :

List I		List II	
A	Tetany	I	Inflammation of joints
B	Arthritis	II	Autoimmune disorder affecting neuromuscular junction
C	Myasthenia gravis	III	Wild contraction in muscle due to low Ca^{++} in body fluid
D	Muscular dystrophy	IV	Progressive degeneration of skeletal muscle

Choose the **correct** answer from the options given below:

Option 1:

A-IV, B-III, C-II, D-I

Option 2:

A-III, B-I, C-II, D-IV

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-III, B-II, C-I, D-IV

Correct Answer:

A-III, B-I, C-II, D-IV

Solution:

Tetany is due to low Ca^{2+} causing spasms (III), arthritis causes inflammation of joints (I), myasthenia gravis is an autoimmune disease and affects the muscles (II), and muscular dystrophy shows degeneration (IV).

Hence, the correct answer is option 2) A-III, B-I, C-II, D-IV

- B. Diffusion of O_2 and CO_2 across alveolar membrane *m*
- C. Pulmonary ventilation by which atmospheric air is drawn in and CO_2 rich alveolar air is released out
- D. Cellular respiration
- E. Transport of gases by the blood

Choose the **correct** answer from the options given below:

Option 1:

A, B, C, D, E

Option 2:

C, A, B, E, D

Option 3:

C, B, E, A, D

Option 4:

E, A, C, D, B

Correct Answer:

C, B, E, A, D

Solution:

Respiration is the process of obtaining energy from the breakdown of food molecules. It usually starts with pulmonary ventilation (C), followed by diffusion across alveoli (B), transport via blood (E), diffusion into tissues (A), and finally cellular respiration (D).

Hence, the correct answer is option 3) C, B, E, A, D

Q. 71 Arrange the following cell layers/structures around the female gamete, from outer to inner side:

- A. Zona pellucida
- B. Perivitelline space
- C. Corona radiata
- D. Plasma membrane of ovum

Choose the **correct** answer from the options given below:

Option 1:

D, B, A, C

Option 3:

C, A, D, B

Option 4:

C, A, B, D

Correct Answer:

C, A, B, D

Solution:

From outer to inner: corona radiata → zona pellucida → perivitelline space → plasma membrane of ovum.

- Corona radiata - the outermost layer of the ovum
- Zona pellucida - protective glycoprotein layer.
- Perivitelline space - is the plasma membrane of the ovum

Hence, the correct answer is option 4) C, A, B, D

Q. 72 The human protein named α -1-antitrypsin, obtained from transgenic animals, is used for the treatment of ____

Option 1:

Alzheimer's disease

Option 2:

Rheumatoid arthritis

Option 3:

Emphysema

Option 4:

Cystic fibrosis

Correct Answer:

Emphysema

Solution:

α -1-antitrypsin deficiency leads to lung damage due to unchecked protease activity, causing emphysema. Transgenic animals produce this protein for therapy. Alpha antitrypsin is a human protein that is produced by using transgenic animals. It is used to treat emphysema. Emphysema is a lung disease caused by a deficiency of this protein. Alpha antitrypsin protects lung tissues from damage by certain enzymes.

Select the correct statements regarding cell membrane in eukaryotic cell.

- A. Membrane of human , RBCs has approximately 52% protein.
- B. Major phospholipids are arranged in a bilayer.
- C. Extensions of the plasma membrane into the cell form mesosomes.
- D. Tails towards the inner part of lipids are hydrophobic and - thus protected from aqueous medium
- E. Glycocalyx is present on the outer surface of the plasma membrane.

Choose the **correct** answer from the options given below :

Option 1:

C, D and E only

Option 2:

B, C and E only

Option 3:

A, B and D only

Option 4:

A, C and E only

Correct Answer:

A, B and D only

Solution:

RBC membrane has 52% protein (A true), phospholipids form a bilayer (B true), hydrophobic(water fearing) tails face inward (D true). Mesosomes are found in prokaryotes (C false), glycocalyx(a carbohydrate-rich covering) is present, but statement combos make (3) correct.

Hence, the correct answer is option 3) A, B, and D only

Q. 74 Male frogs can be distinguished from female frogs due to the presence of :

- A. Bulging eyes
- B. Vocal sacs
- C. Webbed digits in feet
- D. Copulatory pad on first digit of fore limbs
- E. Olive green-coloured skin with dark irregular spots

Choose the **correct** answer from the options given below:

Option 2:

C and E only

Option 3:

B and D only

Option 4:

B and C only

Correct Answer:

B and D only

Solution:

Male frogs possess vocal sacs for croaking and copulatory (nuptial) pads on forelimbs. Other features that are common in both male and female frogs are bulging eyes, webbed feet, and skin color. Therefore, we can not use these features to distinguish the sexes.

Hence, the correct answer is option 3) B and D only

Q. 75 Which of the following equations depicts Verhulst–Pearl logistic population growth?

Option 1:

$$\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$$

Option 2:

$$\frac{dN}{dt} = rN \left(\frac{K}{K-N} \right)$$

Option 3:

$$\frac{dN}{dt} = rN \left(\frac{K-N}{N} \right)$$

Option 4:

$$\frac{dN}{dt} = rN \left(\frac{K+N}{K} \right)$$

Correct Answer:

$$\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$$

Solution:

The logistic growth equation explains that the population grows rapidly at first when resources are provided in abundance.

The logistic growth equation is $\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$, where K is the carrying capacity and growth slows as the population approaches K .

Hence, the correct answer is option 1) $\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$

- A. Hepatic portal system is the special venous connection between liver and intestine. .
- B. There are twelve pairs of cranial nerves arising from the brain.
- C. The ureters and oviducts open separately into the cloaca in female frogs ×
- D. Hind-brain consists of cerebellum, medulla oblongata and optic lobes.
- E. Sinus venosus joins the right atrium of heart.

Choose the **correct** answer from the options given below:

Option 1:

B and D only

Option 2:

A, B and C only

Option 3:

B and C only

Option 4:

A, C and E only

Correct Answer:

A, C and E only

Solution:

Hepatic portal system connects intestine to liver (A true), ureters and oviducts open separately into cloaca (C true), sinus venosus joins right atrium (E true). Frogs have 10 pairs of cranial nerves (B false), and optic lobes belong to the midbrain (D false).

Hence, the correct answer is option 4) A, C, and E only

Q. 77 Select the **incorrect** statements with reference to Rh grouping.

- A. Erythroblastosis foetalis is a condition observed having foetus with Rh^{-ve} blood and mother with Rh^{+ve} blood.
- B. Rh antigen is observed on RBCs in the majority of human beings.
- C. Before blood transfusion, Rh group should also be matched.
- D. Rh incompatibility is observed when a pregnant mother is Rh^{-ve} and the foetus is Rh^{+ve} .
- E. Erythroblastosis foetalis can be avoided by administering anti-Rh antibodies to the mother immediately after the delivery of the second child.

Choose the answer from the options given below :

Option 2:

C and D only

Option 3:

A and E only

Option 4:

B and C only

Correct Answer:

A and E only

Solution:

Erythroblastosis occurs when mother is Rh^- and foetus Rh^+ , not vice versa. Anti-Rh antibodies are given after the first delivery, not the second. These Anti-RH antibodies destroy any RH-positive foetal cells in the blood of the mother before her immune system can react and form permanent antibodies.

Hence, the correct answer is option 3) A and E only

Q. 78 Which of the following statements are correct with reference to human endoskeleton?

- A. Human skull is monocondylic.
- B. The joint between any two adjoining vertebrae is a cartilaginous joint.
- C. In human beings, the number of cervical vertebrae is seven
- D. All ribs except the last 2 pairs are bicephalic.
- E. The occipital bone of skull is articulated with atlas vertebra.

Choose the **correct** answer from the options given below :

Option 1:

C, D and E only

Option 2:

B, C and E only

Option 3:

A, B and D only

Option 4:

B and E only

Solution:

Humans are dicondylic (not monocondylic). Vertebrae joints are cartilaginous, cervical vertebrae = 7, and the occipital bone articulates with the atlas. The joints between the adjacent vertebrae are cartilaginous because of the presence of the intervertebral disc. Humans have seven cervical vertebrae in the neck region.

Hence, the correct answer is option 2) B, C, and E only

Q. 79 Match **List I** with **List II** :

List I		List II	
A	Cortisol	I	Stimulates the formation of alveoli in mammary glands
B	Aldosterone	II	Produces anti-inflammatory reactions
C	Cholecystokinin	III	Stimulates reabsorption of Na^+ and water from renal tubule
D	Progesterone	IV	Stimulates secretion of pancreatic enzymes and bile juice

Choose the **correct** answer from the options given below:

Option 1:

A-II, B-III, C-IV, D-I

Option 2:

A-II, B-III, C-I, D-IV

Option 3:

A-IV, B-II, C-I, D-III

Option 4:

A-III, B-II, C-IV, D-I

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

Cortisol shows anti-inflammatory action (II), aldosterone increases Na^+ reabsorption (III), cholecystokinin stimulates pancreatic enzyme and bile secretion (IV), and progesterone helps in mammary gland development (I).

- A. The parasites reproduce asexually in RBCs, bursting the cells.
- B. The parasites reproduce asexually in liver cells, bursting the cells and releasing into blood.
- C. Gametocytes develop in RBCs.
- D. Sporozoites reach the liver through the blood
- E. Female mosquito injects sporozoites into humans during bite (1)

Choose the **correct** answer from the options given below :

Option 1:

A, B, C, D, E

Option 2:

E, C, D, B, A

Option 3:

E, D, B, A, C

Option 4:

C, A, B, D, E

Correct Answer:

E, D, B, A, C

Solution:

Sequence: mosquito injects sporozoites → reach liver → multiply in liver → infect RBCs → gametocytes form. The Plasmodium sporozoites multiply in the liver cells and then infect RBCs. They reproduce further there and develop into gametocytes.

Hence, the correct answer is option 3) E, D, B, A, C

Q. 81 Select the **incorrect** statements from the following :

- A. Digestive system in Platyhelminthes is incomplete.
- B. Bilateral symmetry is a characteristic feature of adult Echinoderms?
- C. Pseudocoelom is possessed by Aschelminthes.
- D. Notochord is persistent throughout life in the class Chondrichthyes.
- E. Members of class Reptilia maintain a constant body temperature.

Choose the answer from the options given below :

Option 2:

B and E only

Option 3:

B and D only

Option 4:

C and D only

Correct Answer:

B and E only

Solution:

Echinoderms are radially symmetrical (not bilateral), and reptiles are ectothermic (not constant temperature). Others are correct. The adult echinoderms have radial symmetry; during their larval stage, they possess bilateral symmetry. Reptiles, being cold-blooded animals, can not maintain a constant body temperature.

Hence, the correct answer is option 2) B and E only

Q. 82 The specific receptors for neurotransmitters synapse are present on?

Option 1:

Pre-synaptic membrane

Option 2:

Post-synaptic membrane

Option 3:

Myelin sheath

Option 4:

Schwann cell

Correct Answer:

Post-synaptic membrane

Solution:

Hence, the correct answer is option 2) Post-synaptic membrane

Q. 83 Choose the correct statements regarding muscle contraction.

- A. A motor neuron carries a signal sent by the Central Nervous System (CNS) to the sarcolemma of the muscle fibre.
- B. The neural signal generates an action potential which cause the release of Ca^{++} into sarcoplasm.
- C. Increase in Ca^{++} inactivates the actin for breaking cross bridges.
- D. Actin binds to the myosin head to form a cross bridge.
- E. Shortening of sarcomere takes place, by pulling actin filaments towards the centre of 'A' band.

Choose the **correct** answer from the options given below :

Option 1:

A and B only

Option 2:

C and E only

Option 3:

C and D only

Option 4:

A, B, D and E only

Correct Answer:

A, B, D and E only

Solution:

Ca^{2+} activates actin by removing troponin inhibition, not inactivates it. Cross-bridge formation and sliding of actin toward A-band center cause contraction.

Hence, the correct answer is option 4).A, B, D and E only

Q. 84 Which of the following is not an example of convergent evolution?

Option 1:

Fore limbs of whales and bats .

Option 3:

Eyes of octopuses and mammals

Option 4:

Wings of butterflies and birds

Correct Answer:

Fore limbs of whales and bats .

Solution:

Forelimbs of whales and bats are homologous (divergent evolution), not convergent.

Hence, the correct answer is option 1) Fore limbs of whales and bats.

Q. 85 The JGA (Juxta Glomerular Apparatus) is a special sensitive region formed by cellular modifications in _____related to the same nephron.

Option 1:

Proximal convoluted tubule and efferent renal arteriole

Option 2:

Distal convoluted tubule and efferent renal arteriole

Option 3:

Distal convoluted tubule and afferent renal arteriole

Option 4:

Proximal convoluted tubule and afferent renal arteriole

Correct Answer:

Distal convoluted tubule and afferent renal arteriole

Solution:

JGA is formed by modified cells of the distal convoluted tubule (macula densa) and the afferent arteriole, regulating blood pressure and filtration. Together, both the distal convoluted tubule and afferent arteriole regulate blood pressure and glomerular filtration rate.

Hence, the correct answer is option 3) Distal convoluted tubule and afferent renal arteriole

(substrate) (product) (product)

Identify the enzyme class 'E' from the following options:

Option 1:
Isomerases

Option 2:
Ligases

Option 3:
Transferases

Option 4:
Lyases

Correct Answer:
Lyases

Solution:
Lyases catalyze the breaking of bonds without hydrolysis or ATP, often forming double bonds (like $C = C$).

Hence, the correct answer is option 4) Lyases

Q. 87 Match **List I** with **List II** :

List I		List II	
A	Molluscs	I	Pulmonary respiration only
B	Reptiles	II	Branchial respiration
C	Adult amphibians	III	Cellular respiration
D	Amoeba	IV	Pulmonary and Cutaneous respiration amphibians

Choose the **correct** answer from the options given below :

Option 2:

A-II, B-I, C-III, D-IV

Option 3:

A-I, B-II, C-IV, D-III

Option 4:

A-II, B-I, C-IV, D-III

Correct Answer:

A-II, B-I, C-IV, D-III

Solution:

Molluscs → gills (II); Reptiles → lungs only (I); Adult amphibians → lungs + skin (IV); Amoeba → cellular respiration (III).
Hence, the correct answer is option (4).

Q. 88 What is the reason behind production of large holes in 'Swiss Cheese'?

Option 1:

The production of large amount of CO_2 and H_2 by *Trichoderma polysporum*

Option 2:

The production of large amount of CO_2 and H_2 by lactic acid bacteria called *Lactobacillus*

Option 3:

The production of large amount of CO_2 by *Propionibacterium sharmanii*

Option 4:

The production of large amount of CO_2 by *Clostridium butylicum*

Correct Answer:

The production of large amount of CO_2 by *Propionibacterium sharmanii*

Solution:

Large holes are due to CO_2 production by *Propionibacterium sharmanii* during fermentation. This normally happens because the gas can not escape easily from the semi-solid cheese. And hence, creates large holes in the cheese.

Hence, the correct answer is option 3) The production of large amount of CO_2 by *Propionibacterium sharmanii*

LIST I		LIST II	
A	About 65 mya	I	Jawless fish probably evolved
B	About 500 mya	II	The dinosaurs suddenly disappeared from the earth
C	About 350 mya	III	Seaweeds and few plants probably existed
D	About 320 mya	IV	Invertebrates were formed and became active

Choose the **correct** answer from the options given below :

Option 1:

A-III, B-IV, C-I, D-II

Option 2:

A-II, B-IV, C-III, D-I

Option 3:

A-II, B-IV, C-I, D-III

Option 4:

A-I, B-II, C-III, D-IV

Correct Answer:

A-II, B-IV, C-I, D-III

Solution:

About 65 million years ago, the dinosaurs disappeared from the Earth. Around 500 million years ago, the invertebrates were active in the oceans. Jawless fishes probably evolved about 350 million years ago, while seaweeds and a few plants existed somewhere around 320 million years ago.

Hence, the correct answer is option 3) A-II, B-IV, C-I, D-III

- B. Both species benefit in mutualism
- C. Both species benefit in commensalism.
- D. In parasitism, only one species benefits and the other species is harmed.
- E. In amensalism, one species is harmed and the other is unaffected.

Choose the **correct** answer from the options given below:

Option 1:

A and B only

Option 2:

B and E only

Option 3:

B, D and E only

Option 4:

A and D only X

Correct Answer:

B, D and E only

Solution:

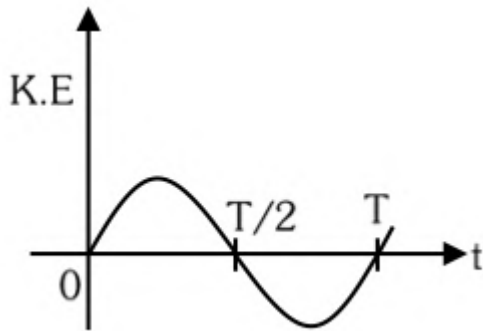
In mutualism, both species benefit (B true). In parasitism, one benefits and the other is harmed (D true). In amensalism, one species is harmed while the other remains unaffected (E true). Statement A is incorrect because in commensalism, the other species is unaffected, not harmed. Statement C is incorrect as only one species benefits in commensalism.

Hence, the correct answer is option 3) B, D, and E only

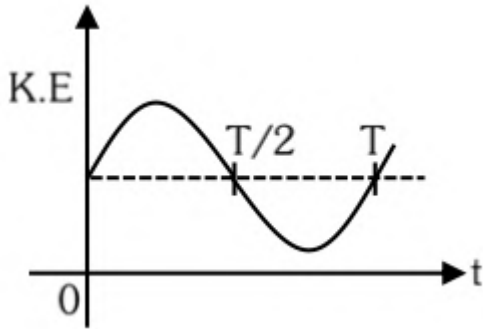
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- Q. 1 For a simple pendulum, having time period T , the variation of kinetic energy (K.E.) with time (t) is represented by :

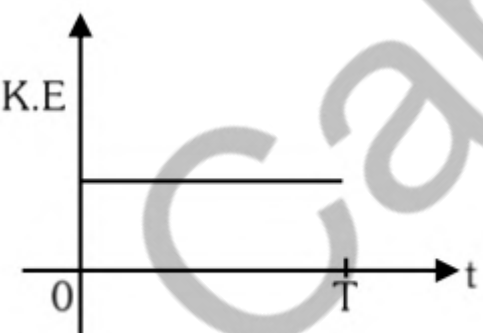
Option 1:



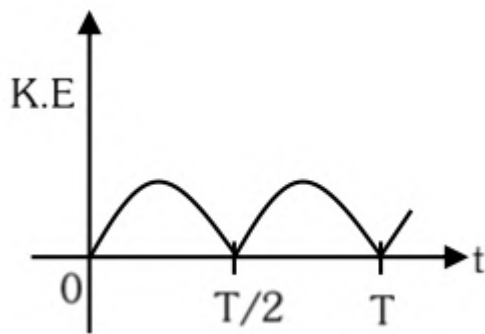
Option 2:



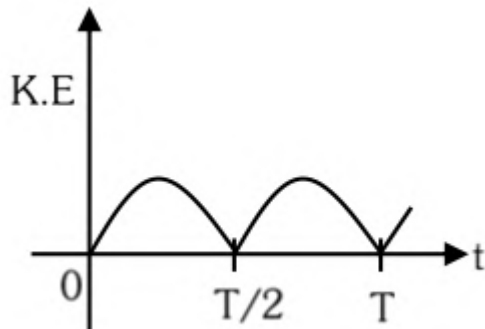
Option 3:



Option 4:



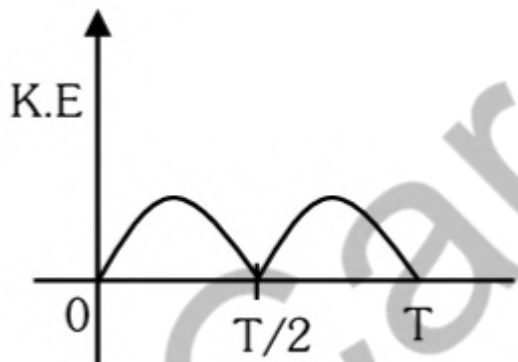
Correct Answer:



Solution:

$$K.E = \frac{1}{2} m \omega^2 A^2 \cos^2 \omega t$$

oscillation starts from extreme position



Hence, the correct answer is option (4).

Q. 2 A room heater is rated 400 W, 220 V. If the supply voltage drops to 200 V, what will be the power consumed (approximately)?

Option 1:

400 W

Option 2:

121 W

Option 3:

331 W

Option 4:

200 W

Correct Answer:

331 W

Solution:

Given that:

Printed voltage = 220 V

Printed Power = 400 W

Supply voltage = 200 V

Resistance of heater $R_H = \frac{(220)^2}{400}$

So Actual power gained

$$\begin{aligned} P' &= \frac{V_s^2}{R_H} = \frac{(200)^2}{(220)^2} (400) \\ &= 330.57 \text{ W} \\ &\simeq 331 \text{ W} \end{aligned}$$

Hence, the correct option is (3) 331 W.

Q. 3 The angular speed of a flywheel is increased from 600 rpm to 1200 rpm in 10 s . The number of revolutions completed by the flywheel during this times is:

Option 1:

600

Option 2:

300

Option 3:

900

Option 4:

150

Correct Answer:

150

Solution:

$$\omega_1 = 600\text{rpm} = 20\pi\text{rad/sec}$$

$$\omega_f = 1200\text{rpm} = 40\pi\text{rad/sec}$$

$$t = 10\text{sec}$$

$$\theta = \left(\frac{\omega_i + \omega_f}{2} \right) t$$

$$\theta = \left(\frac{40\pi + 20\pi}{2} \right) 10$$

$$\theta = 300\pi$$

$$\text{number of revolution} = \frac{300\pi}{2\pi} = 150 \text{ rotation}$$

Hence, the correct option is (4).

Q. 4 The sum of kinetic energy and potential energy of a simple pendulum bob is 0.02 joule. The speed of the simple pendulum bob at equilibrium position is approximately :
(Consider mass of the bob = 20 g)

Option 1:

2.0 m/s

Option 2:

0.2 m/s

Option 3:

14 · 1 m/s

Option 4:

1.41 m/s

Correct Answer:

1.41 m/s

Solution:

$$\text{Total energy} \Rightarrow \frac{1}{2}KA^2 = 0.02 \text{ J}$$

$$\frac{1}{2} m^2 A^2 = 0.02 \text{ J}$$

$$\frac{1}{2} mV_{\max}^2 = 0.02 \text{ J}$$

$$V_{\max}^2 = \frac{0.04}{20 \times 10^{-3}} = 2$$

$$V_{\max} = 1.41 \text{ m/s}$$

Hence, the correct option is (4).

- Q. 5** A 100-turn closely wound circular coil of radius 5 cm has a magnetic field of $3.14 \times 10^{-3} \text{ T}$ at its centre. The current flowing through the coil, and the magnitude of the magnetic moment of this coil are, respectively :
(Take $\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$)

Option 1:
2.5 A, 20 A m²

Option 2:
2 A, 4Am²

Option 3:
2.5 A, 2 A m²

Option 4:
2 A, 10 A m²

Correct Answer:
2.5 A, 2 A m²

Solution:

$$B = \frac{\mu_0 NI}{2R} \Rightarrow I = \frac{2BR}{\mu_0 N}$$

$$\text{Current } I = \frac{2 \times 3.14 \times 10^{-3} \times 5 \times 10^{-2}}{4\pi \times 10^{-7} \times 100}$$

$$\text{Current } I = 2.5 \text{ A}$$

$$\text{Magnetic moment } M = NIA$$

$$= 100 \times 2.5 \times \pi r^2$$

$$= 100 \times 2.5 \times 3.14 \times 25 \times 10^{-4}$$

$$= 1.96$$

$$= 2 \text{ A} \times \text{m}^2$$

Hence, the correct option is (3).

-
- Q. 6** A submarine is designed to withstand an absolute pressure of 100 atm . How deep can it go below the water surface?
(Consider the density of water = 1000 kg m^{-3} . $1 \text{ atm} = 1 \times 10^5 \text{ Pa}$ and gravitational acceleration $g = 10 \text{ m/s}^2$)

Option 1:
9900 m

Option 2:
990 m

Option 3:
9000 m

Option 4:

99 m

Correct Answer:

990 m

Solution:

Absolute pressure at depth h in water

$$P = P_0 + h\rho g \quad P_0 = \text{atmospheric pressure}$$

$$P - P_0 = h\rho g$$

$$h = \frac{P - P_0}{\rho g}$$
$$= \frac{(100 - 1) \times 10^5}{10^3 \times 10}$$

$$h = 990 \text{ m}$$

Hence, the correct option is (2).

Q. 7 Match List I with List II :

List I		List II	
A.	$E = h\nu$	I.	de Broglie wavelength
B.	Diffraction and Interference	II.	Particle nature of light
C.	$\lambda = h/p$	III.	Wave nature of light
D.	Compton effect	IV.	Energy of photon

Choose the **correct** answer from the options given below :

Option 1:

A-IV, B-III, C-II, D-I

Option 2:

A-IV, B-III, C-I, D-II

Option 3:

A-I, B-IV, C-III, D-II

Option 4:

A-IV, B-I, C-II, D-III

Correct Answer:

A-IV, B-III, C-I, D-II

Solution:

- A. $E = h\nu \rightarrow$ Energy of photon $\rightarrow$ IV
 B. Diffraction and Interference $\rightarrow$ Wave nature of light $\rightarrow$ III
 C. $\lambda = \frac{h}{p} \rightarrow$ de Broglie wavelength $\rightarrow$ I
 D. Compton effect $\rightarrow$ Particle nature of light $\rightarrow$ II

Hence, the correct option is (2).

Q. 8 Match **List I** with **List II** :

List I		List II	
A.	Young's Modulus	I.	$\frac{\Delta d}{\Delta L} \left(\frac{L}{d} \right)$
B.	Compressibility	II.	$\frac{FL}{A(\Delta L)}$
C.	Bulk Modulus	III.	$-\frac{1}{\Delta P} \left(\frac{\Delta V}{V} \right)$
D.	Poisson's Ratio	IV.	$-P \left(\frac{V}{\Delta V} \right)$

Choose the **correct** answer from the options given below:

Option 1:

A-III, B-II, C-I, D-IV

Option 2:

A-II, B-III, C-IV, D-I

Option 3:

A-I, B-IV, C-III, D-II

Option 4:

A-IV, B-I, C-II, D-III

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

A. Young's Modulus $Y = \frac{FL}{A(\Delta L)}$

B. Compressibility $\text{Compressibility} = -\frac{1}{\Delta P} \left(\frac{\Delta V}{V} \right)$

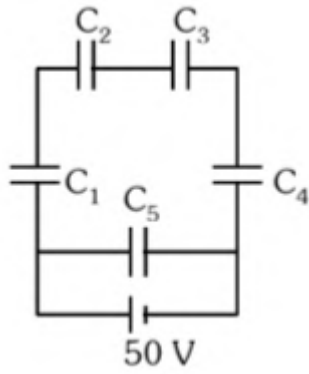
C. Bulk Modulus $K = -P \left(\frac{V}{\Delta V} \right)$

D. Poisson's Ratio

Poisson's ratio $= \frac{\Delta d}{\Delta L} \left(\frac{L}{d} \right)$

Hence, the correct option is (2).

- Q. 9** Five capacitors of capacitances $C_1 = C_2 = C_3 = C_4 = 40\mu\text{ F}$ and $C_5 = 2 \cdot 5\mu\text{ F}$ are connected as shown, along with a battery of 50 V. The equivalent capacitance and the charges on each capacitor respectively are :



Option 1:

$4\mu\text{ F}$, $250\mu\text{C}$ on C_1 to C_4 and $125\mu\text{C}$ on C_5

Option 2:

$5\mu\text{ F}$, $250\mu\text{C}$ on all capacitors

Option 3:

$5\mu\text{ F}$, $125\mu\text{C}$ on C_1 to C_4 and $25\mu\text{C}$ on C_5

Option 4:

$5\mu\text{ F}$, $125\mu\text{C}$ on all capacitors

Correct Answer:

$5\mu\text{ F}$, $125\mu\text{C}$ on all capacitors

Solution:

Top branch: C_2 and C_3 in series

$$\frac{1}{C_{23}} = \frac{1}{10} + \frac{1}{10} \Rightarrow C_{23} = 5\mu\text{F}$$

Now C_1 , C_{23} , C_4 are in series:

$$\frac{1}{C_{eq1}} = \frac{1}{10} + \frac{1}{5} + \frac{1}{10} = \frac{1+2+1}{10} = \frac{4}{10} \Rightarrow C_{eq1} = 2.5\mu F$$

Parallel with C_5

$$C_{total} = C_{eq1} + C_5 = 2.5 + 2.5 = 5\mu F$$

Total charge

$$Q = C_{total} V = 5 \times 50 = 250\mu C$$

Each branch capacitance = $2.5\mu F \rightarrow$ equal charge sharing

$$Q_{\text{each branch}} = 125\mu C$$

Hence, the correct option is (4).

Q. 10 The amount of work done to raise a mass 'm' from the surface of the Earth to a height equal to the radius of the Earth ' R ', will be :

Option 1:
 $mg\frac{R}{2}$

Option 2:
 mgR

Option 3:
 $mg\frac{R}{4}$

Option 4:
 $2 mg R$

Correct Answer:
 $mg\frac{R}{2}$

Solution:
 Change in potential energy:

$$W = U_2 - U_1 = -\frac{GMm}{2R} - \left(-\frac{GMm}{R}\right)$$

$$W = \frac{GMm}{R} - \frac{GMm}{2R} = \frac{GMm}{2R}$$

$$g = \frac{GM}{R^2}$$

So,

$$W = \frac{mgR}{2}$$

Hence, the correct option is (1).

Q. 11 When a ruler falls vertically, 5 different person catch it with different reaction times.
($g = 9.8 \text{ m s}^{-2}$)

- A. Person A has reaction time of 0.20 s .
- B. Person B has reaction time of 0.22 s .
- C. Person C has reaction time of 0.18 s .
- D. Person D has reaction time of 0.19 s .
- E. Person E has reaction time of 0.21 s .

What is the **correct** order of the distance travelled by the ruler for each person?

Option 1:

$$B > E > A > C > D$$

Option 2:

$$C > D > A > E > B$$

Option 3:

$$C > D > A > B > E$$

Option 4:

$$B > E > A > D > C$$

Correct Answer:

$$B > E > A > D > C$$

Solution:

$$\text{Distance fallen: } s = \frac{1}{2}gt^2$$

Distance $\propto t^2$, so compare reaction times:

Largest time $\rightarrow$ largest distance:

Hence, the correct option is (4).

Q. 12 The power of a crane, which lifts a mass of 1000 kg to a height of 20 m in 10 s is :

$$(g = 9.8 \text{ m/s}^2)$$

Option 1:

19.6 kW

Option 2:

19 · 6 W

Option 3:

39 · 2 kW

Option 4:

39 · 2 W

Correct Answer:

19.6 kW

Solution:

$$\begin{aligned}\text{Power of crane} &= \frac{mgh}{t} \\ &= \frac{1000 \times 9.8 \times 20}{10} \\ &= 19.6 \times 1000 \\ &= 19.6 \text{ kW}\end{aligned}$$

Hence, the correct option is (1).

Q. 13 Consider two uncharged capacitors of equal capacitance 200 pF . One of them is charged by a 100 V supply and disconnected. Now this capacitor is connected to the uncharged capacitor. The amount of electrostatic energy lost in the process is :

Option 1:

0.5 J

Option 2:

$1.0 \times 10^{-6} \text{ J}$

Option 3:

$0.5 \times 10^{-6} \text{ J}$

Option 4:

1.0 J

Correct Answer:

$0.5 \times 10^{-6} \text{ J}$

Solution:

Initial energy:

$$U_i = \frac{1}{2} CV^2$$

Final voltage becomes $V/2$ (equal capacitors sharing charge)

Final energy:

$$U_f = \frac{1}{2}(2C)\left(\frac{V}{2}\right)^2 = \frac{1}{4}CV^2$$

Energy lost:

$$\Delta U = U_i - U_f = \frac{1}{2}CV^2 - \frac{1}{4}CV^2 = \frac{1}{4}CV^2$$

$$C = 200\text{pF} = 200 \times 10^{-12}$$

$$\begin{aligned}\Delta U &= \frac{1}{4} \times 200 \times 10^{-12} \times (100)^2 \\ &= 0.5 \times 10^{-6} \text{ J}\end{aligned}$$

Hence, the correct option is (3).

Q. 14 An ac circuit contains a resistance of $1\text{k}\Omega$, a capacitor of $0.1\mu\text{F}$ and an inductor of 1 mH connected in series. The resonance frequency of the circuit is approximately :

Option 1:
15.9 kHz

Option 2:
20.7 kHz

Option 3:
 $10 \cdot 1\text{kHz}$

Option 4:
 $13 \cdot 5\text{kHz}$

Correct Answer:
15.9 kHz

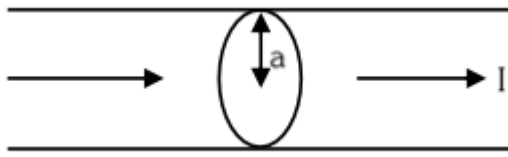
Solution:

For a series LCR circuit, resonance frequency is:

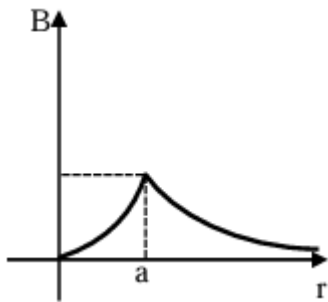
$$\begin{aligned}f &= \frac{1}{2\pi\sqrt{LC}} \\ f &= \frac{1}{2\pi\sqrt{10^{-3} \times 10^{-7}}} = \frac{1}{2\pi\sqrt{10^{-10}}} = \frac{1}{2\pi \times 10^{-5}} \\ f &\approx \frac{1}{6.28 \times 10^{-5}} \approx 1.59 \times 10^4 \text{ Hz} \\ f &\approx 15.9\text{kHz}\end{aligned}$$

Hence, the correct option is (1).

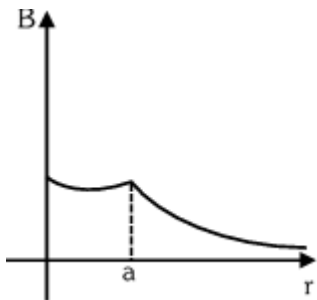
- Q. 15** The figure given below shows a long straight solid wire of circular cross-section of radius 'a' carrying steady current I . The current I is uniformly distributed across its cross-section. The plot which correctly represents the variation of magnetic field (B) with distance (r) from the axis of the conductor in the region is :



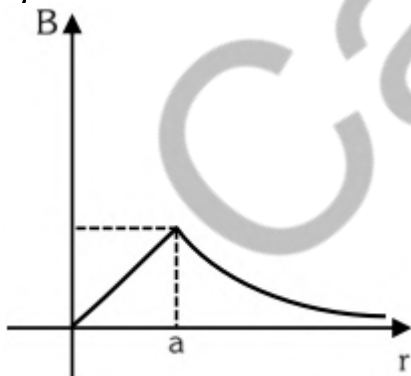
Option 1:



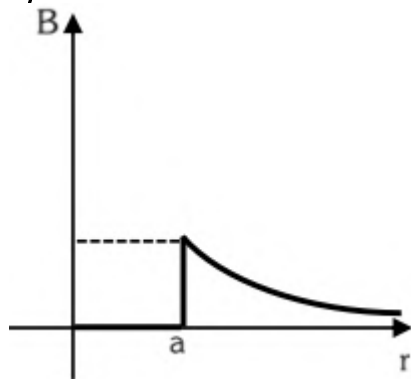
Option 2:



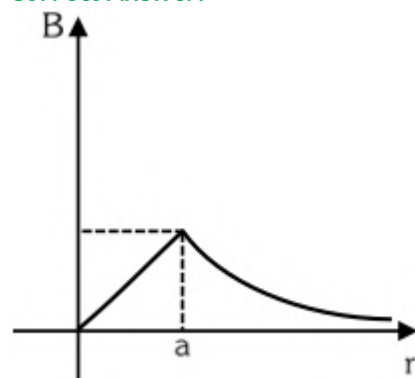
Option 3:



Option 4:



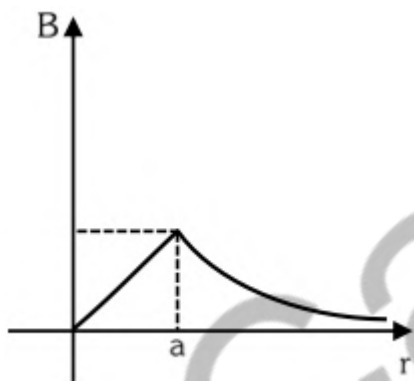
Correct Answer:



Solution:

$$B_{\text{in}} = \frac{\mu_0 I r}{2\pi R^2} \propto r$$

$$B_{\text{out}} = \frac{\mu_0 I}{2\pi r} \propto \frac{1}{r}$$



Hence, the correct option is (3).

Q. 16 An electric heater supplies heat to a system at a rate of 100 W . If the system performs work at a rate of 75 J/s, then the rate at which internal energy increases will be :

Option 1:

75 W

Option 2:

25 W

Option 3:

100 W

Option 4:

125 W

Correct Answer:

25 W

Solution:

Rate of increase of internal energy Using first law of thermodynamics:

$$\frac{dU}{dt} = \text{Heat supplied} - \text{Work done} = 100 - 75 = 25 \text{ W}$$

Hence, the correct option is (2).

Q. 17 The peak value of an alternating current is 5 A and frequency is 60 Hz . How long will the current, starting from zero, take to reach the peak value ?

Option 1:

$\frac{1}{120}$ s

Option 2:

$\frac{1}{240}$ s

Option 3:

$\frac{1}{30}$ s

Option 4:

$\frac{1}{60}$ s

Correct Answer:

$\frac{1}{240}$ s

Solution:

Time to reach peak current

For AC:

$$i = I_0 \sin(\omega t)$$

Peak occurs when $\omega t = \frac{\pi}{2}$

$$t = \frac{\pi/2}{\omega} = \frac{\pi/2}{2\pi f} = \frac{1}{4f}$$

$$t = \frac{1}{4 \times 60} = \frac{1}{240} \text{ s}$$

Hence, the correct option is (2).

Q. 18 In Young's double slit experiment, using monochromatic light of wavelength λ , the intensity of light at a point on the screen where the path difference is λ , is K units. The intensity of light at a point where the path difference is $\frac{\lambda}{3}$ will be :

Option 1:

K

Option 2:

2K

Option 3:

$\frac{K}{2}$

Option 4:

$\frac{K}{4}$

Correct Answer:

$\frac{K}{4}$

Solution:

$$\begin{aligned} I_R &= K \cos^2 \frac{\Delta\phi}{2} \\ &= K \cos^2 \left(\frac{120^\circ}{2} \right) \\ &= K \cos^2 60^\circ = \frac{K}{4} \end{aligned}$$

Hence, the correct option is (4).

Q. 19 Four statements are given (A is mass number) :

A. The volume of a nucleus is proportional to $A^{1/3}$.

B. The volume of a nucleus is proportional to A.

C. The difference in mass of an atom and its nucleus is called the mass defect.

D. The difference in mass of a nucleus and its constituents is called the mass defect.

Choose the **correct** answer from the options given below :

Option 1:

A and D are true, but B and C are false

Option 2:

B and D are true, but A and C are false

Option 3:

B and C are true, but A and D are false

Option 4:

A and C are true, but B and D are false

Correct Answer:

B and D are true, but A and C are false

Solution:

Correct statements

A: Volume $\propto R^3$, and $R \propto A^{1/3} \rightarrow \text{Volume} \propto A \rightarrow \text{false}$

B: Volume $\propto \rightarrow \text{true}$

C: Mass defect = difference between atom and nucleus $\rightarrow \text{false}$

D: Mass defect = difference between nucleus and its nucleons $\rightarrow \text{true}$

Hence, the correct option is (2).

Q. 20 In interference and diffraction, the light energy is redistributed. If it reduces in one region, producing a dark fringe, it increases in another region, producing a bright fringe.

A. As there is no gain or loss of energy, these phenomena are consistent with the principle of conservation of energy.

B. Diffraction and interference are characteristics exhibited only by light waves.

Choose the **correct** answer from the options given below :

Option 1:

A is false, but B is true

Option 2:

A is true, but B is false

Option 3:

A is true and B is also true

Option 4:

Both A and B are false

Correct Answer:

A is true, but B is false

Solution:

Statement A: Energy is redistributed (bright & dark fringes), no net loss $\rightarrow$ consistent with conservation of energy $\rightarrow$ True

Statement B: Interference and diffraction are wave phenomena, but not only for light (also seen in sound, water waves, etc.) $\rightarrow$ False

Hence, the correct option is (2).

Q. 21 A resistor is connected to a battery of 12 V emf and internal resistance 2Ω . If the current in the circuit is 0.6 A, the terminal voltage of the battery is :

Option 1:

12 V

Option 2:

1.2 V

Option 3:

10 V

Option 4:

10.8 V

Correct Answer:

10.8 V

Solution:

Given that:

$$\text{EMF} = 12 \text{ V}$$

$$r = 2\Omega$$

$$I = 0.6 \text{ A}$$

So terminal voltage

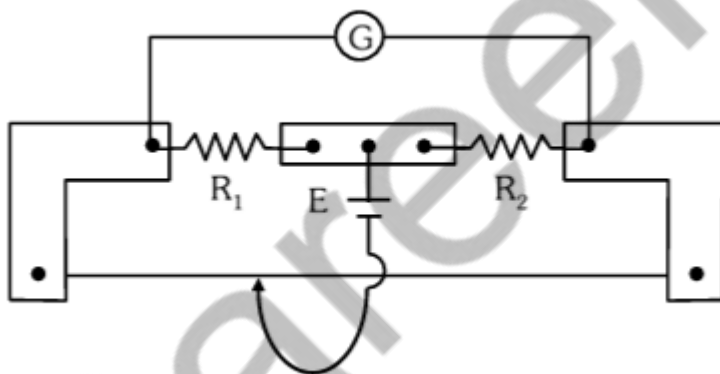
$$V = E - Ir$$

$$= 12 - (.6)^2$$

$$V = 10.8 \text{ volt}$$

Hence, the correct option is (4).

- Q. 22** In a metre bridge experiment (see figure), the positions of the cell, E, and galvanometer, G, are interchanged. We shall observe in the galvanometer :



Option 1:

Only the right-sided deflection

Option 2:

Only the left-sided deflection

Option 3:

There will be no deflection irrespective of the position of the jockey

Option 4:

Both right-sided and left-sided deflection and at balance point, no deflection

Correct Answer:

Both right-sided and left-sided deflection and at balance point, no deflection

Solution:

In the given circuit, it is not mentioned whether the meter bridge is balanced or not. So on interchanging the positions of battery and galvanometer, deflection may or may not be zero. In case of unbalanced condition, deflection may be left or right sided in galvanometer.

Hence, the correct option is (4).

Q. 23 Savitha, a XI standard student, while conducting an experiment to determine the effective length of a simple pendulum L , notes down the data of time taken to complete 30 oscillations as 60 s and hence calculates the length of the simple pendulum as :
(Take $\pi^2 = 9.8$, and $g = 9.8 \text{ m/s}^2$)

Option 1:

0.75 m

Option 2:

1 m

Option 3:

1.5 m

Option 4:

2 m

Correct Answer:

1 m

Solution:

$$T = \frac{\text{Total Time}}{\text{Number of Oscillations}} = \frac{60 \text{ s}}{30} = 2 \text{ s}$$

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$T^2 = 4\pi^2 \left(\frac{L}{g} \right)$$

$$L = \frac{T^2 \cdot g}{4\pi^2}$$

$$L = \frac{2^2 \cdot 9.8}{4 \cdot 9.8} = \frac{4 \cdot 9.8}{4 \cdot 9.8} = 1 \text{ m}$$

Hence, the correct option is (2).

- Q. 24** Which of the following statements are correct?
- A. Inside a conductor, the electrostatic field is zero.
 - B. Electric field at the surface of a charged conductor does not depend on its surface charge density.
 - C. The interior of a charged conductor can have no excess charge in the static situation.
 - D. At the surface of a charged conductor, the electrostatic field must be normal to the surface at every point.
 - E. The electrostatic potential is zero everywhere inside a charged conductor.
- Choose the **correct** answer from the options given below :

Option 1:

A, C and D only

Option 2:

A, C and E only

Option 3:

C, D and E only

Option 4:

A, B and D only

Correct Answer:

A, C and D only

Solution:

- A. Inside a conductor, the electrostatic field is zero. (True)
- B. Electric field at the surface does not depend on surface charge density. (False: $E = \sigma/\epsilon_0$)
- C. The interior of a charged conductor can have no excess charge in the static situation. (True)
- D. At the surface, the electrostatic field must be normal to the surface at every point. (True)
- E. The electrostatic potential is zero everywhere inside a charged conductor. (False)

Analysis: Statements A, C, and D are correct.

Hence, the correct option is (1).

- Q. 25** Two statements are given below :

- A. When the forward bias voltage across a p-n junction diode increases above a certain threshold voltage, the diode current increases significantly.
- B. This current is called reverse saturation current.

Choose the **correct** answer from the options given below :

Option 1:

Both Statements A and B are true

Option 2:

Both Statements A and B are false

Option 3:

Statement *A* is true, but Statement *B* is false

Option 4:

Statement A is false, but Statement B is true

Correct Answer:

Statement **A** is true, but Statement **B** is false

Solution:

Statement A: When forward bias voltage increases above a certain threshold, the diode current increases significantly. (True)

Statement B: This current is called reverse saturation current. (False)

Analysis: Statement A is true, but Statement B is false.

Hence, the correct option is (3).

Q. 26 In a concave lens, a ray of light emanating from the object parallel to the principal axis of the lens, after refraction :

Option 1:

passes through the second principal focus.

Option 2:

appears to diverge from the first principal focus.

Option 3:

passes through $2F$, which is the radius of curvature of the lens.

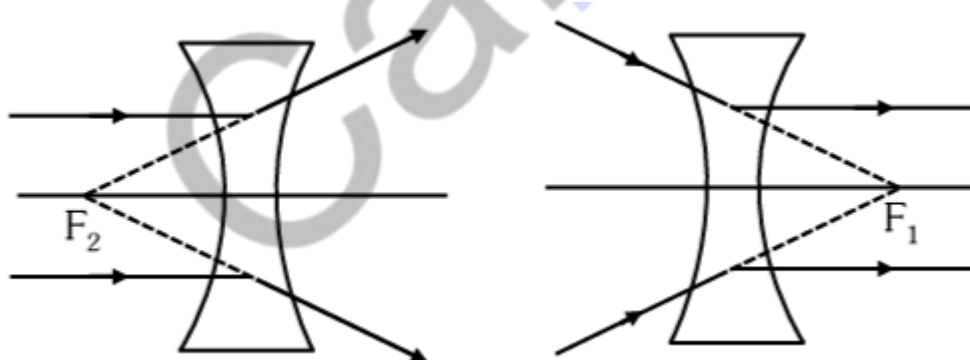
Option 4:

emerges parallel to the principal axis.

Correct Answer:

appears to diverge from the first principal focus.

Solution:



When incident light is parallel to principal axis, after refraction image will be at focus. This focus will be second principal focus (F_2).

In this question as lens is concave (diverging) after refraction light will diverge and appears to be coming from second principal focus (F_2).

Hence, the correct option is (2).

Q. 27 An unknown nucleus has nuclear density $2.29 \times 10^{17} \text{ kg/m}^3$ and mass of $19.926 \times 10^{-27} \text{ kg}$. Its mass number A is approximately :
(Take $R_0 = 1.2 \times 10^{-15} \text{ m}$, $4\pi = 12.56$)

Option 1:
16

Option 2:
20

Option 3:
12

Option 4:
19

Correct Answer:
12

Solution:

$$\text{Density } \rho = \frac{\text{Mass}}{\text{Volume}} = \frac{M}{\frac{4}{3}\pi R^3}$$

$$\text{Nuclear Radius: } R = R_0 A^{1/3}, \text{ so } R^3 = R_0^3 A$$

$$A = \frac{3M}{4\pi R_0^3 \rho} = \frac{3 \times 19.926 \times 10^{-27}}{12.56 \times (1.2 \times 10^{-15})^3 \times 2.29 \times 10^{17}}$$
$$A \approx \frac{59.778 \times 10^{-27}}{12.56 \times 1.728 \times 10^{-45} \times 2.29 \times 10^{17}} \approx 12$$

Hence, the correct option is (3).

Q. 28 A galvanometer of resistance 100Ω gives full scale deflection for a current of 1 mA . It is converted into an ammeter of range $0 - 10 \text{ A}$. The shunt required is :

Option 1:
 0.001Ω

Option 2:
 0.10Ω

Option 3:
 1.0Ω

Option 4:
 0.01Ω

Correct Answer:
 0.01Ω

Solution:

Given that

$$R_g = 100\Omega$$

$$I_g = 1 \text{ mA}$$

$$I = 10 \text{ A}$$

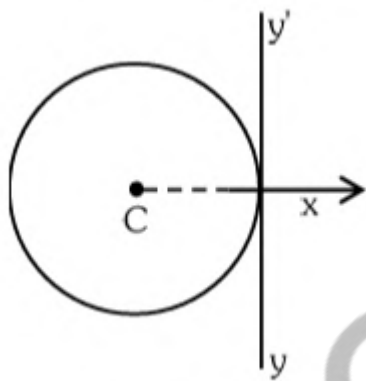
$$S = \frac{I_g R_g}{(I - I_g)} = \frac{10^{-3}(100)}{(10 - 10^{-3})}$$

Since $I_g \ll I$ so $I - I_g \simeq I$

$$S = 10^{-2} = 0.01\Omega$$

Hence, the correct option is (4).

- Q. 29** A thin wire of length ' L ' and linear mass density ' m ' is bent into a circular ring (in $x - y$ plane) with centre ' C ' as shown in figure. The moment of inertia of the ring about an axis yy' will be :



Option 1:

$$\frac{3 mL^3}{8\pi}$$

Option 2:

$$\frac{3 mL^2}{8\pi^2}$$

Option 3:

$$\frac{3 mL^3}{8\pi^2}$$

Option 4:

$$\frac{3 mL^2}{8}$$

Correct Answer:

$$\frac{3 mL^3}{8\pi^2}$$

Solution:

Let mass of wire is M
length = L
given m = Linear mass density

$$m = \frac{M}{L} \quad L = 2\pi R$$

$$M = mL \quad R = \frac{L}{2\pi}$$

$$I_{yy'} = \frac{MR^2}{2} + MR^2$$
$$= \frac{3}{2}MR^2 = \frac{3}{2}(mL) \left(\frac{L^2}{4\pi^2} \right)$$

$$I_{yy'} = \frac{3}{8} \frac{mL^3}{\pi^2}$$

Hence, the correct option is (3).

Q. 30 For a travelling harmonic wave $y(x, t) = 2.0 \cos 2\pi(10t - 0.0080x + 0.35)$, where x and y are in cm and t in s . The phase difference between oscillatory motion of two points separated by a distance of 0.5m is :

Option 1:
 $0.08\pi\text{rad}$

Option 2:
 $0.008\pi\text{rad}$

Option 3:
 $0.8\pi\text{rad}$

Option 4:
 $8\pi\text{rad}$

Correct Answer:
 $0.8\pi\text{rad}$

Solution:

Wave Number (k) : From the equation, $k = 2\pi \times 0.0080$

Phase Difference: $\Delta\phi = k\Delta x = (2\pi \times 0.0080) \times 0.5 = 0.008\pi\text{rad}$

Hence, the correct option is (3).

Q. 31 A box of mass 15 kg is kept on the floor of a stationary trolley. The coefficient of static friction between the box and the trolley is 0.12 . Keeping the box in stationary state over the trolley, the maximum acceleration with which the trolley can be moved horizontally in ms^{-2} is :

$$(g = 10 \text{ m/s}^2)$$

Option 1:
 1.8

Option 2:

1.2

Option 3:

1.5

Option 4:

2.1

Correct Answer:

1.2

Solution:

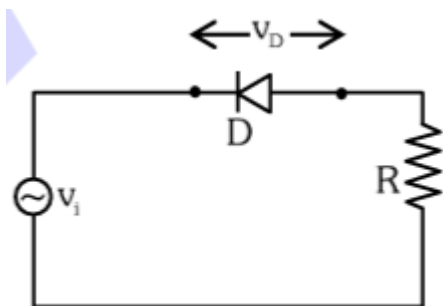
$$ma_{\max} = F_{lm}$$

$$ma_{\max} = \mu mg$$

$$a_{\max} = \mu g = 1.2 \text{ m/s}^2$$

Hence, the correct option is (2).

Q. 32 In the circuit shown below, the voltage appearing across the diode D will be of the form :



Option 1:

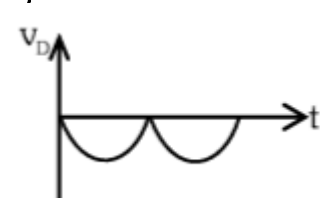


Option 2:



v_D

Option 3:



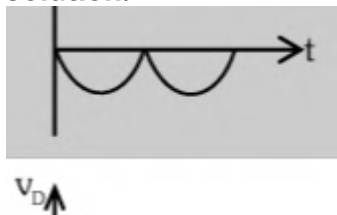
Option 4:



Correct Answer:



Solution:



Logic: In forward bias (positive half cycle), the diode conducts, and voltage across it is nearly 0 V . In reverse bias (negative half cycle), the diode does not conduct, and the entire input voltage appears across the diode.

Graph: The output V_D will show the negative half cycles.

Hence, the correct option is (4).

- Q. 33** A flask contains argon and chlorine in the ratio of 2 : 1 by mass. The temperature of the mixture is 27°C . The ratio of root mean square speed of the molecules of the two gases $\left(\frac{v_{\text{rms}}^{\text{Ar}}}{v_{\text{rms}}^{\text{Cl}}}\right)$ is :
(Atomic mass of argon = 40.0u and molecular mass of chlorine 70.0u)

Option 1:

$$\frac{7}{4}$$

Option 2:

$$\frac{-2}{\sqrt{7}}$$

Option 3:

$$\frac{\sqrt{7}}{2}$$

Option 4:

$$\frac{7}{2}$$

Correct Answer:

$$\frac{\sqrt{7}}{2}$$

Solution:

$$\text{As } V_{\text{RMS}} = \sqrt{\frac{3RT}{M_w}}$$

Given temperature $\rightarrow$ constant

$$\text{Then } V_{\text{RMS}} \propto \frac{1}{\sqrt{M_w}}$$

$$\Rightarrow \frac{V_{\text{RMS}}^{\text{Ar}}}{V_{\text{RMS}}^{\text{C}}} = \sqrt{\frac{M_{wQ}}{M_{wAr}}} = \sqrt{\frac{70}{40}} = \frac{\sqrt{7}}{2}$$

Hence, the correct option (3)

Q. 34 Match List I with List II :

List I (Electromagnetic wave)		List II (Production)	
A.	Microwave	I.	Electrons in atoms emit light when they move from a higher energy level to a lower energy level
B.	Visible light	II.	Radioactive decay of nucleus
C.	Gamma rays	III.	Vibration of atoms and molecules
D.	Infra-red rays	IV.	Klystron valve or magnetron valve

Choose the **correct** answer from the options given below:

Option 1:

A-IV, B-III, C-II, D-I

Option 2:

A-III, B-IV, C-I, D-IX

Option 3:

A-III, B-I, C-II, D-IV

Option 4:

A-IV, B-I, C-II, D-III

Correct Answer:

A-IV, B-I, C-II, D-III

Solution:

A. Microwave: Produced by Klystron valve or magnetron valve

B. Visible light: Electrons in atoms emit light moving to lower energy levels

C. Gamma rays: Radioactive decay of nucleus

D. Infra-red rays: Vibration of atoms and molecules

Hence, the correct option is (4).

Q. 35 The magnitude and direction of the acceleration produced in a body of mass 5 kg when two mutually perpendicular forces 8 N and 6 N act on it, are respectively

Option 1:

2 m s^{-2} ; $\tan^{-1}(3/4)$ with 8 N force

Option 2:

2 m s^{-2} ; $\tan^{-1} 2/3$ with 8 N force

Option 3:

2 m s^{-2} ; $\tan^{-1}(3/4)$ with 6 N force

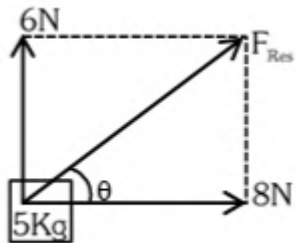
Option 4:

20 m s^{-2} ; $\tan^{-1}(4/3)$ with 8 N force

Correct Answer:

2 m s^{-2} ; $\tan^{-1}(3/4)$ with 8 N force

Solution:



$$F_{\text{Res}} = \sqrt{6^2 + 8^2}$$

$$F_{\text{Res}} = 10 \text{ N}$$

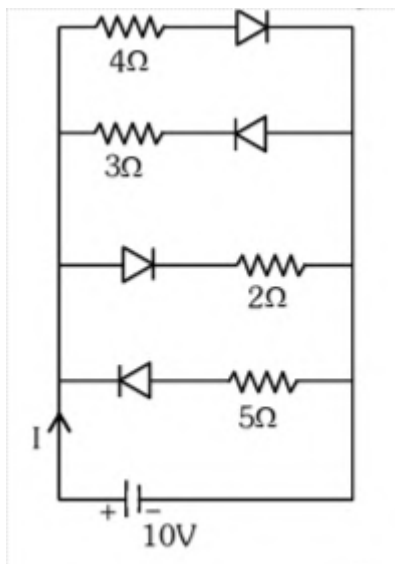
$$a = \frac{F_{\text{Res}}}{m} = \frac{10}{5} = 2 \text{ m/s}^2$$

$$\tan \theta = \frac{6}{8} = \frac{3}{4}$$

$$a = 2 \text{ m/s}^2 \tan^{-1} \left(\frac{3}{4} \right) \text{ with 8 N force}$$

Hence, the correct the option is (1).

Q. 36 The current I in the circuit shown below is :
(All diodes are ideal and identical)



Option 1:

$$\frac{1}{3} \text{ A}$$

Option 2:

$$\frac{15}{2} \text{ A}$$

Option 3:

$$\frac{5}{3} \text{ A}$$

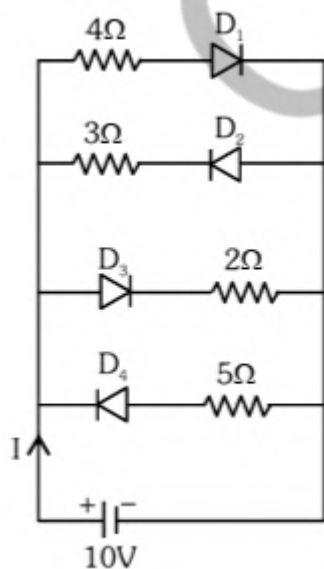
Option 4:

$$\frac{5}{9} \text{ A}$$

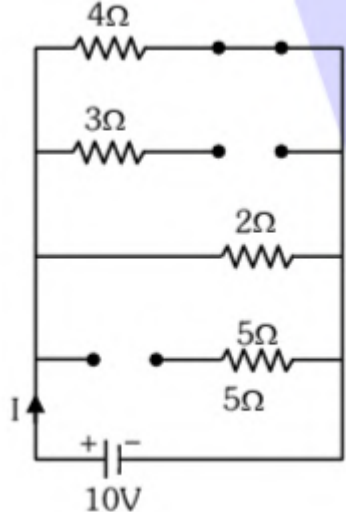
Correct Answer:

$$\frac{15}{2} \text{ A}$$

Solution:



Here D_1 and D_3 are F.B and D_2 and D_4 are R.B. then equivalent circuit



$$R_{eq} = \frac{4 \times 2}{4 + 2} = \frac{4}{3} \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{10}{4/3} = \frac{15}{2} \text{ A}$$

Hence, the correct option is (2).

- Q. 37** For a metal of work function 6.6 eV, which of the following wavelengths of incident radiation does not give rise to the photoelectric effect?
(Take Planck's constant as $6.6 \times 10^{-34} \text{ J s}$)

Option 1:

200 nm

Option 2:

100 nm

Option 3:

50 nm

Option 4:

150 nm

Correct Answer:

200 nm

Solution:

$$\phi = 6.6 \text{ eV}$$

$$\frac{hc}{\lambda} = 6.6 \text{ eV}$$

$$\lambda_0 = \frac{12400}{6.6}$$

$$\lambda_0 = 1878.7 = 187.87 \text{ nm}$$

For photo electric effect

$$\lambda \leq \lambda_0$$

$\therefore$ The wavelength of radiation does not give rise to photoelectric effect is 200 nm .

Hence, the correct option is (1).

Q. 38 The speed of light in vacuum is taken as unity. If light takes 6 min 40 s to reach the Earth from the Sun, the distance between the Sun and the Earth in new unit is :

Option 1:

$$3. \times 10^8$$

Option 2:

$$500$$

Option 3:

$$3 \times 10^{10}$$

Option 4:

$$400$$

Correct Answer:

$$400$$

Solution:

Distance = speed $\times$ time

$$d = 1 \times 400 = 400 \text{ units}$$

Hence, the correct option is (4).

Q. 39 A rectangular wire loop of sides 8 cm and 3 cm with a small cut, is moving out of a region of uniform magnetic field of magnitude 0.3 T directed normal to the plane of the loop. The emf developed across the cut, if the velocity of the loop is 2 cm s^{-1} , in a direction normal to the shorter side of the loop, will be :

Option 1:

$$1.8 \times 10^{-4} \text{ volt}$$

Option 2:

$$1.3 \times 10^{-4} \text{ volt}$$

Option 3:

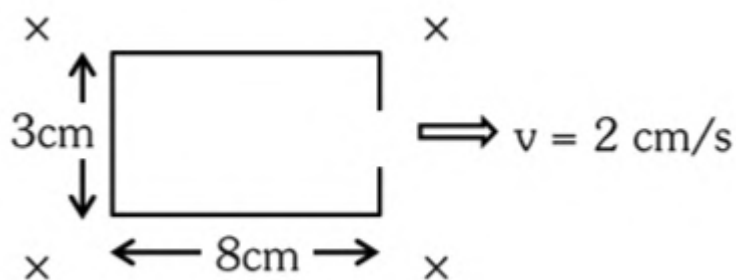
$$1.2 \times 10^{-4} \text{ volt}$$

Option 4:

$$4.8 \times 10^{-4} \text{ volt}$$

Correct Answer:
 1.8×10^{-4} volt

Solution:

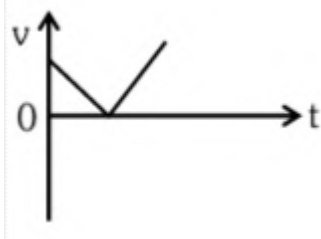


$$e = Blv = 0.3 \times 0.03 \times 0.02 = 1.8 \times 10^{-4} \text{ V}$$

Hence, the correct option is (1).

Q. 40 The following plots show variation of velocity (v) with time (t), of a ball thrown vertically upward, and falling back. Which of the following plots is/are correct ?

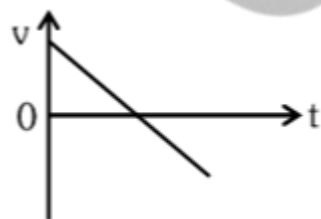
Option 1:



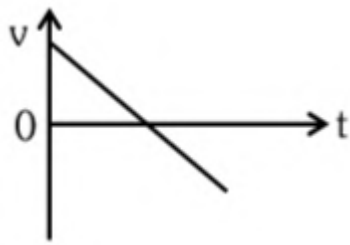
Option 2:



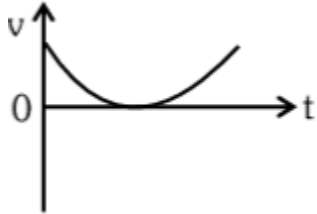
Option 3:



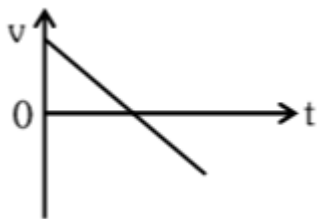
Option 4:



Option 5:



Correct Answer:



Solution:

Physics: Velocity decreases linearly to zero, then becomes negative Graph: Straight line with negative slope crossing zero

Hence, the correct option is (3).

Q. 41 In a vernier callipers, 200 VSD comerde 16 MSD (each division $\bar{p}$ f length 1 mm). The least count of the vernier callipers is :

Option 1:

0.2 cm

Option 2:

0.1 cm

Option 3:

0.02 cm

Option 4:

0.01 cm

Correct Answer:

0.02 cm

Solution:

$$20\text{VSD} = 16\text{MSD}$$

$$\text{VSD} = \frac{16}{20}(1 \text{ mm})$$

$$= 0.8 \text{ mm}$$

$$\text{LC} = \text{MSD} - \text{VSD}$$

$$= 1 - 0.8$$

$$= 0.2 \text{ mm}$$

$$= 0.02 \text{ cm}$$

Hence, the correct option is (3).

Q. 42 Each side of a metallic cube of mass 5.580 kg is measured to be 9.0 cm . Keeping the significant figures in view, the density of the material of the cube can be best expressed as $X \times 10^3 \text{ kg m}^{-3}$, where the value of X is :

Option 1:

7.654

Option 2:

7 . 7

Option 3:

7.65

Option 4:

7 . 6

Correct Answer:

7 . 7

Solution:

$$\text{Density} = \frac{\text{mass}}{\text{volume}} = \frac{5.580 \text{ kg}}{(9.0 \times 10^{-2})^3}$$

$$= 7.654 \times 10^3 \text{ kg/m}^3$$

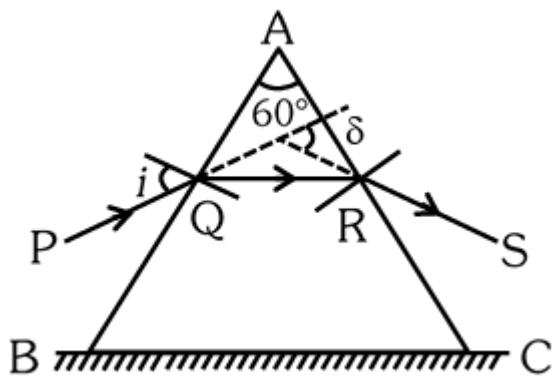
Rounding off to 2 significant digits

$$\text{Result} = 7.7 \times 10^3$$

$$\Rightarrow x = 7.7$$

Hence, the correct option is (2).

- Q. 43** A ray of monochromatic light is passing through an equilateral prism (ABC) as shown in the figure. The refracted ray (QR) is parallel to its base (BC) and the angle of incidence (i) is 50° . Then the angle of deviation (δ) is :



Option 1:

45°

Option 2:

55°

Option 3:

35°

Option 4:

40°

Correct Answer:

40°

Solution:
For an equilateral prism, $A = 60^\circ$ since refracted ray is parallel to base so it is condition of minimum deviation.

$$r = \frac{A}{2} = \frac{60}{2} = 30^\circ$$

$$\text{and } i = e = 50^\circ$$

$$\delta = i + e - A$$

$$\begin{aligned} \text{But } &= 50 + 50 - 60 \\ &= 40^\circ \end{aligned}$$

Hence, the correct option is (4).

- Q. 44** In the first excited state of hydrogen atom, the energy of its electron is -3.4 eV . The radial distance of the electron from the hydrogen nucleus in this case is approximately :

(Take $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$, $e = 1.6 \times 10^{-19} \text{ C}$ and $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$)

Option 1:

$$2.1 \times 10^{-8} \text{ m}$$

Option 2:

$$2.1 \times 10^{-10} \text{ m}$$

Option 3:

$$2.1 \times 10^{-11} \text{ m}$$

Option 4:

$$2.1 \times 10^{-9} \text{ m}$$

Correct Answer:

$$2.1 \times 10^{-10} \text{ m}$$

Solution:

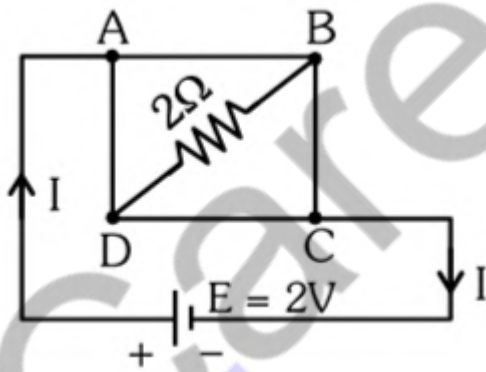
$$E = \frac{kq_1q_2}{2r}$$

$$3.4 = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{2 \times r}$$

$$r = 2.1 \times 10^{-10} \text{ m}$$

Hence, the correct option is (2).

- Q. 45** A uniform metallic wire having resistance 4Ω is bent to form a square loop (ABCD) (see figure). A resistance of 2Ω is connected between points B and D and a battery of 2 V is connected across points A and C as shown in the figure. Now the value of current (I) is :



Option 1:

$$4\text{A}$$

Option 2:

$$4.5\text{A}$$

Option 3:

$$8\text{A}$$

Option 4:

$$2\text{A}$$

Correct Answer:

2A

Solution:

Resistance of each wire of

$$W.S.B. = \frac{4}{4} = 1\Omega$$

This is balance W.S.B.

$$\text{So, } R_{\text{Net}} = 1\Omega$$

So, current

$$I = \frac{E}{R_{\text{Net}}} = \frac{2}{1} = 2 \text{ A}$$

Hence, the correct option is (4).

Chemistry

Q. 1 Match List I with List II

List - I (Complex/ion)	List -II (Shape/geometry)
A. $[\text{Pt}(\text{Cl}_2)(\text{NH}_3)_2]$	I. Octahedral
B. $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$	II. Trigonal bipyramidal
C. $[\text{NiCl}_4]^{2-}$	III. Square planar
D. $[\text{Fe}(\text{CO})_5]$	IV. Tetrahedral

Choose the **correct** answer from the options given below:

Option 1:

A-I, B-III, C-IV, D-II

Option 2:

A-III, B-IV, C-I, D-IX

Option 3:

A-III, B-I, C-IV, D-IV-

Option 4:

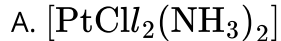
A-IV, B-I, C-III, D-IX

Correct Answer:

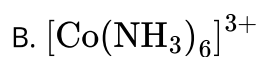
A-III, B-I, C-IV, D-IV-

Solution:

(3) A-III, B-I, C-IV, D-II



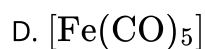
Pt^{2+} is a d^8 system, which typically forms square planar complexes $\rightarrow$ III



Co^{3+} with 6 ligands gives coordination number 6, so octahedral $\rightarrow$ I



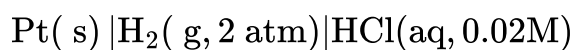
Ni^{2+} with weak field ligand Cl^- forms a tetrahedral complex $\rightarrow$ IV



This is a classic trigonal bipyramidal structure $\rightarrow$ II

Hence, the correct answer is option (3)

Q. 2 Calculate emf of the half cell given below :



$$E_{\text{H}_2/\text{H}^+}^\circ = 0 \text{ V}$$

$$(\text{Given : } \frac{2.303RT}{F} 0.059,$$

$$\log 2 = 0.3010)$$

Option 1:

$$-0.109 \text{ V}$$

Option 2:

$$0.109 \text{ V}$$

Option 3:

$$0.035 \text{ V}$$

Option 4:

$$-0.035 \text{ V}$$

Correct Answer:

$$0.109 \text{ V}$$

Solution:



$$E_{\text{ox}} = E_{\text{ox}}^{\circ} - \frac{0.059}{n} \log Q$$

$$E_{\text{ox}} = 0 - \frac{0.059}{2} \log \left(\frac{(0.02)^2}{2} \right)$$

$$E_{\text{ox}} = -0.0295 \log \left(\frac{0.0004}{2} \right)$$

$$E_{\text{ox}} = -0.0295 \log(0.0002)$$

$$\log(2 \times 10^{-4}) = \log 2 + \log 10^{-4}$$

$$\log(2 \times 10^{-4}) = 0.3010 - 4 = -3.699$$

$$E_{\text{ox}} = -0.0295 \times (-3.699)$$

$$E_{\text{ox}} \approx +0.109 \text{ V}$$

Hence, the correct answer is option (2)

Q. 3 At 298 K, a certain buffer solution contains equal concentrations of X^- and HX , K_b for X^- is 10^{-10} . What is the pH of this buffer solution?

Option 1:

10

Option 2:

4

Option 3:

2

Option 4:

6

Correct Answer:

4

Solution:

For a basic buffer, we use:

$$\text{pOH} = \text{p}K_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

Since concentrations are equal:

$$\log \frac{[\text{salt}]}{[\text{base}]} = \log 1 = 0$$

So,

$$\text{pOH} = \text{p}K_b$$

Given:

$$K_b = 10^{-10} \Rightarrow \text{p}K_b = 10$$

$$\text{pOH} = 10$$

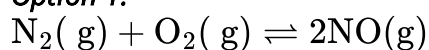
Now:

$$\text{pH} = 14 - \text{pOH} = 14 - 10 = 4$$

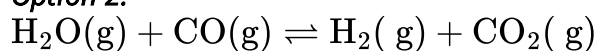
Hence, the correct answer is option (2)

Q. 4 Given below are certain reactions. Identify the reaction for which $K_p \neq K_c$.

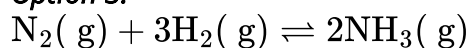
Option 1:



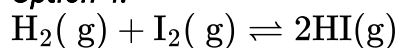
Option 2:



Option 3:



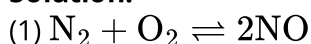
Option 4:



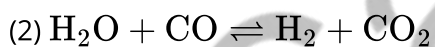
Correct Answer:



Solution:



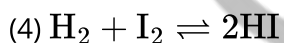
$$\Delta n = 2 - (1 + 1) = 0 \rightarrow K_p = K_c$$



$$\Delta n = 2 - 2 = 0 \rightarrow K_p = K_c$$



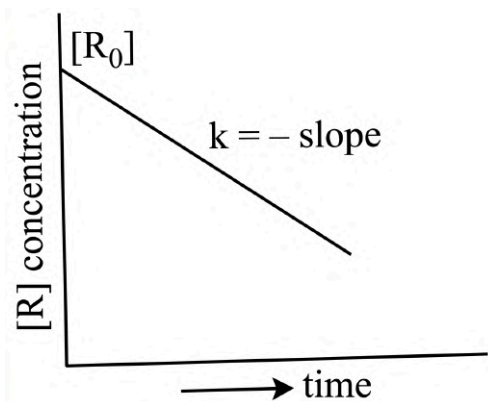
$$\Delta n = 2 - 4 = -2 \rightarrow K_p \neq K_c$$



$$\Delta n = 2 - 2 = 0 \rightarrow K_p = K_c$$

Hence, the correct answer is option (3)

- Q. 5** For a certain reaction $R \rightarrow \text{Product}$, the plot of concentration $[R]$ vs time has a negative slope as shown. The order of reaction is :



Option 1:

1

Option 2:

2.5

Option 3:

2

Option 4:

0

Correct Answer:

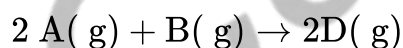
0

Solution:

Zero order: $[R] = [R_0] - kt \rightarrow \text{straight line (slope } = -k)$

Hence, the correct answer is option (4)

- Q. 6** Consider the following reaction :



$$\Delta U^\ominus = -10 \text{ kJ mol}^{-1} \text{ and } \Delta S^\ominus = -44 \text{ J K}^{-1} \text{ at } 298 \text{ K} .$$

Identify the correct option with $\Delta G^\ominus$ for the reaction and spontaneity of the reaction at 298 K .

(Given : $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$)

Option 1:

$+0.63568 \text{ kJmol}^{-1}$, non-spontaneous

Option 2:

$-0.63568 \text{ kJ mol}^{-1}$, spontaneous

Option 3:

$-1.635 \text{ kJ mol}^{-1}$ spontaneous

Option 4:

+1.635 kJ mol⁻¹, non-spontaneous

Correct Answer:

+0.63568 kJmol⁻¹, non-spontaneous

Solution:

$$\Delta S^\circ = -44 \text{ J} = -0.044 \text{ kJ K}^{-1}$$

$$\Delta H = \Delta U + \Delta n_g RT$$

$$\Delta n_g = 2 - 3 = -1$$

$$\Delta H = -10 + (-1)(8.31 \times 298/1000)$$

$$= -10 - 2.476 \approx -12.476 \text{ kJ}$$

$$\Delta G = \Delta H - T\Delta S$$

$$= -12.476 - (298 \times -0.044)$$

$$= -12.476 + 13.112$$

$$= +0.636 \text{ kJ mol}^{-1}$$

$$\Delta G > 0 \Rightarrow \text{N-spontaneous}$$

Hence, the correct answer is option (1)

Q. 7 Given below is an expression for the rate constant of a first order reaction occurring at a certain temperature, T(K).

$$\ln k = 14.34 - \frac{1.25 \times 10^4}{T}$$

The energy of activation in kcalmol⁻¹ for the reaction is :
(Given : k in s⁻¹, R = 1.987 calmol⁻¹ K⁻¹)

Option 1:

12.42

Option 2:

18.63

Option 3:

14.34

Option 4:

24.84

Correct Answer:

24.84

Solution:

$$52. \ln k = \ln A - \frac{E_a}{RT}$$

$$\frac{E_a}{R} = 1.25 \times 10^4$$

$$E_a = (1.25 \times 10^4) \times R$$

$$R = 1.987 \text{ cal mol}^{-1} \text{ K}^{-1}$$

$$E_a = 12500 \times 1.987 = 24837.5 \text{ cal mol}^{-1}$$

$$E_a = \frac{24837.5}{1000} \approx 24.84 \text{ kcal mol}^{-1}$$

Hence, the correct answer is option (4)

Q. 8 Select the reagents that reduce nitriles to primary amines:

- A. (i) LiAlH_4 ;
(ii) H_2O
- B. $\text{Sn} + \text{HCl}$
- C. H_2/Ni
- D. $\text{Na}(\text{Hg})/\text{C}_2\text{H}_5\text{OH}$
- E. $\text{Br}_2/\text{aq. NaOH}$

Choose the correct answer from the options given below

Option 1:

A, B and C only

Option 2:

A, D and E only

Option 3:

A, C and D only

Option 4:

B, D and E only

Correct Answer:

A, C and D only

Solution:

A. $\text{LiAlH}_4/\text{H}_2\text{O}$

Strong reducing agent $\rightarrow$ converts nitrile $\rightarrow$ primary amine

B. $\text{Sn} + \text{HCl}$

Used for nitro $\rightarrow$ amine, not for nitriles $\rightarrow$ Incorrect

C. H_2/Ni

Catalytic hydrogenation $\rightarrow$ nitrile $\rightarrow$ primary amine

D. $\text{Na}(\text{Hg})/\text{C}_2\text{H}_5\text{OH}$

Mild reduction $\rightarrow$ can reduce nitriles $\rightarrow$ primary amine

E. $\text{Br}_2/\text{aq. NaOH}$

Hoffmann bromamide reaction (amide $\rightarrow$ amine with one less carbo

Hence, the correct answer is option (3)

Q. 9 The correct statement with regard to the secondary structure of DNA/RNA is :

Option 1:

DNA possesses a double strand helix structure and contains thymine as one of the four bases.

Option 2:

DNA possesses a single strand helix structure and contains uracil as one of the four bases.

Option 3:

RNA possesses a double strand helix structure and contains uracil as one of the four bases.

Option 4:

RNA possesses a single strand helix structure and contains thymine as one of the four bases.

Correct Answer:

DNA possesses a double strand helix structure and contains thymine as one of the four bases.

Solution:

DNA is **double-stranded (double helix)**

Bases: A, T, G, C $\rightarrow$ **thymine present**

Hence, the correct answer is option (1)

Q. 10 During Lassaigne's test, the elements present in an organic compound are converted from :

Option 1:

ionic form to ionic form

Option 2:

covalent form to ionic form

Option 3:

ionic form to covalent form

Option 4:

covalent form to covalent form

Correct Answer:

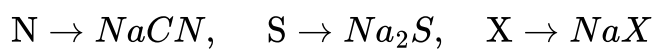
covalent form to ionic form

Solution:

In organic compounds, elements like nitrogen, sulphur and halogens are present in covalent form, so they are not easily detectable by ordinary chemical tests.

During Lassaigne's test, the organic compound is fused with sodium metal. Sodium is highly reactive and converts these covalently bonded elements into ionic (water-soluble) compounds, which can be easily detected.

Examples:



Thus, in Lassaigne's test, elements are converted from:

Covalent form $\rightarrow$ Ionic form

Hence, the correct answer is option (2)

Q. 11 Mixture of chloroform and acetone forms a solution with negative deviation from Raoult's law due to :

Option 1:

stronger intermolecular forces between chloroform molecules than those between chloroform and acetone molecules.

Option 2:

formation of hydrogen bonding between acetone and chloroform.

Option 3:

repulsive forces.

Option 4:

increase in escaping tendency of molecules of each component.

Correct Answer:

formation of hydrogen bonding between acetone and chloroform.

Solution:

- Chloroform (CHCl_3) has a slightly acidic hydrogen
- Acetone (CH_3COCH_3) has a lone pair on oxygen

This leads to hydrogen bonding between chloroform and acetone molecules which results:

- Strong A-B interactions $\rightarrow$ molecules are held more tightly
- Vapour pressure decreases $\rightarrow$ negative deviation

Hydrogen bonding between chloroform and acetone

Hence, the correct answer is option (2)

Q. 12 In a test tube containing a salt, a few drops of dilute H_2SO_4 was added, which gave colourless vapours having the smell of vinegar. The vapours turned the blue litmus paper red. Identify the correct anion from the following :

Option 1:

Carbonate, CO_3^{2-}

Option 2:

Sulphide, S^{2-}

Option 3:

Acetate, CH_3COO^-

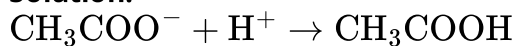
Option 4:

Sulphate, SO_4^{2-}

Correct Answer:

Acetate, CH_3COO^-

Solution:



Acetic acid vapours:

- Are **colourless**
- Have **vinegar smell**
- Turn **blue litmus red (acidic nature)**

Hence, the correct answer is option (3)

Q. 13 Identify the correct statement about ClF_3 from the following options :

Option 1:

It has T-shaped geometry with three lone pairs on Cl atom.

Option 2:

It has T-shaped geometry with two lone pairs on Cl atom.

Option 3:

It has a trigonal pyramidal geometry with two lone pairs on Cl atom.

Option 4:

It has a planar trigonal geometry with two lone pairs on Cl atom.

Correct Answer:

It has T-shaped geometry with two lone pairs on Cl atom.

Solution:

- Central atom: Cl (7 valence electrons)
- Bonded atoms: 3 F atoms
- Total electron pairs = 5
- Electron geometry $\rightarrow$ Trigonal bipyramidal
- Two lone pairs occupy equatorial positions (to minimise repulsion)
- T-shaped geometry
- Lone pairs = 2
- Bond pairs = 3

It has T-shaped geometry with two lone pairs on the Cl atom

Hence, the correct answer is option (2)

Q. 14 Match List - I with List -II

List- I (Complex)	List- II (Type of isomerism)
A. $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$	I. Optical
B. $[\text{Co}(\text{en})_3]^{3+}$	II. 'Solvate
C. $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$	III. Geometrical
D. $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$	IV. Linkage

Choose the correct answer from the options given below:

Option 1:

A-I, B-III, C-II, D=IV

Option 2:

A-II, B-IV, C-III, D-I

Option 3:

A-III, B-I, C-IV, D-II

Option 4:

A-III, B-I, C-II, D-IV

Correct Answer:

A-III, B-I, C-IV, D-II

Solution:

A. $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

- Square planar complex (Pt^{2+} , d^8)
- Shows cis-trans isomerism

Geometrical isomerism $\rightarrow$ III

B. $[\text{Co}(\text{en})_3]^{3+}$

- Octahedral with bidentate ligand (en)
- Forms non-superimposable mirror images

Optical isomerism $\rightarrow$ I

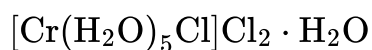
C. $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$

- NO_2^- can bind via N or O

Linkage isomerism $\rightarrow$ IV

D. $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$

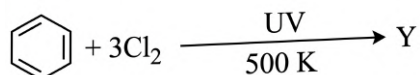
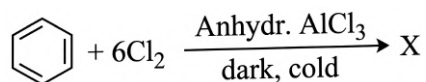
- Can form hydrate/solvate isomers like:



Solvate isomerism $\rightarrow$ II

Hence, the correct answer is option (3)

Q. 15 The number of chlorine atoms present in the organic products X and Y of the following reactions, respectively, are :



Option 1:

3 and 3

Option 2:

6 and 6

Option 3:

6 and 3

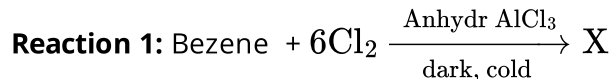
Option 4:

3 and 6

Correct Answer:

6 and 6

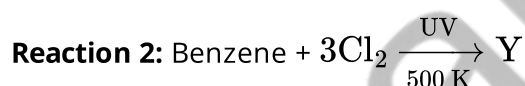
Solution:



Conditions: Lewis acid (AlCl_3) + dark $\rightarrow$ Electrophilic substitution

Benzene undergoes complete chlorination $\rightarrow$ Hexachlorobenzene (C_6Cl_6)

Number of Cl atoms in **X** = **6**



High temperature ($\sim 500 \text{ K}$) $\rightarrow$ Addition reaction

Forms Benzene hexachloride ($\text{BHC}/\text{C}_6\text{H}_6\text{Cl}_6$)

Number of Cl atoms in **Y** = 6

Hence, the correct answer is option (2)

Q. 16 Identify the incorrect statement from the following:

Option 1:

Phosphorus, arsenic and antimony show catenation property.

Option 2:

$\text{P}(\text{C}_2\text{H}_5)_3$ and $\text{As}(\text{C}_6\text{H}_5)_3$ form $d\pi - d\pi$ bond with transition metals.

Option 3:

Nitrogen can form $d\pi$ - $p\pi$ bond with oxygen.

Option 4:

Nitrogen can form $p\pi - p\pi$ multiple bonds with itself.

Correct Answer:

Nitrogen can form $d\pi$ - $p\pi$ bond with oxygen.

Solution:

(1) Correct: Catenation (the linking of atoms of the same element into chains) is a property shown by Phosphorus, Arsenic, and Antimony. Phosphorus shows it most significantly (as seen in P_4 or red phosphorus chains). While Nitrogen's catenation is limited due to high inter-electronic repulsion between lone pairs, the heavier elements do exhibit it.

(2) INCORRECT: Organophosphorus and organoarsenic compounds (like phosphines and arsines) act as ligands with transition metals. They form bonds through π -backbonding, but the specific interaction involves the donation of electrons from the filled d-orbitals of the transition metal into the vacant d-orbitals of Phosphorus or Arsenic. Therefore, the bond is a $d\pi(\text{metal}) - d\pi(\text{ligand})$ interaction. However, the statement is often considered incorrect in standard textbooks because the p -block elements use their vacant d-orbitals to accept electrons, but the primary sigma bond is formed by the lone pair. More importantly, in many competitive contexts, the focus is on the fact that Nitrogen cannot do this because it lacks d-orbitals.

(3) Correct: Nitrogen forms $p\pi - p\pi$ multiple bonds with Oxygen (as seen in oxides like NO_2 or NO_3^-). Since both N and O are in the second period, their p-orbitals are small enough to overlap effectively sideways.

(4) Correct: Nitrogen has a very small atomic size and high electronegativity, allowing it to form very strong $p\pi - p\pi$ multiple bonds with itself, resulting in the stable $\text{N} \equiv \text{N}$ triple bond. Heavier elements like Phosphorus cannot do this effectively because their p-orbitals are too large and diffuse to have effective sideways overlap.

Hence, the correct answer is option (3)

Q. 17 At a certain temperature, $T(\text{K})$, during a process, 500 J is absorbed by the system and work of 200 J is done by the system. Then change in internal energy of the system is :

Option 1:

700 J

Option 2:

400 J

Option 3:

300 J

Option 4:

500 J

Correct Answer:

300 J

Solution:

$$\Delta U = q - w$$

$q = +500J$ (heat absorbed $\rightarrow$ positive)

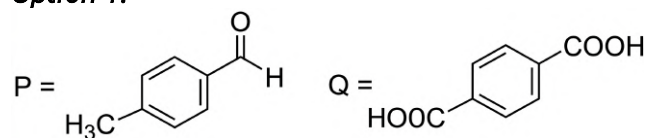
$w = +200J$ (work done by system)

$$\Delta U = 500 - 200 = 300 \text{ J}$$

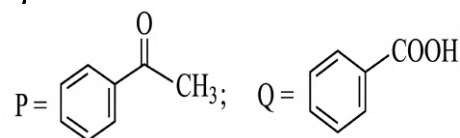
Hence, the correct answer is option (3)

Q. 18 Compound P (C_8H_8O) gives a red orange precipitate with 2,4-DNP reagent and it does not reduce Fehling's reagent. On drastic oxidation with chromic acid, P gives an aromatic product Q that produces effervescence on treating with aq. $NaHCO_3$. Compounds P and Q, respectively, are:

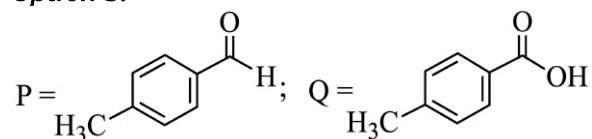
Option 1:



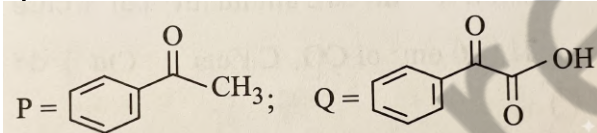
Option 2:



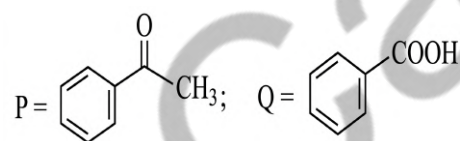
Option 3:



Option 4:



Correct Answer:



Solution:

1. Identify the functional group of P (2,4-DNP Test)

Observation: P gives a red-orange precipitate with 2,4-DNP reagent.

Inference: P contains a carbonyl group ($>C=O$), meaning it is either an aldehyde or a ketone.

2. Narrow down the carbonyl type (Fehling's Test)

Observation: P does not reduce Fehling's reagent.

Inference: P is a ketone (specifically an aromatic ketone here) or an aromatic aldehyde (which also don't reduce Fehling's), but the formula C_8H_8O and the next step point toward a ketone.

3. Analyze Product Q ($NaHCO_3$ Test)

Observation: Drastic oxidation of P with chromic acid yields Q, which produces effervescence with aq. $NaHCO_3$.

Inference: Q is a carboxylic acid (effervescence is due to CO_2 gas release).

4. Final Structure Identification

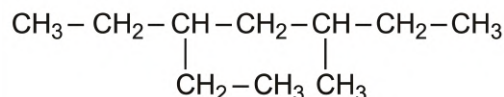
For $P(C_8H_8O)$: Acetophenone ($C_6H_5COCH_3$) fits the formula. It is a ketone, so it gives a positive 2,4-DNP test but a negative Fehling's test.

Drastic Oxidation: When alkyl benzenes or aromatic ketones like acetophenone are treated with strong oxidising agents (like chromic acid or alkaline $KMnO_4$), the entire side chain is oxidised down to a carboxyl group ($-COOH$) attached directly to the ring.

Product Q : Benzoic acid (C_6H_5COOH).

Hence, the correct answer is option (2)

Q. 19 The correct IUPAC name of the following compound is :



Option 1:

3-ethyl-5-methylheptane

Option 2:

3-methyl-5-ethylheptane

Option 3:

2,4-diethylhexane x

Option 4:

3,5-diethylhexane

Correct Answer:

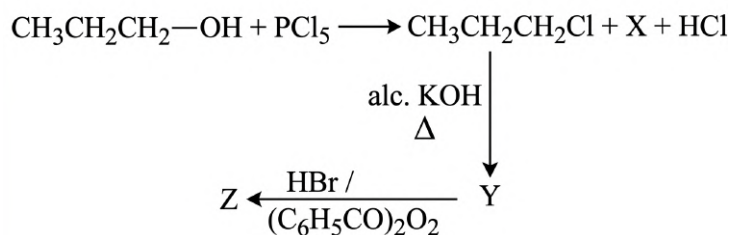
3-ethyl-5-methylheptane

Solution:

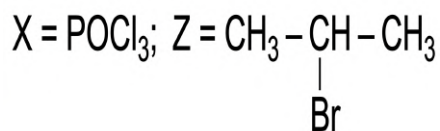
3-ethyl-5-methylheptane

Hence, the correct answer is option (1)

Q. 20 In the following reaction sequence, X and Z , respectively are :



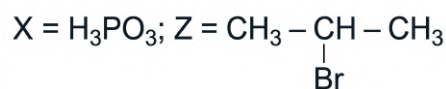
Option 1:



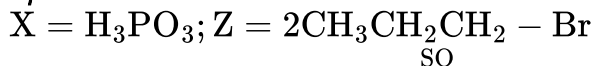
Option 2:



Option 3:



Option 4:



Correct Answer:



Solution:

1. Formation of **X** (Reaction with PCl_5)

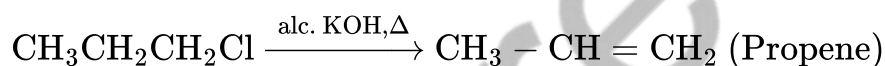
The reaction of propan-1-ol with phosphorus pentachloride (PCl_5) is as follows:



- The byproduct **X** is phosphorus oxychloride (POCl_3).

2. Formation of **Y** (Dehydrohalogenation)

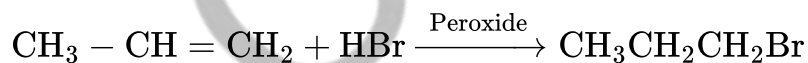
Treating the 1-chloropropane with alcoholic KOH and heat (Δ) causes an **E2** elimination reaction:



- Product **Y** is Propene.

3. Formation of **Z** (Peroxide Effect)

The final step is the addition of HBr in the presence of benzoyl peroxide $(\text{C}_6\text{H}_5\text{CO})_2\text{O}_2$. This is the AntiMarkovnikov addition (also known as the Kharasch effect or Peroxide effect).



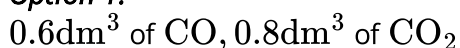
The bromine atom attaches to the terminal carbon (the one with more hydrogens).

Product **Z** is 1-bromopropane.

Hence, the correct answer is option (2)

Q. 21 When 1dm^3 of CO_2 gas is passed over hot coke, the volume of gaseous mixture after complete reaction at STP becomes 1.4dm^3 . The composition of the gaseous mixture at STP is :

Option 1:



Option 2:

0.8dm^3 of CO, 0.8dm^3 of CO_2

Option 3:

0.6dm^3 of CO, 0.4dm^3 of CO_2

Option 4:

0.8dm^3 of CO, 0.6dm^3 of CO_2

Correct Answer:

0.8dm^3 of CO, 0.6dm^3 of CO_2

Solution:

Reaction of CO_2 with hot coke:



Let:

dm^3 of CO_2 reacts

Initial $\text{CO}_2 = 1\text{dm}^3$

After reaction:

Remaining $\text{CO}_2 = 1 - x$

CO formed $= 2x$

Total final volume:

$$(1 - x) + 2x = 1 + x$$

Given:

$$1 + x = 1.4 \Rightarrow x = 0.4$$

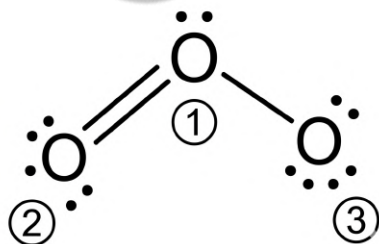
Now calculate composition:

$$\text{CO}_2 = 1 - 0.4 = 0.6\text{dm}^3$$

$$\text{CO} = 2 \times 0.4 = 0.8\text{dm}^3$$

Hence, the correct answer is option (4)

Q. 22



The correct formal charges on oxygen atoms numbered 2, 1 and 3 respectively are :

Option 1:

0, 0, 0

Option 2:
-1, 0, +1

Option 3:
+1, 0, -1

Option 4:
0, +1, -1

Correct Answer:
0, +1, -1

Solution:

O(2) (double bonded terminal O):

- Lone pairs = $4e^-$

- Bonding = $4e^-$

$$FC = 6 - 4 - 2 = 0$$

O(1) (central O):

- Lone pairs = $2e^-$

- Bonding = $6e^-$

$$FC = 6 - 2 - 3 = +1$$

O(3) (single bonded terminal O):

Lone pairs = $6e^-$

Bonding = $2e^-$

$$FC = 6 - 6 - 1 = -1$$

Hence, the correct answer is option (4)

Q. 23 Match List - I with List - II

List - I (order of reaction)	List - II (unit of rate constant)
A. Zero order	I. $\text{mol}^{-1} \text{L s}^{-1}$
B. First order	II. $\text{mol}^{-2} \text{L}^2 \text{s}^{-1}$
C. Second orde	III. s^{-1}
D. Third order	IV. $\text{molL}^{-1} \text{s}^{-1}$

Choose the correct answer from the options given below:

Option 1:
A-IV, B-III, C- II,D- I

Option 2:

A-IV, B-III, C-I, D-II

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-IV, B-II, C-I, D-III

Correct Answer:

A-IV, B-III, C-I, D-II

Solution:

List - I	List - II
A. Zero order	IV. $\text{mol L}^{-1} \text{s}^{-1}$
B. First order	III. s^{-1}
C. Second order	I. $\text{mol}^{-1} \text{L s}^{-1}$
D. Third order	II. $\text{mol}^{-2} \text{L}^2 \text{s}^{-1}$

Hence, the correct answer is option 2

Q. 24 The correct order of increasing metallic character of Na, Be, P, Mg and Si is :

Option 1:

$\text{Be} < \text{Si} < \text{P} < \text{Mg} < \text{Na}$

Option 2:

$\text{P} < \text{Si} < \text{Na} < \text{Mg} < \text{Be}$

Option 3:

$\text{P} < \text{Si} < \text{Be} < \text{Mg} < \text{Na}$

Option 4:

$\text{P} < \text{Mg} < \text{Be} < \text{Si} < \text{Na}$

Correct Answer:

$\text{P} < \text{Si} < \text{Be} < \text{Mg} < \text{Na}$

Solution:

Left to Right metallic character decreases

Top to Bottom metallic character increases

Na, Mg, Si, P: These are all in Period

$\text{P} < \text{Si} < \text{Mg} < \text{Na}$

- Be is in Group 2, located directly above Magnesium. Since metallic character increases downward, Be is less metallic than Mg.
- Beryllium is a metal, while Silicon is a metalloid. Therefore, Si is less metallic than Be.

$\text{P} < \text{Si} < \text{Be} < \text{Mg} < \text{Na}$

Hence, correct answer is option (3)

Q. 25 The number of hydrogen atoms present in 5.4 g of urea is :
(Given : Molar mass of urea : 60 g mol^{-1} ,
 $N_A : 6.022 \times 10^{23} \text{ particles mol}^{-1}$)

Option 1:
 2.168×10^{22}

Option 2:
 2.168×10^{23}

Option 3:
 1.084×10^{22}

Option 4:
 1.084×10^{23}

Correct Answer:
 2.168×10^{23}

Solution:

$$\text{Moles} = \frac{\text{Given Mass}}{\text{Molar Mass}} = \frac{5.4 \text{ g}}{60 \text{ g/mol}} = 0.09 \text{ moles}$$

The formula for urea is NH_2CONH_2 .

Each molecule of urea contains 4 hydrogen atoms.

$$\text{Total H atoms} = \text{Moles} \times \text{H atoms per molecule} \times N_A$$

$$\text{Total H atoms} = 0.09 \times 4 \times 6.022 \times 10^{23}$$

$$\text{Total H atoms} = 0.36 \times 6.022 \times 10^{23} \approx \mathbf{2.168 \times 10^{23}}$$

Hence, the correct answer is option (2)

Q. 26 In a qualitative analysis, Bi^{3+} is detected by appearance of precipitate of $\text{BiO}(\text{OH})(\text{s})$. Calculate pH when the following equilibrium exists at 298 K :
 $\text{BiO}(\text{OH})(\text{s}) \rightleftharpoons \text{BiO}^+(\text{aq}) + \text{OH}^-(\text{aq})$,
 $K = 4 \times 10^{-10}$
(Given : $\log 2 = 0.3010$)

Option 1:
4.699

Option 2:
5.286

Option 3:
8.714

Option 4:

9.301

Correct Answer:

9.301

Solution:

For the reaction: $\text{BiO}(\text{OH})(s) \rightleftharpoons \text{BiO}^+(aq) + \text{OH}^-(aq)$

The equilibrium constant is $K = [\text{BiO}^+][\text{OH}^-]$.

Let $[\text{BiO}^+] = [\text{OH}^-] = s$.

$$s^2 = K = 4 \times 10^{-10}$$

$$s = [\text{OH}^-] = \sqrt{4 \times 10^{-10}} = 2 \times 10^{-5} \text{ M}$$

pOH

$$\text{pOH} = -\log[\text{OH}^-]$$

$$\text{pOH} = -\log(2 \times 10^{-5}) = 5 - \log 2$$

$$\text{pOH} = 5 - 0.3010 = 4.699$$

pH

At 298 K, $\text{pH} + \text{pOH} = 14$.

$$\text{pH} = 14 - 4.699 = \mathbf{9.301}$$

Hence, the correct answer is option (4)

Q. 27 Match the LIST - I with LIST - II

List - I	List - II
A. $\text{H}_3\text{C}-\text{CH}(\text{C}_6\text{H}_5)-\text{CH}_3 \rightarrow \text{C}_6\text{H}_5\text{OH}$	I. (i) oleum; (ii) NaOH, Δ; (iii) H^+
B. $\text{CH}_3\text{COOH} \rightarrow \text{CH}_3\text{CH}_2\text{OH}$	II. (i) O_2; (ii) $\text{H}_2\text{O}/\text{H}^+$
C. $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \rightarrow \text{CH}_3-\text{CH}(\text{OH})-\text{CH}_3$	III. (i) CH_3OH, H^+; (ii) H_2, catalyst
D. $\text{C}_6\text{H}_6 \rightarrow \text{C}_6\text{H}_5\text{OH}$	IV. (i) conc. H_2SO_4, Δ; (ii) $\text{H}^+/\text{H}_2\text{O}$

Choose the correct answer from the options given below :

Option 1:

A-I, B-III, C-IV, D-II

Option 2:

A-II, B-IV, C-III, D-I

Option 3:

A-II, B-III, C-IV, D-I

Option 4:

A-II, B-III, C-I, D-IV

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

A. Cumene to Phenol: This is the industrial preparation of phenol via cumene hydroperoxide. It requires oxidation with air (O_2) followed by acid hydrolysis.

A $\rightarrow$ II

B. Ethanoic Acid to Ethanol: This is a reduction of a carboxylic acid to a primary alcohol. While LiAlH_4 is common, another route involves esterification with methanol (CH_3OH , H^+) followed by catalytic hydrogenation.

B $\rightarrow$ III.

C. Propan-1-ol to Propan-2-ol: This requires dehydration to form propene (using conc. H_2SO_4 , Δ) followed by hydration (acid-catalyzed addition of water) to form the more stable secondary alcohol.

C -> IV.

D. Benzene to Phenol: This follows the classical sulfonating route: sulfonation with oleum to form benzene sulfonic acid, fusion with NaOH, and final acidification.

D -> I

Hence, the correct answer is option (3)

Q. 28 A bulb is rated at 150 watt, converting 8% energy into light. If eneergy of one photon is 4.42×10^{-19} J, how many photons are emitted by the bulb per second?

Option 1:
 27.2×10^{19}

Option 2:
 4.06×10^{19}

Option 3:
 1.35×10^{19}

Option 4:
 2.71×10^{19}

Correct Answer:
 2.71×10^{19}

Solution:

The bulb is rated at 150 W , but only 8% is converted to light.

Light Power = $150 \times \frac{8}{100} = 12$ W (or J/s)

Using the formula $P = n \times E_{\text{photon}}$:

$$n = \frac{\text{Power}}{\text{Energy of one photon}}$$
$$n = \frac{12}{4.42 \times 10^{-19}} \approx 2.7149 \times 10^{19}$$

Hence, the correct answer is option (4)

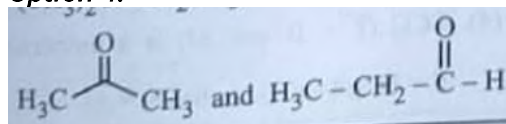
Q. 29 The pair of molecules that are metamers among the following is :

Option 1:
 $\text{CH}_3\text{OCH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$

Option 2:
 $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ and $\text{CH}_3 - \text{CH}(\text{OH}) - \text{CH}_3$

Option 3:
 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_3$

Option 4:



Correct Answer:



Solution:

Metamers are isomers that have the same molecular formula but differ in the distribution of carbon atoms (alkyl groups) on either side of a functional group (like $-\text{O}-$, $-\text{S}-$, or $-\text{NH}-$).

Option (1): $\text{CH}_3\text{OCH}_2\text{CH}_2\text{CH}_3$ (Methyl propyl ether) and $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$ (Diethyl ether). Both have the same formula ($\text{C}_4\text{H}_{10}\text{O}$) but different alkyl groups around the oxygen atom. These are metamers.

Option (2): Propan-1-ol and Propan-2-ol are position isomers.

Option (3): n-pentane and isopentane are chain isomers.

Option (4): Propanone and Propanal are functional isomers.

Hence, the correct answer is option (1)

Q. 30 Identify the correct statements :

A. The molality of 2.5 g of ethanoic acid (Molar mass : 60 g mol^{-1}) in 75 g of benzene solution is 0.556.

B. The molarity of a solution containing 5 g of NaOH (molar mass : 40 g mol^{-1}) in 450 mL of solution is 0.278 M at 298 K .

C. Aquatic species are more comfortable in cold water.

D. The solubility of gas increases with decrease in pressure.

E. For a binary mixture of A and B , the number of moles of A and B are n_A and n_B respectively.

The mole fraction of B will be $x_B = \frac{n_B}{n_A + n_B} =$

Choose the correct answer from the options given below:

Option 1:

A, B and C only

Option 2:

A, D and E only

Option 3:

A and B only

Option 4:

A and C only

Correct Answer:

A, B and C only

Solution:

Statement A (True): Molality (m) = $\frac{\text{moles of solute}}{\text{mass of solvent in kg}}$.

Moles of ethanoic acid = $\frac{2.5}{60} = 0.0417$.

$$m = \frac{0.0417}{0.075 \text{ kg}} = 0.556 \text{ m}$$

Statement B (True): Molarity (M) = $\frac{\text{moles of solute}}{\text{Volume of solution in L}}$.

$$\text{Moles of NaOH} = \frac{5}{40} = 0.125.$$

$$M = \frac{0.125}{0.450 \text{ L}} = 0.278 \text{ M}$$

Statement C (True): The solubility of gases (like O_2) increases as temperature decreases. Therefore, aquatic species are more comfortable in cold water due to higher dissolved oxygen levels.

Statement D (False): According to Henry's Law, the solubility of a gas increases with an increase in pressure.

Statement E (False): The mole fraction of B is $x_B = \frac{n_B}{n_A + n_B}$. The provided formula is for x_A .

Hence, the correct answer is option (1)

Q. 31 Which one of the following is an ambidentate ligand?

Option 1:

Oxalate

Option 2:

Ethane-1,2-diamine

Option 3:

Thiocyanate

Option 4:

Ethylenediaminetetraacetate ion

Correct Answer:

Thiocyanate

Solution:

An ambidentate ligand is a ligand that can coordinate to a central metal atom through two different atoms but only one at a time.

Thiocyanate (SCN^-) can bind through the Sulfur (S) or the Nitrogen (N) atom.

Hence, the correct answer is option (3)

Q. 32 The functional group that can be identified through phthalein dye test is :

Option 1:

Carboxylic acid

Option 2:

Alcohol

Option 3:
Aldehyde

Option 4:
Phenolic

Correct Answer:
Phenolic

Solution:
The phthalein dye test is used to identify the phenolic group. When a phenol is heated with phthalic anhydride in the presence of concentrated H_2SO_4 , it forms a phthalein dye (like phenolphthalein) which turns pink/red in alkaline solution.

Hence, the correct answer is option (4)

Q. 33 Match List- I with List - II:

List-I	List -II
A. C_2H_4	I. 3σ bonds, 2π bonds
B. C_2H_2	II. 3σ bonds, one lone pair
C. CH_4	III. 4σ bonds
D. NH_3	IV. 5σ bonds, 1π bond

Choose the correct answer from the options given below :

Option 1:
A-IV, B-I, C-III, D-II

Option 2:
A-III, B-IV, C-II, D-I

Option 3:
A-II, B-III, C-I, D-IV

Option 4:
A-I, B-II, C-IV, D-III

Correct Answer:
A-IV, B-I, C-III, D-II

Solution:
A. C_2H_4 (Ethene): Contains a $C = C$ double bond and four $C - H$ single bonds. Total: 5 σ bonds, 1 π bond. (A-IV)
B. C_2H_2 (Ethyne): Contains a $C \equiv C$ triple bond and two $C - H$ single bonds. Total: 3 σ bonds, 2 π bonds. (B-I)
C. CH_4 (Methane): Contains four $C - H$ single bonds. Total: 4 σ bonds. (C-III)

D. NH_3 (Ammonia): Contains three $N-H$ single bonds and one lone pair on Nitrogen. Total: 3 σ bonds, one lone pair. (D-II)

Hence, the correct answer is option (1).

- Q. 34** A solution of copper sulphate is electrolysed for 10 minutes with a current of 1.5 amperes. The mass of copper deposited at cathode is :
(Given : Molar mass of Cu = 63 g mol^{-1} ; $1 F = 96487 \text{ C mol}^{-1}$)

Option 1:

0.2938 g

Option 2:

0.5876 g

Option 3:

2.4036 g

Option 4:

1.7018 g

Correct Answer:

0.2938 g

Solution:

79: Total charge (Q) :

$$Q = I \times t = 1.5 \text{ A} \times (10 \times 60) \text{ s} = 900 \text{ C}$$

The reaction at the cathode is $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}(s)$.

$$\text{Mass deposited } (m) = \frac{\text{Molar mass} \times Q}{n \times F}$$

$$m = \frac{63 \times 900}{2 \times 96487} \approx \mathbf{0.2938 \text{ g}}$$

Hence, the correct answer is option (1)

Q. 35 Match List - I with List - II

	List - I (Quantum Numbers)			List - II (Orbital)
	"n"	'l'		
A	2	1	I	3d
B	4	0	II	2p
C	5	3	III	4s
D	3	2	IV	5f

Choose the correct answer from the options given below

Option 1:

A-II, B-III, C-IV, D-I

Option 2:

A-I, B-II, C-III, D-IV

Option 3:

A-II, B-III, C-I, D-IV

Option 4:

A-IV, B-II, C-III, D-I

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

The orbital is designated as nl , where n is the principal quantum number and l corresponds to the subshell ($0 = s, 1 = p, 2 = d, 3 = f$).

A. $n = 2, l = 1 : 2p$ (A – II)

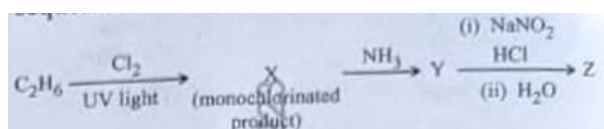
B. $n = 4, l = 0 : 4s$ (B-III)

C. $n = 5, l = 3 : 5f$ (C – IV)

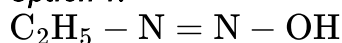
D. $n = 3, l = 2 : 3d$ (D – I)

Hence, the correct answer is option (1)

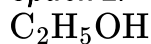
Q. 36 The major product Z formed in the following sequence of reactions is :



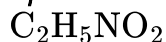
Option 1:



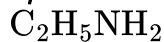
Option 2:



Option 3:



Option 4:



Correct Answer:



Solution:

Step 1: Ethane (C_2H_6) undergoes free radical chlorination with Cl_2/UV light to form Chloroethane (X): $\text{C}_2\text{H}_5\text{Cl}$.

Step 2: Chloroethane reacts with Ammonia (NH_3) via nucleophilic substitution ($\text{S}_\text{N}2$) to form Ethylamine (Y) : $\text{C}_2\text{H}_5\text{NH}_2$.

Step 3: Ethylamine reacts with nitrous acid ($\text{NaNO}_2 + \text{HCl}$) to form an unstable diazonium salt, which immediately reacts with water (H_2O) to form Ethanol (Z): $\text{C}_2\text{H}_5\text{OH}$.

Hence, the correct answer is option (2)

Q. 37 Although +3 oxidation state is most common in lanthanoids, cerium still shows +4 oxidation state because :

Option 1:

After losing one, more electron, it acquires $4f^{14}$ electronic configuration.

Option 2:

Its nearest inert gas is Radon.

Option 3:

After losing one more electron, it acquires $4f^0$ electronic configuration.

Option 4:

Its atomic number is 61 .

Correct Answer:

After losing one more electron, it acquires $4f^0$ electronic configuration.

Solution:

Cerium (Ce) has the atomic number 58 . Its ground-state electronic configuration is $[\text{Xe}]4f^15d^16s^2$.

By losing 4 electrons (to reach the +4 state), it achieves the configuration of the noble gas Xenon ($[\text{Xe}]4f^0$).

This empty $4f^0$ orbital provides extra stability to the +4 state.

Hence, the correct answer is option (3)

Q. 38 Methane reacts with steam at 1273 K in the presence of nickel catalyst to form:

Option 1:

CO and H_2

Option 2:

CO and H_2O

Option 3:

CO_2 and H_2

Option 4:

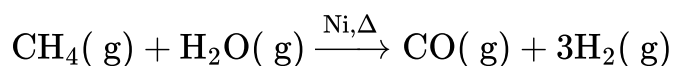
CO_2 and H_2O

Correct Answer:

CO and H_2

Solution:

When methane reacts with steam at high temperatures (1273 K) in the presence of a nickel catalyst, it undergoes a process called steam reforming to produce Syngas (Synthesis Gas).



Hence, the correct answer is option (1)

Q. 39 Identify the incorrect statement from the following :

Option 1:

Oxygen exhibits only -2 oxidation state.

Option 2:

The order of catenation property of Group 14 elements is $C \gg Si > Ge \approx Sn$.

Option 3:

Carbon has the ability to form $p\pi - p\pi$ multiple bond with itself.

Option 4:

ECl_3 ($E = B$ and Al) is a monomer when $E = B$ and a dimer when $E = Al$.

Correct Answer:

Oxygen exhibits only -2 oxidation state.

Solution:

Statement (1) is Incorrect: Oxygen does not exhibit only the -2 oxidation state. It shows +2 in OF_2 , +1 in O_2F_2 , -1 in peroxides (H_2O_2), and $-1/2$ in superoxides (KO_2).

Statement (2) is Correct: Catenation ability decreases down the group as bond enthalpy decreases.

Statement (3) is Correct: Carbon frequently forms $p\pi - p\pi$ multiple bonds ($C = C, C \equiv C$).

Statement (4) is Correct: BCl_3 is a monomer (trigonal planar), while $AlCl_3$ achieves stability by forming a dimer (Al_2Cl_6).

Hence, the correct answer is option (1)

Q. 40 Phenolphthalein is used as an indicator for the titration of sodium hydroxide solution against a standard solution of oxalic acid. The colour change that is observed at an alkaline pH close to the equivalence point during this titration is :

Option 1:

pink to colourless

Option 2:

pinkish red to yellow

Option 3:

colourless to pink

Option 4:

yellow to pinkish red

Correct Answer:

colourless to pink

Solution:

In a titration of Oxalic acid (standard solution in the flask) against Sodium hydroxide (base in the burette). The solution starts as acidic (colorless). As the equivalence point is reached and the solution becomes slightly alkaline, the phenolphthalein changes color. The color change observed is colourless to pink.

Hence, the correct answer is option (3)

Q. 41 Identify the incorrect statement from the following:

Option 1:

The largest and the smallest species among Mg , Mg^{2+} , Al and Al^{3+} are Al and Mg^{2+} , respectively.

Option 2:

The IUPAC name of the element with atomic number 107 is Unnilseptium.

Option 3:

The similarity in behaviour of Li with Mg is referred to as 'diagonal relationship'.

Option 4:

The oxidation state and covalency of Al in $[AlCl(H_2O)_5]^{2+}$ are 3 and 6 respectively.

Correct Answer:

The largest and the smallest species among Mg , Mg^{2+} , Al and Al^{3+} are Al and Mg^{2+} , respectively.

Solution:

Statement (1) is Incorrect: Between Mg and Al , Mg is larger because atomic size decreases across a period. Between Mg^{2+} and Al^{3+} , Al^{3+} is the smallest because it has more protons pulling the same number of electrons. Therefore, the largest is Mg and the smallest is Al^{3+} .

Statement (2) is Correct: Atomic number 107 (1 = Un, 0=nil, 7=sept) is Unnilseptium.

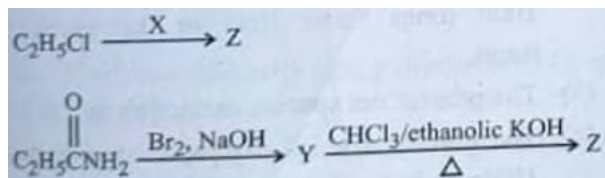
Statement (3) is Correct: Li and Mg show diagonal relationship due to similar ionic sizes and electronegativities.

Statement (4) is Correct: In $[\text{AlCl}(\text{H}_2\text{O})_5]^{2+}$, let x be the oxidation state of Al :

$x + (-1) + 5(0) = +2 \Rightarrow x = +3$. The coordination number (covalency) is 6 ($1\text{Cl} + 5\text{H}_2\text{O}$ ligands).

Hence, the correct answer is option (1)

Q. 42 The following two reactions give e^x the same foul smelling product Z .



X and Z respectively, are:

Option 1:

X = AgCN; Z = $\text{C}_2\text{H}_5\text{CN}$

Option 2:

X = AgCN ; Z = $\text{C}_2\text{H}_5\text{NC}$

Option 3:

X = KCN; Z = $\text{C}_2\text{H}_5\text{CN}$

Option 4:

X = KCN; Z = $\text{C}_2\text{H}_5\text{NC}$

Correct Answer:

X = AgCN ; Z = $\text{C}_2\text{H}_5\text{NC}$

Solution:

$\text{C}_2\text{H}_5\text{Cl} + \text{AgCN} \rightarrow \text{C}_2\text{H}_5\text{NC}$ (Ethyl isocyanide).

AgCN is covalent, so the nitrogen atom acts as the nucleophile, forming an isocyanide.

$\text{C}_2\text{H}_5\text{CONH}_2 \xrightarrow{\text{Br}_2, \text{NaOH}} \text{C}_2\text{H}_5\text{NH}_2$ (Hofmann Bromamide Degradation).

Hence, the correct answer is option (2).

Q. 43 The calculated 'spin-only' magnetic moment of Ti^{2+} ($3 d^2$) is :

Option 1:

3.87 BM

Option 2:

4.90 BM

Option 3:

2.84 BM

Option 4:

5.92 BM

Correct Answer:

2.84 BM

Solution:

The formula for 'spin-only' magnetic moment (μ) is:

$$\mu = \sqrt{n(n+2)}\text{BM}$$

where n is the number of unpaired electrons.

For Ti^{2+} (configuration $3d^2$), there are 2 unpaired electrons ($n = 2$).

$$\mu = \sqrt{2(2+2)} = \sqrt{2 \times 4} = \sqrt{8}$$

$$\mu \approx 2.83\text{BM}$$

Hence, the correct answer is option (3)

Q. 44 Match List - I with List - II

List - I (Transition metal /compound /complex)	List - II (Catalytic Role)
A. V_2O_5	I. Preparation of ammonia from N_2/H_2 mixture
B. Fe	II. Polymerisation of alkynes
C. $PdCl_2$	III. Preparation of H_2SO_4 from SO_2
D. Ni complex	IV. Oxidation of ethyne to ethanal

Choose the correct answer from the options given below:

Option 1:

A-III, B-IV, C-I, D-II

Option 2:

A-III, B-I, C-IV, D-II

Option 3:

A-II, B-I, C-IV, D-III

Option 4:

A-IV, B-I, C-III, D-II

Correct Answer:

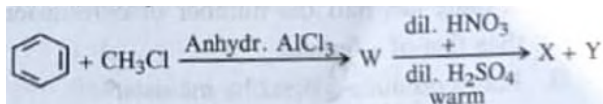
A-III, B-I, C-IV, D-II

Solution:

- A. V_2O_5 : Used in the Contact Process for the preparation of H_2SO_4 from SO_2 . (A-III)
B. Fe (Fine powder): Used in the Haber's Process for the preparation of ammonia from N_2 and H_2 . (B-I)
C. $PdCl_2$: Specifically used in the Wacker Process for the oxidation of ethyne (or ethene) to ethanal. (C-IV)
D. Ni complex: Specifically organometallic Nickel complexes (like Zeigler-Natta relatives) are used in the polymerisation of alkynes. (D-II)

Hence, the correct answer is option (2)

Q. 45 Two products X and Y are formed in the following reaction sequence.



The suitable method that can be used for the separation of products X and Y is :

Option 1:

Continuous extraction

Option 2:

Differential extraction

Option 3:

Sublimation

Option 4:

Fractional distillation

Correct Answer:

Fractional distillation

Solution:

Benzene + CH_3Cl with $\text{AlCl}_3 \rightarrow$ Toluene (W).

Toluene + nitrating mixture ($\text{HNO}_3 + \text{H}_2\text{SO}_4$) $\rightarrow$ o-nitrotoluene (X) and p-nitrotoluene (Y).

Separation Method:

X and Y are structural isomers with different boiling points. While steam distillation is commonly used for similar separations (like o- and p-nitrophenols), fractional distillation is the most suitable general industrial method for separating these two components based on their boiling point difference.

Hence, the correct answer is option (4)

Biology

Q. 1 Match List I with List II :

	List I		List II
A.	Genetically modified organism	I.	Agrobacterium tumefaciens
B.	Thermostable DNA polymerase	II.	Bt cotton
C.	Ti plasmid	III.	Thermus aquaticus
D.	pBR322	IV.	Escherichia coli

Choose the correct answer from the options given below:

Option 1:

A-II, B-I, C-I , D-III

Option 2:

A-II, B-III, G-I, D-IV

Option 3:

A-I, B-IV, C-III, D-II

Option 4:

A-I, B-II, C-IV, D-III

Correct Answer:

A-II, B-III, G-I, D-IV

Solution:

- Bt cotton is a genetically modified organism (A → II)
- Thermostable DNA polymerase comes from Thermus aquaticus (B → III)
- Ti plasmid is from Agrobacterium tumefaciens (C → I)
- pBR322 is a cloning vector (D → IV)

Hence, the correct answer is option 2) A-II, B-III, G-I, D-IV.

Q. 2 Exploring molecular, genetic and species-level diversity for products of economic importance is called :

Option 1:

Bioprospecting

Option 2:

Biofortification

Option 3:
Biomagnification

Option 4:
Bioremediation

Correct Answer:
Bioprospecting

Solution:

Bioprospecting involves searching biological diversity at molecular, genetic, and species levels to identify useful products of economic importance.

Term	Definition	Example Use
Bioprospecting	Exploring biodiversity for economic products	Drugs from medicinal plants
Biomagnification	Toxin concentration increases in the food chain	Mercury in fish
Bioremediation	Microbes clean pollutants	Oil spill cleanup
Biofortification	Nutrient-rich crop breeding	Iron-fortified rice

Hence, the correct answer is option 1) Bioprospecting.

- Q. 3** Which of the following statements are true with reference to sex determination in honeybees?
- A. An offspring formed from the union of a sperm and an egg develops as a female (queen or worker).
 - B. An unfertilized egg develops as a male by parthenogenesis.
 - C. A male has half the number of chromosomes as a female.
 - D. Males produce sperm by meiosis
 - E. Honeybees have a haplodiploid sex-determination system.
- Choose the correct answer from the options given below:

Option 1:
A, B, C and E only

Option 2:
A, B, C and D only

Option 3:
B, C, D and E only

Option 4:
A, B, D and E only

Correct Answer:
A, B, C and E only

Solution:

In honeybees, fertilised eggs (2n) form females, unfertilised eggs (n) form males. Males are haploid and produce sperm by mitosis, not meiosis, under haplodiploidy.

- A. True - Fertilised eggs (sperm + egg) develop into females (queen or worker).
- B. True - Unfertilized eggs develop into males (drones) by parthenogenesis.
- C. True - Males (haploid) have half the number of chromosomes compared to females (diploid).

D. False - Males are haploid, so they cannot undergo meiosis to produce gametes. They produce sperm by mitosis.

E. True - Honeybees indeed have a haplodiploid sex-determination system.

Hence, the correct answer is option 1) A, B, C and E only.

Q. 4 Match List I with List II :

	List I		List II
	(Growth Regulator)		(Function/Effect)
A.	2,4-D	I.	Brewing industry
B.	GA ₃	II.	Stimulation of stomatal closure
C.	Kinetin	III.	Herbicide
D.	ABA	IV.	Nutrient mobilisation

Choose the correct answer from the options given below:

Option 1:

A-I, B-II, C-IV, D-III

Option 2:

A-I, B-IV, C-III, D-II

Option 3:

A-IV, B-III, C-II, D-I

Option 4:

A-III, B-I, C-IV, D-II

Correct Answer:

A-III, B-I, C-IV, D-II

Solution:

- 2, 4-*D* acts as a herbicide ($A \rightarrow III$)
- GA₃ is used in the brewing industry ($B \rightarrow I$)
- Kinetin promotes nutrient mobilisation ($C \rightarrow IV$)
- ABA induces stomatal closure ($D \rightarrow II$)

Hence, the correct answer is option 4) A-III, B-I, C-IV, D-II.

Q. 5 In racemose inflorescence, ____ .

Option 1:

The growth is limited

Option 2:

Flowers are solitary

Option 3:

Flowers are borne in an acropetal succession

Option 4:

The main axis terminates in a flower

Correct Answer:

Flowers are borne in an acropetal succession

Solution:

In a racemose inflorescence, the main axis shows unlimited growth and flowers develop in acropetal succession (older at base, younger towards apex).

Option 1: The growth is limited - False, because in a racemose inflorescence, the main axis shows indefinite growth.

Option 2: Flowers are solitary - False, because racemose inflorescence has multiple flowers arranged along the axis, not solitary.

Option 3: Flowers are borne in an acropetal succession - True, because in a racemose inflorescence, younger flowers are at the apex, older at the base.

Option 4: The main axis terminates in a flower - False, because that is a feature of cymose inflorescence, not racemose.

Hence, the correct answer is option 3) Flowers are borne in an acropetal succession.

Q. 6 Since the origin and diversification of life on Earth, there have been five episodes of mass extinction of species. How is the sixth extinction, which is in progress, different from the previous episodes?

Option 1:

The current species extinction rates are far lower than those in previous episodes.

Option 2:

The present species extinction rates are 100 to 1000 times faster than in pre-human times.

Option 3:

The present net species extinction rate is zero.

Option 4:

The current species extinction rate is nearly 10 times faster than that in previous episodes.

Correct Answer:

The present species extinction rates are 100 to 1000 times faster than in pre-human times.

Solution:

Earth has witnessed five major mass extinctions in the past, caused by natural events (volcanic eruptions, asteroid impacts, climate shifts).

The ongoing (sixth) extinction is largely human-driven, and the extinction rate is estimated to be $100\text{--}1000\times$ higher than the natural background rate. It is human-driven due to deforestation, habitat loss, pollution, overexploitation, and climate change.

Hence, the correct answer is option 2) The present species extinction rates are 100 to 1000 times faster than in pre-human times.

Q. 7 The alpha-helix is found in which level of protein structure?

Option 1:
Secondary structure

Option 2:
Primary structure

Option 3:
Tertiary structure

Option 4:
Quaternary structure

Correct Answer:
Secondary structure

Solution:

The alpha helix is a regular folding pattern stabilised by hydrogen bonds between the backbone atoms of the polypeptide chain. This makes it a hallmark of the secondary structure of proteins.

- Primary - Amino acid sequence
- Secondary - alpha-helix & beta-sheet (local folding)
- Tertiary - Overall 3D folding of one polypeptide
- Quaternary - Association of multiple polypeptides

Hence, the correct answer is option 1) Secondary structure.

Q. 8 The enzyme required for carboxylation in the Calvin cycle is :

Option 1:
PEP carboxylase

Option 2:
RuBP carboxylase - oxygenase

Option 3:
Carboxypeptidase

Option 4:
Hexokinase

Correct Answer:

RuBP carboxylase - oxygenase

Solution:

Carboxylation in the Calvin cycle is catalysed by RuBisCO, which fixes CO_2 to RuBP, forming 3-PGA.

It fixes CO_2 to RuBP, producing 3-phosphoglycerate (3-PGA) as the first stable compound.

This step is the key entry point of inorganic carbon into the biosphere, driving photosynthetic productivity.

Hence, the correct answer is option 2) RuBP carboxylase - oxygenase.

Q. 9 Arrange the following in the correct developmental sequence related to microsporogenesis :

- A. Microspore tetrads
- B. Sporogenous tissue
- C. Pollen grains
- D. Pollen mother cells

Choose the correct answer from the options given below:

Option 1:

A, D, C, B

Option 2:

D, A, C, B

Option 3:

B, D, C, A

Option 4:

B, D, A, C

Correct Answer:

B, D, A, C

Solution:

Sporogenous tissue pollen mother cells → microspore tetrads → pollen grains.

This sequence highlights the progression from diploid sporogenous tissue to haploid pollen grains through meiosis.

The pollen mother cells act as the key transitional stage, giving rise to microspore tetrads. Finally, each microspore matures into a functional pollen grain

Hence, the correct answer is option 4) B, D, A, C.

Q. 10 Which of the following statements are not true regarding restriction endonucleases?

- A. They are called molecular scissors.
- B. These are the enzymes responsible for restricting the growth of bacteriophages in *E. coli*.
- C. They cut the DNA only at the centre of the palindromic sites.
- D. They remove nucleotides only from the ends of DNA fragments.
- E. They recognise specific palindromic base-pair sequences.

Choose the answer from the options given below :

Option 1:

C and D only

Option 2:

A and E only

Option 3:

D and E only

Option 4:

A and B only

Correct Answer:

C and D only

Solution:

Restriction enzymes cut at specific sites but not necessarily at the centre (*C false*), and they do not remove nucleotides from ends (*D false*).

Restriction endonucleases cut DNA at specific recognition sites, usually palindromic sequences, making them vital tools in genetic engineering. Unlike exonucleases, they do not remove nucleotides from the ends, but act within the DNA strand at defined positions.

Hence, the correct answer is option 1) C and D only.

Q. 11 In the lac operon, the *z* gene codes for :

Option 1:

the repressor of lac operon

Option 2:

transacetylase

Option 3:

permease

Option 4:

beta-galactosidase

Correct Answer:

beta-galactosidase

Solution:

The lacZ gene encodes β -galactosidase, which hydrolyses lactose into glucose and galactose.

This enzymatic activity enables bacteria to utilise lactose as a carbon source when glucose is scarce. Together with lacY (permease) and lacA (transacetylase), lacZ forms the structural gene set of the lac operon.

Hence, the correct answer is option 4) beta-galactosidase.

Q. 12 Match List I with List II :

	List I		List II
	(Phase of cell cycle)		(Activity)
A.	G_1 phase	I.	Actual cell division occurs
B.	S phase	II.	The cell is metabolically active and continuously grows, but does not replicate its DNA
C.	G_2 phase	III.	Synthesis of DNA occurs, and the amount of DNA per cell doubles
D.	M phase	IV.	Proteins are synthesised while cell growth continues

Choose the correct answer from the options given below:

Option 1:

A-III, B-IV, C-I, D-II

Option 2:

A-IV, B-I, C-II, D-III

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-II, B-III, C-IV, D-1

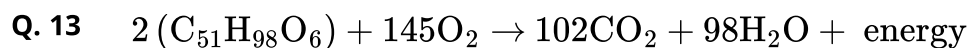
Correct Answer:

A-II, B-III, C-IV, D-1

Solution:

- G_1 is growth without DNA replication (*II*)
- S phase involves DNA synthesis (*III*)
- G_2 prepares for division with protein synthesis (*IV*)
- M phase is the actual cell division (*I*).

Hence, the correct answer is option 4) A-II, B-III, C-IV, D-1.



The Respiratory Quotient (RQ) of a biomolecule used for respiration, as per the above equation, would be :

Option 1:

Less than 0.5

Option 2:

Between 1.25 and 2

Option 3:

1 · 0

Option 4:

Between 0.5 and 0.95

Correct Answer:

Between 0.5 and 0.95

Solution:

$$RQ = \frac{CO_2 \text{ produced}}{O_2 \text{ consumed}} = \frac{102}{145} \approx 0.7, \text{ thus lying between 0.5 and 0.95.}$$

Since the RQ value is less than 1, it clearly indicates the substrate is a fat rather than a carbohydrate. Fats require more oxygen for complete oxidation, hence their RQ typically lies between 0.7 and 0.8, matching the calculated value here.

Hence, the correct answer is option 4) Between 0.5 and 0.95.

Q. 14 Which one of the following is not a characteristic of plant cells in the phase of elongation?

Option 1:

New cell wall deposition

Option 2:

Cell enlargement

Option 3:

Large conspicuous nuclei

Option 4:

Increased vacuolation

Correct Answer:

Large conspicuous nuclei

Solution:

During the elongation phase, cells primarily focus on increasing size through vacuolation and deposition of new cell wall material.

The nuclei become less conspicuous as the cytoplasm stretches, unlike meristematic cells, where nuclei are large and prominent.

Thus, elongation is marked by growth in volume and structure, not active nuclear division.

Hence, the correct answer is option 3) Large conspicuous nuclei.

Q. 15 Arrange the following steps of somatic hybridisation in a correct sequence.

- A. Digestion of cell walls.
- B. Isolation of naked protoplasts.
- C. Fusion of protoplasts to get a hybrid protoplast.
- D. Isolation of single cells from two different varieties of plants.
- E. Growing of hybrid protoplast to form a new plant.

Choose the correct answer from the options given below :

Option 1:

D, B, A, E, C

Option 2:

E, A, B, C, D

Option 3:

E, B, A, D, C

Option 4:

D, A, B, C, E

Correct Answer:

D, A, B, C, E

Solution:

First cells are isolated → cell walls digested → protoplasts obtained → fusion → regeneration into a plant.

This process allows the fusion of protoplasts from different plant varieties, bypassing sexual reproduction barriers.

The regenerated hybrid protoplast develops into a new plant with combined genetic traits of both parents. Thus, somatic hybridisation is a powerful tool in plant biotechnology and crop improvement.

Hence, the correct answer is option 4) D, A, B, C, E

Q. 16 Match List I with List II :

	List I		List II
A.	Conjunctive tissue	I.	Specialised cells in the vicinity of guard cells
B.	Casparian strips	II.	Endodermal cells are rich in starch
C.	Subsidiary cells	III.	Tissue between xylem and phloem
D.	Starch sheath	IV.	Endodermal cells with suberin deposition

Choose the correct answer from the options given below :

Option 1:

A-IV, B-III, C-II, D-I

Option 2:

A-III, B-IV, C-I, D-II

Option 3:

A-IV, B-III, C-I, D-II

Option 4:

A-III, B-IV, G-II, D-I

Correct Answer:

A-III, B-IV, C-I, D-II

Solution:

- Conjunctive tissue lies between xylem & phloem (III)
- Casparian strips have suberin (IV)
- Subsidiary cells surround guard cells (I)
- The starch sheath has starch-rich endodermis (II).

Hence, the correct answer is option 2) A-III, B-IV, C-I, D-II

Q. 17 In angiosperms, root hairs arise from which one of the following regions of the root?

Option 1:

The region of elongation

Option 2:

The region of meristematic activity

Option 3:

The region of maturation

Option 4:

The root cap zone

Correct Answer:

The region of maturation

Solution:

Root hairs develop from epidermal cells in the zone of maturation (differentiation).

This greatly increases the absorptive surface area of roots, helping in the efficient uptake of water and minerals. They are absent in the meristematic and elongation zones, which are primarily for cell division and growth.

Hence, the correct answer is option 3) The region of maturation.

Q. 18 Which of the following floral formula is the correct floral formula of Solanaceae family ?

Option 1:

$$\oplus K_{(5)} C_{(5)} A_5 \underline{G}_{(2)}$$

Option 2:

$$\oplus K_5 C_5 A_5 \underline{G}_{(2)}$$

Option 3:

$$\oplus \overset{\nearrow}{\underset{\searrow}{\text{K}}}_{(5)} \overset{\frown}{C}_{(5)} A_5 \underline{G}_{(2)}$$

Option 4:

$$\overset{\nearrow}{\underset{\searrow}{\text{K}}}_5 \overset{\frown}{C}_{(5)} A_5 \underline{G}_{(2)}$$

Correct Answer:

$$\oplus \overset{\nearrow}{\underset{\searrow}{\text{K}}}_{(5)} \overset{\frown}{C}_{(5)} A_5 \underline{G}_{(2)}$$

Solution:

Flowers of the Solanaceae family are actinomorphic and bisexual. Their calyx consists of five fused sepals, and the corolla has five fused petals. There are five stamens with a bicarpellary gynoecium and a superior ovary.

Hence, the correct answer is option 3) $\oplus \overset{\nearrow}{\underset{\searrow}{\text{K}}}_{(5)} \overset{\frown}{C}_{(5)} A_5 \underline{G}_{(2)}$

Q. 19 Which one of the following is a triploid cell?

Option 1:

Synergid

Option 2:

Primary endosperm cell

Option 3:

Central cell

Option 4:

Zygote

Correct Answer:

Primary endosperm cell

Solution:

The primary endosperm cell is triploid ($3n$), formed by the fusion of one male gamete with two polar nuclei ($n + n + n$).

This triploid cell develops into the endosperm, which nourishes the developing embryo in angiosperms. Thus, it is the only triploid structure among the given options.

Hence, the correct answer is option 2) Primary endosperm cell.

Q. 20 Match List I with List II :

	List I		List II
A.	Decomposition	I.	Accumulation of dark Coloured amorphous colloidal substance
B.	Detritus	II.	Release of inorganic nutrients by the activity of microbes in soil
C.	Mineralisation	III.	Breaking down complex organic matter into inorganic substances
D.	Humification	IV.	Dead remains of plants and animals, including faecal matter

Choose the correct answer from the options given below:

Option 1:

A-III, B-II, C-I, D-IV

Option 2:

A-IV, B-III, C-I, D-II

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-III, B-IV, C-II, D-I

Correct Answer:
A-III, B-IV, C-II, D-I

Solution:

- Decomposition = breakdown (III)
- Detritus = dead remains (IV)
- Mineralisation = release of inorganic nutrients (II)
- Humification = formation of humus (I).

This matching shows the sequential processes of decomposition in soil ecology. Dead organic matter (detritus) is broken down, nutrients are mineralised, and humus is formed, enriching soil fertility.

Hence, the correct answer is option 4) A-III, B-IV, C-II, D-I.

- Q. 21** The main criteria used for Five Kingdom Classification proposed by R.H. Whittaker (1969) included :
- A. Cell structure
 - B. Body organisation
 - C. Presence of flagellum
 - D. Reproductions
 - E. Phylogenetic relationships

Choose the correct answer from the options given below :

Option 1:

A, B and E only

Option 2:

A, B, C, D and E

Option 3:

B, C and D only

Option 4:

A, B, D and E only

Correct Answer:
A, B, D and E only

Solution:

R. H. Whittaker used cell structure, thallus/body organisation, mode of reproduction, and phylogenetic relationships. The presence of a flagellum was not one of the main criteria. This system was revolutionary as it combined morphological, reproductive, and evolutionary criteria for classification. It resolved the limitations of earlier two-kingdom and three-kingdom systems by recognising fungi separately and including Monera and Protista.

Hence, the correct answer is option 4) A, B, D and E only.

- Q. 22** "The Evil Quartet" of biodiversity loss includes which of the following ?

Option 1:

Over-exploitation; Alien species invasions; Soil pollution; Co-extinctions

Option 2:

Habitat loss and fragmentation; Air pollution; Water pollution; Co-extinctions

Option 3:

Habitat loss and fragmentation; over-exploitation; Alien species invasions; Co-extinctions

Option 4:

Over-exploitation; Alien species invasions; Air pollution; Co-extinctions

Correct Answer:

Habitat loss and fragmentation; over-exploitation; Alien species invasions; Co-extinctions

Solution:

The “evil quartet” consists of habitat loss and fragmentation, over-exploitation, alien species invasions, and co-extinctions. These are the major causes of biodiversity decline.

- Habitat loss and fragmentation reduce living space and isolate populations.
- Over-exploitation depletes species faster than they can recover.
- Alien species invasions disrupt native ecosystems and outcompete local species.
- Co-extinctions occur when the extinction of one species leads to the loss of another dependent species.

Hence, the correct answer is option 3) Habitat loss and fragmentation; over-exploitation; Alien species invasions; Co-extinctions.

Q. 23 Arrange the following steps of DNA fingerprinting in a correct sequence.

- A. Isolation of DNA and its digestion by restriction endonucleases
- B. Hybridisation using a labelled VNTR probe
- C. Transferring of separated DNA fragments to synthetic membranes
- D. Detection of hybridised DNA fragments by autoradiography
- E. Separation of DNA fragments by electrophoresis.

Choose the correct answer from the options given below :

Option 1:

A, E, B, C, D

Option 2:

A,D,B,E,C

Option 3:

A, B, D, C, E

Option 4:

A, E, C, B, D

Correct Answer:

A, E, C, B, D

Solution:

In DNA fingerprinting, DNA is isolated and cut, then fragments are separated by electrophoresis, transferred to a membrane, hybridised with a probe, and finally detected by autoradiography.

This technique relies on Variable Number Tandem Repeats (VNTRs), which are unique to individuals, making DNA fingerprinting highly specific. The autoradiography step finally reveals the distinct banding pattern, serving as a genetic identity marker.

Hence, the correct answer is option 4) A, E, C, B, D.

- Q. 24** Which of the following statements are correct with reference to a transcription unit?
- A. A transcription unit in DNA is defined primarily by three regions: promoter, structural gene and terminator
 - B. The promoter is said to be located towards the 5'-end of the structural gene
 - C. The promoter is a DNA sequence that provides binding site for RNA polymerase
 - D. The promoter defines the template and coding strands
 - E. The terminator is located towards the 3'-end of the coding strand and it defines the end of the process of transcription
- Choose the correct answer from the options given below:

Option 1:

B, C, D and E only

Option 2:

A, B, C, D and E

Option 3:

A, B, C and D only

Option 4:

A, C, D and E only

Correct Answer:

A, B, C, D and E

Solution:

A transcription unit has a promoter, a structural gene, and a terminator.

The promoter lies upstream at the 5' end and binds RNA polymerase. It determines template/coding strands, and the terminator marks the transcription end. The promoter region ensures correct initiation of transcription by RNA polymerase, while the terminator signals its end. Together with the structural gene, these elements define the functional transcription unit in both prokaryotes and eukaryotes.

Hence, the correct answer is option 2) A, B, C, D and E.

- Q. 25** Which one of the following types of pollination brings genetically different types of pollen grains to the stigma?

Option 1:

Geitonogamy

Option 2:
Xenogamy

Option 3:
Cleistogamy

Option 4:
Autogamy

Correct Answer:
Xenogamy

Solution:

In xenogamy, pollen is transferred between flowers of different plants of the same species. Xenogamy ensures maximum genetic variation because pollen comes from a different plant of the same species. This type of pollination enhances the adaptability and evolutionary fitness of the population.

Unlike autogamy or geitonogamy, it prevents inbreeding and maintains healthy gene pools. It is often facilitated by agents like insects, wind, or water, depending on the plant species. Thus, xenogamy is the only pollination type that guarantees genetically different pollen grains on the stigma.

Hence, the correct answer is option 2) Xenogamy.

Q. 26 Which of the following is an in situ conservation method?

Option 1:
Seed Banks

Option 2:
Sacred Groves

Option 3:
Botanical Gardens

Option 4:
Wildlife Safari Parks

Correct Answer:
Sacred Groves

Solution:

In situ conservation means protecting organisms in their natural habitat. Sacred groves are protected forest patches where biodiversity is preserved due to cultural and religious beliefs.

Unlike seed banks or botanical gardens (ex situ), this method allows species to evolve naturally in their environment. It is considered the most effective way to conserve endangered plants and animals.

Hence, the correct answer is option 2) Sacred Groves.

Q. 27 Heterophyllous development in response to the environment is an example of which of the following phenomena?

Option 1:
Redifferentiation

Option 2:
Dedifferentiation

Option 3:
Elasticity

Option 4:
Plasticity

Correct Answer:
Plasticity

Solution:

Heterophylly shows that the same genotype can produce different leaf forms under different environmental conditions. This is called phenotypic plasticity. In heterophylly, plants like Cotton, Coriander, and Buttercup show varied leaf shapes in response to light, water, or temperature.

This adaptability helps plants optimise photosynthesis and survival under changing environments. It is not due to genetic change but due to flexible expression of the same genetic makeup. Thus, heterophylly is a classic example of plasticity in plants.

Hence, the correct answer is option 4) Plasticity.

Q. 28 Match List I with List II :

	List I		List II
A.	Productivity	I.	Gross primary productivity minus respiration losses
B.	Net primary productivity	II.	Rate of formation of new organic matter by consumers
C.	Gross primary productivity	III.	Rate of biomass production
D.	Secondary productivity	IV.	The rate of production of organic matter during photosynthesis

Choose the correct answer from the options given below :

Option 1:
A-III, B-I, C-II, D-IV

Option 2:

A-I, B-II, C-III, D-IV

Option 3:

A-III, B-I, C-IV, D-II

Option 4:

A-I, B-III, C-IV, D-II

Correct Answer:

A-III, B-I, C-IV, D-II

Solution:

- Productivity = biomass production rate (III)
- $NPP = GPP - R$ (I)
- GPP is the total photosynthetic production (IV)
- Secondary productivity is production by consumers (II).

Hence, the correct answer is option 3) A-III, B-I, C-IV, D-II.

Q. 29 Which of the following statements are correct regarding amino acids?

- A. They are substituted methanes.
- B. Serine is an aromatic amino acid.
- C. Valine is a neutral amino acid.
- D. Lysine is an acidic amino acid.

Choose the correct answer from the options given below:

Option 1:

A and B only

Option 2:

C and D only

Option 3:

B and C only

Option 4:

A and C only

Correct Answer:

A and C only

Solution:

Amino acids are considered substituted methane derivatives because the central carbon is bonded to an amino, carboxyl, hydrogen, and variable side chains.

Amino acids are substituted methane derivatives (A correct); valine is neutral (C correct). Serine is not aromatic, and lysine is basic, not acidic.

Hence, the correct answer is option 4) A and C only.

Q. 30 In which one of the following is the ovules are not enclosed by an ovary wall and remains exposed?

Option 1:

Pinus

Option 2:

Wolffia

Option 3:

Funaria

Option 4:

Selaginella

Correct Answer:

Pinus

Solution:

In gymnosperms like Pinus, ovules are naked (not enclosed in an ovary) because they are borne directly on megasporophylls without being enclosed by an ovary wall.

This is a key distinction between gymnosperms and angiosperms, where ovules are protected inside the ovary. The exposed ovules of Pinus later develop into seeds after fertilisation.

Hence, the correct answer is option 1) Pinus.

Q. 31 Which of the following statements are correct with reference to the packaging of the DNA helix?

- A. Histones are organised to form a unit of eight molecules, called a histone octamer
- B. Histones are negatively charged basic proteins
- C. Histones are rich in the basic amino acid residues - lysine and arginine.
- D. The positively charged DNA is wrapped around the histone octamer to form a nucleosome.
- E. The packaging of chromatin at higher levels requires an additional set of proteins called non-histone chromosomal proteins.

Choose the correct answer from the options given below :

Option 1:

B, D and E only

Option 2:

A, B and D only

Option 3:

C, D and E only

Option 4:

A, C and E only

Correct Answer:
A, C and E only

Solution:
Histones are positively charged proteins due to lysine and arginine residues, which help them bind tightly to the negatively charged DNA. They form an octamer core around which DNA wraps, creating nucleosomes - the fundamental units of chromatin.

Higher-order chromatin packaging requires non-histone chromosomal proteins for further compaction. Statements B and D are incorrect because histones are not negatively charged, and DNA is negatively charged, not positive.

Hence, the correct answer is option 4) A, C and E only.

Q. 32 Match List I with List II :

	List I (Placentation)		List II (Example)
A.	Marginal	I.	Mustard
B.	Axile	II.	Pea
C.	Parietal	III.	Marigold
D.	Basal	IV.	Lemon

Choose the correct answer from the options given below :

Option 1:
A-II, B-IV, C-I, D-III

Option 2:
A-IV, B-II, C-I, D-III

Option 3:
A-III, B-I, C-IV, D-II

Option 4:
A-I, B-III, C-II, D-IV

Correct Answer:
A-II, B-IV, C-I, D-III

Solution:
Placentation refers to the arrangement of ovules within the ovary, and different plants exhibit distinct types.

- In pea, ovules are borne along the margin (marginal placentation).
- In a lemon, ovules are attached to the central axis (axile placentation).
- In mustard, ovules develop on the inner wall (parietal placentation).
- In marigolds, ovules are attached at the base of the ovary (basal placentation).

Marginal → Pea (II); Axile → Lemon (IV); Parietal → Mustard (I); Basal → Marigold (III).

Hence, the correct answer is option 1) A-II, B-IV, C-I, D-III.

Q. 33 Which one of the following is the site for active ribosomal RNA synthesis?

Option 1:
Nucleolus

Option 2:
Kinetochore

Option 3:
Centrosome

Option 4:
Chromatin

Correct Answer:
Nucleolus

Solution:

rRNA is synthesised in the nucleolus, where ribosome assembly begins. It also serves as the site for assembly of ribosomal subunits, which later move to the cytoplasm for protein synthesis.

Other nuclear structures, like the kinetochore, centrosome, and chromatin, have different roles and are not involved in rRNA synthesis.

Hence, the correct answer is option 1) Nucleolus.

Q. 34 The main function of bulliform cells in grasses is:

Option 1:
to perform photosynthesis.

Option 2:
to minimise water loss during water stress.

Option 3:
to make the leaf impermeable to fungal spores.

Option 4:
to transport water.

Correct Answer:
to minimise water loss during water stress.

Solution:

Bulliform cells help in leaf rolling during water stress, reducing transpiration and conserving water. During water stress, they lose turgidity, causing the leaf to roll inward and reduce the exposed surface area. This rolling mechanism helps minimise transpiration and conserve water.

They do not perform photosynthesis or transport water, but act as a protective adaptation against drought.

Hence, the correct answer is option 2) to minimise water loss during water stress.

- Q. 35** Which of the following statements are correct?
- A. The Amazon rainforest being cut and cleared for the cultivation of soybeans is an example of habitat loss.
 - B. Steller's sea cow and passenger pigeon became extinct due to over-exploitation by humans.
 - C. The Nile perch introduced into Lake Victoria in East Africa helped in the population growth of cichlid fish in the lake.
 - D. Water hyacinth is an invasive species.
 - E. When a species becomes extinct, the plant and animal species associated with it are not affected.

Choose the correct answer from the options given below:

Option 1:

A, B and D only

Option 2:

B, C and D only

Option 3:

A, B and E only

Option 4:

C, D and E only

Correct Answer:

A, B and D only

Solution:

- The Amazon rainforest example highlights habitat loss and fragmentation, the most severe driver of biodiversity decline.
- Species like Steller's sea cow and passenger pigeon vanished due to human over-exploitation.
- The introduction of alien species, such as water hyacinth, disrupts native ecosystems and reduces biodiversity.
- Nile perch actually caused the decline of cichlid fish, and co-extinctions occur when dependent species are lost together.

A, B, and D are correct; Nile perch reduced cichlid populations (C false), and extinction affects associated species (E false).

Hence, the correct answer is option 1) A, B and D only.

Q. 36 Which one of the following statements is not true about the universal rules of binomial nomenclature?

Option 1:

The first word in the biological name represents the specific epithet, while the second component denotes the genus.

Option 2:

The specific epithet in the biological name starts with a small letter.

Option 3:

Both the words in a biological name, when handwritten, are separately underlined or printed in italics.

Option 4:

Biological names are generally in Latin.

Correct Answer:

The first word in the biological name represents the specific epithet, while the second component denotes the genus.

Solution:

According to the universal rules, the first word denotes the genus and always begins with a capital letter, while the second word is the specific epithet written in small letters. Both words are either italicised in print or underlined separately when handwritten to maintain uniformity. Biological names are generally derived from Latin or are Latinized, ensuring global recognition.

The first word represents the genus and the second is the specific epithet, so statement (1) is incorrect.

Hence, the correct answer is option 1) The first word in the biological name represents the specific epithet, while the second component denotes the genus.

Q. 37 Which one of the following disorders is caused by the substitution of Glutamic acid (Glu) by Valine (Val) at the sixth position of the beta globin chain of the haemoglobin molecule?

Option 1:

Phenylketomuria

Option 2:

Haemophilif

Option 3:

Sickle-cell anaemia

Option 4:

Thalassemia

Correct Answer:

Sickle-cell anaemia

Solution:

Sickle-cell anaemia is caused by a point mutation where *Glu* → *Val* at position 6 of β -globin.

This point mutation changes the hydrophilic glutamic acid to hydrophobic valine, altering the haemoglobin structure. As a result, red blood cells become sickle-shaped under low oxygen conditions, leading to blockages in blood vessels.

The disorder causes anaemia, pain crises, and organ damage due to impaired oxygen transport. It is a classic example of a genetic disease caused by a substitution mutation.

Hence, the correct answer is option 3) Sickle-cell anaemia.

Q. 38 Find the incorrect statement(s) about photosynthesis from the following:

- A. The water splitting complex is associated with PS I.
- B. C_4 plants use the C_3 pathway of CO_2 fixation as the main biosynthetic pathway.
- C. In C_4 plants, photorespiration does not occur.
- D. C_3 plants exhibit 'Kranz' anatomy.
- E. ATP synthesis in chloroplasts occurs through chemiosmosis.

Choose the answer from the options given below :

Option 1:

B and C only

Option 2:

B and E only

Option 3:

B only

Option 4:

A and D only

Correct Answer:

A and D only

Solution:

Water splitting is linked to PS II (A false) and Kranz anatomy is seen in C_4 plants, not C_3 (D false).

- The water splitting complex is linked to PS II, not PS I, and it provides electrons by splitting water into oxygen and protons.
- Kranz anatomy is a special leaf structure found only in C_4 plants, where bundle sheath cells surround vascular bundles.
- C_4 plants still use the C_3 cycle as the main biosynthetic pathway, but photorespiration is minimised due to spatial separation of steps.
- ATP synthesis in chloroplasts indeed occurs via chemiosmosis, similar to mitochondria.

Hence, the correct answer is option 4) A and D only.

Q. 39 Match the List - I with List - II

List - I	List - II
A. Trypsin	I. Intercellular ground substance
B. Morphine	II. Lectin
C. Concanavalin	III. Enzyme
D. Collagen	IV. Alkaloid

Choose the correct answer from the options given below:

Option 1:

A-III, B-IV, C-II, D-I

Option 2:

A-IV, B-III, C-II, D-I

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-III, B-II, C-IV, D-I

Correct Answer:

A-III, B-IV, C-II, D-I

Solution:

Trypsin → enzyme (III); Morphine → alkaloid (IV); Concanavalin A → lectin (II); Collagen → intercellular ground substance (I).

- Trypsin is a digestive enzyme that hydrolyses proteins into peptides.
- Morphine is a plant-derived alkaloid used medicinally as an analgesic.
- Concanavalin A is a lectin that binds specifically to certain sugars, often used in biochemical studies.
- Collagen is the main structural protein of the intercellular ground substance, providing strength and elasticity to connective tissues.
- **Hence, the correct answer is option 1) A-III, B-IV, C-II, D-I.**

Q. 40 Identify the correct statements about biomolecules.

- A. Lipids are generally water-soluble.
- B. Proteins are polypeptides
- C. Polysaccharides are long chains of sugars.
- D. Adenine and guanine are substituted pyrimidines.
- E. Almost all enzymes are proteins.

Choose the correct answer from the options given below:

Option 1:

A, B and C only

Option 2:

B, D and E only

Option 3:

B, C and E only

Option 4:

C, D and E only

Correct Answer:

B, C and E only

Solution:

Lipids are water-insoluble (A false), and adenine and guanine are purines (D false). Statements B, C and E are correct.

- Proteins are long chains of amino acids linked by peptide bonds, hence called polypeptides.
- Polysaccharides such as starch, glycogen, and cellulose are complex carbohydrates made of repeated sugar units.
- Almost all enzymes are proteins, functioning as biological catalysts that speed up reactions.
- Lipids, however, are water-insoluble molecules, and adenine and guanine are purines, not pyrimidines.

Hence, the correct answer is option 3) B, C and E only.

Q. 41 Match List I with List II :

List - I	List - II
A. Incomplete dominance	I. Human skin colour
B. Co-dominance	II. Inheritance of flower colour in Antirrhinum sp.
C. Pleiotropy	III. Phenylketonuria disease in humans
D. Polygenic inheritance	IV. ABO blood groups

Choose the correct answer from the options given below :

Option 1:

A-I, B-IV, C-III, D-II

Option 2:

A-I, B-III, C-II, D-IV

Option 3:

A-II, B-IV, C-III, D-I

Option 4:

A-II, B-I, C-III, D-IV

Correct Answer:

A-II, B-IV, C-III, D-I

Solution:

Incomplete dominance → II; Co-dominance → IV; Pleiotropy → III; Polygenic inheritance → I.

- In incomplete dominance (Antirrhinum), the heterozygote shows an intermediate phenotype, e.g., pink flowers from red × white.
- Co-dominance (ABO blood groups) allows both alleles to be expressed equally, such as I^A and I^B producing AB blood type.
- Pleiotropy (Phenylketonuria) occurs when a single gene affects multiple traits, leading to diverse symptoms.
- Polygenic inheritance (human skin colour) involves multiple genes contributing additively to a continuous variation.

Hence, the correct answer is option 3) A-II, B-IV, C-III, D-I.

Q. 42 Identify the correct sequence of steps in each cycle of Polymerase Chain Reaction :

Option 1:

Denaturation → Extension → Annealing

Option 2:

Denaturation → Annealing → Extension

Option 3:

Annealing → Denaturation → Extension

Option 4:

Extension → Annealing → Denaturation

Correct Answer:

Denaturation → Annealing → Extension

Solution:

PCR follows: denaturation (strand separation) → annealing (primer binding) → extension (DNA synthesis).

These together amplify DNA exponentially. This cycle is repeated multiple times (typically 25–35 cycles) to generate millions of copies of the target DNA sequence.

Hence, the correct answer is option 2) Denaturation → Annealing → Extension.

Q. 43 How many ATP and NADPH molecules are required to make one molecule of glucose through the Calvin pathway?

Option 1:

12 ATP and 18 NADPH

Option 2:

18 ATP and 12 NADPH

Option 3:

6 ATP and 12 NADPH

Option 4:

24 ATP and 18 NADPH

Correct Answer:

18 ATP and 12 NADPH

Solution:

For the synthesis of one glucose, the Calvin cycle consumes 18 *ATP* and 12 *NADPH*.

To fix one molecule of CO_2 , the Calvin cycle requires 3 ATP and 2 NADPH. Since glucose formation needs 6 CO_2 molecules, the total requirement becomes 18 ATP and 12 NADPH.

ATP provides the energy, while NADPH supplies the reducing power for carbon fixation. This highlights the high energy demand of photosynthesis, linking the light reactions to the dark reactions.

Hence, the correct answer is option 2) 18 ATP and 12 NADPH.

Q. 44 Match List I with List II :

List - I (Process)	List -II (Location)
A. Glycolysis	I. Inner mitochondrial membrane
B. ETS	II. Mitochondrial matrix
C. Accumulation of protons	III. Cytoplasm
D. Krebs' cycle	IV. Intermembrane space

Choose the correct answer from the options given below:

Option 1:

A-IV, B-II, C-I, D-III

Option 2:

A-I, B-IV, C-III, D-II

Option 3:

A-II, B-III, C-IV, D-I

Option 4:

A-III, B-I, C-IV, D-II

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:

- Glycolysis occurs in the cytoplasm (III)
- ETS on inner membrane (I)
- Protons accumulate in the intermembrane space (IV)
- Krebs cycle in matrix (II).

Hence, the correct answer is option 4) A-III, B-I, C-IV, D-II

- Q. 45** Which of the following statements are correct with respect to DNA separation, isolation and visualisation?
- A. The cutting of DNA is done by molecular scissors.
 - B. The DNA fragments separate according to their size in an agarose gel upon electrophoresis
 - C. The separated DNA fragments can be seen without staining when exposed to UV light
 - D. The separated PNA fragments, when stained with ethidium bromide, can be seen in visible light.

Choose the correct answer from the options given below:

Option 1:

A and B only

Option 2:

B and D only

Option 3:

A and D only

Option 4:

B and C only

Correct Answer:
A and B only

Solution:

Restriction enzymes act as molecular scissors (A true), and electrophoresis separates DNA by size (B true). DNA needs staining and is visualised under UV, not visible light (C, D false).

- Restriction endonucleases act as molecular scissors, cutting DNA at specific recognition sites.
- During agarose gel electrophoresis, DNA fragments migrate according to their size, with smaller fragments moving faster.
- DNA cannot be seen directly; it must be stained (commonly with ethidium bromide) and visualised under UV light.

Hence, the correct answer is option 1) A and B only.

- Q. 46** What is the probability of having children with 'O' blood group, where both mother and father are heterozygous for 'A' and 'B' blood group, respectively?

Option 1:
0%

Option 2:
50%

Option 3:
25%

Option 4:
75%

Correct Answer:
25%

Solution:

Cross: $I^A i \times I^B i \Rightarrow$ offspring genotypes: $I^A I^B, I^A i, I^B i, ii$.

Only ii gives blood group O, so the probability = $\frac{1}{4} = 25$.

Hence, the correct answer is option 3) 25%

Q. 47 Match **List I** with **List II** :

List I		List II	
A	Molluscs	I	Pulmonary respiration only
B	Reptiles	II	Branchial respiration
C	Adult amphibians	III	Cellular respiration
D	Amoeba	IV	Pulmonary and Cutaneous respiration amphibians

Choose the **correct** answer from the options given below :

Option 1:
A-III, B-II, C-I, D-IV

Option 2:
A-II, B-I, C-III, D-IV

Option 3:
A-I, B-II, C-IV, D-III

Option 4:
A-II, B-I, C-IV, D-III

Correct Answer:
A-II, B-I, C-IV, D-III

Solution:

Molluscs → gills (II); Reptiles → lungs only (I); Adult amphibians → lungs + skin (IV); Amoeba → cellular respiration (III).

Hence, the correct answer is option (4).

Q. 48 Insertion of a foreign DNA at BamHI site in an E. coli cloning vector pBR322 results in the loss of antibiotic resistance towards :

Option 1:

Ampicillin and tetracycline

Option 2:

Ampicillin

Option 3:

Tetracycline

Option 4:

Gentamycin

Correct Answer:

Tetracycline

Solution:

The BamHI site lies within the tetracycline resistance gene; insertion disrupts it, causing loss of tetracycline resistance. The insertion normally disrupts it because it interrupts the normal coding sequence of the tetR gene. This turns it non-functional and causes the transformed bacteria to lose tetracycline resistance.

Hence, the correct answer is option 3) Tetracycline

Q. 49 What is the reason behind production of large holes in 'Swiss Cheese'?

Option 1:

The production of large amount of CO_2 and H_2 by Trichoderma polysporum

Option 2:

The production of large amount of CO_2 and H_2 by lactiges acid bacteria called Lactobacillus

Option 3:

The production' of large amount of CO_2 by Propionibacterium sharmanii

Option 4:

The production of large amount of CO₂ by Clostridium butylicum

Correct Answer:

The production' of large amount of CO₂ by Propionibacterium sharmanii

Solution:

Large holes are due to CO₂ production by Propionibacterium sharmanii during fermentation. This normally happens because the gas can not escape easily from the semi-solid cheese. And hence, creates large holes in the cheese.

Hence, the correct answer is option 3) The production' of large amount of CO₂ by Propionibacterium sharmanii

Q. 50 Which of the following is not an example of convergent evolution?

Option 1:

Fore limbs of whales and bats .

Option 2:

Flippers of penguins and dolphins

Option 3:

Eyes of octopuses and mammals

Option 4:

Wings of butterflies and birds

Correct Answer:

Fore limbs of whales and bats .

Solution:

Forelimbs of whales and bats are homologous (divergent evolution), not convergent.

Hence, the correct answer is option 1) Fore limbs of whales and bats.

Q. 51 Non-membrane bound cell organelles found in both prokaryotic and eukaryotic cells are _____ .

Option 1:

Lysosomes

Option 2:

Centrosomes

Option 3:
Mitochondria x

Option 4:
Ribosomes

Correct Answer:
Ribosomes

Solution:
Ribosomes are cellular organelles that synthesize proteins. These are non-membrane-bound and present in both prokaryotic and eukaryotic cells.

Hence, the correct answer is option 4) Ribosomes

Q. 52 Ecological pyramids represent the relationship between the organisms at different trophic levels and they are generally inverted for :

Option 1:
Pyramid of number in grassland

Option 2:
Pyramid of energy in pond ecosystem

Option 3:
Pyramid of biomass in grassland

Option 4:
Pyramid of biomass in sea

Correct Answer:
Pyramid of biomass in sea

Solution:
In aquatic ecosystems, the biomass pyramid is inverted as phytoplankton biomass is lower than that of consumers.

Hence, the correct answer is option 4) Pyramid of biomass in sea

Q. 53 Arrange the following events occurring in Renin-Angiotensin mechanism in the correct order :

- A. Increase in blood pressure and Glomerular filtration rate.
- B. Reabsorption of Na^+ and water from distal parts of tubule due to Aldosterone.
- C. Fall in Glomerular filtration rate.
- D. Vasoconstriction by Angiotensin II and release of Aldosterone.
- E. Renin converts Angiotensinogen into Angiotensin I, followed by Angiotensin II.

Choose the **correct** answer from the options given below:

Option 1:

A, C, E, B, D

Option 2:

C, A, B, D, E

Option 3:

A, D, B, E, C

Option 4:

C, E, D, B, A

Correct Answer:

C, E, D, B, A

Solution:

The glomerular filtration rate usually refers to the amount of filtrate formed by both kidneys per minute.

Fall in GFR → renin release → Angiotensin II formation → vasoconstriction & aldosterone → Na⁺/water reabsorption → increased BP & GFR.

Hence, the correct answer is option 4) C, E, D, B, A

Q. 54 Choose the correct statements regarding population interactions between two species.

- A. In both parasitism and commensalism, only one species benefits and the other species is harmed.
- B. Both species benefit in mutualism
- C. Both species benefit in commensalism.
- D. In parasitism, only one species benefits and the other species is harmed.
- E. In amensalism, one species is harmed and the other is unaffected.

Choose the **correct** answer from the options given below:

Option 1:

A and B only

Option 2:

B and E only

Option 3:

B, D and E only

Option 4:

A and D only X

Correct Answer:

B, D and E only

Solution:

In mutualism, both species benefit (B true). In parasitism, one benefits and the other is harmed (D true). In amensalism, one species is harmed while the other remains unaffected (E true). Statement A is incorrect because in commensalism, the other species is unaffected, not harmed. Statement C is incorrect as only one species benefits in commensalism.

Hence, the correct answer is option 3) B, D, and E only

Q. 55 In which animal do haploid cells mitotically to produce gametes ?

Option 1:

Male honeybees

Option 2:

Male grasshoppers

Option 3:

Male earthworms

Option 4:

Male frogs

Correct Answer:

Male honeybees

Solution:

Male honeybees are haploid organisms formed from unfertilized eggs. Since they are already haploid, gamete formation occurs by mitosis instead of meiosis. Meiosis reduces chromosome number by half, which is why it becomes an unnecessary process in drones.

Hence, the correct answer is option 1) Male honeybees

Q. 56 In humans, respiration occurs in the following steps. Arrange these steps in the correct order.

- A. Diffusion of O_2 and CO_2 between blood and tissues
- B. Diffusion of O_2 and CO_2 across alveolar membrane
- C. Pulmonary ventilation by which atmospheric air is drawn in and CO_2 rich alveolar air is released out
- D. Cellular respiration
- E. Transport of gases by the blood

Choose the **correct** answer from the options given below:

Option 1:

A, B, C, D, E

Option 2:

C, A, B, E, D

Option 3:

C, B, E, A, D

Option 4:

E, A, C, D, B

Correct Answer:

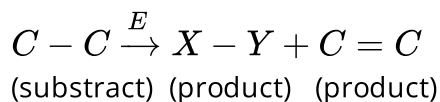
C, B, E, A, D

Solution:

Respiration is the process of obtaining energy from the breakdown of food molecules. It usually starts with pulmonary ventilation (C), followed by diffusion across alveoli (B), transport via blood (E), diffusion into tissues (A), and finally cellular respiration (D).

Hence, the correct answer is option 3) C, B, E, A, D

Q. 57 The following reaction depicts the activity of a particular class of enzymes:



Identify the enzyme class 'E' from the following options:

Option 1:

Isomerases

Option 2:

Ligases

Option 3:

Transferases

Option 4:

Lyases

Correct Answer:

Lyases

Solution:

Lyases catalyze the breaking of bonds without hydrolysis or ATP, often forming double bonds (like $C = C$).

Hence, the correct answer is option 4) Lyases

Q. 58 Match **List I** with **List II**:

List I Bioactive molecules)		List II (Importance)	
A	Streptokinase	I	Immunosuppressive agent
B	Statins	II	Removal of clots from the blood vessels
C	Lipases	III	Blood cholesterol-lowering agent
D	Cyclosporin A	IV	Detergent formulations

Choose the **correct** answer from the options given below:

Option 1:

A-II, B-III, C-I, D-IV

Option 2:

A-III, B-II, C-IV, D-I

Option 3:

A-II, B-III, C-IV, D-I

Option 4:

A-IV, B-III, C-II, D-I

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

Streptokinase dissolves clots and is used during heart attacks (II), statins lower cholesterol (III), lipases are used in detergents (IV), and cyclosporin A is an immunosuppressant that is used during organ transplantation (I).

Hence, the correct answer is option 3) A-II, B-III, C-IV, D-I

Q. 59 Which of the following equations depicts Verhulst–Pearl logistic population growth?

Option 1:

$$\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$$

Option 2:

$$\frac{dN}{dt} = rN \left(\frac{K}{K-N} \right)$$

Option 3:

$$\frac{dN}{dt} = rN \left(\frac{K-N}{N} \right)$$

Option 4:

$$\frac{dN}{dt} = rN \left(\frac{K+N}{K} \right)$$

Correct Answer:

$$\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$$

Solution:

The logistic growth equation explains that the population grows rapidly at first when resources are provided in abundance.

The logistic growth equation is $\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$, where K is the carrying capacity and growth slows as the population approaches K .

Hence, the correct answer is option 1) $\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$

Q. 60 Arrange the following cell layers/structures around the female gamete, from outer to inner side:

- A. Zona pellucida
- B. Perivitelline space
- C. Corona radiata
- D. Plasma membrane of ovum

Choose the **correct** answer from the options given below:

Option 1:

D, B, A, C

Option 2:

A, C, B, D

Option 3:

C, A, D, B

Option 4:

C, A, B, D

Correct Answer:

C, A, B, D

Solution:

From outer to inner: corona radiata → zona pellucida → perivitelline space → plasma membrane of ovum.

- Corona radiata - the outermost layer of the ovum
- Zona pellucida - protective glycoprotein layer.
- Perivitelline space - is the plasma membrane of the ovum

Hence, the correct answer is option 4) C, A, B, D

Q. 61 Which one of the following is an appropriate example of 'sexual deceit'?

Option 1:

Sea anemone and clown fish

Option 2:

Ophrys and bumblebee

Option 3:

Female wasp and fig

Option 4:

Cuckoo and crow

Correct Answer:

Ophrys and bumblebee

Solution:

In sexual deceit, the orchid Ophrys mimics female insects to attract male pollinators, leading to pollination without reward. It is because the male insects try to mate with the flower, during which the pollen transfer and pollination occur without getting any reward.

Hence, the correct answer is option 2) Ophrys and bumblebee

Q. 62 Match List I with List II related to muscular/skeletal system :

List I		List II	
A	Tetany	I	Inflammation of joints
B	Arthritis	II	Autoimmune disorder affecting neuromuscular Junction
C	Myasthenia gravis	III	Wild contraction in muscle due to low Ca^{++} in body fluid
D	Muscular dystrophy	IV	Progressive degeneration of skeletal muscle

Choose the **correct** answer from the options given below:

Option 1:

A-IV, B-III, C-II, D-I

Option 2:

A-III, B-I, C-II, D-IV

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-III, B-II, C-I, D-IV

Correct Answer:

A-III, B-I, C-II, D-IV

Solution:

Tetany is due to low Ca^{2+} causing spasms (III), arthritis causes inflammation of joints (I), myasthenia gravis is an autoimmune disease and affects the muscles (II), and muscular dystrophy shows degeneration (IV).

Hence, the correct answer is option 2) A-III, B-I, C-II, D-IV

Q. 63 Select the correct statements regarding cell membrane in eukaryotic cell.

- A. Membrane of human , RBCs has approximately 52% protein.
- B. Major phospholipids are arranged in a bilayer.
- C. Extensions of the plasma membrane into the cell form mesosomes.
- D. Tails towards the inner part of lipids are hydrophobic and - thus protected from aqueous medium
- E. Glycocalyx is present on the outer surface of the plasma membrane.

Choose the **correct** answer from the options given below :

Option 1:

C, D and E only

Option 2:

B, C and E only

Option 3:

A, B and D only

Option 4:

A, C and E only

Correct Answer:

A, B and D only

Solution:

RBC membrane has 52% protein (A true), phospholipids form a bilayer (B true), hydrophobic(water fearing) tails face inward (D true). Mesosomes are found in prokaryotes (C false), glycocalyx(a carbohydrate-rich covering) is present, but statement combos make (3) correct.

Hence, the correct answer is option 3) A, B, and D only

Q. 64 Choose the correct statements regarding cell organelles and their inclusions.

- A. The endomembrane system includes Golgi complex, endoplasmic reticulum and mitochondria.
- B. Rough endoplasmic reticulum bears ribosomes on its surface.
- C. Both mitochondria and plastids have circular DNA.
- D. A network of microtubules, microfilaments and intermediate filaments present in the cytoplasm is called cytoskeleton.
- E. Mitochondrion is a single membrane-bound structure.

Choose the **correct** answer from the options given below :

Option 1:

A, B and C only

Option 2:

C, D and E only

Option 3:

A and B only

Option 4:

B, C and D only

Correct Answer:

B, C and D only

Solution:

Mitochondria are not part of the endomembrane system (A false). RER has ribosomes (B true), mitochondria & plastids have circular DNA (C true), cytoskeleton definition is correct (D true), mitochondria are double membrane-bound (E false).

Hence, the correct answer is option 4) B, C, and D only

Q. 65 The toxin proteins isolated from *Bacillus huringiensis*, coded by which of the following genes would control cotton bollworms and corn borer, respectively ?

Option 1:

cryIAc and *cryIIb*

Option 2:

cryIAc and *cryIIIAb*

Option 3:

cryIAc and *cryIAB*

Option 4:

cryIIAb and *cryIAc*

Correct Answer:

cryIAc and *cryIAB*

Solution:

The *cryIAC* gene produces a toxin against the cotton bollworms, whereas on the other hand, the *cryIAb* gene produces a toxin that controls corn borers.

Hence, the correct answer is option 3) *cryIAc* and *cryIAB*

Q. 66 The JGA (Juxta Glomerular Apparatus) is a special sensitive region formed by cellular modifications in _____ related to the same rephron.

Option 1:

Proximal convoluted tubule and efferent renal arteriole

Option 2:

Distal convoluted tubule and efferent renal arteriole

Option 3:

Distal convoluted tubule and afferent renal arteriole

Option 4:

Proximal convoluted tubule and afferent renal arteriole

Correct Answer:

Distal convoluted tubule and afferent renal arteriole

Solution:

JGA is formed by modified cells of the distal convoluted tubule (macula densa) and the afferent arteriole, regulating blood pressure and filtration. Together, both the distal convoluted tubule and afferent arteriole regulate blood pressure and glomerular filtration rate.

Hence, the correct answer is option 3) Distal convoluted tubule and afferent renal arteriole

Q. 67 Choose the correct statements regarding frogs anatomy :

- A. Hepatic portal system is the special venous connection between liver and intestine. .
- B. There are twelve pairs of cranial nerves arising from the brain.
- C. The ureters and oviducts open separately into the cloaca in female frogs ×
- D. Hind-brain consists of cerebellum, medulla oblongata and optic lobes.
- E. Sinus venosus joins the right atrium of heart.

Choose the **correct** answer from the options given below:

Option 1:

B and D only

Option 2:

A, B and C only

Option 3:
B and C only

Option 4:
A, C and E only

Correct Answer:
A, C and E only

Solution:
Hepatic portal system connects intestine to liver (A true), ureters and oviducts open separately into cloaca (C true), sinus venosus joins right atrium (E true). Frogs have 10 pairs of cranial nerves (B false), and optic lobes belong to the midbrain (D false).

Hence, the correct answer is option 4) A, C, and E only

Q. 68 Match **List I** with **List II** :

List I		List II	
A	Cortisol	I	Stimulates the formation of alveoli in mammary glands
B	Aldosterone	II	Produces anti-inflammatory reactions
C	Cholecystokinin	III	Stimulates reabsorption of Na^+ and water from renal tubule
D	Progesterone	IV	Stimulates secretion of pancreatic enzymes and bile juice

Choose the **correct** answer from the options given below:

Option 1:
A-II, B-III, C-IV, D-I

Option 2:
A-II, B-III, C-I, D-IV

Option 3:
A-IV, B-II, C-I, D-III

Option 4:
A-III, B-II, C-IV, D-I

Correct Answer:
A-II, B-III, C-IV, D-I

Solution:

Cortisol shows anti-inflammatory action (II), aldosterone increases Na⁺ reabsorption (III), cholecystokinin stimulates pancreatic enzyme and bile secretion (IV), and progesterone helps in mammary gland development (I).

Hence, the correct answer is option 1) A-II, B-III, C-IV, D-I

Q. 69 The sixth mutant codon of beta globin gene causing polymerization of Haemoglobin and change in RBC shape is ____

Option 1:
CAG

Option 2:
AUG

Option 3:
GUG

Option 4:
GAG

Correct Answer:
GUG

Solution:

In Sickle cell anemia disease, the 6th codon of the beta globin gene mutates from GAG to GUG. This substitution results in the production of Hemoglobin S. This Hemoglobin S polymerizes at low oxygen levels and is responsible for causing sickle-shaped RBCs.

Hence, the correct answer is option 3) GUG

Q. 70 Male frogs can be distinguished from female frogs due to the presence of :

- A. Bulging eyes
- B. Vocal sacs
- C. Webbed digits in feet
- D. Copulatory pad on first digit of fore limbs
- E. Olive green-coloured skin with dark irregular spots

Choose the **correct** answer from the options given below:

Option 1:

A and B only

Option 2:

C and E only

Option 3:

B and D only

Option 4:

B and C only

Correct Answer:

B and D only

Solution:

Male frogs possess vocal sacs for croaking and copulatory (nuptial) pads on forelimbs. Other features that are common in both male and female frogs are bulging eyes, webbed feet, and skin color. Therefore, we can not use these features to distinguish the sexes.

Hence, the correct answer is option 3) B and D only

Q. 71 The human protein named α -1-antitrypsin, obtained from transgenic animals, is used for the treatment of ____

Option 1:

Alzheimer's disease

Option 2:

Rheumatoid arthritis

Option 3:

Emphysema

Option 4:

Cystic fibrosis

Correct Answer:

Emphysema

Solution:

α -1-antitrypsin deficiency leads to lung damage due to unchecked protease activity, causing emphysema. Transgenic animals produce this protein for therapy. Alpha antitrypsin is a human protein that is produced by using transgenic animals. It is used to treat emphysema. Emphysema is a lung disease caused by a deficiency of this protein. Alpha antitrypsin protects lung tissues from damage by certain enzymes.

Hence, the correct answer is option 3).Emphysema

Q. 72 Match **List I** and **List II**

List I (Drug)		List II (Effect)	
A	Nicotine	I	Causes sense of euphoria and increased energy
B	Morphine	II	Stimulates adrenal gland to release catecholamines into blood circulation
C	Herois	III	Effective sedative and painkiller
D	Cocaine	IV	A depressant; slows down body functio

Choose the **correct** answer from the options given below :

Option 1:

A-II, B-III, C-I, D-IV

Option 2:

A-III, B-II, C-IV, D-I

Option 3:

A-III, B-II, C-I, D-IV

Option 4:

A-II, B-III, C-IV, D-I

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

Nicotine → II (stimulates catecholamine release), Morphine → III (sedative), Heroin → IV (depressant), Cocaine → I (euphoria).

Hence, the correct answer is option 4) A-II, B-III, C-IV, D-I

Q. 73 The WBC count of apperson's blood sample 8000/cu. mm. How many eosinophils and lymphocytes would bein the same blood sample approximately ?

Option 1:

300 – 500/cu. mm and 500 – 700/cu. mm, respectively

Option 2:

300 – 500/cu. mm and 1200 – 1500/cu. mm, respectively

Option 3:

100 – 120/cu. mm and 160 – 200/cu. mm, respectively

Option 4:

160 – 240/cu. mm and 1600 – 2000/cu. mm, respectively

Correct Answer:

160 – 240/cu. mm and 1600 – 2000/cu. mm, respectively

Solution:

Eosinophils $\approx$ 2–3 of 8000 $\Rightarrow$ 160–240.

Lymphocytes $\approx$ 20–25 of 8000 $\Rightarrow$ 1600–2000.

Eosinophils constitute about 2-3% and lymphocytes about 20-25% of total WBCs.

Hence, the correct answer is option 4) 160 – 240/cu. mm and 1600 – 2000/cu. mm, respectively

Q. 74 Match **List I** with **List II** with respect to chronology of evolution of life forms :

List I		List II	
A	About 65 mya	I	Jawless fish probably evolved
B	About 500 mya	II	The dinosaurs suddenly disappeared from the earth
C	About 350 mya	III	Seaweeds and few plants probably existed
D	About 320 mya	IV	Invertebrates were formed and became active

Choose the **correct** answer from the options given below :

Option 1:

A-III, B-IV, C-I, D-II

Option 2:

A-II, B-IV, C-III, D-I

Option 3:

A-II, B-IV, C-I, D-III

Option 4:

A-I, B-II, C-III, D-IV

Correct Answer:

A-II, B-IV, C-I, D-III

Solution:

About 65 million years ago, the dinosaurs disappeared from the Earth. Around 500 million years ago, the invertebrates were active in the oceans. Jawless fishes probably evolved about 350 million years ago, while seaweeds and a few plants existed somewhere around 320 million years ago.

Hence, the correct answer is option 3) A-II, B-IV, C-I, D-III

Q. 75 Match List I with List II :

List I		List II	
A	Progestasert	I	Barrier made of rubber used by females
B	Multiload 375	II	Oral contraceptive
C	Diaphragm	III	Hormone releasing IUD
D	Saheli	IV	Copper releasing IUD

Choose the **correct** answer from the options given below :

Option 1:

A-IV, B-II, C-I, D-III

Option 2:

A-IV, B-III, C-I, D-II

Option 3:

A-III, B-IV, C-II, D-I

Option 4:

A-III, B-IV, C-I, D-II

Correct Answer:

A-III, B-IV, C-I, D-II

Solution:

Progestasert → hormone IUD (III), Multiload 375 → copper IUD (IV), Diaphragm → barrier (I), Saheli → oral pill (II).

Hence, the correct answer is option 4) A-III, B-IV, C-I, D-II

Q. 76 The following are the stages of life cycle of Plasmodium. Arrange the stages in the proper order.

- A. The parasites reproduce asexually in RBCs, bursting the cells.
- B. The parasites reproduce asexually in liver cells, bursting the cells and releasing into blood.
- C. Gametocytes develop in RBCs.
- D. Sporozoites reach the liver through the blood
- E. Female mosquito injects sporozoites into humans during bite (1)

Choose the **correct** answer from the options given below :

Option 1:

A, B, C, D, E

Option 2:

E, C, D, B, A

Option 3:

E, D, B, A, C

Option 4:

C, A, B, D, E

Correct Answer:

E, D, B, A, C

Solution:

Sequence: mosquito injects sporozoites → reach liver → multiply in liver → infect RBCs → gametocytes form. The Plasmodium sporozoites multiply in the liver cells and then infect RBCs. They reproduce further there and develop into gametocytes.

Hence, the correct answer is option 3) E, D, B, A, C

Q. 77 Match List I and List II related to embryonic development at various months of pregnancy :

List I		List II	
A	The foetus movement starts and hair appears on the head	I	24 weeks of pregnancy
B	The foetus develops limbs and digits	II	20 weeks of pregnancy
C	The foetus develops external genital organs	III	8 weeks of pregnancy
D	The foetus body is covered with fine hair; eyelids N separate and eyelashes are formed	IV	12 weeks of pregnancy

Choose the **correct** answer from the options given below:

Option 1:

A-II, B-IV, C-III, D-I

Option 2:

A-II, B-III, C-IV, D-I

Option 3:

A-IV, B-II, C-III, D-I

Option 4:

A-III, B-II, C-IV, D-I

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

20 weeks → movement & hair, 8 weeks → limbs, 12 weeks → genital organs, 24 weeks → eyelashes & fine hair.

During the development of a foetus, the limbs and digits appear by 8 weeks, external genital organs appear by 12 weeks of pregnancy. In the same way, foetal movement starts around 20 weeks.

Hence, the correct answer is option 2) A-II, B-III, C-IV, D-I

Q. 78 The flightless bird with forelimbs modified as paddle-like structures suited for swimming is known as :

Option 1:

Psittacula

Option 2:

Aptenodytes

Option 3:

Neophron

Option 4:

Struthio

Correct Answer:

Aptenodytes

Solution:

Aptenodytes (penguins) are flightless birds with flipper-like wings adapted for swimming.

Hence, the correct answer is option 2) Aptenodytes

Q. 79 Select the **incorrect** statements from the following :

- A. Digestive system in Platyhelminthes is incomplete.
- B. Bilateral symmetry is a characteristic feature of adult Echinoderms?
- C. Pseudocoelom is possessed by Aschelminthes.
- D. Notochord is persistent throughout life in the class Chondrichthyes.
- E. Members of class Reptilia maintain a constant body temperature.

Choose the answer from the options given below :

Option 1:

A and C only

Option 2:

B and E only

Option 3:
B and D only

Option 4:
C and D only

Correct Answer:
B and E only

Solution:
Echinoderms are radially symmetrical (not bilateral), and reptiles are ectothermic (not constant temperature). Others are correct. The adult echinoderms have radial symmetry; during their larval stage, they possess bilateral symmetry. Reptiles, being cold-blooded animals, can not maintain a constant body temperature.

Hence, the correct answer is option 2) B and E only

Q. 80 A group of researchers procured some fish-like animals and upon investigation the following characters were observed :

- A. Endoskeleton was made of cartilage.
- B. Ectoparasitic; as they were found attached on fish skin with their circular sucking mouth.
- C. Paired fins and scales were absent, but 7 pairs of gill slits were present.

Which of the following species of animals did they consider to fit best with these characters?

Option 1:
Scoliodon sp.

Option 2:
Exocoetus sp

Option 3:
Petromyzon sp

Option 4:
Branchiostoma sp

Correct Answer:
Petromyzon sp

Solution:
Petromyzon (lamprey) shows cartilaginous skeleton, ectoparasitic habit, circular mouth, no paired fins, and multiple gill slits.

Hence, the correct answer is option 3) *Petromyzon* sp

Q. 81 Choose the correct statements regarding muscle contraction.

- A. A motor neuron carries a signal sent by the Central Nervous System (CNS) to the sarcolemma of the muscle fibre.
- B. The neural signal generates an action potential which cause the release of Ca^{++} into sarcoplasm.
- C. Increase in Ca^{++} inactivates the actin for breaking cross bridges.
- D. Actin binds to the myosin head to form a cross bridge.
- E. Shortening of sarcomere takes place, by pulling actin filaments towards the centre of 'A' band.

Choose the **correct** answer from the options given below :

Option 1:

A and B only

Option 2:

C and E only

Option 3:

C and D only

Option 4:

A, B, D and E only

Correct Answer:

A, B, D and E only

Solution:

Ca^{2+} activates actin by removing troponin inhibition, not inactivates it. Cross-bridge formation and sliding of actin toward A-band center cause contraction.

Hence, the correct answer is option 4). A, B, D and E only

Q. 82 Choose the correct statement regarding GIFT to overcome infertility.

Option 1:

Ova collected from a female donor are transferred to the uterus of an infertile female.

Option 2:

Early embryos with up to 8 blastomeres are transferred into the fallopian tube of an infertile female

Option 3:

Early embryos with up to 8 blastomeres are transferred to the uterus of an infertile female.

Option 4:

It is the transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce ovum but can provide suitable environment for fertilization and development.

Correct Answer:

It is the transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce ovum but can provide suitable environment for fertilization and development.

Solution:

GIFT is Gamete Intra-Fallopian Transfer. GIFT involves transfer of an ovum into the fallopian tube for in vivo fertilization, not embryo transfer (which is IVF-ET). Embryo transfer is a technique in which the fertilised egg or embryo in the early stage is transferred into the female reproductive tract.

Hence, the correct answer is option 4).It is the transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce ovum but can provide suitable environment for fertilization and development.

Q. 83 Which of the following statements are correct with reference to human endoskeleton?

- A. Human skull is monocondylic.
- B. The joint between any two adjoining vertebrae is a cartilaginous joint.
- C. In human beings, the number of cervical vertebrae is seven
- D. All ribs except the last 2 pairs are bicephalic.
- E. The occipital bone of skull is articulated with atlas vertebra.

Choose the **correct** answer from the options given below :

Option 1:

C, D and E only

Option 2:

B, C and E only

Option 3:

A, B and D only

Option 4:

B and E only

Correct Answer:

B, C and E only

Solution:

Humans are dicondylic (not monocondylic). Vertebrae joints are cartilaginous, cervical vertebrae = 7, and the occipital bone articulates with the atlas. The joints between the adjacent vertebrae are cartilaginous because of the presence of the intervertebral disc. Humans have seven cervical vertebrae in the neck region.

Hence, the correct answer is option 2) B, C, and E only

Q. 84 Spermatogonia undergo a series of cell divisions to produce sperms. Select the correct statements from the following:

- A. Spermatogonia always undergo meiotic cell division.
- B. Primary spermatocytes divide mitotically to produce secondary spermatocytes
- C. Secondary spermatocytes, through their second meiotic division, produce haploid spermatids.
- D. Spermatids produce spermatozoa through mitosis.
- E. Spermatids transform into spermatozoa by spermiogenesis

Choose the **correct** answer from the options given below :

Option 1:

A, C and E only

Option 2:

C and E only

Option 3:

A and E only

Option 4:

B, C and D only

Correct Answer:

C and E only

Solution:

Spermatogonia divide mitotically, not always meiotically. Meiosis II forms haploid spermatids, which transform into spermatozoa by spermiogenesis. Spermatids are immature sperm cells, while the spermatozoa are the mature and functional sperm.

Hence, the correct answer is option 2) C and E only

Q. 85 Select the **incorrect** statements with reference to Rh grouping.

- A. Erythroblastosis foetalis is a condition observed having foetus with Rh^{-ve} blood and mother with Rh^{+ve} blood.
- B. Rh antigen is observed on RBCs in the majority of human beings.
- C. Before blood transfusion, Rh group should also be matched.
- D. Rh incompatibility is observed when a pregnant mother is Rh^{-ve} and the foetus is Rh^{+ve} .
- E. Erythroblastosis foetalis can be avoided by administering anti-Rh antibodies to the mother immediately after the delivery of the second child.

Choose the answer from the options given below :

Option 1:

A and B only

Option 2:

C and D only

Option 3:

A and E only

Option 4:

B and C only

Correct Answer:

A and E only

Solution:

Erythroblastosis occurs when mother is Rh^{-} and foetus Rh^{+} , not vice versa. Anti-Rh antibodies are given after the first delivery, not the second. These Anti-RH antibodies destroy any RH-positive foetal cells in the blood of the mother before her immune system can react and form permanent antibodies.

Hence, the correct answer is option 3) A and E only

Q. 86 Select the set of fishes which belong to class Osteichthyes?

Option 1:

Saw fish, Fighting fish and Dog fish

Option 2:

Devil fish, Cuttlefish and Hagfish

Option 3:

Starfish, Hagfish and Cuttlefish

Option 4:

Flying fish, Angel fish and Fighting fish

Correct Answer:

Flying fish, Angel fish and Fighting fish

Solution:

Osteichthyes are bony fishes; Flying fish, Angel fish, and fighting fish belong to this class, while others include cartilaginous or non-fish organisms. This is because the Flying fish, angel fish, and fighting fish are true bony fish. Osteichthyes are bony fishes with a skeleton that is made mainly of bone. Sharks and rays, on the other hand, belong to Class Chondrichthyes.

Hence, the correct answer is option 4) Flying fish, Angel fish, and Fighting fish

Q. 87 In a population of a grasshopper species, the chromosome number of some members is 23 and some other members possess 24 chromosomes. The 23 and 24 chromosome-bearing members in this species are ____.

Option 1:

females and males, respectively

Option 2:

males and females, respectively

Option 3:

all males

Option 4:

all females

Correct Answer:

males and females, respectively

Solution:

Grasshoppers show XO type sex determination: males = 23 (XO), females = 24 (XX).

The grasshopper shows XO-type sex determination. In the XO type, females have two X chromosomes, and males have only one X chromosome. Both males and females have the same number of autosomes. Therefore, grasshoppers with 24 chromosomes are females, and those with 23 chromosomes are males.

Hence, the correct answer is option 2) males and females, respectively

Q. 88 Evolution of human appears parallel to the progressive development of brain and language skills. As such, the evolution of individual species in the order of sequence of their appearance is :

Option 1:

Ramapithecus → Homo habilis → Homo erectus → Neanderthal → Homo sapiens

Option 2:

Neanderthal → Ramapithecus →
Homo habilis → Homo erectus →
Homo sapiens

Option 3:

Homo habilis → Homo erectus →
Ramapithecus → Neanderthal →
Homo sapiens

Option 4:

Homo sapiens → Ramapithecus →
Homo habilis → Neanderthal
Homo erectus

Correct Answer:

Ramapithecus → Homo habilis → Homo erectus → Neanderthal → Homo sapiens

Solution:

Correct evolutionary sequence: Ramapithecus → Homo habilis → Homo erectus → Neanderthal → Homo sapiens.

- Ramapithecus was the earliest ape-man ancestor.
- Homo habilis - first tool maker
- Homo erectus - upright man
- Neanderthal man - these were more advanced
- Homo sapiens - modern humans

Hence, the correct answer is option 1) Ramapithecus → Homo habilis → Homo erectus → Neanderthal → Homo sapiens

Q. 89 The specific receptors for neurotransmitters synapse are present on?

Option 1:

Pre-synaptic membrane

Option 2:

Post-synaptic membrane

Option 3:
Myelin sheath

Option 4:
Schwann cell

Correct Answer:
Post-synaptic membrane

Solution:
The membranes of post-synaptic neurons contain receptor proteins which can bind neurotransmitters. Therefore, the Neurotransmitters bind to receptors located on the post-synaptic membrane to transmit nerve impulses after diffusing through the synaptic cleft. This binding is what initiates new electrical impulses in the next neuron.

Hence, the correct answer is option 2) Post-synaptic membrane

Q. 90 Match List I with List II :

List I (Respiratory Volume)		List II (Capacity)	
A	ERV (Expiratory Reserve Volume)	I	2500 - 3000 mL
B	RV (Residual Volume)	II	500 mL
C	IRV (Inspiratory Reserve Volume)	III	1000-1100mL
D	TV (Tidal Volume)	IV	1100-1200mL

Choose the **correct** answer from the options below:

Option 1:
A-III, B-IV, C-I, D-II

Option 2:
A-III, B-I, C-IV, D-II

Option 3:
A-I, B-II, C-III, D-IV

Option 4:
A-I, B-III, C-II, D-IV

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:

$ERV \approx 1000-1100$, $RV \approx 1100-1200$, $IRV \approx 2500-3000$, $TV \approx 500$ mL.

TV = Tidal Volume is the amount of air inhaled or exhaled during normal breathing.

IRV, = IRV on the other hand, is the extra amount of air a person can inhale forcibly after a normal inspiration.

ERV = It is the extra amount of air that a person can exhale forcibly after a normal expiration.

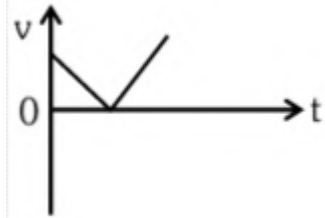
Hence, the correct answer is option 1. A-III, B-IV, C-I, D-II

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Q. 1 The following plots show variation of velocity (v) with time (t), of a ball thrown vertically upward, and falling back. Which of the following plots is/are correct ?

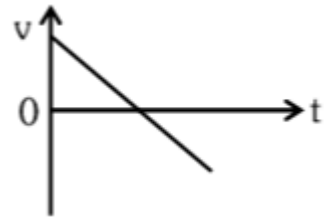
Option 1:



Option 2:



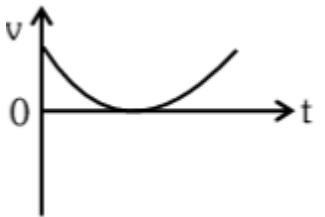
Option 3:



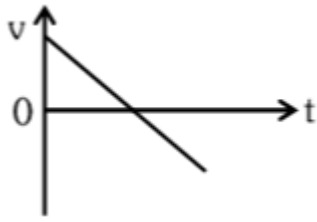
Option 4:



Option 5:



Correct Answer:



Solution:

Physics: Velocity decreases linearly to zero, then becomes negative Graph: Straight line with negative slope crossing zero

Hence, the correct option is (3).

Q. 2 For a metal of work function 6.6 eV , which of the following wavelengths of incident radiation does not give rise to the photoelectric effect?
(Take Planck's constant as $6.6 \times 10^{-34} \text{ J s}$)

Option 1:

200 nm

Option 2:

100 nm

Option 3:

50 nm

Option 4:

150 nm

Correct Answer:

200 nm

Solution:

$$\phi = 6.6 \text{ eV}$$

$$\frac{hc}{\lambda} = 6.6 \text{ eV}$$

$$\lambda_0 = \frac{12400}{6.6}$$

$$\lambda_0 = 1878.7 \quad = 187.87 \text{ nm}$$

For photo electric effect

$$\lambda \leq \lambda_0$$

$\therefore$ The wavelength of radiation does not give rise to photoelectric effect is 200 nm .

Hence, the correct option is (1).

Q. 3 The power of a crane, which lifts a mass of 1000 kg to a height of 20 m in 10 s is :
($g = 9.8 \text{ m/s}^2$)

Option 1:
19.6 kW

Option 2:
 $19 \cdot 6 \text{ W}$

Option 3:
 $39 \cdot 2 \text{ kW}$

Option 4:
 $39 \cdot 2 \text{ W}$

Correct Answer:
19.6 kW

Solution:

$$\begin{aligned}\text{Power of crane} &= \frac{mgh}{t} \\ &= \frac{1000 \times 9.8 \times 20}{10} \\ &= 19.6 \times 1000 \\ &= 19.6 \text{ kW}\end{aligned}$$

Hence, the correct option is (1).

Q. 4 Match List I with List II :

List I		List II	
A.	$E = h\nu$	I.	de Broglie wavelength
B.	Diffraction and Interference	II.	Particle nature of light
C.	$\lambda = h/p$	III.	Wave nature of light
D.	Compton effect	IV.	Energy of photon

Choose the **correct** answer from the options given below :

Option 1:
A-IV, B-III, C-II, D-I

Option 2:
A-IV, B-III, C-I, D-II

Option 3:

A-I, B-IV, C-III, D-II

Option 4:

A-IV, B-I, C-II, D-III

Correct Answer:

A-IV, B-III, C-I, D-II

Solution:

A. $E = h\nu \rightarrow$ Energy of photon $\rightarrow$ IV

B. Diffraction and Interference $\rightarrow$ Wave nature of light $\rightarrow$ III

C. $\lambda = \frac{h}{p} \rightarrow$ de Broglie wavelength $\rightarrow$ I

D. Compton effect $\rightarrow$ Particle nature of light $\rightarrow$ II

Hence, the correct option is (2).

Q. 5 The magnitude and direction of the acceleration produced in a body of mass 5 kg when two mutually perpendicular forces 8 N and 6 N act on it, are respectively

Option 1:

2 m s^{-2} ; $\tan^{-1}(3/4)$ with 8 N force

Option 2:

2 ms^{-2} ; $\tan^{-1} 2 \text{ s/3}$) with 8 N force

Option 3:

2 m s^{-2} ; $\tan^{-1}(3/4)$ with 6 N force

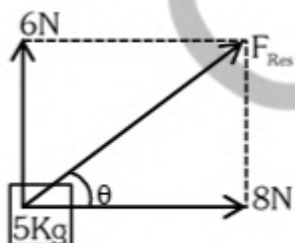
Option 4:

20 m s^{-2} ; $\tan^{-1}(4/3)$ with 8 N force

Correct Answer:

2 m s^{-2} ; $\tan^{-1}(3/4)$ with 8 N force

Solution:



$$F_{\text{Res}} = \sqrt{6^2 + 8^2}$$

$$F_{\text{Res}} = 10 \text{ N}$$

$$a = \frac{F_{\text{Res}}}{m} = \frac{10}{5} = 2 \text{ m/s}^2$$

$$\tan \theta = \frac{6}{8} = \frac{3}{4}$$

$$a = 2 \text{ m/s}^2 \tan^{-1} \left(\frac{3}{4} \right) \text{ with } 8 \text{ N force}$$

Hence, the correct the option is (1).

Q. 6 The sum of kinetic energy and potential energy of a simple pendulum bob is 0.02 joule. The speed of the simple pendulum bob at equilibrium position is approximately :
(Consider mass of the bob = 20 g)

Option 1:
2.0 m/s

Option 2:
0.2 m/s

Option 3:
14 · 1 m/s

Option 4:
1.41 m/s

Correct Answer:
1.41 m/s

Solution:

$$\text{Total energy} \Rightarrow \frac{1}{2}KA^2 = 0.02 \text{ J}$$

$$\frac{1}{2} m^2 A^2 = 0.02 \text{ J}$$

$$\frac{1}{2} mV_{\text{max}}^2 = 0.02 \text{ J}$$

$$V_{\text{max}}^2 = \frac{0.04}{20 \times 10^{-3}} = 2$$

$$V_{\text{max}} = 1.41 \text{ m/s}$$

Hence, the correct option is (4).

Q. 7 A box of mass 15 kg is kept on the floor of a stationary trolley. The coefficient of static friction between the box and the trolley is 0.12 . Keeping the box in stationary state over the trolley, the maximum acceleration with which the trolley can be moved horizontally in ms^{-2} is :

$$(g = 10 \text{ m/s}^2)$$

Option 1:

1.8

Option 2:

1.2

Option 3:

1.5

Option 4:

2.1

Correct Answer:

1.2

Solution:

$$ma_{\max} = F_{lm}$$

$$ma_{\max} = \mu mg$$

$$a_{\max} = \mu g = 1.2 \text{ m/s}^2$$

Hence, the correct option is (2).

Q. 8 The speed of light in vacuum is taken as unity. If light takes 6 min 40 s to reach the Earth from the Sun, the distance between the Sun and the Earth in new unit is :

Option 1:

3×10^8

Option 2:

500

Option 3:

3×10^{10}

Option 4:

400

Correct Answer:

400

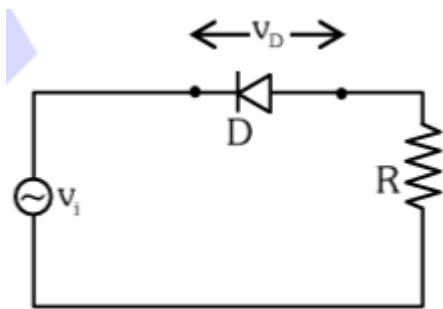
Solution:

Distance = speed $\times$ time

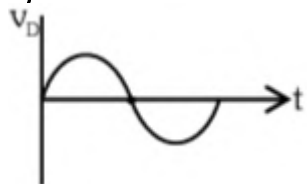
$$d = 1 \times 400 = 400 \text{ units}$$

Hence, the correct option is (4).

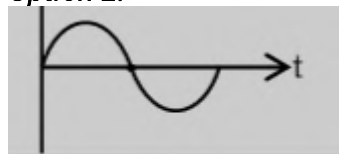
Q. 9 In the circuit shown below, the voltage appearing across the diode D will be of the form :



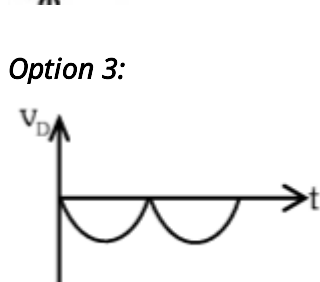
Option 1:



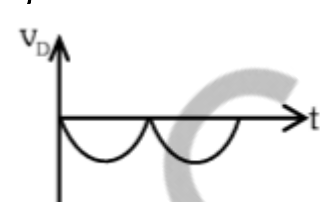
Option 2:



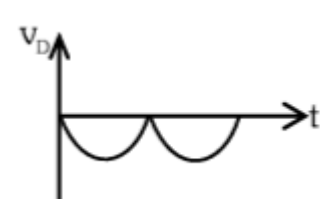
Option 3:



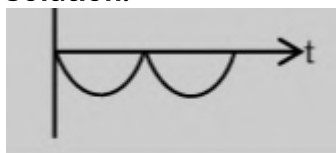
Option 4:



Correct Answer:



Solution:



Option 3:

Logic: In forward bias (positive half cycle), the diode conducts, and voltage across it is nearly 0 V . In reverse bias (negative half cycle), the diode does not conduct, and the entire input voltage appears across the diode.
Graph: The output V_D will show the negative half cycles.

Hence, the correct option is (4).

Q. 10 A submarine is designed to withstand an absolute pressure of 100 atm . How deep can it go below the water surface?

(Consider the density of water = 1000 kg m^{-3} . $1 \text{ atm} = 1 \times 10^5 \text{ Pa}$ and gravitational acceleration $g = 10 \text{ m/s}^2$)

Option 1:

9900 m

Option 2:

990 m

Option 3:

9000 m

Option 4:

99 m

Correct Answer:

990 m

Solution:

Absolute pressure at depth h in water

$$P = P_0 + h\rho g \quad P_0 = \text{atmospheric pressure}$$

$$P - P_0 = h\rho g$$

$$h = \frac{P - P_0}{\rho g}$$
$$= \frac{(100 - 1) \times 10^5}{10^3 \times 10}$$

$$h = 990 \text{ m}$$

Hence, the correct option is (2).

Q. 11 An electric heater supplies heat to a system at a rate of 100 W . If the system performs work at a rate of 75 J/s, then the rate at which internal energy increases will be :

Option 1:

75 W

Option 2:

25 W

Option 3:

100 W

Option 4:

125 W

Correct Answer:

25 W

Solution:

Rate of increase of internal energy Using first law of thermodynamics:

$$\frac{dU}{dt} = \text{Heat supplied} - \text{Work done} = 100 - 75 = 25 \text{ W}$$

Hence, the correct option is (2).

- Q. 12** A 100-turn closely wound circular coil of radius 5 cm has a magnetic field of $3.14 \times 10^{-3} \text{ T}$ at its centre. The current flowing through the coil, and the magnitude of the magnetic moment of this coil are, respectively :
(Take $\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$)

Option 1:

2.5 A, 20 A m²

Option 2:

2 A, 4Am²

Option 3:

2.5 A, 2 A m²

Option 4:

2 A, 10 A m²

Correct Answer:

2.5 A, 2 A m²

Solution:

$$B = \frac{\mu_0 NI}{2R} \Rightarrow I = \frac{2BR}{\mu_0 N}$$

$$\text{Current } I = \frac{2 \times 3.14 \times 10^{-3} \times 5 \times 10^{-2}}{4\pi \times 10^{-7} \times 100}$$

$$\text{Current } I = 2.5 \text{ A}$$

$$\text{Magnetic moment } M = NIA$$

$$= 100 \times 2.5 \times \pi r^2$$

$$= 100 \times 2.5 \times 3.14 \times 25 \times 10^{-4}$$

$$= 1.96$$

$$= 2 \text{ A} \times \text{m}^2$$

Hence, the correct option is (3).

Q. 13 In Young's double slit experiment, using monochromatic light of wavelength λ , the intensity of light at a point on the screen where the path difference is λ , is K units. The intensity of light at a point where the path difference is $\frac{\lambda}{3}$ will be :

Option 1:

K

Option 2:

2K

Option 3:

$\frac{K}{2}$

Option 4:

$\frac{K}{4}$

Correct Answer:

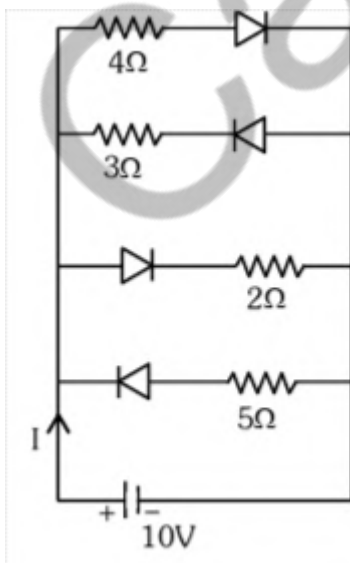
$\frac{K}{4}$

Solution:

$$\begin{aligned} I_R &= K \cos^2 \frac{\Delta\phi}{2} \\ &= K \cos^2 \left(\frac{120^\circ}{2} \right) \\ &= K \cos^2 60^\circ = \frac{K}{4} \end{aligned}$$

Hence, the correct option is (4).

Q. 14 The current I in the circuit shown below is :
(All diodes are ideal and identical)



Option 1:

$$\frac{1}{3} \text{ A}$$

Option 2:

$$\frac{15}{2} \text{ A}$$

Option 3:

$$\frac{5}{3} \text{ A}$$

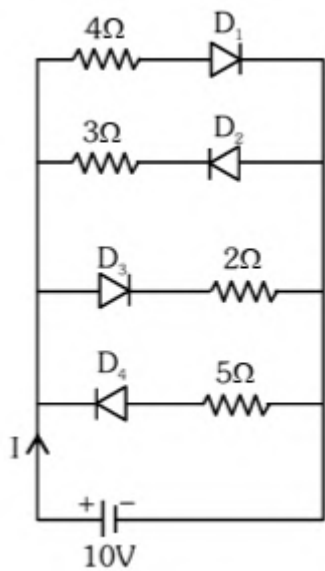
Option 4:

$$\frac{5}{9} \text{ A}$$

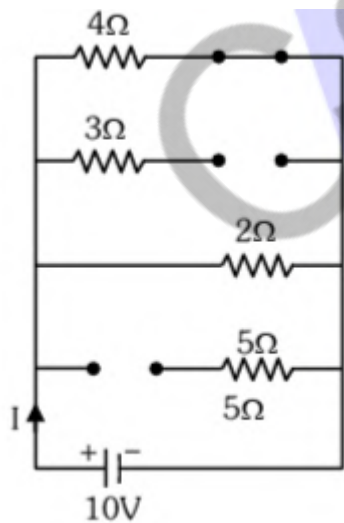
Correct Answer:

$$\frac{15}{2} \text{ A}$$

Solution:



Here D_1 and D_3 are F.B and D_2 and D_4 are R.B. then equivalent circuit



$$R_{eq} = \frac{4 \times 2}{4 + 2} = \frac{4}{3} \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{10}{4/3} = \frac{15}{2} \text{ A}$$

Hence, the correct option is (2).

Q. 15 In a concave lens, a ray of light emanating from the object parallel to the principal axis of the lens, after refraction :

Option 1:

passes through the second principal focus.

Option 2:

appears to diverge from the first principal focus.

Option 3:

passes through $2F$, which is the radius of curvature of the lens.

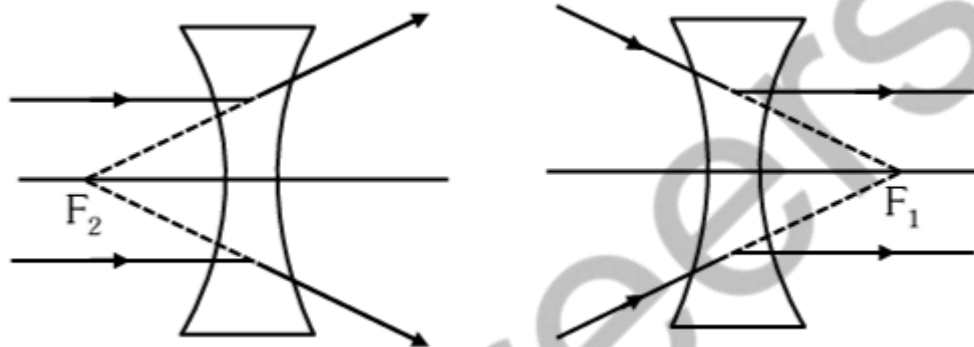
Option 4:

emerges parallel to the principal axis.

Correct Answer:

appears to diverge from the first principal focus.

Solution:



When incident light is parallel to principal axis, after refraction image will be at focus. This focus will be second principal focus (F_2).

In this question as lens is concave (diverging) after refraction light will diverge and appears to be coming from second principal focus (F_2).

Hence, the correct option is (2).

Q. 16 A galvanometer of resistance 100Ω gives full scale deflection for a current of 1 mA . It is converted into an ammeter of range $0 - 10\text{ A}$. The shunt required is :

Option 1:

0.001Ω

Option 2:

0.10Ω

Option 3:

$$1.0\Omega$$

Option 4:

$$0.01\Omega$$

Correct Answer:

$$0.01\Omega$$

Solution:

Given that

$$R_g = 100\Omega$$

$$I_g = 1 \text{ mA}$$

$$I = 10 \text{ A}$$

$$S = \frac{I_g R_g}{(I - I_g)} = \frac{10^{-3}(100)}{(10 - 10^{-3})}$$

Since $I_g \ll I$ so $I - I_g \simeq I$

$$S = 10^{-2} = 0.01\Omega$$

Hence, the correct option is (4).

Q. 17 In the first excited state of hydrogen atom, the energy of its electron is -3.4 eV . The radial distance of the electron from the hydrogen nucleus in this case is approximately :

(Take $1\text{eV} = 1.6 \times 10^{-19} \text{ J}$, $e = 1.6 \times 10^{-19} \text{ C}$ and $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$)

Option 1:

$$2.1 \times 10^{-8} \text{ m}$$

Option 2:

$$2.1 \times 10^{-10} \text{ m}$$

Option 3:

$$2.1 \times 10^{-11} \text{ m}$$

Option 4:

$$2.1 \times 10^{-9} \text{ m}$$

Correct Answer:

$$2.1 \times 10^{-10} \text{ m}$$

Solution:

$$E = \frac{kq_1q_2}{2r}$$

$$3.4 = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{2 \times r}$$

$$r = 2.1 \times 10^{-10} \text{ m}$$

Hence, the correct option is (2).

- Q. 18** The amount of work done to raise a mass ' m ' from the surface of the Earth to a height equal to the radius of the Earth ' R ', will be :

Option 1:

$$mg \frac{R}{2}$$

Option 2:

$$mgR$$

Option 3:

$$mg \frac{R}{4}$$

Option 4:

$$2 mg R$$

Correct Answer:

$$mg \frac{R}{2}$$

Solution:

Change in potential energy:

$$W = U_2 - U_1 = -\frac{GMm}{2R} - \left(-\frac{GMm}{R}\right)$$

$$W = \frac{GMm}{R} - \frac{GMm}{2R} = \frac{GMm}{2R}$$

$$g = \frac{GM}{R^2}$$

So,

$$W = \frac{mgR}{2}$$

Hence, the correct option is (1).

- Q. 19** An ac circuit contains a resistance of $1k\Omega$, a capacitor of $0.1\mu F$ and an inductor of 1 mH connected in series. The resonance frequency of the circuit is approximately :

Option 1:

$$15.9 \text{ kHz}$$

Option 2:

$$20.7 \text{ kHz}$$

Option 3:

$$10 \cdot 1 \text{ kHz}$$

Option 4:
13 · 5kHz

Correct Answer:
15.9 kHz

Solution:

For a series LCR circuit, resonance frequency is:

$$f = \frac{1}{2\pi\sqrt{LC}}$$
$$f = \frac{1}{2\pi\sqrt{10^{-3} \times 10^{-7}}} = \frac{1}{2\pi\sqrt{10^{-10}}} = \frac{1}{2\pi \times 10^{-5}}$$
$$f \approx \frac{1}{6.28 \times 10^{-5}} \approx 1.59 \times 10^4 \text{ Hz}$$
$$f \approx 15.9\text{kHz}$$

Hence, the correct option is (1).

Q. 20 Consider two uncharged capacitors of equal capacitance 200 pF . One of them is charged by a 100 V supply and disconnected. Now this capacitor is connected to the uncharged capacitor. The amount of electrostatic energy lost in the process is :

Option 1:
0.5 J

Option 2:
 $1.0 \times 10^{-6} \text{ J}$

Option 3:
 $0.5 \times 10^{-6} \text{ J}$

Option 4:
1.0 J

Correct Answer:
 $0.5 \times 10^{-6} \text{ J}$

Solution:

Initial energy:

$$U_i = \frac{1}{2}CV^2$$

Final voltage becomes $V/2$ (equal capacitors sharing charge)

Final energy:

$$U_f = \frac{1}{2}(2C)\left(\frac{V}{2}\right)^2 = \frac{1}{4}CV^2$$

Energy lost:

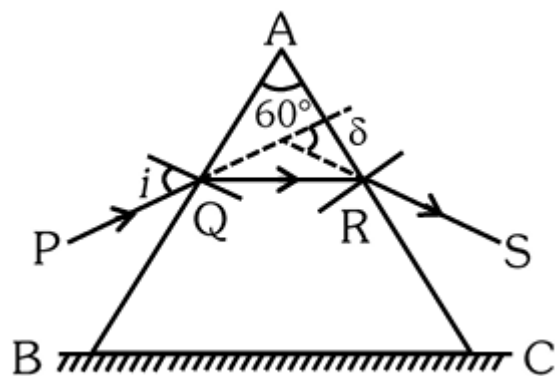
$$\Delta U = U_i - U_f = \frac{1}{2}CV^2 - \frac{1}{4}CV^2 = \frac{1}{4}CV^2$$

$$C = 200pF = 200 \times 10^{-12}$$

$$\begin{aligned}\Delta U &= \frac{1}{4} \times 200 \times 10^{-12} \times (100)^2 \\ &= 0.5 \times 10^{-6} \text{ J}\end{aligned}$$

Hence, the correct option is (3).

- Q. 21** A ray of monochromatic light is passing through an equilateral prism (ABC) as shown in the figure. The refracted ray (QR) is parallel to its base (BC) and the angle of incidence (i) is 50° . Then the angle of deviation (δ) is :



Option 1:
 45°

Option 2:
 55°

Option 3:
 35°

Option 4:
 40°

Correct Answer:
 40°

Solution:
For an equilateral prism, $A = 60^\circ$ since refracted ray is parallel to base so it is condition of minimum deviation.

$$r = \frac{A}{2} = \frac{60}{2} = 30^\circ$$

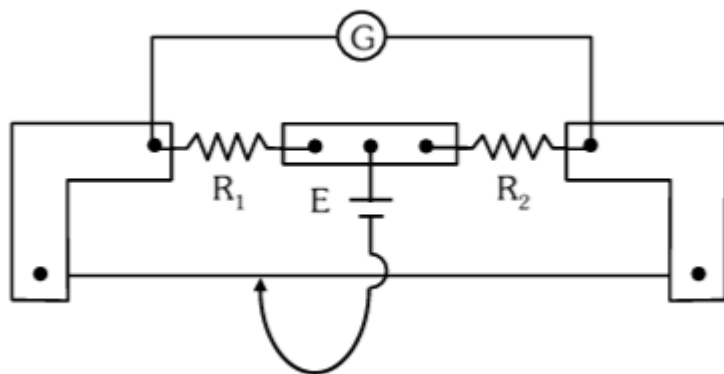
$$\text{and } i = e = 50^\circ$$

$$\delta = i + e - A$$

$$\begin{aligned}\text{But } &= 50 + 50 - 60 \\ &= 40^\circ\end{aligned}$$

Hence, the correct option is (4).

- Q. 22** In a metre bridge experiment (see figure), the positions of the cell, E, and galvanometer, G, are interchanged. We shall observe in the galvanometer :



Option 1:

Only the right-sided deflection

Option 2:

Only the left-sided deflection

Option 3:

There will be no deflection irrespective of the position of the jockey

Option 4:

Both right-sided and left-sided deflection and at balance point, no deflection

Correct Answer:

Both right-sided and left-sided deflection and at balance point, no deflection

Solution:

In the given circuit, it is not mentioned whether the meter bridge is balanced or not. So on interchanging the positions of battery and galvanometer, deflection may or may not be zero. In case of unbalanced condition, deflection may be left or right sided in galvanometer.

Hence, the correct option is (4).

- Q. 23** A flask contains argon and chlorine in the ratio of 2 : 1 by mass. The temperature of the mixture is 27°C . The ratio of root mean square speed of the molecules of the two gases $\left(\frac{v_{\text{rms}}^{\text{Ar}}}{v_{\text{rms}}^{\text{Cl}}}\right)$ is :
(Atomic mass of argon = 40.0u and molecular mass of chlorine 70.0u)

Option 1:

$$\frac{7}{4}$$

Option 2:

$$\frac{-2}{\sqrt{7}}$$

Option 3:

$$\frac{\sqrt{7}}{2}$$

Option 4:

$$\frac{7}{2}$$

Correct Answer:

$$\frac{\sqrt{7}}{2}$$

Solution:

$$\text{As } V_{\text{RMS}} = \sqrt{\frac{3RT}{M_w}}$$

Given temperature $\rightarrow$ constant

$$\text{Then } V_{\text{RMS}} \propto \frac{1}{\sqrt{M_w}}$$

$$\Rightarrow \frac{V_{\text{RMS}}^{\text{Ar}}}{V_{\text{RMS}}^{\text{C}}} = \sqrt{\frac{M_{\text{wQ}}}{M_{\text{wAr}}}} = \sqrt{\frac{70}{40}} = \frac{\sqrt{7}}{2}$$

Hence, the correct option (3)

Q. 24 Two statements are given below :

A. When the forward bias voltage across a p-n junction diode increases above a certain threshold voltage, the diode current increases significantly.

B. This current is called reverse saturation current.

Choose the **correct** answer from the options given below :

Option 1:

Both Statements A and B are true

Option 2:

Both Statements A and B are false

Option 3:

Statement *A* is true, but Statement *B* is false

Option 4:

Statement A is false, but Statement B is true

Correct Answer:

Statement *A* is true, but Statement *B* is false

Solution:

Statement A: When forward bias voltage increases above a certain threshold, the diode current increases significantly. (True)

Statement B: This current is called reverse saturation current. (False)

Analysis: Statement A is true, but Statement B is false.

Hence, the correct option is (3).

Q. 25 For a travelling harmonic wave $y(x, t) = 2.0 \cos 2\pi(10t - 0.0080x + 0.35)$, where x and y are in cm and t in s. The phase difference between oscillatory motion of two points separated by a distance of 0.5m is :

Option 1:
 $0.08\pi\text{rad}$

Option 2:
 $0.008\pi\text{rad}$

Option 3:
 $0.8\pi\text{rad}$

Option 4:
 $8\pi\text{rad}$

Correct Answer:
 $0.8\pi\text{rad}$

Solution:

Wave Number (k) : From the equation, $k = 2\pi \times 0.0080$

Phase Difference: $\Delta\phi = k\Delta x = (2\pi \times 0.0080) \times 0.5 = 0.008\pi\text{rad}$

Hence, the correct option is (3).

Q. 26 A rectangular wire loop of sides 8 cm and 3 cm with a small cut, is moving out of a region of uniform magnetic field of magnitude 0.3 T directed normal to the plane of the loop. The emf developed across the cut, if the velocity of the loop is 2 cm s^{-1} , in a direction normal to the shorter side of the loop, will be :

Option 1:
 $1.8 \times 10^{-4}\text{ volt}$

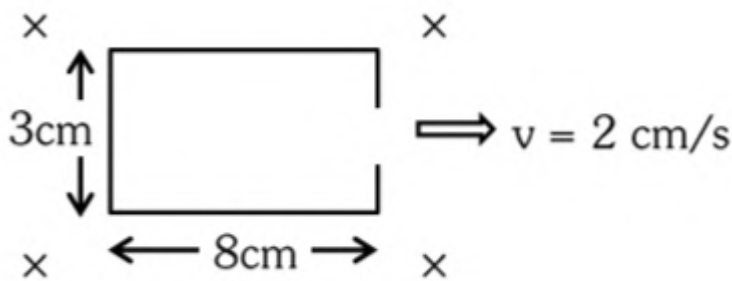
Option 2:
 $1.3 \times 10^{-4}\text{ volt}$

Option 3:
 $1.2 \times 10^{-4}\text{ volt}$

Option 4:
 $4.8 \times 10^{-4}\text{ volt}$

Correct Answer:
 $1.8 \times 10^{-4}\text{ volt}$

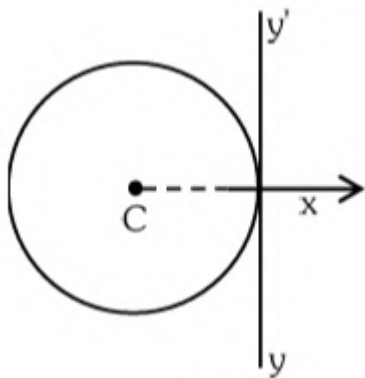
Solution:



$$e = Blv = 0.3 \times 0.03 \times 0.02 = 1.8 \times 10^{-4} \text{ V}$$

Hence, the correct option is (1).

- Q. 27** A thin wire of length ' L ' and linear mass density ' m ' is bent into a circular ring (in $x - y$ plane) with centre ' C ' as shown in figure. The moment of inertia of the ring about an axis yy' will be :



Option 1:

$$\frac{3 mL^3}{8\pi}$$

Option 2:

$$\frac{3 mL^2}{8\pi^2}$$

Option 3:

$$\frac{3 mL^3}{8\pi^2}$$

Option 4:

$$\frac{3 mL^2}{8}$$

Correct Answer:

$$\frac{3 mL^3}{8\pi^2}$$

Solution:

Let mass of wire is M

length = L

given m = Linear mass density

$$m = \frac{M}{L} \quad L = 2\pi R$$

$$M = mL \quad R = \frac{L}{2\pi}$$

$$I_{yy'} = \frac{MR^2}{2} + MR^2$$

$$= \frac{3}{2}MR^2 = \frac{3}{2}(mL) \left(\frac{L^2}{4\pi^2} \right)$$

$$I_{yy'} = \frac{3}{8} \frac{mL^3}{\pi^2}$$

Hence, the correct option is (3).

Q. 28 A resistor is connected to a battery of 12 V emf and internal resistance 2Ω . If the current in the circuit is 0.6 A , the terminal voltage of the battery is :

Option 1:

12 V

Option 2:

1.2 V

Option 3:

10 V

Option 4:

10.8 V

Correct Answer:

10.8 V

Solution:

Given that:

$$\text{EMF} = 12 \text{ V}$$

$$r = 2\Omega$$

$$I = 0.6 \text{ A}$$

So terminal voltage

$$V = E - Ir$$

$$= 12 - (.6)^2$$

$$V = 10.8 \text{ volt}$$

Hence, the correct option is (4).

Q. 29 When a ruler falls vertically, 5 different person catch it with different reaction times.
($g = 9.8 \text{ m s}^{-2}$)

- A. Person A has reaction time of 0.20 s .
- B. Person B has reaction time of 0.22 s .
- C. Person C has reaction time of 0.18 s .
- D. Person D has reaction time of 0.19 s .
- E. Person E has reaction time of 0.21 s .

What is the **correct** order of the distance travelled by the ruler for each person?

Option 1:

$B > E > A > C > D$

Option 2:

$C > D > A > E > B$

Option 3:

$C > D > A > B > E$

Option 4:

$B > E > A > D > C$

Correct Answer:

$B > E > A > D > C$

Solution:

Distance fallen: $s = \frac{1}{2}gt^2$

Distance $\propto t^2$, so compare reaction times:

Largest time $\rightarrow$ largest distance:

Hence, the correct option is (4).

Q. 30 The angular speed of a flywheel is increased from 600 rpm to 1200 rpm in 10 s . The number of revolutions completed by the flywheel during this times is:

Option 1:

600

Option 2:

300

Option 3:

900

Option 4:

150

Correct Answer:

150

Solution:

$$\omega_1 = 600\text{rpm} = 20\pi\text{rad/sec}$$

$$\omega_f = 1200\text{rpm} = 40\pi\text{rad/sec}$$

$$t = 10\text{sec}$$

$$\theta = \left(\frac{\omega_i + \omega_f}{2} \right) t$$

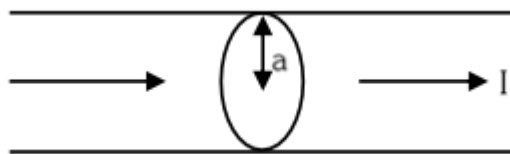
$$\theta = \left(\frac{40\pi + 20\pi}{2} \right) 10$$

$$\theta = 300\pi$$

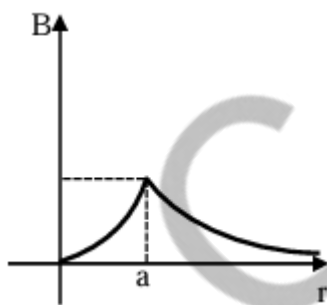
$$\text{number of revolution} = \frac{300\pi}{2\pi} = 150 \text{ rotation}$$

Hence, the correct option is (4).

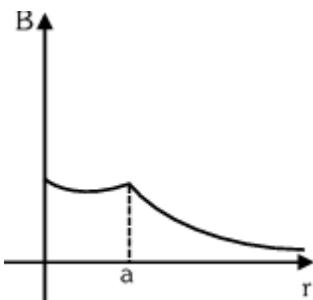
- Q. 31** The figure given below shows a long straight solid wire of circular cross-section of radius 'a' carrying steady current I . The current I is uniformly distributed across its cross-section. The plot which correctly represents the variation of magnetic field (B) with distance (r) from the axis of the conductor in the region is :



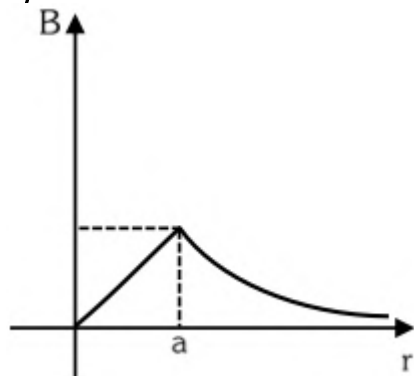
Option 1:



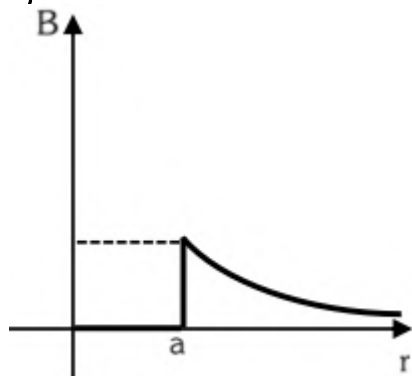
Option 2:



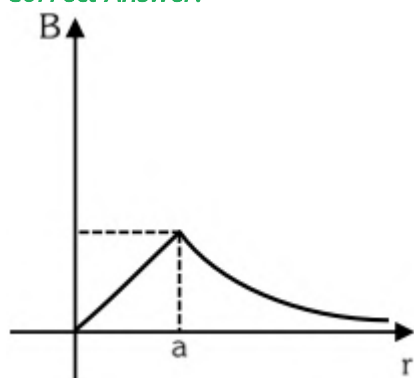
Option 3:



Option 4:



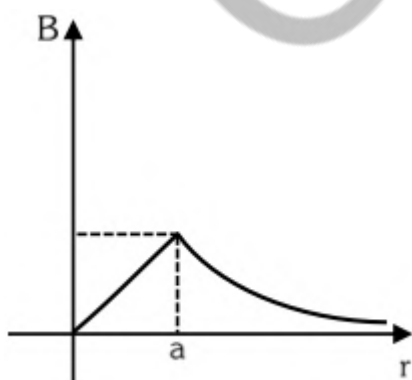
Correct Answer:



Solution:

$$B_{\text{in}} = \frac{\mu_0 I r}{2\pi R^2} \propto r$$

$$B_{\text{out}} = \frac{\mu_0 I}{2\pi r} \propto \frac{1}{r}$$



Hence, the correct option is (3).

- Q. 32** Four statements are given (A is mass number) :
- A. The volume of a nucleus is proportional to $A^{1/3}$.
 - B. The volume of a nucleus is proportional to A.
 - C. The difference in mass of an atom and its nucleus is called the mass defect.
 - D. The difference in mass of a nucleus and its constituents is called the mass defect.
- Choose the **correct** answer from the options given below :

Option 1:

A and D are true, but B and C are false

Option 2:

B and D are true, but A and C are false

Option 3:

B and C are true, but A and D are false

Option 4:

A and C are true, but B and D are false

Correct Answer:

B and D are true, but A and C are false

Solution:

Correct statements

A: Volume $\propto R^3$, and $R \propto A^{1/3} \rightarrow \text{Volume} \propto A \rightarrow \text{false}$

B: Volume $\propto A \rightarrow \text{true}$

C: Mass defect = difference between atom and nucleus $\rightarrow \text{false}$

D: Mass defect = difference between nucleus and its nucleons $\rightarrow \text{true}$

Hence, the correct option is (2).

-
- Q. 33** Savitha, a XI standard student, while conducting an experiment to determine the effective length of a simple pendulum L, notes down the data of time taken to complete 30 oscillations as 60 s and hence calculates the length of the simple pendulum as :
- (Take $\pi^2 = 9.8$, and $g = 9.8 \text{ m/s}^2$)

Option 1:

0.75 m

Option 2:

1 m

Option 3:

1.5 m

Option 4:

2 m

Correct Answer:

1 m

Solution:

$$T = \frac{\text{Total Time}}{\text{Number of Oscillations}} = \frac{60 \text{ s}}{30} = 2 \text{ s}$$

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$T^2 = 4\pi^2 \left(\frac{L}{g} \right)$$

$$L = \frac{T^2 \cdot g}{4\pi^2}$$

$$L = \frac{2^2 \cdot 9.8}{4 \cdot 9.8} = \frac{4 \cdot 9.8}{4 \cdot 9.8} = 1 \text{ m}$$

Hence, the correct option is (2).

Q. 34 In a vernier callipers, 200 VSD comerde 16 MSD (each division $\bar{p}$ f length 1 mm). The least count of the vernier callipers is :

Option 1:

0.2 cm

Option 2:

0.1 cm

Option 3:

0.02 cm

Option 4:

0.01 cm

Correct Answer:

0.02 cm

Solution:

$$20\text{VSD} = 16\text{MSD}$$

$$\text{VSD} = \frac{16}{20}(1 \text{ mm})$$

$$= 0.8 \text{ mm}$$

$$\text{LC} = \text{MSD} - \text{VSD}$$

$$= 1 - 0.8$$

$$= 0.2 \text{ mm}$$

$$= 0.02 \text{ cm}$$

Hence, the correct option is (3).

- Q. 35** Each side of a metallic cube of mass 5.580 kg is measured to be 9.0 cm . Keeping the significant figures in view, the density of the material of the cube can be best expressed as $X \times 10^3 \text{ kg m}^{-3}$, where the value of X is :

Option 1:

7.654

Option 2:

7 . 7

Option 3:

7.65

Option 4:

7 . 6

Correct Answer:

7 . 7

Solution:

$$\text{Density} = \frac{\text{mass}}{\text{volume}} = \frac{5.580 \text{ kg}}{(9.0 \times 10^{-2})^3}$$

$$= 7.654 \times 10^3 \text{ kg/m}^3$$

Rounding off to 2 significant digits

$$\text{Result} = 7.7 \times 10^3$$

$$\Rightarrow x = 7.7$$

Hence, the correct option is (2).

- Q. 36** In interference and diffraction, the light energy is redistributed. If it reduces in one region, producing a dark fringe, it increases in another region, producing a bright fringe.
- A. As there is no gain or loss of energy, these phenomena are consistent with the principle of conservation of energy.
- B. Diffraction and interference are characteristics exhibited only by light waves.
- Choose the **correct** answer from the options given below :

Option 1:

A is false, but B is true

Option 2:

A is true, but B is false

Option 3:

A is true and B is also true

Option 4:

Both A and B are false

Correct Answer:

A is true, but B is false

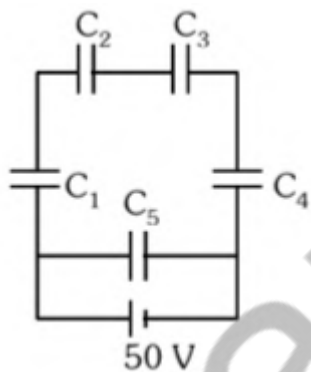
Solution:

Statement A: Energy is redistributed (bright & dark fringes), no net loss $\rightarrow$ consistent with conservation of energy $\rightarrow$ True

Statement B: Interference and diffraction are wave phenomena, but not only for light (also seen in sound, water waves, etc.) $\rightarrow$ False

Hence, the correct option is (2).

Q. 37 Five capacitors of capacitances $C_1 = C_2 = C_3 = C_4 = 40\mu\text{ F}$ and $C_5 = 2 \cdot 5\mu\text{ F}$ are connected as shown, along with a battery of 50 V. The equivalent capacitance and the charges on each capacitor respectively are :



Option 1:

$4\mu\text{ F}$, $250\mu\text{C}$ on C_1 to C_4 and $125\mu\text{C}$ on C_5

Option 2:

$5\mu\text{ F}$, $250\mu\text{C}$ on all capacitors

Option 3:

$5\mu\text{ F}$, $125\mu\text{C}$ on C_1 to C_4 and $25\mu\text{C}$ on C_5

Option 4:

$5\mu\text{ F}$, $125\mu\text{C}$ on all capacitors

Correct Answer:

$5\mu\text{ F}$, $125\mu\text{C}$ on all capacitors

Solution:

Top branch: C_2 and C_3 in series

$$\frac{1}{C_{23}} = \frac{1}{10} + \frac{1}{10} \Rightarrow C_{23} = 5\mu\text{F}$$

Now C_1 , C_{23} , C_4 are in series:

$$\frac{1}{C_{eq1}} = \frac{1}{10} + \frac{1}{5} + \frac{1}{10} = \frac{1+2+1}{10} = \frac{4}{10} \Rightarrow C_{eq1} = 2.5\mu\text{F}$$

Parallel with C_5

$$C_{total} = C_{eq1} + C_5 = 2.5 + 2.5 = 5\mu\text{F}$$

Total charge

$$Q = C_{total} V = 5 \times 50 = 250\mu\text{C}$$

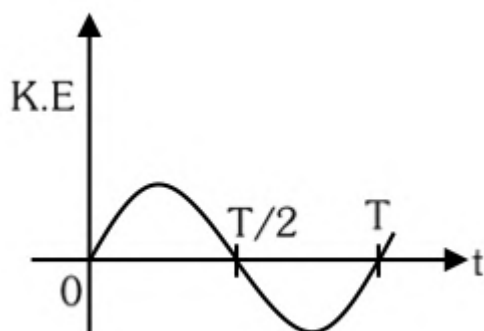
Each branch capacitance = $2.5\mu\text{ F}$ $\rightarrow$ equal charge sharing

$$Q_{\text{each branch}} = 125\mu\text{C}$$

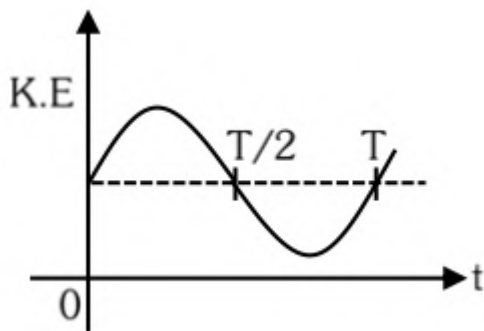
Hence, the correct option is (4).

Q. 38 For a simple pendulum, having time period T , the variation of kinetic energy (K.E.) with time (t) is represented by :

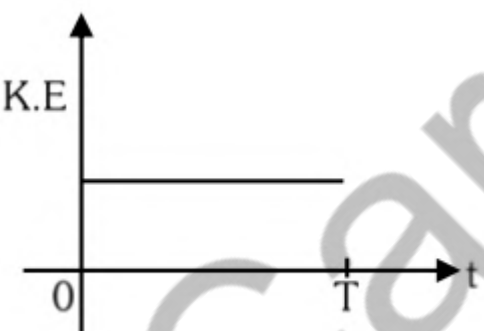
Option 1:



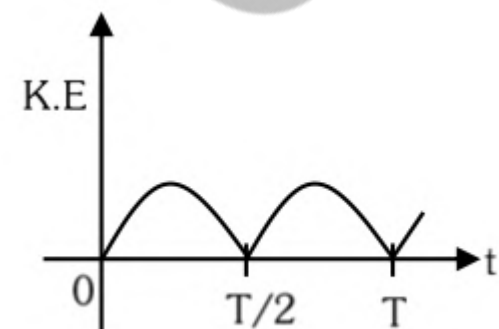
Option 2:



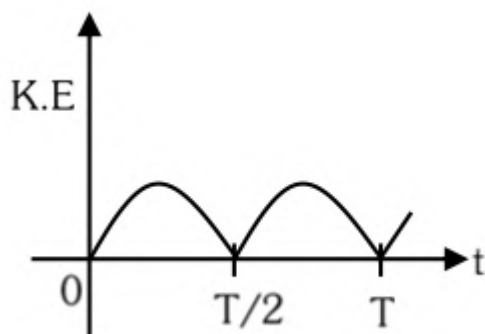
Option 3:



Option 4:



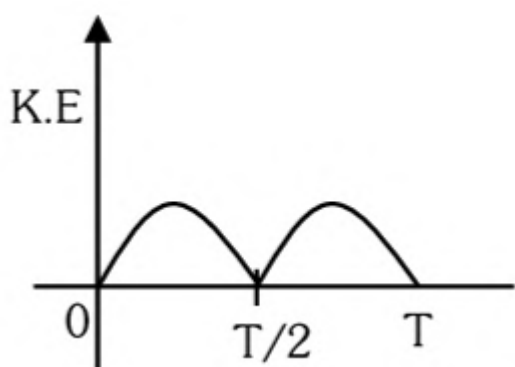
Correct Answer:



Solution:

$$K.E = \frac{1}{2} m \omega^2 A^2 \cos^2 \omega t$$

oscillation starts from extreme position



Hence, the correct answer is option (4).

Q. 39 A room heater is rated 400 W, 220 V. If the supply voltage drops to 200 V, what will be the power consumed (approximately)?

Option 1:

400 W

Option 2:

121 W

Option 3:

331 W

Option 4:

200 W

Correct Answer:

331 W

Solution:

Given that:

Printed voltage = 220 V

Printed Power = 400 W

Supply voltage = 200 V

Resistance of heater $R_H = \frac{(220)^2}{400}$

So Actual power gained

$$\begin{aligned} P' &= \frac{V_s^2}{R_H} = \frac{(200)^2}{(220)^2} (400) \\ &= 330.57 \text{ W} \\ &\simeq 331 \text{ W} \end{aligned}$$

Hence, the correct option is (3) 331 W.

Q. 40 The peak value of an alternating current is 5 A and frequency is 60 Hz . How long will the current, starting from zero, take to reach the peak value ?

Option 1:

$$\frac{1}{120} \text{ s}$$

Option 2:

$$\frac{1}{240} \text{ s}$$

Option 3:

$$\frac{1}{30} \text{ s}$$

Option 4:

$$\frac{1}{60} \text{ s}$$

Correct Answer:

$$\frac{1}{240} \text{ s}$$

Solution:

Time to reach peak current

For AC:

$$i = I_0 \sin(\omega t)$$

Peak occurs when $\omega t = \frac{\pi}{2}$

$$t = \frac{\pi/2}{\omega} = \frac{\pi/2}{2\pi f} = \frac{1}{4f}$$

$$t = \frac{1}{4 \times 60} = \frac{1}{240} \text{ s}$$

Hence, the correct option is (2).

- Q. 41** Which of the following statements are correct?
- A. Inside a conductor, the electrostatic field is zero.
 - B. Electric field at the surface of a charged conductor does not depend on its surface charge density.
 - C. The interior of a charged conductor can have no excess charge in the static situation.
 - D. At the surface of a charged conductor, the electrostatic field must be normal to the surface at every point.
 - E. The electrostatic potential is zero everywhere inside a charged conductor.
- Choose the **correct** answer from the options given below :

Option 1:

A, C and D only

Option 2:

A, C and E only

Option 3:

C, D and E only

Option 4:

A, B and D only

Correct Answer:

A, C and D only

Solution:

- A. Inside a conductor, the electrostatic field is zero. (True)
- B. Electric field at the surface does not depend on surface charge density. (False: $E = \sigma/\epsilon_0$)
- C. The interior of a charged conductor can have no excess charge in the static situation. (True)
- D. At the surface, the electrostatic field must be normal to the surface at every point. (True)
- E. The electrostatic potential is zero everywhere inside a charged conductor. (False)

Analysis: Statements A, C, and D are correct.

Hence, the correct option is (1).

Q. 42 Match List I with List II :

List I (Electromagnetic wave)		List II (Production)	
A.	Microwave	I.	Electrons in atoms emit light when they move from a higher energy level to a lower energy level
B.	Visible light	II.	Radioactive decay of nucleus
C.	Gamma rays	III.	Vibration of atoms and molecules
D.	Infra-red rays	IV.	Klystron valve or magnetron valve

Choose the **correct** answer from the options given below:

Option 1:

A-IV, B-III, C-II, D-I

Option 2:

A-III, B-IV, C-I, D-IX

Option 3:

A-III, B-I, C-II, D-IV

Option 4:

A-IV, B-I, C-II, D-III

Correct Answer:

A-IV, B-I, C-II, D-III

Solution:

A. Microwave: Produced by Klystron valve or magnetron valve

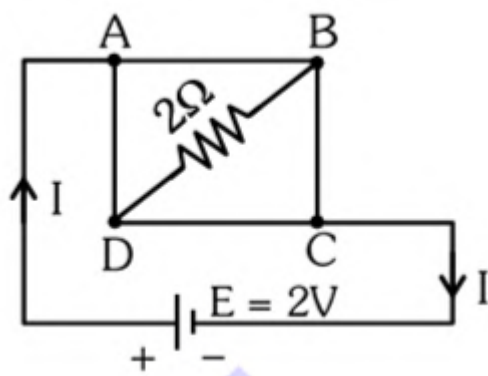
B. Visible light: Electrons in atoms emit light moving to lower energy levels

C. Gamma rays: Radioactive decay of nucleus

D. Infra-red rays: Vibration of atoms and molecules

Hence, the correct option is (4).

- Q. 43** A uniform metallic wire having resistance 4Ω is bent to form a square loop (ABCD) (see figure). A resistance of 2Ω is connected between points B and D and a battery of 2 V is connected across points A and C as shown in the figure. Now the value of current (I) is :



Option 1:

4A

Option 2:

4.5A

Option 3:

8A

Option 4:

2A

Correct Answer:

2A

Solution:

Resistance of each wire of

$$\text{W.S.B.} = \frac{4}{4} = 1\Omega$$

This is balance W.S.B.

$$\text{So, } R_{\text{Net}} = 1\Omega$$

So, current

$$I = \frac{E}{R_{\text{Net}}} = \frac{2}{1} = 2 \text{ A}$$

Hence, the correct option is (4).

- Q. 44** An unknown nucleus has nuclear density $2.29 \times 10^{17} \text{ kg/m}^3$ and mass of $19.926 \times 10^{-27} \text{ kg}$. Its mass number A is approximately :
(Take $R_0 = 1.2 \times 10^{-15} \text{ m}$, $4\pi = 12.56$)

Option 1:

16

Option 2:

20

Option 3:

12

Option 4:

19

Correct Answer:

12

Solution:

$$\text{Density } \rho = \frac{\text{Mass}}{\text{Volume}} = \frac{M}{\frac{4}{3}\pi R^3}$$

$$\text{Nuclear Radius: } R = R_0 A^{1/3}, \text{ so } R^3 = R_0^3 A$$

$$A = \frac{3M}{4\pi R_0^3 \rho} = \frac{3 \times 19.926 \times 10^{-27}}{12.56 \times (1.2 \times 10^{-15})^3 \times 2.29 \times 10^{17}}$$

$$A \approx \frac{59.778 \times 10^{-27}}{12.56 \times 1.728 \times 10^{-45} \times 2.29 \times 10^{17}} \approx 12$$

Hence, the correct option is (3).

Q. 45 Match **List I** with **List II** :

List I		List II	
A.	Young's Modulus	I.	$\frac{\Delta d}{\Delta L} \left(\frac{L}{d} \right)$
B.	Compressibility	II.	$\frac{FL}{A(\Delta L)}$
C.	Bulk Modulus	III.	$-\frac{1}{\Delta P} \left(\frac{\Delta V}{V} \right)$
D.	Poisson's Ratio	IV.	$-P \left(\frac{V}{\Delta V} \right)$

Choose the **correct** answer from the options given below:

Option 1:

A-III, B-II, C-I, D-IV

Option 2:

A-II, B-III, C-IV, D-I

Option 3:

A-I, B-IV, C-III, D-II

Option 4:

A-IV, B-I, C-II, D-III

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

A. Young's Modulus $Y = \frac{FL}{A(\Delta L)}$

B. Compressibility $\text{Compressibility} = -\frac{1}{\Delta P} \left(\frac{\Delta V}{V} \right)$

C. Bulk Modulus $K = -P \left(\frac{V}{\Delta V} \right)$

D. Poisson's Ratio

Poisson's ratio $= \frac{\Delta d}{\Delta L} \left(\frac{L}{d} \right)$

Hence, the correct option is (2).

Chemistry

Q. 1 The correct statement with regard to the secondary structure of DNA/RNA is :

Option 1:

DNA possesses a double strand helix structure and contains thymine as one of the four bases.

Option 2:

DNA possesses a single strand helix structure and contains uracil as one of the four bases.

Option 3:

RNA possesses a double strand helix structure and contains uracil as one of the four bases.

Option 4:

RNA possesses a single strand helix structure and contains thymine as one of the four bases.

Correct Answer:

DNA possesses a double strand helix structure and contains thymine as one of the four bases.

Solution:

DNA is **double-stranded (double helix)**

Bases: A, T, G, C → **thymine present**

Hence, the correct answer is option (1)

Q. 2 Match List - I with List - II

	List - I (Quantum Numbers)			List - II (Orbital)
	"n"	'l'		
A	2	1	I	3d
B	4	0	II	2p
C	5	3	III	4s
D	3	2	IV	5f

Choose the correct answer from the options given below

Option 1:

A-II, B-III, C-IV, D-I

Option 2:

A-I, B-II, C-III, D-IV

Option 3:

A-II, B-III, C-I, D-IV

Option 4:

A-IV, B-II, C-III, D-I

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

The orbital is designated as nl , where n is the principal quantum number and l corresponds to the subshell ($0 = s, 1 = p, 2 = d, 3 = f$).

A. $n = 2, l = 1 : 2p$ (A – II)

B. $n = 4, l = 0 : 4s$ (B-III)

C. $n = 5, l = 3 : 5f$ (C – IV)

D. $n = 3, l = 2 : 3d$ (D – I)

Hence, the correct answer is option (1)

Q. 3 During Lassaigne's test, the elements present in an organic compound are converted from :

Option 1:

ionic form to ionic form

Option 2:

covalent form to ionic form

Option 3:

ionic form to covalent form

Option 4:

covalent form to covalent form

Correct Answer:

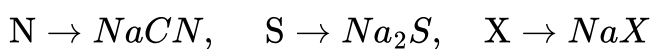
covalent form to ionic form

Solution:

In organic compounds, elements like nitrogen, sulphur and halogens are present in covalent form, so they are not easily detectable by ordinary chemical tests.

During Lassaigne's test, the organic compound is fused with sodium metal. Sodium is highly reactive and converts these covalently bonded elements into ionic (water-soluble) compounds, which can be easily detected.

Examples:

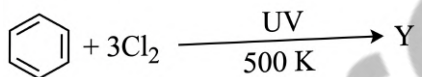
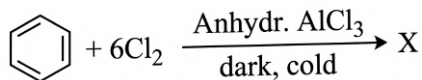


Thus, in Lassaigne's test, elements are converted from:

Covalent form $\rightarrow$ Ionic form

Hence, the correct answer is option (2)

Q. 4 The number of chlorine atoms present in the organic products X and Y of the following reactions, respectively, are :



Option 1:

3 and 3

Option 2:

6 and 6

Option 3:

6 and 3

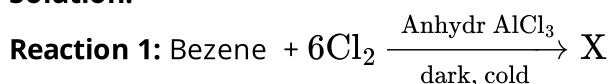
Option 4:

3 and 6

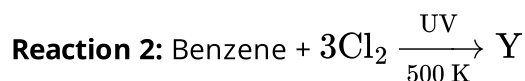
Correct Answer:

6 and 6

Solution:



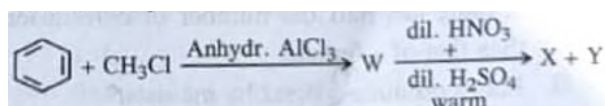
Conditions: Lewis acid (AlCl_3) + dark $\rightarrow$ Electrophilic substitution
Benzene undergoes complete chlorination $\rightarrow$ Hexachlorobenzene (C_6Cl_6)
Number of Cl atoms in **X** = **6**



High temperature ($\sim 500\text{ K}$) $\rightarrow$ Addition reaction
Forms Benzene hexachloride (BHC/ $\text{C}_6\text{H}_6\text{Cl}_6$)
Number of Cl atoms in **Y** = **6**

Hence, the correct answer is option (2)

Q. 5 Two products X and Y are formed in the following reaction sequence.



The suitable method that can be used for the separation of products X and Y is :

Option 1:

Continuous extraction

Option 2:

Differential extraction

Option 3:

Sublimation

Option 4:

Fractional distillation

Correct Answer:

Fractional distillation

Solution:

Benzene + CH_3Cl with $\text{AlCl}_3 \rightarrow$ Toluene (W).

Toluene + nitrating mixture ($\text{HNO}_3 + \text{H}_2\text{SO}_4$) $\rightarrow$ o-nitrotoluene (X) and p-nitrotoluene (Y).

Separation Method:

X and Y are structural isomers with different boiling points. While steam distillation is commonly used for similar separations (like o- and p-nitrophenols), fractional distillation is the most suitable general industrial method for separating these two components based on their boiling point difference.

Hence, the correct answer is option (4)

Q. 6 Identify the correct statement about ClF_3 from the following options :

Option 1:

It has T-shaped geometry with three lone pairs on Cl atom.

Option 2:

It has T-shaped geometry with two lone pairs on Cl atom.

Option 3:

It has a trigonal pyramidal geometry with two lone pairs on Cl atom.

Option 4:

It has a planar trigonal geometry with two lone pairs on Cl atom.

Correct Answer:

It has T-shaped geometry with two lone pairs on Cl atom.

Solution:

- Central atom: Cl (7 valence electrons)
- Bonded atoms: 3 F atoms
- Total electron pairs = 5
- Electron geometry → Trigonal bipyramidal
- Two lone pairs occupy equatorial positions (to minimise repulsion)
- T-shaped geometry
- Lone pairs = 2
- Bond pairs = 3

It has T-shaped geometry with two lone pairs on the Cl atom

Hence, the correct answer is option (2)

Q. 7 The functional group that can be identified through phthalein dye test is :

Option 1:

Carboxylic acid

Option 2:

Alcohol

Option 3:

Aldehyde

Option 4:

Phenolic

Correct Answer:

Phenolic

Solution:

The phthalein dye test is used to identify the phenolic group. When a phenol is heated with phthalic anhydride in the presence of concentrated H_2SO_4 , it forms a phthalein dye (like phenolphthalein) which turns pink/red in alkaline solution.

Hence, the correct answer is option (4)

Q. 8 A solution of copper sulphate is electrolysed for 10 minutes with a current of 1.5 amperes. The mass of copper deposited at cathode is :
(Given : Molar mass of Cu = 63 g mol⁻¹; 1 F = 96487 C mol⁻¹)

Option 1:
0.2938 g

Option 2:
0.5876 g

Option 3:
2 · 4036 g

Option 4:
1.7018 g

Correct Answer:
0.2938 g

Solution:
79: Total charge (Q) :

$$Q = I \times t = 1.5 \text{ A} \times (10 \times 60) \text{ s} = 900 \text{ C}$$

The reaction at the cathode is $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}(s)$.

$$\text{Mass deposited } (m) = \frac{\text{Molar mass} \times Q}{n \times F}$$

$$m = \frac{63 \times 900}{2 \times 96487} \approx \mathbf{0.2938 \text{ g}}$$

Hence, the correct answer is option (1)

Q. 9 Match List - I with List - II

List - I (Transition metal /compound /complex)	List - II (Catalytic Role)
A. V ₂ O ₅	I. Preparation of ammonia from N ₂ /H ₂ mixture
B. Fe	II. Polymerisation of alkynes
C. PdCl ₂	III. Preparation of H ₂ SO ₄ from SO ₂
D. Ni complex	IV. Oxidation of ethyne to ethanal

Choose the correct answer from the options given below:

Option 1:
A-III, B-IV, C-I, D-II

Option 2:

A-III, B-I, C-IV, D-II

Option 3:

A-II, B-I, C-IV, D-III

Option 4:

A-IV, B-I, C-III, D-II

Correct Answer:

A-III, B-I, C-IV, D-II

Solution:

- A. V_2O_5 : Used in the Contact Process for the preparation of H_2SO_4 from SO_2 . (A-III)
- B. Fe (Fine powder): Used in the Haber's Process for the preparation of ammonia from N_2 and H_2 . (B-I)
- C. $PdCl_2$: Specifically used in the Wacker Process for the oxidation of ethyne (or ethene) to ethanal. (C-IV)
- D. Ni complex: Specifically organometallic Nickel complexes (like Zeigler-Natta relatives) are used in the polymerisation of alkynes. (D-II)

Hence, the correct answer is option (2)

Q. 10 Match List I with List II

List - I (Complex/ion)	List -II (Shape/geometry)
A. $[Pt(Cl_2)(NH_3)_2]$	I. Octahedral
B. $[Co(NH_3)_6]Cl_3$	II. Trigonal bipyramidal
C. $[NiCl_4]^{2-}$	III. Square planar
D. $[Fe(CO)_5]$	IV. Tetrahedral

Choose the **correct** answer from the options given below:

Option 1:

A-I, B-III, C-IV, D-II

Option 2:

A-III, B-IV, C-I, D-IIX

Option 3:

A-III, B-I, C-IV, D-IV-

Option 4:

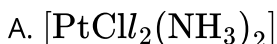
A-IV, B-I, C-III, D-IIX

Correct Answer:

A-III, B-I, C-IV, D-IV-

Solution:

(3) A-III, B-I, C-IV, D-II



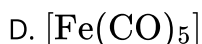
Pt^{2+} is a d^8 system, which typically forms square planar complexes $\rightarrow$ III



Co^{3+} with 6 ligands gives coordination number 6, so octahedral $\rightarrow$ I



Ni^{2+} with weak field ligand Cl^- forms a tetrahedral complex $\rightarrow$ IV



This is a classic trigonal bipyramidal structure $\rightarrow$ II

Hence, the correct answer is option (3)

Q. 11 Identify the incorrect statement from the following:

Option 1:

The largest and the smallest species among Mg , Mg^{2+} , Al and Al^{3+} are Al and Mg^{2+} , respectively.

Option 2:

The IUPAC name of the element with atomic number 107 is Unnilseptium.

Option 3:

The similarity in behaviour of Li with Mg is referred to as 'diagonal relationship'.

Option 4:

The oxidation state and covalency of Al in $[\text{AlCl}(\text{H}_2\text{O})_5]^{2+}$ are 3 and 6 respectively.

Correct Answer:

The largest and the smallest species among Mg , Mg^{2+} , Al and Al^{3+} are Al and Mg^{2+} , respectively.

Solution:

Statement (1) is Incorrect: Between Mg and Al , Mg is larger because atomic size decreases across a period. Between Mg^{2+} and Al^{3+} , Al^{3+} is the smallest because it has more protons pulling the same number of electrons. Therefore, the largest is Mg and the smallest is Al^{3+} .

Statement (2) is Correct: Atomic number 107 (1 = Un, 0=nil, 7=sept) is Unnilseptium.

Statement (3) is Correct: Li and Mg show diagonal relationship due to similar ionic sizes and electronegativities.

Statement (4) is Correct: In $[\text{AlCl}(\text{H}_2\text{O})_5]^{2+}$, let x be the oxidation state of Al :

$x + (-1) + 5(0) = +2 \Rightarrow x = +3$. The coordination number (covalency) is 6 (1 Cl + 5 H_2O ligands).

Hence, the correct answer is option (1)

Q. 12 Given below is an expression for the rate constant of a first order reaction occurring at a certain temperature, $T(K)$.

$$\ln k = 14.34 - \frac{1.25 \times 10^4}{T}$$

The energy of activation in kcal mol^{-1} for the reaction is :
(Given : k in s^{-1} , $R = 1.987 \text{ cal mol}^{-1} \text{ K}^{-1}$)

Option 1:

12.42

Option 2:

18.63

Option 3:

14.34

Option 4:

24.84

Correct Answer:

24.84

Solution:

$$52. \ln k = \ln A - \frac{E_a}{RT}$$

$$\frac{E_a}{R} = 1.25 \times 10^4$$

$$E_a = (1.25 \times 10^4) \times R$$

$$R = 1.987 \text{ cal mol}^{-1} \text{ K}^{-1}$$

$$E_a = 12500 \times 1.987 = 24837.5 \text{ cal mol}^{-1}$$

$$E_a = \frac{24837.5}{1000} \approx 24.84 \text{ kcal mol}^{-1}$$

Hence, the correct answer is option (4)

Q. 13 Phenolphthalein is used as an indicator for the titration of sodium hydroxide solution against a standard solution of oxalic acid. The colour change that is observed at an alkaline pH close to the equivalence point during this titration is :

Option 1:

pink to colourless

Option 2:

pinkish red to yellow

Option 3:

colourless to pink

Option 4:

yellow to pinkish red

Correct Answer:

colourless to pink

Solution:

In a titration of Oxalic acid (standard solution in the flask) against Sodium hydroxide (base in the burette). The solution starts as acidic (colorless). As the equivalence point is reached and the solution becomes slightly alkaline, the phenolphthalein changes color. The color change observed is colourless to pink.

Hence, the correct answer is option (3)

Q. 14 Methane reacts with steam at 1273 K in the presence of nickel catalyst to form:

Option 1:

CO and H₂

Option 2:

CO and H₂O

Option 3:

CO₂ and H₂

Option 4:

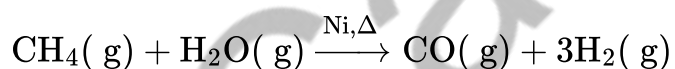
CO₂ and H₂O

Correct Answer:

CO and H₂

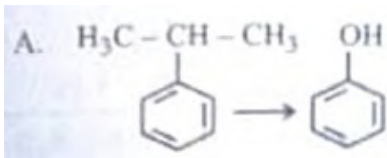
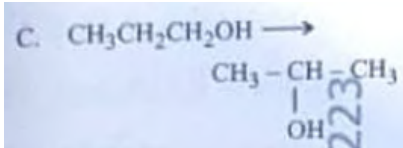
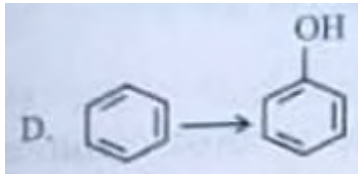
Solution:

When methane reacts with steam at high temperatures (1273 K) in the presence of a nickel catalyst, it undergoes a process called steam reforming to produce Syngas (Synthesis Gas).



Hence, the correct answer is option (1)

Q. 15 Match the LIST - I with LIST - II

List - I	List - II
A. 	I. (i) oleum; (ii) NaOH, Δ; (iii) H^+
B. $\text{CH}_3\text{COOH} \longrightarrow \text{CH}_3\text{CH}_2\text{OH}$	II. (i) O_2; (ii) $\text{H}_2\text{O}/\text{H}^+$
C. 	III. (i) CH_3OH, H^+; (ii) H_2, catalyst
D. 	IV. (i) conc. H_2SO_4, Δ; (ii) $\text{H}^+/\text{H}_2\text{O}$

Choose the correct answer from the options given below :

Option 1:

A-I, B-III, C-IV, D-II

Option 2:

A-II, B-IV, C-III, D-I

Option 3:

A-II, B-III, C-IV, D-I

Option 4:

A-II, B-III, C-I, D-IV

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

A. Cumene to Phenol: This is the industrial preparation of phenol via cumene hydroperoxide. It requires oxidation with air (O_2) followed by acid hydrolysis.

A $\rightarrow$ II

B. Ethanoic Acid to Ethanol: This is a reduction of a carboxylic acid to a primary alcohol. While LiAlH_4 is common, another route involves esterification with methanol (CH_3OH , H^+) followed by catalytic hydrogenation.

B $\rightarrow$ III.

C. Propan-1-ol to Propan-2-ol: This requires dehydration to form propene (using conc. H_2SO_4 , Δ) followed by hydration (acid-catalyzed addition of water) to form the more stable secondary alcohol.

C -> IV.

D. Benzene to Phenol: This follows the classical sulfonating route: sulfonation with oleum to form benzene sulfonic acid, fusion with NaOH , and final acidification.

D -> I

Hence, the correct answer is option (3)

Q. 16 The pair of molecules that are metamers among the following is :

Option 1:

$\text{CH}_3\text{OCH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$

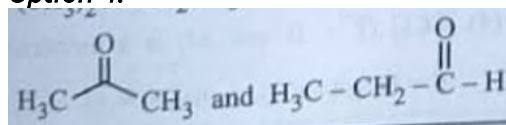
Option 2:

$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ and $\text{CH}_3 - \text{CH}(\text{OH}) - \text{CH}_3$

Option 3:

$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_3$

Option 4:



Correct Answer:

$\text{CH}_3\text{OCH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$

Solution:

Metamers are isomers that have the same molecular formula but differ in the distribution of carbon atoms (alkyl groups) on either side of a functional group (like $-\text{O}-$, $-\text{S}-$, or $-\text{NH}-$).

Option (1): $\text{CH}_3\text{OCH}_2\text{CH}_2\text{CH}_3$ (Methyl propyl ether) and $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$ (Diethyl ether). Both have the same formula ($\text{C}_4\text{H}_{10}\text{O}$) but different alkyl groups around the oxygen atom. These are metamers.

Option (2): Propan-1-ol and Propan-2-ol are position isomers.

Option (3): n-pentane and isopentane are chain isomers.

Option (4): Propanone and Propanal are functional isomers.

Hence, the correct answer is option (1)

Q. 17 Match List - I with List -II

List- I (Complex)	List- II (Type of isomerism)
A. $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$	I. Optical
B. $[\text{Co}(\text{en})_3]^{3+}$	II. 'Solvate
C. $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$	III. Geometrical
D. $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$	IV. Linkage

Choose the correct answer from the options given below:

Option 1:

A-I, B-III, C-II, D-IV

Option 2:

A-II, B-IV, C-III, D-I

Option 3:

A-III, B-I, C-IV, D-II

Option 4:

A-III, B-I, C-II, D-IV

Correct Answer:

A-III, B-I, C-IV, D-II

Solution:

A. $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

- Square planar complex (Pt^{2+} , d^8)
- Shows cis-trans isomerism

Geometrical isomerism $\rightarrow$ III

B. $[\text{Co}(\text{en})_3]^{3+}$

- Octahedral with bidentate ligand (en)
- Forms non-superimposable mirror images

Optical isomerism $\rightarrow$ I

C. $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$

- NO_2^- can bind via N or O

Linkage isomerism $\rightarrow$ IV

D. $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$

- Can form hydrate/solvate isomers like:



Solvate isomerism $\rightarrow$ II

Hence, the correct answer is option (3)

Q. 18 The number of hydrogen atoms present in 5.4 g of urea is :
(Given : Molar mass of urea : 60 g mol^{-1} ,
 $N_A : 6.022 \times 10^{23} \text{ particles mol}^{-1}$)

Option 1:

$$2.168 \times 10^{22}$$

Option 2:

$$2.168 \times 10^{23}$$

Option 3:

$$1.084 \times 10^{22}$$

Option 4:

$$1.084 \times 10^{23}$$

Correct Answer:

$$2.168 \times 10^{23}$$

Solution:

$$\text{Moles} = \frac{\text{Given Mass}}{\text{Molar Mass}} = \frac{5.4 \text{ g}}{60 \text{ g/mol}} = 0.09 \text{ moles}$$

The formula for urea is NH_2CONH_2 .

Each molecule of urea contains 4 hydrogen atoms.

$$\text{Total H atoms} = \text{Moles} \times \text{H atoms per molecule} \times N_A$$

$$\text{Total H atoms} = 0.09 \times 4 \times 6.022 \times 10^{23}$$

$$\text{Total H atoms} = 0.36 \times 6.022 \times 10^{23} \approx \mathbf{2.168 \times 10^{23}}$$

Hence, the correct answer is option (2)

Q. 19 The calculated 'spin-only' magnetic moment of Ti^{2+} (3 d^2) is :

Option 1:

3.87 BM

Option 2:

4.90 BM

Option 3:

2.84 BM

Option 4:

5.92 BM

Correct Answer:

2.84 BM

Solution:

The formula for 'spin-only' magnetic moment (μ) is:

$$\mu = \sqrt{n(n+2)}\text{BM}$$

where n is the number of unpaired electrons.

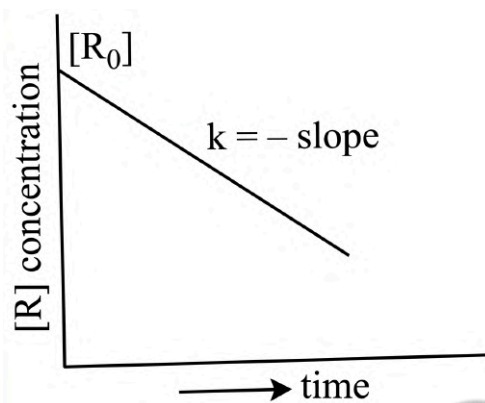
For Ti^{2+} (configuration $3d^2$), there are 2 unpaired electrons ($n = 2$).

$$\mu = \sqrt{2(2+2)} = \sqrt{2 \times 4} = \sqrt{8}$$

$$\mu \approx 2.83\text{BM}$$

Hence, the correct answer is option (3)

- Q. 20** For a certain reaction $R \rightarrow \text{Product}$, the plot of concentration $[R]$ vs time has a negative slope as shown. The order of reaction is :



Option 1:

1

Option 2:

2.5

Option 3:

2

Option 4:

0

Correct Answer:

0

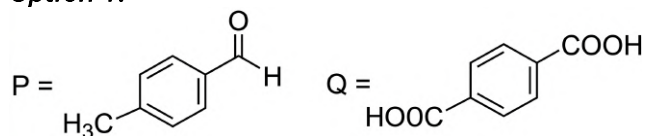
Solution:

Zero order: $[R] = [R_0] - kt \rightarrow$ straight line (slope $= -k$)

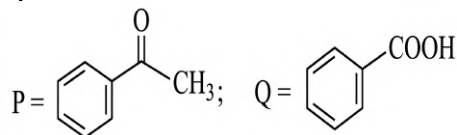
Hence, the correct answer is option (4)

Q. 21 Compound **P** (C_8H_8O) gives a red orange precipitate with 2,4-DNP reagent and it does not reduce Fehling's reagent. On drastic oxidation with chromic acid, **P** gives an aromatic product **Q** that produces effervescence on treating with aq. $NaHCO_3$. Compounds **P** and **Q**, respectively, are:

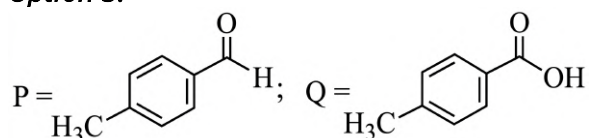
Option 1:



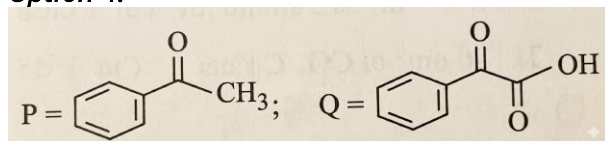
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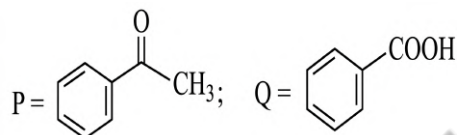
Option 3:



Option 4:



Correct Answer:



Solution:

1. Identify the functional group of **P** (2,4-DNP Test)

Observation: **P** gives a red-orange precipitate with 2,4-DNP reagent.

Inference: **P** contains a carbonyl group ($>C=O$), meaning it is either an aldehyde or a ketone.

2. Narrow down the carbonyl type (Fehling's Test)

Observation: **P** does not reduce Fehling's reagent.

Inference: **P** is a ketone (specifically an aromatic ketone here) or an aromatic aldehyde (which also don't reduce Fehling's), but the formula C_8H_8O and the next step point toward a ketone.

3. Analyze Product **Q** ($NaHCO_3$ Test)

Observation: Drastic oxidation of **P** with chromic acid yields **Q**, which produces effervescence with aq. $NaHCO_3$.

Inference: **Q** is a carboxylic acid (effervescence is due to CO_2 gas release).

4. Final Structure Identification

For **P** (C_8H_8O): Acetophenone ($C_6H_5COCH_3$) fits the formula. It is a ketone, so it gives a positive 2,4-DNP test but a negative Fehling's test.

Drastic Oxidation: When alkyl benzenes or aromatic ketones like acetophenone are treated with strong oxidising agents (like chromic acid or alkaline $KMnO_4$), the entire side chain is oxidised down to a carboxyl group ($-COOH$) attached directly to the ring.

Product **Q**: Benzoic acid (C_6H_5COOH).

Hence, the correct answer is option (2)

Q. 22 Match List - I with List - II

List - I (order of reaction)	List - II (unit of rate constant)
A. Zero order	I. $\text{mol}^{-1} \text{L s}^{-1}$
B. First order	II. $\text{mol}^{-2} \text{L}^2 \text{s}^{-1}$
C. Second orde	III. s^{-1}
D. Third order	IV. $\text{molL}^{-1} \text{s}^{-1}$

Choose the correct answer from the options given below:

Option 1:

A-IV, B-III, C- II,D- I

Option 2:

A-IV, B-III, C-I, D-II

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-IV, B-II, C-I, D-III

Correct Answer:

A-IV, B-III, C-I, D-II

Solution:

List -I	List - II
A. Zero order	IV. $\text{molL}^{-1} \text{s}^{-1}$
B. First order	III. s^{-1}
C. Second order	I. $\text{mol}^{-1} \text{L s}^{-1}$
D. Third order	II. $\text{mol}^{-2} \text{L}^2 \text{s}^{-1}$

Hence, the corect answer is option 2

Q. 23 Although +3 oxidation state is most common in lanthanoids, cerium still shows +4 oxidation state because :

Option 1:

After losing one, more electron, it acquires $4f^{14}$ electronic configuration.

Option 2:

Its nearest inert gas is Radon.

Option 3:

After losing one more electron, it acquires $4f^0$ electronic configuration.

Option 4:

Its atomic number is 61 .

Correct Answer:

After losing one more electron, it acquires $4f^0$ electronic configuration.

Solution:

Cerium (Ce) has the atomic number 58 . Its ground-state electronic configuration is $[Xe]4f^15d^16s^2$.
By losing 4 electrons (to reach the +4 state), it achieves the configuration of the noble gas Xenon ($[Xe]4f^0$).
This empty $4f^0$ orbital provides extra stability to the +4 state.

Hence, the correct answer is option (3)

Q. 24 In a test tube containing a salt, a few drops of dilute H_2SO_4 was added, which gave colourless vapours having the smell of vinegar. The vapours turned the blue litmus paper red.
Identify the correct anion from the following :

Option 1:

Carbonate, CO_3^{2-}

Option 2:

Sulphide, S^{2-}

Option 3:

Acetate, CH_3COO^-

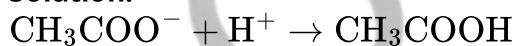
Option 4:

Sulphate, SO_4^{2-}

Correct Answer:

Acetate, CH_3COO^-

Solution:

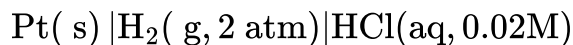


Acetic acid vapours:

- Are **colourless**
- Have **vinegar smell**
- Turn **blue litmus red (acidic nature)**

Hence, the correct answer is option (3)

Q. 25 Calculate emf of the half cell given below :



$$E_{\text{H}_2/\text{H}^+}^\circ = 0 \text{ V}$$

$$\left(\text{Given : } \frac{2.303RT}{F} 0.059, \right. \\ \left. \log 2 = 0.3010 \right)$$

Option 1:

$$-0.109 \text{ V}$$

Option 2:

$$0.109 \text{ V}$$

Option 3:

$$0.035 \text{ V}$$

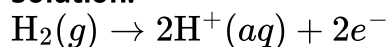
Option 4:

$$-0.035 \text{ V}$$

Correct Answer:

$$0.109 \text{ V}$$

Solution:



$$E_{\text{ox}} = E_{\text{ox}}^\circ - \frac{0.059}{n} \log Q$$

$$E_{\text{ox}} = 0 - \frac{0.059}{2} \log \left(\frac{(0.02)^2}{2} \right)$$

$$E_{\text{ox}} = -0.0295 \log \left(\frac{0.0004}{2} \right)$$

$$E_{\text{ox}} = -0.0295 \log(0.0002)$$

$$\log(2 \times 10^{-4}) = \log 2 + \log 10^{-4}$$

$$\log(2 \times 10^{-4}) = 0.3010 - 4 = -3.699$$

$$E_{\text{ox}} = -0.0295 \times (-3.699)$$

$$E_{\text{ox}} \approx +0.109 \text{ V}$$

Hence, the correct answer is option (2)

Q. 26 At 298 K , a certain buffer solution contains equal concentrations of X^- and HX , K_b for X^- is 10^{-10} . What is the pH of this buffer solution?

Option 1:

10

Option 2:

4

Option 3:

2

Option 4:

6

Correct Answer:

4

Solution:

For a basic buffer, we use:

$$pOH = pK_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

Since concentrations are equal:

$$\log \frac{[\text{salt}]}{[\text{base}]} = \log 1 = 0$$

So,

$$pOH = pK_b$$

Given:

$$K_b = 10^{-10} \Rightarrow pK_b = 10$$

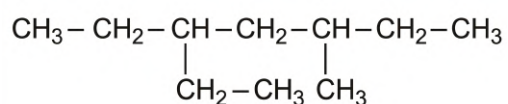
$$pOH = 10$$

Now:

$$pH = 14 - pOH = 14 - 10 = 4$$

Hence, the correct answer is option (2)

Q. 27 The correct IUPAC name of the following compound is :



Option 1:

3-ethyl-5-methylheptane

Option 2:

3-methyl-5-ethylheptane

Option 3:

2,4-diethylhexane

Option 4:

3,5-diethylhexane

Correct Answer:

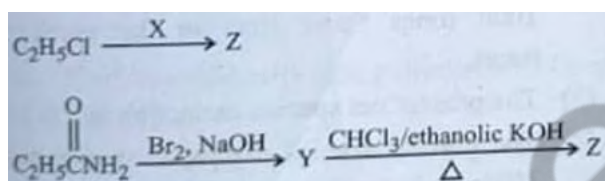
3-ethyl-5-methylheptane

Solution:

3-ethyl-5-methylheptane

Hence, the correct answer is option (1)

Q. 28 The following two reactions give e^x the same foul smelling product Z .



X and Z respectively, are:

Option 1:

$\text{X} = \text{AgCN}$; $\text{Z} = \text{C}_2\text{H}_5\text{CN}$

Option 2:

$\text{X} = \text{AgCN}$; $\text{Z} = \text{C}_2\text{H}_5\text{NC}$

Option 3:

$\text{X} = \text{KCN}$; $\text{Z} = \text{C}_2\text{H}_5\text{CN}$

Option 4:

$\text{X} = \text{KCN}$; $\text{Z} = \text{C}_2\text{H}_5\text{NC}$

Correct Answer:

$\text{X} = \text{AgCN}$; $\text{Z} = \text{C}_2\text{H}_5\text{NC}$

Solution:

$\text{C}_2\text{H}_5\text{Cl} + \text{AgCN} \rightarrow \text{C}_2\text{H}_5\text{NC}$ (Ethyl isocyanide).

AgCN is covalent, so the nitrogen atom acts as the nucleophile, forming an isocyanide.

$\text{C}_2\text{H}_5\text{CONH}_2 \xrightarrow{\text{Br}_2, \text{NaOH}} \text{C}_2\text{H}_5\text{NH}_2$ (Hofmann Bromamide Degradation).

Hence, the correct answer is option (2).

Q. 29 Mixture of chloroform and acetone forms a solution with negative deviation from Raoult's law due to :

Option 1:

stronger intermolecular forces between chloroform molecules than those between chloroform and acetone molecules.

Option 2:

formation of hydrogen bonding between acetone and chloroform.

Option 3:

repulsive forces.

Option 4:

increase in escaping tendency of molecules of each component.

Correct Answer:

formation of hydrogen bonding between acetone and chloroform.

Solution:

- Chloroform (CHCl_3) has a slightly acidic hydrogen
- Acetone (CH_3COCH_3) has a lone pair on oxygen

This leads to hydrogen bonding between chloroform and acetone molecules which results:

- Strong A-B interactions $\rightarrow$ molecules are held more tightly
- Vapour pressure decreases $\rightarrow$ negative deviation

Hydrogen bonding between chloroform and acetone

Hence, the correct answer is option (2)

Q. 30 Identify the correct statements :

- A. The molality of 2.5 g of ethanoic acid (Molar mass : 60 g mol^{-1}) in 75 g of benzene solution is 0.556.
- B. The molarity of a solution containing 5 g of NaOH (molar mass : 40 g mol^{-1}) in 450 mL of solution is 0.278 M at 298 K .
- C. Aquatic species are more comfortable in cold water.
- D. The solubility of gas increases with decrease in pressure.
- E. For a binary mixture of A and B , the number of moles of A and B are n_A and n_B respectively. The mole fraction of B will be $x_B = \frac{n_A}{n_A + n_B} =$

Choose the correct answer from the options given below:

Option 1:

A, B and C only

Option 2:

A, D and E only

Option 3:

A and B only

Option 4:

A and C only

Correct Answer:

A, B and C only

Solution:

Statement A (True): Molality (m) = $\frac{\text{moles of solute}}{\text{mass of solvent in kg}}$.

Moles of ethanoic acid = $\frac{2.5}{60} = 0.0417$.

$$m = \frac{0.0417}{0.075 \text{ kg}} = 0.556 \text{ m}$$

Statement B (True): Molarity (M) = $\frac{\text{moles of solute}}{\text{Volume of solution in L}}$.

Moles of NaOH = $\frac{5}{40} = 0.125$.

$$M = \frac{0.125}{0.450 \text{ L}} = 0.278 \text{ M}$$

Statement C (True): The solubility of gases (like O_2) increases as temperature decreases. Therefore, aquatic species are more comfortable in cold water due to higher dissolved oxygen levels.

Statement D (False): According to Henry's Law, the solubility of a gas increases with an increase in pressure.

Statement E (False): The mole fraction of B is $x_B = \frac{n_B}{n_A + n_B}$. The provided formula is for x_A .

Hence, the correct answer is option (1)

Q. 31 Identify the incorrect statement from the following :

Option 1:

Oxygen exhibits only -2 oxidation state.

Option 2:

The order of catenation property of Group 14 elements is $\text{C} \gg \text{Si} > \text{Ge} \approx \text{Sn}$.

Option 3:

Carbon has the ability to form $p\pi - p\pi$ multiple bond with itself.

Option 4:

ECl_3 ($\text{E} = \text{B}$ and Al) is a monomer when $\text{E} = \text{B}$ and a dimer when $\text{E} = \text{Al}$.

Correct Answer:

Oxygen exhibits only -2 oxidation state.

Solution:

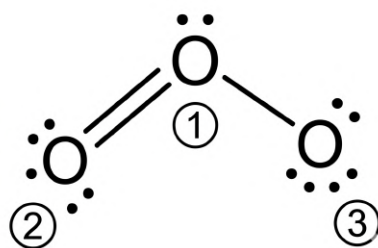
Statement (1) is Incorrect: Oxygen does not exhibit only the -2 oxidation state. It shows +2 in OF_2 , +1 in O_2F_2 , -1 in peroxides (H_2O_2), and $-1/2$ in superoxides (KO_2).

Statement (2) is Correct: Catenation ability decreases down the group as bond enthalpy decreases.

Statement (3) is Correct: Carbon frequently forms $p\pi - p\pi$ multiple bonds ($C = C, C \equiv C$).
Statement (4) is Correct: BCl_3 is a monomer (trigonal planar), while AlCl_3 achieves stability by forming a dimer (Al_2Cl_6).

Hence, the correct answer is option (1)

Q. 32



The correct formal charges on oxygen atoms numbered 2, 1 and 3 respectively are :

Option 1:

0, 0, 0

Option 2:

-1, 0, +1

Option 3:

+1, 0, -1

Option 4:

0, +1, -1

Correct Answer:

0, +1, -1

Solution:

O(2) (double bonded terminal O):

- Lone pairs = $4e^-$

- Bonding = $4e^-$

$$FC = 6 - 4 - 2 = 0$$

O(1) (central O):

- Lone pairs = $2e^-$

- Bonding = $6e^-$

$$FC = 6 - 2 - 3 = +1$$

O(3) (single bonded terminal O):

Lone pairs = $6e^-$

Bonding = $2e^-$

$$FC = 6 - 6 - 1 = -1$$

Hence, the correct answer is option (4)

Q. 33 At a certain temperature, $T(K)$, during a process, 500 J is absorbed by the system and work of 200 J is done by the system. Then change in internal energy of the system is :

Option 1:

700 J

Option 2:

400 J

Option 3:

300 J

Option 4:

500 J

Correct Answer:

300 J

Solution:

$$\Delta U = q - w$$

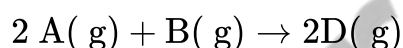
$q = +500J$ (heat absorbed $\rightarrow$ positive)

$w = +200J$ (work done by system)

$$\Delta U = 500 - 200 = 300 \text{ J}$$

Hence, the correct answer is option (3)

Q. 34 Consider the following reaction :



$\Delta U^\ominus = -10 \text{ kJ mol}^{-1}$ and $\Delta S^\ominus = -44 \text{ J K}^{-1}$ at 298 K .

Identify the correct option with $\Delta G^\ominus$ for the reaction and spontaneity of the reaction at 298 K .

(Given : $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$)

Option 1:

$+0.63568 \text{ kJ mol}^{-1}$, non-spontaneous

Option 2:

$-0.63568 \text{ kJ mol}^{-1}$, spontaneous

Option 3:

$-1.635 \text{ kJ mol}^{-1}$ spontaneous

Option 4:

$+1.635 \text{ kJ mol}^{-1}$, non-spontaneous

Correct Answer:

$+0.63568 \text{ kJ mol}^{-1}$, non-spontaneous

Solution:

$$\Delta S^\circ = -44 \text{ J} = -0.044 \text{ kJ K}^{-1}$$

$$\Delta H = \Delta U + \Delta n_g RT$$

$$\Delta n_g = 2 - 3 = -1$$

$$\Delta H = -10 + (-1)(8.31 \times 298/1000)$$

$$= -10 - 2.476 \approx -12.476 \text{ kJ}$$

$$\Delta G = \Delta H - T\Delta S$$

$$= -12.476 - (298 \times -0.044)$$

$$= -12.476 + 13.112$$

$$= +0.636 \text{ kJ mol}^{-1}$$

$$\Delta G > 0 \Rightarrow \text{N-spontaneous}$$

Hence, the correct answer is option (1)

Q. 35 Select the reagents that reduce nitriles to primary amines:

A. (i) LiAlH_4 ;

(ii) H_2O

B. $\text{Sn} + \text{HCl}$

C. H_2/Ni

D. $\text{Na}(\text{Hg})/\text{C}_2\text{H}_5\text{OH}$

E. $\text{Br}_2/\text{aq. NaOH}$

Choose the correct answer from the options given below

Option 1:

A, B and C only

Option 2:

A, D and E only

Option 3:

A, C and D only

Option 4:

B, D and E only

Correct Answer:

A, C and D only

Solution:

A. $\text{LiAlH}_4/\text{H}_2\text{O}$

Strong reducing agent $\rightarrow$ converts nitrile $\rightarrow$ primary amine

B. $\text{Sn} + \text{HCl}$

Used for nitro $\rightarrow$ amine, not for nitriles $\rightarrow$ Incorrect

C. H_2/Ni

Catalytic hydrogenation $\rightarrow$ nitrile $\rightarrow$ primary amine

D. $\text{Na}(\text{Hg})/\text{C}_2\text{H}_5\text{OH}$

Mild reduction $\rightarrow$ can reduce nitriles $\rightarrow$ primary amine

E. $\text{Br}_2 / \text{aq. NaOH}$

Hoffmann bromamide reaction (amide $\rightarrow$ amine with one less carbon)

Hence, the correct answer is option (3)

Q. 36 Which one of the following is an ambidentate ligand?

Option 1:

Oxalate

Option 2:

Ethane-1,2-diamine

Option 3:

Thiocyanate

Option 4:

Ethylenediaminetetraacetate ion

Correct Answer:

Thiocyanate

Solution:

An ambidentate ligand is a ligand that can coordinate to a central metal atom through two different atoms but only one at a time.

Thiocyanate (SCN^-) can bind through the Sulfur (S) or the Nitrogen (N) atom.

Hence, the correct answer is option (3)

Q. 37 A bulb is rated at 150 watt, converting 8% energy into light. If energy of one photon is $4.42 \times 10^{-19} \text{ J}$, how many photons are emitted by the bulb per second?

Option 1:

27.2×10^{19}

Option 2:

4.06×10^{19}

Option 3:

1.35×10^{19}

Option 4:

2.71×10^{19}

Correct Answer:

2.71×10^{19}

Solution:

The bulb is rated at 150 W , but only 8% is converted to light.

$$\text{Light Power} = 150 \times \frac{8}{100} = 12 \text{ W (or J/s)}$$

Using the formula $P = n \times E_{\text{photon}}$:

$$n = \frac{\text{Power}}{\text{Energy of one photon}}$$

$$n = \frac{12}{4.42 \times 10^{-19}} \approx 2.7149 \times 10^{19}$$

Hence, the correct answer is option (4)

Q. 38 Identify the incorrect statement from the following:

Option 1:

Phosphorus, arsenic and antimony show catenation property.

Option 2:

$\text{P}(\text{C}_2\text{H}_5)_3$ and $\text{As}(\text{C}_6\text{H}_5)_3$ form $d\pi - d\pi$ bond with transition metals.

Option 3:

Nitrogen can form $d\pi-p \pi$ bond with oxygen.

Option 4:

Nitrogen can form $\pi - p\pi$ multiple bonds with itself.

Correct Answer:

Nitrogen can form $d\pi-p \pi$ bond with oxygen.

Solution:

(1) Correct: Catenation (the linking of atoms of the same element into chains) is a property shown by Phosphorus, Arsenic, and Antimony. Phosphorus shows it most significantly (as seen in P_4 or red phosphorus chains). While Nitrogen's catenation is limited due to high inter-electronic repulsion between lone pairs, the heavier elements do exhibit it.

(2) INCORRECT: Organophosphorus and organoarsenic compounds (like phosphines and arsines) act as ligands with transition metals. They form bonds through π -backbonding, but the specific interaction involves the donation of electrons from the filled d-orbitals of the transition metal into the vacant d-orbitals of Phosphorus or Arsenic. Therefore, the bond is a $d\pi(\text{metal}) - d\pi(\text{ligand})$ interaction. However, the statement is often considered incorrect in standard textbooks because the p -block elements use their vacant d-orbitals to accept electrons, but the primary sigma bond is formed by the lone pair. More importantly, in many competitive contexts, the focus is on the fact that Nitrogen cannot do this because it lacks d-orbitals.

(3) Correct: Nitrogen forms $p\pi - p\pi$ multiple bonds with Oxygen (as seen in oxides like NO_2 or NO_3^-). Since both N and O are in the second period, their p-orbitals are small enough to overlap effectively sideways.

(4) Correct: Nitrogen has a very small atomic size and high electronegativity, allowing it to form very strong $p\pi - p\pi$ multiple bonds with itself, resulting in the stable $\text{N} \equiv \text{N}$ triple bond. Heavier elements like Phosphorus cannot do this effectively because their p-orbitals are too large and diffuse to have effective sideways overlap.

Hence, the correct answer is option (3)

Q. 39 The correct order of increasing metallic character of Na, Be, P, Mg and Si is :

Option 1:

Be < Si < P < Mg < Na

Option 2:

P < Si < Na < Mg < Be

Option 3:

P < Si < Be < Mg < Na

Option 4:

P < Mg < Be < Si < Na

Correct Answer:

P < Si < Be < Mg < Na

Solution:

Left to Right metallic character decreases

Top to Bottom metallic character increases

Na, Mg, Si, P: These are all in Period

P < Si < Mg < Na

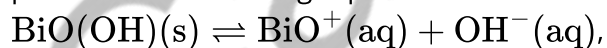
Be is in Group 2, located directly above Magnesium. Since metallic character increases downward, Be is less metallic than Mg.

Beryllium is a metal, while Silicon is a metalloid. Therefore, Si is less metallic than Be.

P < Si < Be < Mg < Na

Hence, correct answer is option (3)

Q. 40 In a qualitative analysis, Bi^{3+} is detected by appearance of precipitate of $\text{BiO}(\text{OH})(\text{s})$. Calculate pH when the following equilibrium exists at 298 K :



$$K = 4 \times 10^{-10}$$

(Given : $\log 2 = 0.3010$)

Option 1:

4.699

Option 2:

5.286

Option 3:

8.714

Option 4:

9.301

Correct Answer:

9.301

Solution:

For the reaction: $\text{BiO}(\text{OH})(s) \rightleftharpoons \text{BiO}^+(aq) + \text{OH}^-(aq)$

The equilibrium constant is $K = [\text{BiO}^+][\text{OH}^-]$.

Let $[\text{BiO}^+] = [\text{OH}^-] = s$.

$$s^2 = K = 4 \times 10^{-10}$$

$$s = [\text{OH}^-] = \sqrt{4 \times 10^{-10}} = 2 \times 10^{-5} \text{ M}$$

pOH

$$\text{pOH} = -\log[\text{OH}^-]$$

$$\text{pOH} = -\log(2 \times 10^{-5}) = 5 - \log 2$$

$$\text{pOH} = 5 - 0.3010 = 4.699$$

pH

At 298 K, $\text{pH} + \text{pOH} = 14$.

$$\text{pH} = 14 - 4.699 = \mathbf{9.301}$$

Hence, the correct answer is option (4)

Q. 41 Match List- I with List - II:

List-I	List -II
A. C_2H_4	I. 3σ bonds, 2π bonds
B. C_2H_2	II. 3σ bonds, one lone pair
C. CH_4	III. 4σ bonds
D. NH_3	IV. 5σ bonds, 1π bond

Choose the correct answer from the options given below :

Option 1:

A-IV, B-I, C-III, D-II

Option 2:

A-III, B-IV, C-II, D-I

Option 3:

A-II, B-III, C-I, D-IV

Option 4:

A-I, B-II, C-IV, D-III

Correct Answer:

A-IV, B-I, C-III, D-II

Solution:

A. C_2H_4 (Ethene): Contains a $C = C$ double bond and four $C - H$ single bonds. Total: 5 σ bonds, 1 π bond. (A-IV)

B. C_2H_2 (Ethyne): Contains a $C \equiv C$ triple bond and two $C - H$ single bonds. Total: 3 σ bonds, 2 π bonds. (B-I)

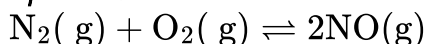
C. CH_4 (Methane): Contains four $C - H$ single bonds. Total: 4 σ bonds. (C-III)

D. NH_3 (Ammonia): Contains three $N - H$ single bonds and one lone pair on Nitrogen. Total: 3 σ bonds, one lone pair. (D-II)

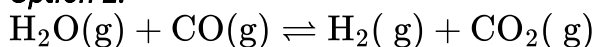
Hence, the correct answer is option (1).

Q. 42 Given below are certain reactions. Identify the reaction for which $K_p \neq K_c$.

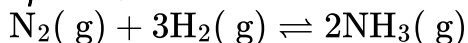
Option 1:



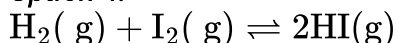
Option 2:



Option 3:



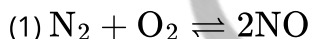
Option 4:



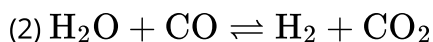
Correct Answer:



Solution:



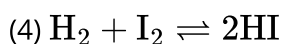
$$\Delta n = 2 - (1 + 1) = 0 \rightarrow K_p = K_c$$



$$\Delta n = 2 - 2 = 0 \rightarrow K_p = K_c$$



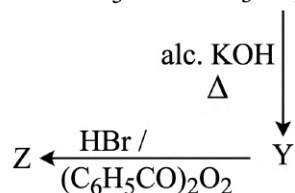
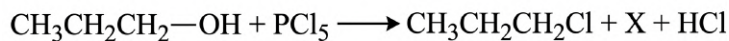
$$\Delta n = 2 - 4 = -2 \rightarrow K_p \neq K_c$$



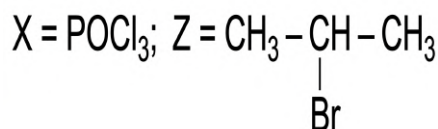
$$\Delta n = 2 - 2 = 0 \rightarrow K_p = K_c$$

Hence, the correct answer is option (3)

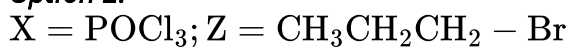
Q. 43 In the following reaction sequence, X and Z , respectively are :



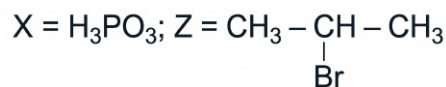
Option 1:



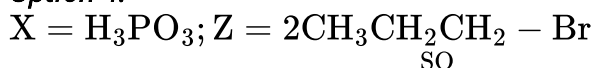
Option 2:



Option 3:



Option 4:



Correct Answer:



Solution:

1. Formation of **X** (Reaction with PCl_5)

The reaction of propan-1-ol with phosphorus pentachloride (PCl_5) is as follows:



- The byproduct **X** is phosphorus oxychloride (POCl_3).

2. Formation of Y (Dehydrohalogenation)

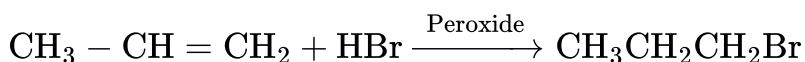
Treating the 1-chloropropane with alcoholic KOH and heat (Δ) causes an **E2** elimination reaction:



- Product Y is Propene.

3. Formation of **Z** (Peroxide Effect)

The final step is the addition of HBr in the presence of benzoyl peroxide ($(\text{C}_6\text{H}_5\text{CO})_2\text{O}_2$). This is the AntiMarkovnikov addition (also known as the Kharasch effect or Peroxide effect).

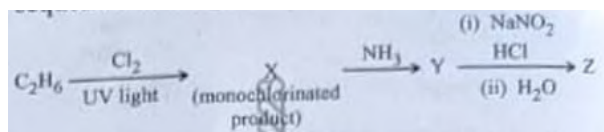


The bromine atom attaches to the terminal carbon (the one with more hydrogens).

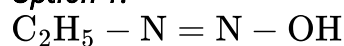
Product **Z** is 1-bromopropane.

Hence, the correct answer is option (2)

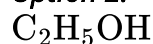
Q. 44 The major product Z formed in the following sequence of reactions is :



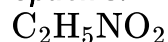
Option 1:



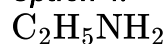
Option 2:



Option 3:



Option 4:



Correct Answer:



Solution:

Step 1: Ethane (C_2H_6) undergoes free radical chlorination with Cl_2/UV light to form Chloroethane (X): $\text{C}_2\text{H}_5\text{Cl}$.

Step 2: Chloroethane reacts with Ammonia (NH_3) via nucleophilic substitution ($\text{S}_\text{N}2$) to form Ethylamine (Y): $\text{C}_2\text{H}_5\text{NH}_2$.

Step 3: Ethylamine reacts with nitrous acid ($\text{NaNO}_2 + \text{HCl}$) to form an unstable diazonium salt, which immediately reacts with water (H_2O) to form Ethanol (Z): $\text{C}_2\text{H}_5\text{OH}$.

Hence, the correct answer is option (2)

Q. 45 When 1dm^3 of CO_2 gas is passed over hot coke, the volume of gaseous mixture after complete reaction at STP becomes 1.4dm^3 . The composition of the gaseous mixture at STP is :

Option 1:

0.6dm^3 of CO, 0.8dm^3 of CO_2

Option 2:

0.8dm^3 of CO, 0.8dm^3 of CO_2

Option 3:

0.6dm^3 of CO, 0.4dm^3 of CO_2

Option 4:

0.8dm³ of CO 0.6dm³ of CO₂

Correct Answer:

0.8dm³ of CO 0.6dm³ of CO₂

Solution:

Reaction of CO₂ with hot coke:



Let:

dm³ of CO₂ reacts

Initial CO₂ = 1dm³

After reaction:

Remaining CO₂ = $1 - x$

CO formed = $2x$

Total final volume:

$$(1 - x) + 2x = 1 + x$$

Given:

$$1 + x = 1.4 \Rightarrow x = 0.4$$

Now calculate composition:

$$\text{CO}_2 = 1 - 0.4 = 0.6\text{dm}^3$$

$$\text{CO} = 2 \times 0.4 = 0.8\text{dm}^3$$

Hence, the correct answer is option (4)

Biology

Q. 1 Match List I with List II :

	List I		List II
	(Phase of cell cycle)		(Activity)
A.	G_1 phase	I.	Actual cell division occurs
B.	S phase	II.	The cell is metabolically active and continuously grows, but does not replicate its DNA
C.	G_2 phase	III.	Synthesis of DNA occurs, and the amount of DNA per cell doubles
D.	M phase	IV.	Proteins are synthesised while cell growth continues

Choose the correct answer from the options given below:

Option 1:

A-III, B-IV, C-I, D-II

Option 2:

A-IV, B-I, C-II, D-III

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-II, B-III, C-IV, D-1

Correct Answer:

A-II, B-III, C-IV, D-1

Solution:

- G_1 is growth without DNA replication (*II*)
- S phase involves DNA synthesis (*III*)
- G_2 prepares for division with protein synthesis (*IV*)
- M phase is the actual cell division (*I*).

Hence, the correct answer is option 4) A-II, B-III, C-IV, D-1.

Q. 2 Match List I with List II :

List - I	List - II
A. Incomplete dominance	I. Human skin colour
B. Co-dominance	II. Inheritance of flower colour in Antirrhinum sp.
C. Pleiotropy	III. Phenylketonuria disease in humans
D. Polygenic inheritance	IV. ABO blood groups

Choose the correct answer from the options given below :

Option 1:

A-I, B-IV, C-III, D-II

Option 2:

A-I, B-III, C-II, D-IV

Option 3:

A-II, B-IV, C-III, D-I

Option 4:

A-II, B-I, C-III, D-IV

Correct Answer:

A-II, B-IV, C-III, D-I

Solution:

Incomplete dominance → II; Co-dominance → IV; Pleiotropy → III; Polygenic inheritance → I.

- In incomplete dominance (Antirrhinum), the heterozygote shows an intermediate phenotype, e.g., pink flowers from red × white.
- Co-dominance (ABO blood groups) allows both alleles to be expressed equally, such as I^A and I^B producing AB blood type.
- Pleiotropy (Phenylketonuria) occurs when a single gene affects multiple traits, leading to diverse symptoms.
- Polygenic inheritance (human skin colour) involves multiple genes contributing additively to a continuous variation.

Hence, the correct answer is option 3) A-II, B-IV, C-III, D-I.

- Q. 3** Which of the following statements are correct?
- A. The Amazon rainforest being cut and cleared for the cultivation of soybeans is an example of habitat loss.
 - B. Steller's sea cow and passenger pigeon became extinct due to over-exploitation by humans.
 - C. The Nile perch introduced into Lake Victoria in East Africa helped in the population growth of cichlid fish in the lake.
 - D. Water hyacinth is an invasive species.
 - E. When a species becomes extinct, the plant and animal species associated with it are not affected.

Choose the correct answer from the options given below:

Option 1:

A, B and D only

Option 2:

B, C and D only

Option 3:

A, B and E only

Option 4:

C, D and E only

Correct Answer:

A, B and D only

Solution:

- The Amazon rainforest example highlights habitat loss and fragmentation, the most severe driver of biodiversity decline.
- Species like Steller's sea cow and passenger pigeon vanished due to human over-exploitation.
- The introduction of alien species, such as water hyacinth, disrupts native ecosystems and reduces biodiversity.
- Nile perch actually caused the decline of cichlid fish, and co-extinctions occur when dependent species are lost together.

A, B, and D are correct; Nile perch reduced cichlid populations (C false), and extinction affects associated species (E false).

Hence, the correct answer is option 1) A, B and D only.

-
- Q. 4** Which of the following statements are correct with reference to a transcription unit?

- A. A transcription unit in DNA is defined primarily by three regions: promoter, structural gene and terminator
- B. The promoter is said to be located towards the 5'-end of the structural gene
- C. The promoter is a DNA sequence that provides binding site for RNA polymerase
- D. The promoter defines the template and coding strands
- E. The terminator is located towards the 3'-end of the coding strand and it defines the end of the process of transcription

Choose the correct answer from the options given below:

Option 1:

B, C, D and E only

Option 2:

A, B, C, D and E

Option 3:

A, B, C and D only

Option 4:

A, C, D and E only

Correct Answer:

A, B, C, D and E

Solution:

A transcription unit has a promoter, a structural gene, and a terminator.

The promoter lies upstream at the 5' end and binds RNA polymerase. It determines template/coding strands, and the terminator marks the transcription end. The promoter region ensures correct initiation of transcription by RNA polymerase, while the terminator signals its end. Together with the structural gene, these elements define the functional transcription unit in both prokaryotes and eukaryotes.

Hence, the correct answer is option 2) A, B, C, D and E.

- Q. 5** Which of the following statements are true with reference to sex determination in honeybees?
- A. An offspring formed from the union of a sperm and an egg develops as a female (queen or worker).
 - B. An unfertilized egg develops as a male by parthenogenesis.
 - C. A male has half the number of chromosomes as a female.
 - D. Males produce sperm by meiosis
 - E. Honeybees have a haplodiploid sex-determination system.
- Choose the correct answer from the options given below:

Option 1:

A, B, C and E only

Option 2:

A, B, C and D only

Option 3:

B, C, D and E only

Option 4:

A, B, D and E only

Correct Answer:

A, B, C and E only

Solution:

In honeybees, fertilised eggs (2n) form females, unfertilised eggs (n) form males. Males are haploid and produce sperm by mitosis, not meiosis, under haplodiploidy.

- A. True - Fertilised eggs (sperm + egg) develop into females (queen or worker).
- B. True - Unfertilized eggs develop into males (drones) by parthenogenesis.
- C. True - Males (haploid) have half the number of chromosomes compared to females (diploid).
- D. False - Males are haploid, so they cannot undergo meiosis to produce gametes. They produce sperm by mitosis.
- E. True - Honeybees indeed have a haplodiploid sex-determination system.

Hence, the correct answer is option 1) A, B, C and E only.

Q. 6 Match List I with List II :

List - I (Process)	List -II (Location)
A. Glycolysis	I. Inner mitochondrial membrane
B. ETS	II. Mitochondrial matrix
C. Accumulation of protons	III. Cytoplasm
D. Krebs' cycle	IV. Intermembrane space

Choose the correct answer from the options given below:

Option 1:

A-IV, B-II, C-I, D-III

Option 2:

A-I, B-IV, C-III, D-II

Option 3:

A-II, B-III, C-IV, D-I

Option 4:

A-III, B-I, C-IV, D-II

Correct Answer:

A-III, B-I, C-IV, D-II

Solution:

- Glycolysis occurs in the cytoplasm (III)
- ETS on inner membrane (I)
- Protons accumulate in the intermembrane space (IV)
- Krebs cycle in matrix (II).

Hence, the correct answer is option 4) A-III, B-I, C-IV, D-II

Q. 7 How many ATP and NADPH molecules are required to make one molecule of glucose through the Calvin pathway?

Option 1:

12 ATP and 18 NADPH

Option 2:

18 ATP and 12 NADPH

Option 3:

6 ATP and 12 NADPH

Option 4:

24 ATP and 18 NADPH

Correct Answer:

18 ATP and 12 NADPH

Solution:

For the synthesis of one glucose, the Calvin cycle consumes 18 *ATP* and 12 *NADPH*.

To fix one molecule of CO_2 , the Calvin cycle requires 3 ATP and 2 NADPH. Since glucose formation needs 6 CO_2 molecules, the total requirement becomes 18 ATP and 12 NADPH.

ATP provides the energy, while NADPH supplies the reducing power for carbon fixation. This highlights the high energy demand of photosynthesis, linking the light reactions to the dark reactions.

Hence, the correct answer is option 2) 18 ATP and 12 NADPH.

Q. 8 Match List I with List II :

	List I		List II
A.	Genetically modified organism	I.	Agrobacterium tumefaciens
B.	Thermostable DNA polymerase	II.	Bt cotton
C.	Ti plasmid	III.	Thermus aquaticus
D.	pBR322	IV.	Escherichia coli

Choose the correct answer from the options given below:

Option 1:

A-II, B-I, C-I , D-III

Option 2:

A-II, B-III, G-I, D-IV

Option 3:

A-I, B-IV, C-III, D-II

Option 4:

A-I, B-II, C-IV, D-III

Correct Answer:

A-II, B-III, G-I, D-IV

Solution:

- Bt cotton is a genetically modified organism (A → II)
- Thermostable DNA polymerase comes from *Thermus aquaticus* (B → III)
- Ti plasmid is from *Agrobacterium tumefaciens* (C → I)
- pBR322 is a cloning vector (D → IV)

Hence, the correct answer is option 2) A-II, B-III, G-I, D-IV.

Q. 9 In which one of the following is the ovules are not enclosed by an ovary wall and remains exposed?

Option 1:

Pinus

Option 2:

Wolffia

Option 3:

Funaria

Option 4:

Selaginella

Correct Answer:

Pinus

Solution:

In gymnosperms like Pinus, ovules are naked (not enclosed in an ovary) because they are borne directly on megasporophylls without being enclosed by an ovary wall.

This is a key distinction between gymnosperms and angiosperms, where ovules are protected inside the ovary. The exposed ovules of Pinus later develop into seeds after fertilisation.

Hence, the correct answer is option 1) Pinus.

Q. 10 The enzyme required for carboxylation in the Calvin cycle is :

Option 1:
PEP carboxylase

Option 2:
RuBP carboxylase - oxygenase

Option 3:
Carboxypeptidase

Option 4:
Hexokinase

Correct Answer:
RuBP carboxylase - oxygenase

Solution:

Carboxylation in the Calvin cycle is catalysed by RuBisCO, which fixes CO_2 to RuBP, forming 3-PGA.

It fixes CO_2 to RuBP, producing 3-phosphoglycerate (3-PGA) as the first stable compound.

This step is the key entry point of inorganic carbon into the biosphere, driving photosynthetic productivity.

Hence, the correct answer is option 2) RuBP carboxylase - oxygenase.

Q. 11 Match the List - I with List - II

List - I	List - II
A. Trypsin	I. Intercellular ground substance
B. Morphine	II. Lectin
C. Concanavalin	III. Enzyme
D. Collagen	IV. Alkaloid

Choose the correct answer from the options given below:

Option 1:
A-III, B-IV, C-II, D-I

Option 2:
A-IV, B-III, C-II, D-I

Option 3:
A-I, B-II, C-III, D-IV

Option 4:
A-III, B-II, C-IV, D-I

Correct Answer:
A-III, B-IV, C-II, D-I

Solution:

Trypsin → enzyme (III); Morphine → alkaloid (IV); Concanavalin A → lectin (II); Collagen → intercellular ground substance (I).

- Trypsin is a digestive enzyme that hydrolyses proteins into peptides.
- Morphine is a plant-derived alkaloid used medicinally as an analgesic.
- Concanavalin A is a lectin that binds specifically to certain sugars, often used in biochemical studies.
- Collagen is the main structural protein of the intercellular ground substance, providing strength and elasticity to connective tissues.
- Hence, the correct answer is option 1) A-III, B-IV, C-II, D-I.**

Q. 12 Match **List I** and **List II**

List I (Drug)		List II (Effect)	
A	Nicotine	I	Causes sense of euphoria and increased energy
B	Morphine	II	Stimulates adrenal gland to release catecholamines into blood circulation
C	Herois	III	Effective sedative and painkiller
D	Cocaine	IV	A depressant; slows down body functio

Choose the **correct** answer from the options given below :

Option 1:
A-II, B-III, C-I, D-IV

Option 2:
A-III, B-II, C-IV, D-I

Option 3:
A-III, B-II, C-I, D-IV

Option 4:
A-II, B-III, C-IV, D-I

Correct Answer:
A-II, B-III, C-IV, D-I

Solution:

Nicotine → II (stimulates catecholamine release), Morphine → III (sedative), Heroin → IV (depressant), Cocaine → I (euphoria).

Hence, the correct answer is option 4) A-II, B-III, C-IV, D-I

Q. 13 Which of the following statements are correct regarding amino acids?

- A. They are substituted methanes.
- B. Serine is an aromatic amino acid.
- C. Valine is a neutral amino acid.
- D. Lysine is an acidic amino acid.

Choose the correct answer from the options given below:

Option 1:

A and B only

Option 2:

C and D only

Option 3:

B and C only

Option 4:

A and C only

Correct Answer:

A and C only

Solution:

Amino acids are considered substituted methane derivatives because the central carbon is bonded to an amino, carboxyl, hydrogen, and variable side chains.

Amino acids are substituted methane derivatives (A correct); valine is neutral (C correct). Serine is not aromatic, and lysine is basic, not acidic.

Hence, the correct answer is option 4) A and C only.

Q. 14 Which one of the following disorders is caused by the substitution of Glutamic acid (Glu) by Valine (Val) at the sixth position of the beta globin chain of the haemoglobin molecule?

Option 1:

Phenylketomuria

Option 2:

Haemophilif

Option 3:

Sickle-cell anaemia

Option 4:

Thalassemia

Correct Answer:
Sickle-cell anaemia

Solution:

Sickle-cell anaemia is caused by a point mutation where *Glu* → *Val* at position 6 of β -globin.

This point mutation changes the hydrophilic glutamic acid to hydrophobic valine, altering the haemoglobin structure. As a result, red blood cells become sickle-shaped under low oxygen conditions, leading to blockages in blood vessels.

The disorder causes anaemia, pain crises, and organ damage due to impaired oxygen transport. It is a classic example of a genetic disease caused by a substitution mutation.

Hence, the correct answer is option 3) Sickle-cell anaemia.

Q. 15 Which one of the following is the site for active ribosomal RNA synthesis?

Option 1:

Nucleolus

Option 2:

Kinetochores

Option 3:

Centrosome

Option 4:

Chromatin

Correct Answer:

Nucleolus

Solution:

rRNA is synthesised in the nucleolus, where ribosome assembly begins. It also serves as the site for assembly of ribosomal subunits, which later move to the cytoplasm for protein synthesis.

Other nuclear structures, like the kinetochores, centrosome, and chromatin, have different roles and are not involved in rRNA synthesis.

Hence, the correct answer is option 1) Nucleolus.

Q. 16 Which one of the following statements is not true about the universal rules of binomial nomenclature?

Option 1:

The first word in the biological name represents the specific epithet, while the second component denotes the genus.

Option 2:

The specific epithet in the biological name starts with a small letter.

Option 3:

Both the words in a biological name, when handwritten, are separately underlined or printed in italics.

Option 4:

Biological names are generally in Latin.

Correct Answer:

The first word in the biological name represents the specific epithet, while the second component denotes the genus.

Solution:

According to the universal rules, the first word denotes the genus and always begins with a capital letter, while the second word is the specific epithet written in small letters. Both words are either italicised in print or underlined separately when handwritten to maintain uniformity. Biological names are generally derived from Latin or are Latinized, ensuring global recognition.

The first word represents the genus and the second is the specific epithet, so statement (1) is incorrect.

Hence, the correct answer is option 1) The first word in the biological name represents the specific epithet, while the second component denotes the genus.

Q. 17 Match List I with List II :

	List I		List II
	(Growth Regulator)		(Function/Effect)
A.	2,4-D	I.	Brewing industry
B.	GA ₃	II.	Stimulation of stomatal closure
C.	Kinetin	III.	Herbicide
D.	ABA	IV.	Nutrient mobilisation

Choose the correct answer from the options given below:

Option 1:

A-I, B-II, C-IV, D-III

Option 2:

A-I, B-IV, C-III, D-II

Option 3:

A-IV, B-III, C-II, D-I

Option 4:

A-III, B-I, C-IV, D-II

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:

- 2, 4-*D* acts as a herbicide ($A \rightarrow III$)
- GA_3 is used in the brewing industry ($B \rightarrow I$)
- Kinetin promotes nutrient mobilisation ($C \rightarrow IV$)
- ABA induces stomatal closure ($D \rightarrow II$)

Hence, the correct answer is option 4) A-III, B-I, C-IV, D-II.

Q. 18 Match List I with List II :

	List I		List II
A.	Conjunctive tissue	I.	Specialised cells in the vicinity of guard cells
B.	Casparian strips	II.	Endodermal cells are rich in starch
C.	Subsidiary cells	III.	Tissue between xylem and phloem
D.	Starch sheath	IV.	Endodermal cells with suberin deposition

Choose the correct answer from the options given below :

Option 1:

A-IV, B-III, C-II, D-I

Option 2:

A-III, B-IV, C-I, D-II

Option 3:

A-IV, B-III, C-I, D-II

Option 4:

A-III, B-IV, G-II, D-I

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:

- Conjunctive tissue lies between xylem & phloem (III)
- Casparian strips have suberin (IV)
- Subsidiary cells surround guard cells (I)
- The starch sheath has starch-rich endodermis (II).

Hence, the correct answer is option 2) A-III, B-IV, C-I, D-II

Q. 19 Which one of the following types of pollination brings genetically different types of pollen grains to the stigma?

Option 1:
Geitonogamy

Option 2:
Xenogamy

Option 3:
Cleistogamy

Option 4:
Autogamy

Correct Answer:
Xenogamy

Solution:

In xenogamy, pollen is transferred between flowers of different plants of the same species. Xenogamy ensures maximum genetic variation because pollen comes from a different plant of the same species. This type of pollination enhances the adaptability and evolutionary fitness of the population.

Unlike autogamy or geitonogamy, it prevents inbreeding and maintains healthy gene pools. It is often facilitated by agents like insects, wind, or water, depending on the plant species. Thus, xenogamy is the only pollination type that guarantees genetically different pollen grains on the stigma.

Hence, the correct answer is option 2) Xenogamy.

Q. 20 Heterophyllous development in response to the environment is an example of which of the following phenomena?

Option 1:
Redifferentiation

Option 2:
Dedifferentiation

Option 3:
Elasticity

Option 4:
Plasticity

Correct Answer:
Plasticity

Solution:

Heterophylly shows that the same genotype can produce different leaf forms under different environmental conditions. This is called phenotypic plasticity. In heterophylly, plants like Cotton, Coriander, and Buttercup show varied leaf shapes in response to light, water, or temperature.

This adaptability helps plants optimise photosynthesis and survival under changing environments. It is not due to genetic change but due to flexible expression of the same genetic makeup. Thus, heterophylly is a classic example of plasticity in plants.

Hence, the correct answer is option 4) Plasticity.

Q. 21 Which of the following statements are not true regarding restriction endonucleases?

- A. They are called molecular scissors.
- B. These are the enzymes responsible for restricting the growth of bacteriophages in *E. coli*.
- C. They cut the DNA only at the centre of the palindromic sites.
- D. They remove nucleotides only from the ends of DNA fragments.
- E. They recognise specific palindromic base-pair sequences.

Choose the answer from the options given below :

Option 1:

C and D only

Option 2:

A and E only

Option 3:

D and E only

Option 4:

A and B only

Correct Answer:

C and D only

Solution:

Restriction enzymes cut at specific sites but not necessarily at the centre (*C false*), and they do not remove nucleotides from ends (*D false*).

Restriction endonucleases cut DNA at specific recognition sites, usually palindromic sequences, making them vital tools in genetic engineering. Unlike exonucleases, they do not remove nucleotides from the ends, but act within the DNA strand at defined positions.

Hence, the correct answer is option 1) C and D only.

- Q. 22** Arrange the following steps of DNA fingerprinting in a correct sequence.
- A. Isolation of DNA and its digestion by restriction endonucleases
 - B. Hybridisation using a labelled VNTR probe
 - C. Transferring of separated DNA fragments to synthetic membranes
 - D. Detection of hybridised DNA fragments by autoradiography
 - E. Separation of DNA fragments by electrophoresis.
- Choose the correct answer from the options given below :

Option 1:

A, E, B, C, D

Option 2:

A,D,B,E,C

Option 3:

A, B, D, C, E

Option 4:

A, E, C, B, D

Correct Answer:

A, E, C, B, D

Solution:

In DNA fingerprinting, DNA is isolated and cut, then fragments are separated by electrophoresis, transferred to a membrane, hybridised with a probe, and finally detected by autoradiography.

This technique relies on Variable Number Tandem Repeats (VNTRs), which are unique to individuals, making DNA fingerprinting highly specific. The autoradiography step finally reveals the distinct banding pattern, serving as a genetic identity marker.

Hence, the correct answer is option 4) A, E, C, B, D.

-
- Q. 23** Find the incorrect statement(s) about photosynthesis from the following:

- A. The water splitting complex is associated with PS I.
- B. C_4 plants use the C_3 pathway of CO_2 fixation as the main biosynthetic pathway.
- C. In C_4 plants, photorespiration does not occur.
- D. C_3 plants exhibit ' Kranz ' anatomy.
- E. ATP synthesis in chloroplasts occurs through chemiosmosis.

Choose the answer from the options given below :

Option 1:

B and C only

Option 2:

B and E only

Option 3:

B only

Option 4:

A and D only

Correct Answer:

A and D only

Solution:

Water splitting is linked to PS II (A false) and Kranz anatomy is seen in C_4 plants, not C_3 (D false).

- The water splitting complex is linked to PS II, not PS I, and it provides electrons by splitting water into oxygen and protons.
- Kranz anatomy is a special leaf structure found only in C_4 plants, where bundle sheath cells surround vascular bundles.
- C_4 plants still use the C_3 cycle as the main biosynthetic pathway, but photorespiration is minimised due to spatial separation of steps.
- ATP synthesis in chloroplasts indeed occurs via chemiosmosis, similar to mitochondria.

Hence, the correct answer is option 4) A and D only.

Q. 24 Arrange the following steps of somatic hybridisation in a correct sequence.

- A. Digestion of cell walls.
- B. Isolation of naked protoplasts.
- C. Fusion of protoplasts to get a hybrid protoplast.
- D. Isolation of single cells from two different varieties of plants.
- E. Growing of hybrid protoplast to form a new plant.

Choose the correct answer from the options given below :

Option 1:

D, B, A, E, C

Option 2:

E, A, B, C, D

Option 3:

E, B, A, D, C

Option 4:

D, A, B, C, E

Correct Answer:

D, A, B, C, E

Solution:

First cells are isolated → cell walls digested → protoplasts obtained → fusion → regeneration into a plant.

This process allows the fusion of protoplasts from different plant varieties, bypassing sexual reproduction barriers.

The regenerated hybrid protoplast develops into a new plant with combined genetic traits of both parents. Thus, somatic hybridisation is a powerful tool in plant biotechnology and crop improvement.

Hence, the correct answer is option 4) D, A, B, C, E

Q. 25 The main function of bulliform cells in grasses is:

Option 1:

to perform photosynthesis.

Option 2:

to minimise water loss during water stress.

Option 3:

to make the leaf impermeable to fungal spores.

Option 4:

to transport water.

Correct Answer:

to minimise water loss during water stress.

Solution:

Bulliform cells help in leaf rolling during water stress, reducing transpiration and conserving water. During water stress, they lose turgidity, causing the leaf to roll inward and reduce the exposed surface area. This rolling mechanism helps minimise transpiration and conserve water.

They do not perform photosynthesis or transport water, but act as a protective adaptation against drought.

Hence, the correct answer is option 2) to minimise water loss during water stress.

Q. 26 Arrange the following in the correct developmental sequence related to microsporogenesis :

- A. Microspore tetrads
- B. Sporogenous tissue
- C. Pollen grains
- D. Pollen mother cells

Choose the correct answer from the options given below:

Option 1:

A, D, C, B

Option 2:

D, A, C, B

Option 3:

B, D, C, A

Option 4:

B, D, A, C

Correct Answer:

B, D, A, C

Solution:

Sporogenous tissue pollen mother cells → microspore tetrads → pollen grains.

This sequence highlights the progression from diploid sporogenous tissue to haploid pollen grains through meiosis.

The pollen mother cells act as the key transitional stage, giving rise to microspore tetrads. Finally, each microspore matures into a functional pollen grain

Hence, the correct answer is option 4) B, D, A, C.

Q. 27 Which of the following is an in situ conservation method?

Option 1:

Seed Banks

Option 2:

Sacred Groves

Option 3:

Botanical Gardens

Option 4:

Wildlife Safari Parks

Correct Answer:

Sacred Groves

Solution:

In situ conservation means protecting organisms in their natural habitat. Sacred groves are protected forest patches where biodiversity is preserved due to cultural and religious beliefs.

Unlike seed banks or botanical gardens (ex situ), this method allows species to evolve naturally in their environment. It is considered the most effective way to conserve endangered plants and animals.

Hence, the correct answer is option 2) Sacred Groves.

Q. 28 Which one of the following is a triploid cell?

Option 1:

Synergid

Option 2:

Primary endosperm cell

Option 3:

Central cell

Option 4:

Zygote

Correct Answer:

Primary endosperm cell

Solution:

The primary endosperm cell is triploid ($3n$), formed by the fusion of one male gamete with two polar nuclei ($n + n + n$).

This triploid cell develops into the endosperm, which nourishes the developing embryo in angiosperms. Thus, it is the only triploid structure among the given options.

Hence, the correct answer is option 2) Primary endosperm cell.

Q. 29 Since the origin and diversification of life on Earth, there have been five episodes of mass extinction of species. How is the sixth extinction, which is in progress, different from the previous episodes?

Option 1:

The current species extinction rates are far lower than those in previous episodes.

Option 2:

The present species extinction rates are 100 to 1000 times faster than in pre-human times.

Option 3:

The present net species extinction rate is zero.

Option 4:

The current species extinction rate is nearly 10 times faster than that in previous episodes.

Correct Answer:

The present species extinction rates are 100 to 1000 times faster than in pre-human times.

Solution:

Earth has witnessed five major mass extinctions in the past, caused by natural events (volcanic eruptions, asteroid impacts, climate shifts).

The ongoing (sixth) extinction is largely human-driven, and the extinction rate is estimated to be $100\text{--}1000\times$ higher than the natural background rate. It is human-driven due to deforestation, habitat loss, pollution, overexploitation, and climate change.

Hence, the correct answer is option 2) The present species extinction rates are 100 to 1000 times faster than in pre-human times.

Q. 30 In the lac operon, the *z* gene codes for :

Option 1:

the repressor of lac operon

Option 2:

transacetylase

Option 3:

permease

Option 4:

beta-galactosidase

Correct Answer:

beta-galactosidase

Solution:

The lacZ gene encodes β -galactosidase, which hydrolyses lactose into glucose and galactose.

This enzymatic activity enables bacteria to utilise lactose as a carbon source when glucose is scarce. Together with lacY (permease) and lacA (transacetylase), lacZ forms the structural gene set of the lac operon.

Hence, the correct answer is option 4) beta-galactosidase.

Q. 31 In racemose inflorescence, ____ .

Option 1:

The growth is limited

Option 2:

Flowers are solitary

Option 3:

Flowers are borne in an acropetal succession

Option 4:

The main axis terminates in a flower

Correct Answer:

Flowers are borne in an acropetal succession

Solution:

In a racemose inflorescence, the main axis shows unlimited growth and flowers develop in acropetal succession (older at base, younger towards apex).

Option 1: The growth is limited - False, because in a racemose inflorescence, the main axis shows indefinite growth.

Option 2: Flowers are solitary - False, because racemose inflorescence has multiple flowers arranged along the axis, not solitary.

Option 3: Flowers are borne in an acropetal succession -True, because in a racemose inflorescence, younger flowers are at the apex, older at the base.

Option 4: The main axis terminates in a flower - False, because that is a feature of cymose inflorescence, not racemose.

Hence, the correct answer is option 3) Flowers are borne in an aerepetal succession.

- Q. 32** The main criteria used for Five Kingdom Classification proposed by R.H. Whittaker (1969) included :
- A. Cell structure
 - B. Body organisation
 - C. Presence of flagellum
 - D. Reproductions
 - E. Phylogenetic relationships

Choose the correct answer from the options given below :

Option 1:

A, B and E only

Option 2:

A, B, C, D and E

Option 3:

B, C and D only

Option 4:

A, B, D and E only

Correct Answer:

A, B, D and E only

Solution:

R. H. Whittaker used cell structure, thallus/body organisation, mode of reproduction, and phylogenetic relationships. The presence of a flagellum was not one of the main criteria. This system was revolutionary as it combined morphological, reproductive, and evolutionary criteria for classification. It resolved the limitations of earlier two-kingdom and three-kingdom systems by recognising fungi separately and including Monera and Protista.

Hence, the correct answer is option 4) A, B, D and E only.

- Q. 33** "The Evil Quartet" of biodiversity loss includes which of the following ?

Option 1:

Over-exploitation; Alien species invasions; Soil pollution; Co-extinctions

Option 2:

Habitat loss and fragmentation; Air pollution; Water pollution; Co-extinctions

Option 3:

Habitat loss and fragmentation; over-exploitation; Alien species invasions; Co-extinctions

Option 4:

Over-exploitation; Alien species invasions; Air pollution; Co-extinctions

Correct Answer:

Habitat loss and fragmentation; over-exploitation; Alien species invasions; Co-extinctions

Solution:

The “evil quartet” consists of habitat loss and fragmentation, over-exploitation, alien species invasions, and co-extinctions. These are the major causes of biodiversity decline.

- Habitat loss and fragmentation reduce living space and isolate populations.
- Over-exploitation depletes species faster than they can recover.
- Alien species invasions disrupt native ecosystems and outcompete local species.
- Co-extinctions occur when the extinction of one species leads to the loss of another dependent species.

Hence, the correct answer is option 3) Habitat loss and fragmentation; over-exploitation; Alien species invasions; Co-extinctions.

Q. 34 Identify the correct sequence of steps in each cycle of Polymerase Chain Reaction :

Option 1:

Denaturation → Extension → Annealing

Option 2:

Denaturation → Annealing → Extension

Option 3:

Annealing → Denaturation → Extension

Option 4:

Extension → Annealing → Denaturation

Correct Answer:

Denaturation → Annealing → Extension

Solution:

PCR follows: denaturation (strand separation) → annealing (primer binding) → extension (DNA synthesis).

These together amplify DNA exponentially. This cycle is repeated multiple times (typically 25–35 cycles) to generate millions of copies of the target DNA sequence.

Hence, the correct answer is option 2) Denaturation → Annealing → Extension.

Q. 35 $2(C_{51}H_{98}O_6) + 145O_2 \rightarrow 102CO_2 + 98H_2O + \text{energy}$

The Respiratory Quotient (RQ) of a biomolecule used for respiration, as per the above equation, would be :

Option 1:
Less than 0.5

Option 2:
Between 1.25 and 2

Option 3:
1 · 0

Option 4:
Between 0.5 and 0.95

Correct Answer:
Between 0.5 and 0.95

Solution:
 $RQ = \frac{CO_2 \text{ produced}}{O_2 \text{ consumed}} = \frac{102}{145} \approx 0.7$, thus lying between 0.5 and 0.95.

Since the RQ value is less than 1, it clearly indicates the substrate is a fat rather than a carbohydrate. Fats require more oxygen for complete oxidation, hence their RQ typically lies between 0.7 and 0.8, matching the calculated value here.

Hence, the correct answer is option 4) Between 0.5 and 0.95.

Q. 36 Exploring molecular, genetic and species-level diversity for products of economic importance is called :

Option 1:
Bioprospecting

Option 2:
Biofortification

Option 3:
Biomagnification

Option 4:
Bioremediation

Correct Answer:
Bioprospecting

Solution:
Bioprospecting involves searching biological diversity at molecular, genetic, and species levels to identify useful products of economic importance.

Term	Definition	Example Use
Bioprospecting	Exploring biodiversity for economic products	Drugs from medicinal plants
Biomagnification	Toxin concentration increases in the food chain	Mercury in fish
Bioremediation	Microbes clean pollutants	Oil spill cleanup

Term	Definition	Example Use
Biofortification	Nutrient-rich crop breeding	Iron-fortified rice

Hence, the correct answer is option 1) Bioprospecting.

Q. 37 Match List I with List II :

	List I		List II
A.	Decomposition	I.	Accumulation of dark Coloured amorphous colloidal substance
B.	Detritus	II.	Release of inorganic nutrients by the activity of microbes in soil
C.	Mineralisation	III.	Breaking down complex organic matter into inorganic substances
D.	Humification	IV.	Dead remains of plants and animals, including faecal matter

Choose the correct answer from the options given below:

Option 1:

A-III, B-II, C-I, D-IV

Option 2:

A-IV, B-III, C-I, D-II

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-III, B-IV, C-II, D-I

Correct Answer:

A-III, B-IV, C-II, D-I

Solution:

- Decomposition = breakdown (III)
- Detritus = dead remains (IV)
- Mineralisation = release of inorganic nutrients (II)
- Humification = formation of humus (I).

This matching shows the sequential processes of decomposition in soil ecology. Dead organic matter (detritus) is broken down, nutrients are mineralised, and humus is formed, enriching soil fertility.

Hence, the correct answer is option 4) A-III, B-IV, C-II, D-I.

- Q. 38** Identify the correct statements about biomolecules.
- A. Lipids are generally water-soluble.
 - B. Proteins are polypeptides
 - C. Polysaccharides are long chains of sugars.
 - D. Adenine and guanine are substituted pyrimidines.
 - E. Almost all enzymes are proteins.

Choose the correct answer from the options given below:

Option 1:

A, B and C only

Option 2:

B, D and E only

Option 3:

B, C and E only

Option 4:

C, D and E only

Correct Answer:

B, C and E only

Solution:

Lipids are water-insoluble (A false), and adenine and guanine are purines (D false). Statements B, C and E are correct.

- Proteins are long chains of amino acids linked by peptide bonds, hence called polypeptides.
- Polysaccharides such as starch, glycogen, and cellulose are complex carbohydrates made of repeated sugar units.
- Almost all enzymes are proteins, functioning as biological catalysts that speed up reactions.
- Lipids, however, are water-insoluble molecules, and adenine and guanine are purines, not pyrimidines.

Hence, the correct answer is option 3) B, C and E only.

-
- Q. 39** In angiosperms, root hairs arise from which one of the following regions of the root?

Option 1:

The region of elongation

Option 2:

The region of meristematic activity

Option 3:

The region of maturation

Option 4:

The root cap zone

Correct Answer:

The region of maturation

Solution:

Root hairs develop from epidermal cells in the zone of maturation (differentiation).

This greatly increases the absorptive surface area of roots, helping in the efficient uptake of water and minerals. They are absent in the meristematic and elongation zones, which are primarily for cell division and growth.

Hence, the correct answer is option 3) The region of maturation.

Q. 40 Which one of the following is not a characteristic of plant cells in the phase of elongation?

Option 1:

New cell wall deposition

Option 2:

Cell enlargement

Option 3:

Large conspicuous nuclei

Option 4:

Increased vacuolation

Correct Answer:

Large conspicuous nuclei

Solution:

During the elongation phase, cells primarily focus on increasing size through vacuolation and deposition of new cell wall material.

The nuclei become less conspicuous as the cytoplasm stretches, unlike meristematic cells, where nuclei are large and prominent.

Thus, elongation is marked by growth in volume and structure, not active nuclear division.

Hence, the correct answer is option 3) Large conspicuous nuclei.

Q. 41 Which of the following floral formula is the correct floral formula of Solanaceae family ?

Option 1:

$\oplus K_{(5)} C_{(5)} A_5 \underline{G}_{(2)}$

Option 2:

$\oplus K_5 C_5 A_5 \underline{G}_{(2)}$

Option 3:

$$\oplus \text{ } \overset{\nearrow}{\underset{\searrow}{\text{♀}}} \text{ K}_{(5)} \text{ } \overset{\frown}{\text{C}_{(5)}} \text{ A}_5 \text{ } \underline{\text{G}}_{(2)}$$

Option 4:

$$\overset{\nearrow}{\underset{\searrow}{\text{♂}}} \text{ } \underset{\searrow}{\text{♀}} \text{ K}_5 \text{ } \overset{\frown}{\text{C}_{(5)}} \text{ A}_5 \text{ } \underline{\text{G}}_{(2)}$$

Correct Answer:

$$\oplus \text{ } \overset{\nearrow}{\underset{\searrow}{\text{♀}}} \text{ K}_{(5)} \text{ } \overset{\frown}{\text{C}_{(5)}} \text{ A}_5 \text{ } \underline{\text{G}}_{(2)}$$

Solution:

Flowers of the Solanaceae family are actinomorphic and bisexual. Their calyx consists of five fused sepals, and the corolla has five fused petals. There are five stamens with a bicarpellary gynoecium and a superior ovary.

Hence, the correct answer is option 3) $\oplus \text{ } \overset{\nearrow}{\underset{\searrow}{\text{♀}}} \text{ K}_{(5)} \text{ } \overset{\frown}{\text{C}_{(5)}} \text{ A}_5 \text{ } \underline{\text{G}}_{(2)}$

- Q. 42** Which of the following statements are correct with reference to the packaging of the DNA helix?
- A. Histones are organised to form a unit of eight molecules, called a histone octamer
 - B. Histones are negatively charged basic proteins
 - C. Histones are rich in the basic amino acid residues - lysine and arginine.
 - D. The positively charged DNA is wrapped around the histone octamer to form a nucleosome.
 - E. The packaging of chromatin at higher levels requires an additional set of proteins called non-histone chromosomal proteins.
- Choose the correct answer from the options given below :

Option 1:

B, D and E only

Option 2:

A, B and D only

Option 3:

C, D and E only

Option 4:

A, C and E only

Correct Answer:

A, C and E only

Solution:

Histones are positively charged proteins due to lysine and arginine residues, which help them bind tightly to the negatively charged DNA. They form an octamer core around which DNA wraps, creating nucleosomes - the fundamental units of chromatin.

Higher-order chromatin packaging requires non-histone chromosomal proteins for further compaction. Statements B and D are incorrect because histones are not negatively charged, and DNA is negatively charged, not positive.

Hence, the correct answer is option 4) A, C and E only.

- Q. 43** Which of the following statements are correct with respect to DNA separation, isolation and visualisation?
- A. The cutting of DNA is done by molecular scissors.
 - B. The DNA fragments separate according to their size in an agarose gel upon electrophoresis
 - C. The separated DNA fragments can be seen without staining when exposed to UV light
 - D. The separated PNA fragments, when stained with ethidium bromide, can be seen in visible light.

Choose the correct answer from the options given below:

Option 1:

A and B only

Option 2:

B and D only

Option 3:

A and D only

Option 4:

B and C only

Correct Answer:

A and B only

Solution:

Restriction enzymes act as molecular scissors (A true), and electrophoresis separates DNA by size (B true). DNA needs staining and is visualised under UV, not visible light (C, D false).

- Restriction endonucleases act as molecular scissors, cutting DNA at specific recognition sites.
- During agarose gel electrophoresis, DNA fragments migrate according to their size, with smaller fragments moving faster.
- DNA cannot be seen directly; it must be stained (commonly with ethidium bromide) and visualised under UV light.

Hence, the correct answer is option 1) A and B only.

- Q. 44** The alpha-helix is found in which level of protein structure?

Option 1:

Secondary structure

Option 2:

Primary structure

Option 3:

Tertiary structure

Option 4:

Quaternary structure

Correct Answer:

Secondary structure

Solution:

The alpha helix is a regular folding pattern stabilised by hydrogen bonds between the backbone atoms of the polypeptide chain. This makes it a hallmark of the secondary structure of proteins.

- Primary - Amino acid sequence
- Secondary - alpha-helix & beta-sheet (local folding)
- Tertiary - Overall 3D folding of one polypeptide
- Quaternary - Association of multiple polypeptides

Hence, the correct answer is option 1) Secondary structure.

Q. 45 Match List I with List II :

	List I		List II
A.	Productivity	I.	Gross primary productivity minus respiration losses
B.	Net primary productivity	II.	Rate of formation of new organic matter by consumers
C.	Gross primary productivity	III.	Rate of biomass production
D.	Secondary productivity	IV.	The rate of production of organic matter during photosynthesis

Choose the correct answer from the options given below :

Option 1:

A-III, B-I, C-II, D-IV

Option 2:

A-I, B-II, C-III, D-IV

Option 3:

A-III, B-I, C-IV, D-II

Option 4:

A-I, B-III, C-IV, D-II

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:

- Productivity = biomass production rate (III)
- $NPP = GPP - R$ (I)
- GPP is the total photosynthetic production (IV)
- Secondary productivity is production by consumers (II).

Hence, the correct answer is option 3) A-III, B-I, C-IV, D-II.

Q. 46 Match List I with List II :

	List I (Placentation)		List II (Example)
A.	Marginal	I.	Mustard
B.	Axile	II.	Pea
C.	Parietal	III.	Marigold
D.	Basal	IV.	Lemon

Choose the correct answer from the options given below :

Option 1:

A-II, B-IV, C-I, D-III

Option 2:

A-IV, B-II, C-I, D-III

Option 3:

A-III, B-I, C-IV, D-II

Option 4:

A-I, B-III, C-II, D-IV

Correct Answer:
A-II, B-IV, C-I, D-III

Solution:

Placentation refers to the arrangement of ovules within the ovary, and different plants exhibit distinct types.

- In pea, ovules are borne along the margin (marginal placentation).
- In a lemon, ovules are attached to the central axis (axile placentation).
- In mustard, ovules develop on the inner wall (parietal placentation).
- In marigolds, ovules are attached at the base of the ovary (basal placentation).

Marginal → Pea (II); Axile → Lemon (IV); Parietal → Mustard (I); Basal → Marigold (III).

Hence, the correct answer is option 1) A-II, B-IV, C-I, D-III.

Q. 47 Choose the correct statements regarding frogs anatomy :

- A. Hepatic portal system is the special venous connection between liver and intestine. .
- B. There are twelve pairs of cranial nerves arising from the brain.
- C. The ureters and oviducts open separately into the cloaca in female frogs ×
- D. Hind-brain consists of cerebellum, medulla oblongata and optic lobes.
- E. Sinus venosus joins the right atrium of heart.

Choose the **correct** answer from the options given below:

Option 1:

B and D only

Option 2:

A, B and C only

Option 3:

B and C only

Option 4:

A, C and E only

Correct Answer:

A, C and E only

Solution:

Hepatic portal system connects intestine to liver (A true), ureters and oviducts open separately into cloaca (C true), sinus venosus joins right atrium (E true). Frogs have 10 pairs of cranial nerves (B false), and optic lobes belong to the midbrain (D false).

Hence, the correct answer is option 4) A, C, and E only

Q. 48 The flightless bird with forelimbs modified as paddle-like structures suited for swimming is known as :

Option 1:

Psittacula

Option 2:

Aptenodytes

Option 3:
Neophron

Option 4:
Struthio

Correct Answer:
Aptenodytes

Solution:

Aptenodytes (penguins) are flightless birds with flipper-like wings adapted for swimming.

Hence, the correct answer is option 2) Aptenodytes

Q. 49 Male frogs can be distinguished from female frogs due to the presence of :

- A. Bulging eyes
- B. Vocal sacs
- C. Webbed digits in feet
- D. Copulatory pad on first digit of fore limbs
- E. Olive green-coloured skin with dark irregular spots

Choose the **correct** answer from the options given below:

Option 1:
A and B only

Option 2:
C and E only

Option 3:
B and D only

Option 4:
B and C only

Correct Answer:
B and D only

Solution:

Male frogs possess vocal sacs for croaking and copulatory (nuptial) pads on forelimbs. Other features that are common in both male and female frogs are bulging eyes, webbed feet, and skin color. Therefore, we can not use these features to distinguish the sexes.

Hence, the correct answer is option 3) B and D only

Q. 50 A group of researchers procured some fish-like animals and upon investigation the following characters were observed :

- A. Endoskeleton was made of cartilage.
- B. Ectoparasitic; as they were found attached on fish skin with their circular sucking mouth.
- C. Paired fins and scales were absent, but 7 pairs of gill slits were present.

Which of the following species of animals did they consider to fit best with these characters?

Option 1:

Scoliodon sp.

Option 2:

Exocoetus sp

Option 3:

Petromyzon sp

Option 4:

Branchiostoma sp

Correct Answer:

Petromyzon sp

Solution:

Petromyzon (lamprey) shows cartilaginous skeleton, ectoparasitic habit, circular mouth, no paired fins, and multiple gill slits.

Hence, the correct answer is option 3) Petromyzon sp

Q. 51 In humans, respiration occurs in the following steps. Arrange these steps in the correct order.

- A. Diffusion of O_2 and CO_2 between blood and tissues
- B. Diffusion of O_2 and CO_2 across alveolar membrane
- C. Pulmonary ventilation by which atmospheric air is drawn in and CO_2 rich alveolar air is released out
- D. Cellular respiration
- E. Transport of gases by the blood

Choose the **correct** answer from the options given below:

Option 1:

A, B, C, D, E

Option 2:

C, A, B, E, D

Option 3:

C, B, E, A, D

Option 4:

E, A, C, D, B

Correct Answer:

C, B, E, A, D

Solution:

Respiration is the process of obtaining energy from the breakdown of food molecules. It usually starts with pulmonary ventilation (C), followed by diffusion across alveoli (B), transport via blood (E), diffusion into tissues (A), and finally cellular respiration (D).

Hence, the correct answer is option 3) C, B, E, A, D

Q. 52 Non-membrane bound cell organelles found in both prokaryotic and eukaryotic cells are ____ .

Option 1:

Lysosomes

Option 2:

Centrosomes

Option 3:

Mitochondria

Option 4:

Ribosomes

Correct Answer:

Ribosomes

Solution:

Ribosomes are cellular organelles that synthesize proteins. These are non-membrane-bound and present in both prokaryotic and eukaryotic cells.

Hence, the correct answer is option 4) Ribosomes

Q. 53 Choose the correct statement regarding GIFT to overcome infertility.

Option 1:

Ova collected from a female donor are transferred to the uterus of an infertile female.

Option 2:

Early embryos with up to 8 blastomeres are transferred into the fallopian tube of an infertile female

Option 3:

Early embryos with up to 8 blastomeres are transferred to the uterus of an infertile female.

Option 4:

It is the transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce ovum but can provide suitable environment for fertilization and development.

Correct Answer:

It is the transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce ovum but can provide suitable environment for fertilization and development.

Solution:

GIFT is Gamete Intra-Fallopian Transfer. GIFT involves transfer of an ovum into the fallopian tube for in vivo fertilization, not embryo transfer (which is IVF-ET). Embryo transfer is a technique in which the fertilised egg or embryo in the early stage is transferred into the female reproductive tract.

Hence, the correct answer is option 4).It is the transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce ovum but can provide suitable environment for fertilization and development.

Q. 54 Choose the correct statements regarding muscle contraction.

- A. A motor neuron carries a signal sent by the Central Nervous System (CNS) to the sarcolemma of the muscle fibre.
- B. The neural signal generates an action potential which cause the release of Ca^{++} into sarcoplasm.
- C. Increase in Ca^{++} inactivates the actin for breaking cross bridges.
- D. Actin binds to the myosin head to form a cross bridge.
- E. Shortening of sarcomere takes place, by pulling actin filaments towards the centre of 'A' band.

Choose the **correct** answer from the options given below :

Option 1:

A and B only

Option 2:
C and E only

Option 3:
C and D only

Option 4:
A, B, D and E only

Correct Answer:
A, B, D and E only

Solution:
 Ca^{2+} activates actin by removing troponin inhibition, not inactivates it. Cross-bridge formation and sliding of actin toward A-band center cause contraction.

Hence, the correct answer is option 4).A, B, D and E only

Q. 55 Insertion of a foreign DNA at BamHI site in an E. coli cloning vvector pBR322 results in the loss of antibiotic resistance towards :

Option 1:
Ampicillin and tetracycline

Option 2:
Ampicillin

Option 3:
Tetracycline

Option 4:
Gentamycin

Correct Answer:
Tetracycline

Solution:
The BamHI site lies within the tetracycline resistance gene; insertion disrupts it, causing loss of tetracycline resistance. The insertion normally disrupts it because it interrupts the normal coding sequence of the tetR gene. This turns it non-functional and causes the transformed bacteria to lose tetracycline resistance.

Hence, the correct answer is option 3) Tetracycline

Q. 56 The specific receptors for neurotransmitters synapse are present on?

Option 1:

Pre-synaptic membrane

Option 2:

Post-synaptic membrane

Option 3:

Myelin sheath

Option 4:

Schwann cell

Correct Answer:

Post-synaptic membrane

Solution:

The membranes of post-synaptic neurons contain receptor proteins which can bind neurotransmitters. Therefore, the Neurotransmitters bind to receptors located on the post-synaptic membrane to transmit nerve impulses after diffusing through the synaptic cleft. This binding is what initiates new electrical impulses in the next neuron.

Hence, the correct answer is option 2) Post-synaptic membrane

Q. 57 Which of the following statements are correct with reference to human endoskeleton?

- A. Human skull is monocondylic.
- B. The joint between any two adjoining vertebrae is a cartilaginous joint.
- C. In human beings, the number of cervical vertebrae is seven
- D. All ribs except the last 2 pairs are bicephalic.
- E. The occipital bone of skull is articulated with atlas vertebra.

Choose the **correct** answer from the options given below :

Option 1:

C, D and E only

Option 2:

B, C and E only

Option 3:

A, B and D only

Option 4:

B and E only

Correct Answer:

B, C and E only

Solution:

Humans are dicondylic (not monocondylic). Vertebrae joints are cartilaginous, cervical vertebrae = 7, and the occipital bone articulates with the atlas. The joints between the adjacent vertebrae are cartilaginous because of the presence of the intervertebral disc. Humans have seven cervical vertebrae in the neck region.

Hence, the correct answer is option 2) B, C, and E only

Q. 58 The human protein named α -1-antitrypsin, obtained from transgenic animals, is used for the treatment of ____

Option 1:

Alzheimer's disease

Option 2:

Rheumatoid arthritis

Option 3:

Emphysema

Option 4:

Cystic fibrosis

Correct Answer:

Emphysema

Solution:

α -1-antitrypsin deficiency leads to lung damage due to unchecked protease activity, causing emphysema. Transgenic animals produce this protein for therapy. Alpha antitrypsin is a human protein that is produced by using transgenic animals. It is used to treat emphysema. Emphysema is a lung disease caused by a deficiency of this protein. Alpha antitrypsin protects lung tissues from damage by certain enzymes.

Hence, the correct answer is option 3).Emphysema

Q. 59 Select the **incorrect** statements with reference to Rh grouping.

- A. Erythroblastosis foetalis is a condition observed having foetus with $\text{Rh}^{-\text{ve}}$ blood and mother with $\text{Rh}^{+\text{ve}}$ blood.
- B. Rh antigen is observed on RBCs in the majority of human beings.
- C. Before blood transfusion, Rh group should also be matched.
- D. Rh incompatibility is observed when a pregnant mother is $\text{Rh}^{-\text{ve}}$ and the foetus is $\text{Rh}^{+\text{ve}}$.
- E. Erythroblastosis foetalis can be avoided by administering anti-Rh antibodies to the mother immediately after the delivery of the second child.

Choose the answer from the options given below :

Option 1:

A and B only

Option 2:

C and D only

Option 3:

A and E only

Option 4:

B and C only

Correct Answer:

A and E only

Solution:

Erythroblastosis occurs when mother is Rh^{-} and foetus Rh^{+} , not vice versa. Anti-Rh antibodies are given after the first delivery, not the second. These Anti-RH antibodies destroy any RH-positive foetal cells in the blood of the mother before her immune system can react and form permanent antibodies.

Hence, the correct answer is option 3) A and E only

Q. 60 Match List I with List II related to muscular/skeletal system :

List I		List II	
A	Tetany	I	Inflammation of joints
B	Arthritis	II	Autoimmune disorder affecting neuromuscular Junction
C	Myasthenia gravis	III	Wild contraction in muscle due to low Ca^{++} in body fluid
D	Muscular dystrophy	IV	Progressive degeneration of skeletal muscle

Choose the **correct** answer from the options given below:

Option 1:

A-IV, B-III, C-II, D-I

Option 2:

A-III, B-I, C-II, D-IV

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-III, B-II, C-I, D-IV

Correct Answer:

A-III, B-I, C-II, D-IV

Solution:

Tetany is due to low Ca^{2+} causing spasms (III), arthritis causes inflammation of joints (I), myasthenia gravis is an autoimmune disease and affects the muscles (II), and muscular dystrophy shows degeneration (IV).

Hence, the correct answer is option 2) A-III, B-I, C-II, D-IV

Q. 61 Match List I with List II :

List I		List II	
A	Progestasert	I	Barrier made of rubber used by females
B	Multiload 375	II	Oral contraceptive
C	Diaphragm	III	Hormone releasing IUD
D	Saheli	IV	Copper releasing IUD

Choose the **correct** answer from the options given below :

Option 1:

A-IV, B-II, C-I, D-III

Option 2:

A-IV, B-III, C-I, D-II

Option 3:

A-III, B-IV, C-II, D-I

Option 4:

A-III, B-IV, C-I, D-II

Correct Answer:

A-III, B-IV, C-I, D-II

Solution:

Progestasert → hormone IUD (III), Multiload 375 → copper IUD (IV), Diaphragm → barrier (I), Saheli → oral pill (II).

Hence, the correct answer is option 4) A-III, B-IV, C-I, D-II

Q. 62 Select the correct statements regarding cell membrane in eukaryotic cell.

- A. Membrane of human , RBCs has approximately 52% protein.
- B. Major phospholipids are arranged in a bilayer.
- C. Extensions of the plasma membrane into the cell form mesosomes.
- D. Tails towards the inner part of lipids are hydrophobic and - thus protected from aqueous medium
- E. Glycocalyx is present on the outer surface of the plasma membrane.

Choose the **correct** answer from the options given below :

Option 1:

C, D and E only

Option 2:

B, C and E only

Option 3:

A, B and D only

Option 4:

A, C and E only

Correct Answer:

A, B and D only

Solution:

RBC membrane has 52% protein (A true), phospholipids form a bilayer (B true), hydrophobic(water fearing) tails face inward (D true). Mesosomes are found in prokaryotes (C false), glycocalyx(a carbohydrate-rich covering) is present, but statement combos make (3) correct.

Hence, the correct answer is option 3) A, B, and D only

Q. 63 Choose the correct statements regarding cell organelles and their inclusions.

- A. The endomembrane system includes Golgi complex, endoplasmic reticulum and mitochondria.
- B. Rough endoplasmic reticulum bears ribosomes on its surface.
- C. Both mitochondria and plastids have circular DNA.
- D. A network of microtubules, microfilaments and intermediate filaments present in the cytoplasm is called cytoskeleton.
- E. Mitochondrion is a single membrane-bound structure.

Choose the **correct** answer from the options given below :

Option 1:

A, B and C only

Option 2:

C, D and E only

Option 3:

A and B only

Option 4:

B, C and D only

Correct Answer:

B, C and D only

Solution:

Mitochondria are not part of the endomembrane system (A false). RER has ribosomes (B true), mitochondria & plastids have circular DNA (C true), cytoskeleton definition is correct (D true), mitochondria are double membrane-bound (E false).

Hence, the correct answer is option 4) B, C, and D only

Q. 64 Match List I and List II related to embryonic development at various months of pregnancy :

List I		List II	
A	The foetus movement starts and hair appears on the head	I	24 weeks of pregnancy
B	The foetus develops limbs and digits	II	20 weeks of pregnancy
C	The foetus develops external genital organs	III	8 weeks of pregnancy
D	The foetus body is covered with fine hair; eyelids N separate and eyelashes are formed	IV	12 weeks of pregnancy

Choose the **correct** answer from the options given below:

Option 1:

A-II, B-IV, C-III, D-I

Option 2:

A-II, B-III, C-IV, D-I

Option 3:

A-IV, B-II, C-III, D-I

Option 4:

A-III, B-II, C-IV, D-I

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

20 weeks → movement & hair, 8 weeks → limbs, 12 weeks → genital organs, 24 weeks → eyelashes & fine hair.

During the development of a foetus, the limbs and digits appear by 8 weeks, external genital organs appear by 12 weeks of pregnancy. In the same way, foetal movement starts around 20 weeks.

Hence, the correct answer is option 2) A-II, B-III, C-IV, D-I

Q. 65 Which one of the following is an appropriate example of 'sexual deceit'?

Option 1:

Sea anemone and clown fish

Option 2:

Ophrys and bumblebee

Option 3:

Female wasp and fig

Option 4:

Cuckoo and crow

Correct Answer:

Ophrys and bumblebee

Solution:

In sexual deceit, the orchid Ophrys mimics female insects to attract male pollinators, leading to pollination without reward. It is because the male insects try to mate with the flower, during which the pollen transfer and pollination occur without getting any reward.

Hence, the correct answer is option 2) Ophrys and bumblebee

Q. 66 Select the set of fishes which belong to class Osteichthyes?

Option 1:

Saw fish, Fighting fish and Dog fish

Option 2:

Devil fish, Cuttlefish and Hagfish

Option 3:

Starfish, Hagfish and Cuttlefish

Option 4:

Flying fish, Angel fish and Fighting fish

Correct Answer:

Flying fish, Angel fish and Fighting fish

Solution:

Osteichthyes are bony fishes; Flying fish, Angel fish, and fighting fish belong to this class, while others include cartilaginous or non-fish organisms. This is because the Flying fish, angel fish, and fighting fish are true bony fish. Osteichthyes are bony fishes with a skeleton that is made mainly of bone. Sharks and rays, on the other hand, belong to Class Chondrichthyes.

Hence, the correct answer is option 4) Flying fish, Angel fish, and Fighting fish

Q. 67 Match **List I** with **List II** with respect to chronology of evolution of life forms :

List I		List II	
A	About 65 mya	I	Jawless fish probably evolved
B	About 500 mya	II	The dinosaurs suddenly disappeared from the earth
C	About 350 mya	III	Seaweeds and few plants probably existed
D	About 320 mya	IV	Invertebrates were formed and became active

Choose the **correct** answer from the options given below :

Option 1:

A-III, B-IV, C-I, D-II

Option 2:

A-II, B-IV, C-III, D-I

Option 3:

A-II, B-IV, C-I, D-III

Option 4:

A-I, B-II, C-III, D-IV

Correct Answer:
A-II, B-IV, C-I, D-III

Solution:

About 65 million years ago, the dinosaurs disappeared from the Earth. Around 500 million years ago, the invertebrates were active in the oceans. Jawless fishes probably evolved about 350 million years ago, while seaweeds and a few plants existed somewhere around 320 million years ago.

Hence, the correct answer is option 3) A-II, B-IV, C-I, D-III

Q. 68 In which animal do haploid cells mitotically to produce gametes ?

Option 1:
Male honeybees

Option 2:
Male grasshoppers

Option 3:
Male earthworms

Option 4:
Male frogs

Correct Answer:
Male honeybees

Solution:

Male honeybees are haploid organisms formed from unfertilized eggs. Since they are already haploid, gamete formation occurs by mitosis instead of meiosis. Meiosis reduces chromosome number by half, which is why it becomes an unnecessary process in drones.

Hence, the correct answer is option 1) Male honeybees

Q. 69 The WBC count of apperson's blood sample 8000/cu. mm. How many eosinophils and lymphocytes would bein the same blood sample approximately ?

Option 1:
300 – 500/cu. mm and 500 – 700/cu. mm, respectively

Option 2:
300 – 500/cu. mm and 1200 – 1500/cu. mm, respectively

Option 3:

100 – 120/cu. mm and 160 – 200/cu. mm, respectively

Option 4:

160 – 240/cu. mm and 1600 – 2000/cu. mm, respectively

Correct Answer:

160 – 240/cu. mm and 1600 – 2000/cu. mm, respectively

Solution:

Eosinophils $\approx$ 2–3 of 8000 $\Rightarrow$ 160–240.

Lymphocytes $\approx$ 20–25 of 8000 $\Rightarrow$ 1600–2000.

Eosinophils constitute about 2-3% and lymphocytes about 20-25% of total WBCs.

Hence, the correct answer is option 4) 160 – 240/cu. mm and 1600 – 2000/cu. mm, respectively

Q. 70 What is the probability of having children with 'O' blood group, where both mother and father are heterozygous for 'A' and 'B' blood group, respectively?

Option 1:

0%

Option 2:

50%

Option 3:

25%

Option 4:

75%

Correct Answer:

25%

Solution:

Cross: $I^A i \times I^B i \Rightarrow$ offspring genotypes: $I^A I^B, I^A i, I^B i, ii$.

Only ii gives blood group O, so the probability = $\frac{1}{4} = 25$.

Hence, the correct answer is option 3) 25%

Q. 71 Arrange the following events occurring in Renin-Angiotensin mechanism in the correct order :

- A. Increase in blood pressure and Glomerular filtration rate.
- B. Reabsorption of Na^+ and water from distal parts of tubule due to Aldosterone.
- C. Fall in Glomerular filtration rate.
- D. Vasoconstriction by Angiotensin II and release of Aldosterone.
- E. Renin converts Angiotensinogen into Angiotensin I, followed by Angiotensin II.

Choose the **correct** answer from the options given below:

Option 1:

A, C, E, B, D

Option 2:

C, A, B, D, E

Option 3:

A, D, B, E, C

Option 4:

C, E, D, B, A

Correct Answer:

C, E, D, B, A

Solution:

The glomerular filtration rate usually refers to the amount of filtrate formed by both kidneys per minute.

Fall in GFR → renin release → Angiotensin II formation → vasoconstriction & aldosterone → Na⁺/water reabsorption → increased BP & GFR.

Hence, the correct answer is option 4) C, E, D, B, A

Q. 72 Choose the correct statements regarding population interactions between two species.

- A. In both parasitism and commensalism, only one species benefits and the other species is harmed.
- B. Both species benefit in mutualism
- C. Both species benefit in commensalism.
- D. In parasitism, only one species benefits and the other species is harmed.
- E. In amensalism, one species is harmed and the other is unaffected.

Choose the **correct** answer from the options given below:

Option 1:

A and B only

Option 2:

B and E only

Option 3:

B, D and E only

Option 4:

A and D only X

Correct Answer:

B, D and E only

Solution:

In mutualism, both species benefit (B true). In parasitism, one benefits and the other is harmed (D true). In amensalism, one species is harmed while the other remains unaffected (E true). Statement A is incorrect because in commensalism, the other species is unaffected, not harmed. Statement C is incorrect as only one species benefits in commensalism.

Hence, the correct answer is option 3) B, D, and E only

Q. 73 Spermatogonia undergo a series of cell divisions to produce sperms. Select the correct statements from the following:

- A. Spermatogonia always undergo meiotic cell division.
- B. Primary spermatocytes divide mitotically to produce secondary spermatocytes
- C. Secondary spermatocytes, through their second meiotic division, produce haploid spermatids.
- D. Spermatids produce spermatozoa through mitosis.
- E. Spermatids transform into spermatozoa by spermiogenesis

Choose the **correct** answer from the options given below :

Option 1:

A, C and E only

Option 2:

C and E only

Option 3:

A and E only

Option 4:

B, C and D only

Correct Answer:

C and E only

Solution:

Spermatogonia divide mitotically, not always meiotically. Meiosis II forms haploid spermatids, which transform into spermatozoa by spermiogenesis. Spermatids are immature sperm cells, while the spermatozoa are the mature and functional sperm.

Hence, the correct answer is option 2) C and E only

Q. 74 The following are the stages of life cycle of Plasmodium. Arrange the stages in the proper order.

- A. The parasites reproduce asexually in RBCs, bursting the cells.
- B. The parasites reproduce asexually in liver cells, bursting the cells and releasing into blood.
- C. Gametocytes develop in RBCs.
- D. Sporozoites reach the liver through the blood
- E. Female mosquito injects sporozoites into humans during bite (1)

Choose the **correct** answer from the options given below :

Option 1:

A, B, C, D, E

Option 2:

E, C, D, B, A

Option 3:

E, D, B, A, C

Option 4:

C, A, B, D, E

Correct Answer:

E, D, B, A, C

Solution:

Sequence: mosquito injects sporozoites → reach liver → multiply in liver → infect RBCs → gametocytes form. The Plasmodium sporozoites multiply in the liver cells and then infect RBCs. They reproduce further there and develop into gametocytes.

Hence, the correct answer is option 3) E, D, B, A, C

Q. 75 Match **List I** with **List II**:

List I Bioactive molecules)		List II (Importance)	
A	Streptokinase	I	Immunosuppressive agent
B	Statins	II	Removal of clots from the blood vessels
C	Lipases	III	Blood cholesterol-lowering agent
D	Cyclosporin A	IV	Detergent formulations

Choose the **correct** answer from the options given below:

Option 1:

A-II, B-III, C-I, D-IV

Option 2:

A-III, B-II, C-IV, D-I

Option 3:

A-II, B-III, C-IV, D-I

Option 4:

A-IV, B-III, C-II, D-I

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

Streptokinase dissolves clots and is used during heart attacks (II), statins lower cholesterol (III), lipases are used in detergents (IV), and cyclosporin A is an immunosuppressant that is used during organ transplantation (I).

Hence, the correct answer is option 3) A-II, B-III, C-IV, D-I

Q. 76 Which of the following is not an example of convergent evolution?

Option 1:

Fore limbs of whales and bats .

Option 2:

Flippers of penguins and dolphins

Option 3:

Eyes of octopuses and mammals

Option 4:

Wings of butterflies and birds

Correct Answer:

Fore limbs of whales and bats .

Solution:

Forelimbs of whales and bats are homologous (divergent evolution), not convergent.

Hence, the correct answer is option 1) Fore limbs of whales and bats.

Q. 77 What is the reason behind production of large holes in 'Swiss Cheese'?

Option 1:

The production of large amount of CO_2 and H_2 by Trichoderma polysporum

Option 2:

The production of large amount of CO_2 and H_2 by lactiges acid bacteria called Lactobacillus

Option 3:

The production' of large amount of CO_2 by Propionibacterium sharmanii

Option 4:

The production of large amount of CO_2 by Clostridium butylicum

Correct Answer:

The production' of large amount of CO_2 by Propionibacterium sharmanii

Solution:

Large holes are due to CO_2 production by Propionibacterium sharmanii during fermentation. This normally happens because the gas can not escape easily from the semi-solid chees. And hence, creates large holes in the cheese.

Hence, the correct answer is option 3) The production' of large amount of CO_2 by Propionibacterium sharmanii

Q. 78 Match **List I** with **List II** :

List I		List II	
A	Cortisol	I	Stimulates the formation of alveoli in mammary glands
B	Aldosterone	II	Produces anti-inflammatory reactions
C	Cholecystokinin	III	Stimulates reabsorption of Na^+ and water from renal tubule
D	Progesterone	IV	Stimulates secretion of pancreatic enzymes and bile juice

Choose the **correct** answer from the options given below:

Option 1:

A-II, B-III, C-IV, D-I

Option 2:

A-II, B-III, C-I, D-IV

Option 3:

A-IV, B-II, C-I, D-III

Option 4:

A-III, B-II, C-IV, D-I

Correct Answer:

A-II, B-III, C-IV, D-I

Solution:

Cortisol shows anti-inflammatory action (II), aldosterone increases Na^+ reabsorption (III), cholecystokinin stimulates pancreatic enzyme and bile secretion (IV), and progesterone helps in mammary gland development (I).

Hence, the correct answer is option 1) A-II, B-III, C-IV, D-I

Q. 79 Arrange the following cell layers/structures around the female gamete, from outer to inner side:

- A. Zona pellucida
- B. Perivitelline space
- C. Corona radiata
- D. Plasma membrane of ovum

Choose the **correct** answer from the options given below:

Option 1:

D, B, A, C

Option 2:

A, C, B, D

Option 3:

C, A, D, B

Option 4:

C, A, B, D

Correct Answer:

C, A, B, D

Solution:

From outer to inner: corona radiata → zona pellucida → perivitelline space → plasma membrane of ovum.

- Corona radiata - the outermost layer of the ovum
- Zona pellucida - protective glycoprotein layer.
- Perivitelline space - is the plasma membrane of the ovum

Hence, the correct answer is option 4) C, A, B, D

Q. 80 Which of the following equations depicts Verhulst–Pearl logistic population growth?

Option 1:

$$\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$$

Option 2:

$$\frac{dN}{dt} = rN \left(\frac{K}{K-N} \right)$$

Option 3:

$$\frac{dN}{dt} = rN \left(\frac{K-N}{N} \right)$$

Option 4:

$$\frac{dN}{dt} = rN \left(\frac{K+N}{K} \right)$$

Correct Answer:

$$\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$$

Solution:

The logistic growth equation explains that the population grows rapidly at first when resources are provided in abundance.

The logistic growth equation is $\frac{dN}{dt} = rN \left(\frac{K - N}{K} \right)$, where K is the carrying capacity and growth slows as the population approaches K .

Hence, the correct answer is option 1) $\frac{dN}{dt} = rN \left(\frac{K - N}{K} \right)$

Q. 81 The toxin proteins isolated from *Bacillus huringiensis*, coded by which of the following genes would control cotton bollworms and corn borerer, respectively ?

Option 1:
cryIAc and *cryIIb*

Option 2:
cryIAc and *cryIIIAb*

Option 3:
cryIAc and *cryIAB*

Option 4:
cryIIAb and *cryIAc*

Correct Answer:
cryIAc and *cryIAB*

Solution:

The *cryIAC* gene produces a toxin against the cotton bollworms, whereas on the other hand, the *cryIAb* gene produces a toxin that controls corn borers.

Hence, the correct answer is option 3) *cryIAc* and *cryIAB*

Q. 82 The JGA (Juxta Glomerular Apparatus) is a special sensitive region formed by cellular modifications in _____ related to the same nephron.

Option 1:
Proximal convoluted tubule and efferent renal arteriole

Option 2:
Distal convoluted tubule and efferent renal arteriole

Option 3:
Distal convoluted tubule and afferent renal arteriole

Option 4:

Proximal convoluted tubule and afferent renal arteriole

Correct Answer:

Distal convoluted tubule and afferent renal arteriole

Solution:

JGA is formed by modified cells of the distal convoluted tubule (macula densa) and the afferent arteriole, regulating blood pressure and filtration. Together, both the distal convoluted tubule and afferent arteriole regulate blood pressure and glomerular filtration rate.

Hence, the correct answer is option 3) Distal convoluted tubule and afferent renal arteriole

Q. 83 Match **List I** with **List II** :

List I		List II	
A	Molluscs	I	Pulmonary respiration only
B	Reptiles	II	Branchial respiration
C	Adult amphibians	III	Cellular respiration
D	Amoeba	IV	Pulmonary and Cutaneous respiration amphibians

Choose the **correct** answer from the options given below :

Option 1:

A-III, B-II, C-I, D-IV

Option 2:

A-II, B-I, C-III, D-IV

Option 3:

A-I, B-II, C-IV, D-III

Option 4:

A-II, B-I, C-IV, D-III

Correct Answer:

A-II, B-I, C-IV, D-III

Solution:

Molluscs → gills (II); Reptiles → lungs only (I); Adult amphibians → lungs + skin (IV); Amoeba → cellular respiration (III).

Hence, the correct answer is option (4).

Q. 84 The sixth mutant codon of beta globin gene causing polymerization of Haemoglobin and change in RBC shape is ____

Option 1:
CAG

Option 2:
AUG

Option 3:
GUG

Option 4:
GAG

Correct Answer:
GUG

Solution:

In Sickle cell anemia disease, the 6th codon of the beta globin gene mutates from GAG to GUG. This substitution results in the production of Hemoglobin S. This Hemoglobin S polymerizes at low oxygen levels and is responsible for causing sickle-shaped RBCs.

Hence, the correct answer is option 3) GUG

Q. 85 In a population of a grasshopper species, the chromosome number of some members is 23 and some other members possess 24 chromosomes. The 23 and 24 chromosome-bearing members in this species are ____.

Option 1:
females and males, respectively

Option 2:
males and females, respectively

Option 3:
all males

Option 4:
all females

Correct Answer:
males and females, respectively

Solution:

Grasshoppers show XO type sex determination: males = 23 (XO), females = 24 (XX).

The grasshopper shows XO-type sex determination. In the XO type, females have two X chromosomes, and males have only one X chromosome. Both males and females have the same number of autosomes. Therefore, grasshoppers with 24 chromosomes are females, and those with 23 chromosomes are males.

Hence, the correct answer is option 2) males and females, respectively

Q. 86 Select the **incorrect** statements from the following :

- A. Digestive system in Platyhelminthes is incomplete.
- B. Bilateral symmetry is a characteristic feature of adult Echinoderms?
- C. Pseudocoelom is possessed by Aschelminthes.
- D. Notochord is persistent throughout life in the class Chondrichthyes.
- E. Members of class Reptilia maintain a constant body temperature.

Choose the answer from the options given below :

Option 1:

A and C only

Option 2:

B and E only

Option 3:

B and D only

Option 4:

C and D only

Correct Answer:

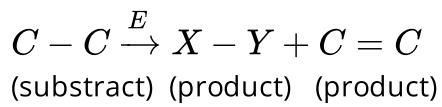
B and E only

Solution:

Echinoderms are radially symmetrical (not bilateral), and reptiles are ectothermic (not constant temperature). Others are correct. The adult echinoderms have radial symmetry; during their larval stage, they possess bilateral symmetry. Reptiles, being cold-blooded animals, can not maintain a constant body temperature.

Hence, the correct answer is option 2) B and E only

Q. 87 The following reaction depicts the activity of a particular class of enzymes:



Identify the enzyme class 'E' from the following options:

Option 1:
Isomerases

Option 2:
Ligases

Option 3:
Transferases

Option 4:
Lyases

Correct Answer:
Lyases

Solution:
Lyases catalyze the breaking of bonds without hydrolysis or ATP, often forming double bonds (like $C = C$).

Hence, the correct answer is option 4) Lyases

Q. 88 Ecological pyramids represent the relationship between the organisms at different trophic levels and they are generally inverted for :

Option 1:
Pyramid of number in gassland

Option 2:
Pyramid of energy in pond ecosystem

Option 3:
Pyramid of biomass in grassland

Option 4:

Pyramid of biomass in sea

Correct Answer:

Pyramid of biomass in sea

Solution:

In aquatic ecosystems, the biomass pyramid is inverted as phytoplankton biomass is lower than that of consumers.

Hence, the correct answer is option 4) Pyramid of biomass in sea

Q. 89 Evolution of human appears parallel to the progressive development of brain and language skills. As such, the evolution of individual species in the order of sequence of their appearance is :

Option 1:

Ramapithecus → Homo habilis → Homo erectus → Neanderthal → Homo sapiens

Option 2:

Neanderthal → Ramapithecus →
Homo habilis → Homo erectus →
Homo sapiens

Option 3:

Homo habilis → Homo erectus →
Ramapithecus → Neanderthal →
Homo sapiens

Option 4:

Homo sapiens → Ramapithecus →
Homo habilis → Neanderthal
Homo erectus

Correct Answer:

Ramapithecus → Homo habilis → Homo erectus → Neanderthal → Homo sapiens

Solution:

Correct evolutionary sequence: Ramapithecus → Homo habilis → Homo erectus → Neanderthal → Homo sapiens.

- Ramapithecus was the earliest ape-man ancestor.
- Homo habilis - first tool maker
- Homo erectus - upright man
- Neanderthal man - these were more advanced
- Homo sapiens - modern humans

Hence, the correct answer is option 1) Ramapithecus → Homo habilis → Homo erectus → Neanderthal → Homo sapiens

Q. 90 Match List I with List II :

List I (Respiratory Volume)		List II (Capacity)	
A	ERV (Expiratory Reserve Volume)	I	2500 - 3000 mL
B	RV (Residual Volume)	II	500 mL
C	IRV (Inspiratory Reserve Volume)	III	1000-1100mL
D	TV (Tidal Volume)	IV	1100-1200mL

Choose the **correct** answer from the options below:

Option 1:

A-III, B-IV, C-I, D-II

Option 2:

A-III, B-I, C-IV, D-II

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-I, B-III, C-II, D-IV

Correct Answer:

A-III, B-IV, C-I, D-II

Solution:

ERV $\approx$ 1000–1100, RV $\approx$ 1100–1200, IRV $\approx$ 2500–3000, TV $\approx$ 500 mL.

TV = Tidal Volume is the amount of air inhaled or exhaled during normal breathing.

IRV, = IRV on the other hand, is the extra amount of air a person can inhale forcibly after a normal inspiration.

ERV = It is the extra amount of air that a person can exhale forcibly after a normal expiration.

Hence, the correct answer is option 1. A-III, B-IV, C-I, D-II